Hida's Universal Ordinary Hecke Algebra at Tame Level One

Components, their intersections, no CM, and the saturation link to Loewy length

Fabian Januszewski
Universität Paderborn
fabian.januszewski@math.uni-paderborn.de

Version 0.7.3 — 2026-11-10 (revision history: Appendix A)

Abstract

Let $\ell \geq 5$. It fixes Hida's ordinary Hecke algebra $\mathbf{T}$ over $\Lambda = \mathbb{Z}_\ell[[X]]$ at tame level 1, split into the even weight classes a modulo $\ell - 1$, and reads it at finite weight: quotients of finitely generated Λ-modules modulo (P_k, ℓ^m) depend only on a weight tower, so characteristic polynomials, mod-ℓ local algebras and truncated congruence lengths are tower invariants, and one squarefree characteristic polynomial modulo ℓ proves a class étale over Λ. At tame level 1 it proves that no component is CM and that, in a class $a \not\equiv 0$, the Eisenstein component meets the cuspidal locus exactly at the irregular pairs, with (through Emerton's theorem) congruence length $L_k = v_\ell(B_k/k)$ and congruence module $\Lambda/(G_a)$; and it links saturation, Gorensteinness and Loewy length. It then determines where CM components can live, and where they can meet other components, at tame level N prime to ℓ with a tame character. Back at tame level 1, it treats truncations of the Λ-adic algebra and modular symbols, recovered from finitely many weights; the Λ-action on the ordinary cohomology of the tower $\Gamma_1(\ell^r)$, at one level and weight 2; and, for $a \not\equiv 0$, the Eisenstein component over Λ, whose ring-theoretic type one weight decides, at cuspidal rank one and at $(\ell, a) = (547, 486)$, the only pair of cuspidal rank two with $\ell < 10^5$. Every non-definitional assertion carries a status tag; the predictions are numbered (U1)–(U24).

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1 Status, conventions, scope

Status. [U] is **binding** at v0.1.0 (owner question Q5 of the slot-14 plan, owner decision 2026-09-28 "Go for publication and release"): like [C] and [E], it is owner-released, cited by label, and code is gated on it wherever a future package reads one of its statements. **Scope**: tame level 1 only; tame $N \geq 3$ (CM ∩ GL(2), CM ∩ Eisenstein) is out of scope until slot 15 (§11). It was refereed by slot-14 WP-9 (Fable, `docs/implementation/slot14/WP-9_referee.md`: accept with repairs); v0.1.0 applies every repair of pre.2 unchanged (no statement moved at the release, `docs/implementation/slot14/WP-13_release_note.md` §6; the owner re-confirmed it as binding on 2026-09-29, audit question Q-A). **Since 2026-10-01** (owner ruling O2) v0.1.1 is signed (the text of 0.1.1-pre.6) and v0.2.0, with part 2, is the owner-signed release. [C] (`gl2_contract.tex`, v1.20.0 since v0.4.0; v1.19.0 at v0.3.0; v1.17.0 at v0.2.1; v1.16.0 at v0.2.0; v1.14.1 through v0.1.1) is cited as [C, *label*], [E] (`hecke_extensions.tex`, 1.20.0 since v0.4.0; 1.19.0 at v0.3.0; 1.17.0 at v0.2.1; v1.16.2 at v0.2.0; v1.15.0 through v0.1.1) as [E, *label*] ([C, `rmk:sattor`], cited here, is a label of [C] from v1.17.0 on; the slot-17 referee F1's m-U3), the slot-14 Loewy note `docs/implementation/slot14/WP-M1_loewy_note.md` as [M1], the slot-14 binding-document proposals `docs/implementation/slot14/WP-D1_binding_doc_proposals.md` as [D1]; statements of [U] as [U, *label*]. **Part 2** (v0.2.0-pre.1, §12) extends the scope to tame level N with a tame character; it cites [C] v1.15.0's [C, `sec:nebentypus`] and [C, `NB6`] (unchanged in [C] v1.16.0), and was a pre-release until the slot-16 referee (WP-14, ACCEPT WITH REPAIRS, repaired) and the owner's signature of v0.2.0 (ruling O2, 2026-10-01); it is binding since v0.2.0. **Part 3** (v0.3.0-pre.1, §13, the truncated Λ-adic algebra) was a pre-release through v0.3.0-pre.4 (refereed by the slot-19 F1, F4 and F1-delta); **since v0.3.0** (owner ruling O3) it is binding with the rest. It cites [C] v1.21.0 and [E] 1.21.0 since v0.5.0 (v1.20.0 / 1.20.0 at v0.4.0; v1.19.0 / 1.19.0 at v0.3.0; every label it cites resolves there). **Part 4** (v0.5.0-pre.1, §14, the truncated Λ-adic modular symbols) was a pre-release, **not signed**, through v0.5.0-pre.8, written for the slot-21 Fable referee before the code of slot-21 P8 (refereed by F4, F4m and the F4 of P5, all repaired); **since v0.5.0** it is released with the slot-21 close, 2026-10-06 (owner decision D16, accepted as recommended), with the rest; it cites [C] v1.21.0 and [E] 1.21.0 (v1.20.0 / 1.20.0 through v0.5.0-pre.8). **Part 5** (v0.6.0-pre.1, §15, the tower $\Gamma_1(\ell^r)$, wild characters and the Ω-type truncations) was a pre-release through v0.6.0-pre.3, written before the code of slot-22 P10 and refereed by the slot-22 Fable referee F-U5 (ACCEPT WITH REPAIRS, all applied in the inserted text); **since v0.6.0** (signed by the owner 2026-10-16) it is released with the rest; it cites [C] v1.22.0-pre.1 ([C, `sec:nebwildsec`]). **Part 6** (v0.7.0-pre.1, §16, the Eisenstein component over Λ at irregular pairs) is a pre-release, **not signed**, written for an adversarial referee before the code of H1 (refereed by the slot-25 U6 referee, ACCEPT WITH MINORS,

repaired at v0.7.0-pre.2); parts 1–5 are as released in 0.6.0 (signed by the owner 2026-10-16). It cites [C] v1.24.0 by label, and [C] 1.25.0-pre.3's [C, `sec:heckeZ`] (a pre-release; the 0.7.0 release must cite a released [C] 1.25.0) by label.

Tags. Every non-definitional assertion carries one tag. [proved here]: proved here, from the statements it names — when those include a [literature:] or [standard:] statement the tag says so ("[proved here] from ..."), and the proof is complete relative to them. [standard: ...]: a standard fact with a citation. [literature: ...]: a theorem of the literature, with its locator; a locator not checked against the text is marked VERIFY (rule A2 of [C]; slot-14 Q6 at its default), one checked against Hida's read-only papers says so. [conjectural]: a prediction; a failure is a finding, not a defect. [verified computationally]: a computation, with its data source.

Conventions. $\ell \geq 5$ is prime, $u := 1+\ell$, $\Lambda := \mathbb{Z}_\ell[[X]]$ (identified with $\mathbb{Z}_\ell[[1+\ell\mathbb{Z}_\ell]]$ by $1+X \leftrightarrow u$), $\mathfrak{m}_\Lambda = (\ell, X)$. Weights $k = n+2$ are even (odd n contribute nothing at $\ell \geq 5$, [C, `def:hidatower`]); a *class* is $a \in 2\mathbb{Z}/(\ell-1)\mathbb{Z}$ and k lies in it when $k \equiv a$. For $k \geq 2$ put $P_k := (1+X)-u^k \in \Lambda$, the weight-$k$ prime of [C, `thm:hidacontrol`](3); $\Lambda/P_k = \mathbb{Z}_\ell$ by $X \mapsto u^k - 1$. In [C]'s notation the tower of level m through n_0 is $\mathrm{Tw}_m(\ell; n_0)$ ([C, `def:hidatower`]); for weights, k, k' lie in one tower of level m iff $k \equiv k' \pmod{(\ell-1)\ell^{m-1}}$. ω is the Teichmüller character, $\langle x \rangle = x\,\omega(x)^{-1}$. $v = v_\ell$ is the ℓ-adic valuation, extended to $\overline{\mathbb{Q}}_\ell$ with $v(\ell) = 1$. Eigenforms are normalised ($a_1 = 1$) and viewed in $\overline{\mathbb{Q}}_\ell$ through a fixed embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$. Wherever $\mathcal{E}_k$, d_k or E_k appears, $k \geq 3$ is assumed (it is, statement by statement): at $k = 2$ Hida's $\mathbf{h}_2$ also contains the ℓ-new forms of $\Gamma_0(\ell)$ ($a_\ell = \pm 1$, ordinary), which nothing here uses ([C, `def:hidatower`] has $n \geq 2$). **Warning:** d_k here (the number of ordinary level-one cuspidal eigenforms, Lemma 2.3) is *not* [C]'s $d_k = \dim M_k$ of [C, `cl:dimH1`]; [C]'s $h = 2d_k - 1$ is [U]'s $h^1(n) = 2\dim S_k + 1$.

Scope (§11). Tame level 1 only. The Λ-adic algebra is not computed; only its specialisations are (§4). **Since v0.3.0-pre.1** (part 3, §13; a pre-release) its *truncations* modulo (ℓ^K, X^m) are in scope, reconstructed from finitely many specialisations with explicit precision; the whole algebra is not. CM $\cap$ GL(2) and CM $\cap$ Eisenstein live at tame level $N \geq 3$ and wait for composite N ([C, `sec:levelN`]; slot 15).

2 The objects

Definition 2.1 (The cuspidal ordinary Hida algebra)**.** For a class a, $\mathbf{h}(a)$ is Hida's cuspidal ordinary Hecke algebra of tame level 1 in the class a — the algebra $h^0(\ell, a)$ of [C, `thm:hidacontrol`](3) (Hida's $N = \ell$, $N_0 = 1$) — a Λ-algebra generated by the images of $T(p)$, $p \neq \ell$, and $U = T(\ell)$ (their $\mathbb{Z}_\ell$-span is dense and Λ acts through the diamond operators). For $k \geq 2$ in the class, $\mathbf{h}_k := h^0_k(\Gamma_0(\ell))$ is the $\mathbb{Z}_\ell$-algebra generated by the same operators on the ordinary cusp forms of weight k on $\Gamma_0(\ell)$.

Theorem 2.2 (Hida: freeness and control)**.** *[literature: [Hid86b, Thm. 3.1 and Cor. 3.2, p. 250], transcribed with verified locators as [C, `thm:hidacontrol`](3) and [C, `thm:hidaweight`](1)]* $\mathbf{h}(a)$ *is free of finite rank* $d(a)$ *over* Λ*, and for every* $k \geq 2$ *in the class the map* $\mathbf{h}(a) \to \mathbf{h}_k$ *induces* $\mathbf{h}(a)/P_k\mathbf{h}(a) \cong \mathbf{h}_k$*, with* $T(p) \mapsto T_p$ *and* $U \mapsto U$*. For* $k \geq 3$*,* $d(a) = \mathrm{rank}_{\mathbb{Z}_\ell} \mathbf{h}_k$ *is the number of* ℓ*-ordinary normalised eigenforms of weight* k *on* $\Gamma_0(\ell)$*.*

Lemma 2.3 (The finite-weight algebra is a level-one algebra)**.** *[proved here] from [C, `prop:stabilisation`](4), [C, `thm:hidaweight`](3) and [standard: [Shi71, Ch. 3]: the* T_n *on* $S_k(\mathrm{SL}_2(\mathbb{Z}))$ *are self-adjoint for the Petersson product and commute; multiplicity one at level* 1*, an eigenform is determined by its* T_n*-eigenvalues]. Let* $k \geq 3$ *be even,* $\mathcal{E}_k$ *the set of normalised eigenforms* $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ *with* $v(a_\ell(f)) = 0$*,* $d_k := |\mathcal{E}_k|$*, and for* $f \in \mathcal{E}_k$ *let* $\alpha_f \in \overline{\mathbb{Q}}_\ell$ *be the unit root of* $Y^2 - a_\ell(f)Y + \ell^{k-1}$*. Then:*

1. $\mathbf{h}_k$ *is isomorphic to the* $\mathbb{Z}_\ell$*-subalgebra of* $\prod_{f\in\mathcal{E}_k}\overline{\mathbb{Q}}_\ell$ *generated by the tuples* $(a_p(f))_f$, $p\neq\ell$, *and* $(\alpha_f)_f$ *(images of* T_p *and* U*); it equals the* $\mathbb{Z}_\ell$*-algebra* $\mathbb{T}_k^{\mathrm{ord}}$ *generated by all* T_n *on* $e\,S_k(\mathrm{SL}_2(\mathbb{Z});\mathbb{Z}_\ell)$, $e=\lim_j T_\ell^{j!}$, *with* $U\mapsto\alpha :=$ *the unique root of* $Y^2-T_\ell Y+\ell^{k-1}$ *in* $\mathbb{T}_k^{\mathrm{ord}}$ *that is a unit.*
2. $\mathbf{h}_k$ *is free over* $\mathbb{Z}_\ell$ *of rank* d_k, $\mathbf{h}_k\otimes\overline{\mathbb{Q}}_\ell\cong\overline{\mathbb{Q}}_\ell^{\mathcal{E}_k}$, *and for every* $p\neq\ell$ *the characteristic polynomial of multiplication by* T_p *on* $\mathbf{h}_k$ *is* $c_{k,p}(X):=\prod_{f\in\mathcal{E}_k}\big(X-a_p(f)\big)\in\mathbb{Z}_\ell[X]$, *the characteristic polynomial of* T_p *on* $e\,S_k$.
3. $h^1_{\mathrm{ord}}(n)=2d_k+1$ *and* $\chi^{\mathrm{ord}}_{n,p}=\big(X-(1+p^{k-1})\big)\,c_{k,p}(X)^2$ *for* $n=k-2$ *and* $p\neq\ell$, *with* h^1_{ord}, χ^{ord} *as in [C,* `def:hidatower`*].*

Proof. (1) By [C, `thm:hidaweight`](3) / [C, `prop:stabilisation`](4) the ordinary part at level ℓ in weight $k\geq 3$ consists of the ordinary ℓ-stabilisations $f_\alpha(z)=f(z)-\ell^{k-1}\alpha_f^{-1}f(\ell z)$, $f\in\mathcal{E}_k$, one for each f, with $a_p(f_\alpha)=a_p(f)$ for $p\neq\ell$ and $Uf_\alpha=\alpha_f f_\alpha$; the ℓ-new forms of weight ≥ 3 have $a_\ell^2=\ell^{k-2}$ and are not ordinary. The f_α are a basis of eigenvectors, and their eigensystems are pairwise distinct (multiplicity one), so the algebra generated by commuting operators diagonal in that basis is its image in $\prod_f\overline{\mathbb{Q}}_\ell$ (a torsion-free algebra acting faithfully on a module on which it is diagonalisable embeds into the product of its eigencharacters). On the level-one side $\mathbb{T}_k^{\mathrm{ord}}$ is diagonal in the basis $\mathcal{E}_k$ with the same a_p; its element T_ℓ is a unit (e projects onto the $v(a_\ell)=0$ part), $Y^2-T_\ell Y+\ell^{k-1}\equiv Y(Y-T_\ell)$ modulo ℓ has the simple root $Y=T_\ell$ in the complete semilocal ring $\mathbb{T}_k^{\mathrm{ord}}$, so Hensel gives a unique unit root α, whose image in each factor is α_f; and $T_\ell=\alpha+\ell^{k-1}\alpha^{-1}$. So both algebras are the subalgebra of $\prod_f\overline{\mathbb{Q}}_\ell$ generated by the same tuples. (2) $\mathbf{h}_k$ is a finitely generated torsion-free $\mathbb{Z}_\ell$-module, of rank $\dim_{\mathbb{Q}_\ell}(\mathbf{h}_k\otimes\mathbb{Q}_\ell)$; the latter is a subalgebra of $\prod_f\overline{\mathbb{Q}}_\ell$ stable under $\mathrm{Gal}(\overline{\mathbb{Q}}_\ell/\mathbb{Q}_\ell)$, which separates the f (distinct eigensystems), so $\mathbf{h}_k\otimes\overline{\mathbb{Q}}_\ell\cong\overline{\mathbb{Q}}_\ell^{\mathcal{E}_k}$; the characteristic polynomial of multiplication by T_p is then $\prod_f(X-a_p(f))$. (3) is [C, `thm:hidaweight`]'s consequence ($h^1_{\mathrm{ord}}=2\,\mathrm{rank}\,S^{\mathrm{ord}}+1$), with Eichler–Shimura ([C, `obl:oracle`]): $e_{\mathrm{ord}}H(n)\otimes\overline{\mathbb{Q}}_\ell$ is the Eisenstein line, with T_p-eigenvalue $1+p^{k-1}$ ([C, `Hi5`]), plus two copies of $e\,S_k\otimes\overline{\mathbb{Q}}_\ell$. □

Definition 2.4 (The Λ-adic Eisenstein character)**.** For a prime $p\neq\ell$ write $\langle p\rangle=u^{s(p)}$, $s(p)\in\mathbb{Z}_\ell$, and put

$$\epsilon_p(X):=1+\omega(p)^{a-1}\langle p\rangle^{-1}(1+X)^{s(p)}\in\Lambda,\qquad \epsilon_\ell(X):=1.$$

Lemma 2.5. *[proved here] For* k *in the class* a *and* $p\neq\ell$: $\epsilon_p(u^k-1)=1+p^{k-1}$, *the* T_p*-eigenvalue of the weight-*k *Eisenstein series; and* 1 *is the* U*-eigenvalue of its ordinary* ℓ*-stabilisation.*

Proof. $(1+X)^{s(p)}$ at $X=u^k-1$ is $u^{ks(p)}=\langle p\rangle^k$, so the value is $1+\omega(p)^{a-1}\langle p\rangle^{k-1}=1+\omega(p)^{k-1}\langle p\rangle^{k-1}=1+p^{k-1}$, as $\omega(p)^{a-1}=\omega(p)^{k-1}$ (ω has order dividing $\ell-1$). The Eisenstein series E_k has $a_\ell=1+\ell^{k-1}$, roots 1 and ℓ^{k-1} of $Y^2-a_\ell Y+\ell^{k-1}$; the unit root is 1 ([C, `rmk:hidacheckable`], [C, `Hi5`]). □

Definition 2.6 (The ordinary Hecke algebra of the class, with its Eisenstein component)**.** $\mathbf{T}(a)$ is the Λ-subalgebra of $\mathbf{h}(a)\times\Lambda$ generated by $(T(p),\epsilon_p)$, $p\neq\ell$, and $(U,1)$; $\mathbf{T}:=\prod_a\mathbf{T}(a)$. For $k\geq 2$ in the class, $\mathbf{T}_k(a)$ is the $\mathbb{Z}_\ell$-subalgebra of $\mathbf{h}_k\times\mathbb{Z}_\ell$ generated by $(T_p,1+p^{k-1})$ and $(U,1)$. $P_{\mathrm{E}}:=\ker(\mathbf{T}(a)\to\Lambda)$ and $P_{\mathrm{S}}:=\ker(\mathbf{T}(a)\to\mathbf{h}(a))$ (the two projections); $I_{\mathrm{E},k}$, $I_{\mathrm{S},k}$ likewise for $\mathbf{T}_k(a)$.

Remark 2.7 (Relation to Hida's algebra of modular forms). Hida's ordinary algebra of Λ-adic *modular* forms of tame level 1 has an Eisenstein component in every class ([literature: [Hid86b, Thm. 7.2, p. 272], verified against the read-only PDF: for primitive χ,ψ of conductors prime to p there is a unique irreducible component isomorphic to the fraction field of Λ carrying the ordinary stabilisations of $E_k(\chi\omega^{a-k},\psi)$, $k\geq 3$]; here $\chi=\psi=1$). Its eigencharacter is $(\epsilon_p)_p$ by Lemma 2.5, and $\mathbf{T}(a)$ is its image in $\mathbf{h}(a)\times\Lambda$. Nothing below uses the identification; $\mathbf{T}(a)$ is defined directly, and by Lemma 2.3 $\mathbf{T}_k(a)$ is the level-one algebra generated by the T_n on $e\,M_k(\mathrm{SL}_2(\mathbb{Z});\mathbb{Z}_\ell)$ (E_k is ordinary, $a_\ell(E_k)=1+\ell^{k-1}$).

3 Components

Lemma 3.1 (The Eisenstein eigencharacter is not cuspidal)**.** *[proved here] from Lemma 2.3 and [standard: Deligne's bound (the Ramanujan–Petersson conjecture, from the Weil conjectures): for a normalised eigenform $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ and every prime p, every complex absolute value of $a_p(f)$ is $\leq 2p^{(k-1)/2}$, [Del71] and Deligne's proof of the Weil conjectures (VERIFY)]. Let $k \geq 3$ lie in the class a. For every $p \neq \ell$, $c_{k,p}(1+p^{k-1}) \neq 0$; hence $(0, c_{k,p}(1+p^{k-1})) \in \mathbf{T}_k(a)$ is non-zero, the ideal $J_k = \{x : (0,x) \in \mathbf{T}_k(a)\}$ is non-zero, $L_k < \infty$ (Proposition 3.5), the Eisenstein eigencharacter of $\mathbf{T}_k(a)$ is distinct from every cuspidal one, and $\mathbf{T}_k(a) \subseteq \mathbf{h}_k \times \mathbb{Z}_\ell$ has finite index, so $\operatorname{rank}_{\mathbb{Z}_\ell} \mathbf{T}_k(a) = d_k + 1$. Passing to Λ: $J := \{x \in \Lambda : (0,x) \in \mathbf{T}(a)\} \neq 0$.*

Proof. $2p^{(k-1)/2} < 1 + p^{k-1}$ as $(p^{(k-1)/2} - 1)^2 > 0$, so no $a_p(f)$, $f \in \mathcal{E}_k$, equals $1 + p^{k-1}$ and $c_{k,p}(1+p^{k-1}) = \prod_f (1 + p^{k-1} - a_p(f)) \neq 0$ (Lemma 2.3(2)). The element $c_{k,p}(T_p) \in \mathbf{T}_k(a)$ is $(0, c_{k,p}(1+p^{k-1}))$ by Cayley–Hamilton on $\mathbf{h}_k$ (the upper bound of Theorem 5.1). So $J_k \ni c_{k,p}(1+p^{k-1}) \neq 0$, $\ell^{L_k}\mathbb{Z}_\ell = J_k$ gives $L_k < \infty$, and $\mathbf{T}_k(a) \supseteq I_{\mathrm{S},k} = \{0\} \times J_k$ and $\mathbf{T}_k(a)/I_{\mathrm{S},k} \cong \mathbf{h}_k$, so the index of $\mathbf{T}_k(a)$ in $\mathbf{h}_k \times \mathbb{Z}_\ell$ is $|\mathbb{Z}_\ell/J_k| < \infty$. For Λ: the element $c(T(p)) \in \mathbf{T}(a)$ with c the characteristic polynomial of multiplication by $T(p)$ on the free $\mathbf{h}(a)$ (Theorem 2.2) is $(0, c(\epsilon_p))$, and $c(\epsilon_p)$ specialises at P_k to $c_{k,p}(1+p^{k-1}) \neq 0$, so $c(\epsilon_p) \neq 0$. (Without Deligne, the weaker "$a_p(f) = 1 + p^{k-1}$ for *all* p is impossible" would not give $c_{k,p}(1+p^{k-1}) \neq 0$ at a single p.) □

Proposition 3.2 (Components of $\mathbf{T}(a)$)**.** *[proved here] from Theorem 2.2, Lemma 2.3 and Lemma 3.1.*

1. *$\mathbf{h}(a)$ and $\mathbf{T}(a)$ are reduced, finite and torsion-free over Λ; $\mathbf{T}(a) \otimes_\Lambda \operatorname{Frac}(\Lambda) = \big(\mathbf{h}(a) \otimes \operatorname{Frac}(\Lambda)\big) \times \operatorname{Frac}(\Lambda)$ is a finite product of fields.*
2. *The minimal primes of $\mathbf{T}(a)$ (its* components*) are P_{E} — the **Eisenstein component**, $\mathbf{T}(a)/P_{\mathrm{E}} = \Lambda$ — and the preimages of the minimal primes of $\mathbf{h}(a)$ (the **cuspidal components**). Each component $\mathbf{T}(a)/P$ is a domain, finite over Λ, containing Λ.*
3. ***(Disjointness at arithmetic weights.)** For $k \geq 3$ in the class, the ring homomorphisms $\mathbf{T}(a) \to \overline{\mathbb{Q}}_\ell$ vanishing on P_k are exactly the $d_k + 1$ eigencharacters of $\mathbf{T}_k(a)$ (Lemma 2.3 and E_k), and each lies on exactly one component.*

Proof. (1) $\mathbf{h}(a)$ is free, so torsion-free. If $x \in \mathbf{h}(a)$ is nilpotent and non-zero, write it in a Λ-basis; a non-zero coordinate $g \in \Lambda$ lies in only finitely many $P_k\Lambda$ (a non-zero power series over $\mathbb{Z}_\ell$ has finitely many zeros in the disc $v(X) > 0$ by Weierstrass preparation, and the $u^k - 1$ are pairwise distinct), so for some $k \geq 3$ in the class the image of x in $\mathbf{h}(a)/P_k\mathbf{h}(a) = \mathbf{h}_k$ is a non-zero nilpotent, which Lemma 2.3(2) excludes. $\mathbf{T}(a)$ is a subalgebra of the reduced torsion-free $\mathbf{h}(a) \times \Lambda$. It contains $\{0\} \times J$ with $J \neq 0$ (Lemma 3.1) and maps onto $\mathbf{h}(a)$, so $\mathbf{T}(a) \otimes \operatorname{Frac}(\Lambda) \supseteq (0,1) \cdot \operatorname{Frac}(\Lambda)$ and hence equals the full product $\big(\mathbf{h}(a) \otimes \operatorname{Frac}(\Lambda)\big) \times \operatorname{Frac}(\Lambda)$; a reduced finite algebra over a field is a product of fields. (2) The minimal primes of a reduced subalgebra $\mathbf{T} \subseteq A \times B$ with both projections surjective ($A = \mathbf{h}(a)$: it is generated by the $T(p)$ and U; $B = \Lambda$: it contains 1) are the preimages of those of A and of B: they correspond to the factors of $\mathbf{T} \otimes \operatorname{Frac}(\Lambda) = (A \otimes \operatorname{Frac}) \times \operatorname{Frac}(\Lambda)$ (a torsion-free Λ-algebra's minimal primes contain no non-zero element of Λ and are the contractions of those of its generic fibre). $\mathbf{T}/P$ is torsion-free and finite over Λ and meets Λ in 0. (3) A homomorphism $\lambda\colon \mathbf{T}(a) \to \overline{\mathbb{Q}}_\ell$ killing P_k factors through $\mathbf{T}(a)/P_k\mathbf{T}(a)$, whose map onto $\mathbf{T}_k(a)$ (the image of the generators) has a torsion kernel — both are finite over $\mathbb{Z}_\ell$ of the same rank $d(a)+1 = d_k+1$ (Theorem 2.2 and (1) for the source, Lemma 3.1 for the target; the $\mathbb{Z}_\ell$-rank of M/P_kM is the generic Λ-rank of a finite torsion-free M, as $M_{(P_k)}$ is free over the discrete valuation ring $\Lambda_{(P_k)}$) — and λ kills torsion; so λ is an eigencharacter of $\mathbf{T}_k(a)$, and all d_k+1 of them occur. For uniqueness, localise at the height-one prime P_k: $\mathbf{T}(a)_{(P_k)}$ is finite and torsion-free, hence free, over the discrete valuation ring $\Lambda_{(P_k)}$, and its special fibre is $\mathbf{T}_k(a) \otimes \mathbb{Q}_\ell$ (the torsion kernel dies), a reduced finite $\mathbb{Q}_\ell$-algebra, hence étale. So $\mathbf{T}(a)_{(P_k)}$ is étale over a regular ring, hence normal, and a normal noetherian ring is a

product of domains: each maximal ideal — each Galois orbit of such λ — contains exactly one minimal prime. □

Definition 3.3 (Congruence modules)**.** For disjoint sets $\mathcal{S}, \mathcal{S}'$ of components with intersections $P_{\mathcal{S}}$, $P_{\mathcal{S}'}$ put $\mathcal{C}(\mathcal{S}, \mathcal{S}') := \mathbf{T}(a)/(P_{\mathcal{S}} + P_{\mathcal{S}'})$ — the ring of the scheme-theoretic intersection of the two unions. The **Eisenstein–cuspidal congruence module** is $\mathcal{C}_{\mathrm{E}}(a) := \mathbf{T}(a)/(P_{\mathrm{E}} + P_{\mathrm{S}})$, and at weight k, $\mathcal{C}_{\mathrm{E},k}(a) := \mathbf{T}_k(a)/(I_{\mathrm{E},k} + I_{\mathrm{S},k})$. Two components *meet* iff their congruence module is non-zero, i.e. they share a maximal ideal. For a component P and $k \geq 3$ in the class,

$$I_{P,k} := \ker\Big(\mathbf{T}_k(a) \to \prod_{\lambda \in P(k)} \overline{\mathbb{Q}}_\ell\Big),$$

$P(k)$ the eigencharacters of $\mathbf{T}_k(a)$ lying on P (Proposition 3.2(3): the weight-k eigenforms of P) — the *kernel* reading, not the image of P in $\mathbf{T}_k(a)$ (the two differ by torsion in general). $I_{\mathrm{E},k}$ and $I_{\mathrm{S},k}$ are the cases $P = P_{\mathrm{E}}$ and $P = P_{\mathrm{S}}$ (the union of the cuspidal components), since $\mathbf{T}_k(a) \subseteq \mathbf{h}_k \times \mathbb{Z}_\ell$ embeds into the product of its eigencharacters.

Lemma 3.4 (Fibre products)**.** *[proved here] Let $\mathbf{T} \subseteq A \times B$ be a subring with both projections π_A, π_B surjective, $I_A = \ker \pi_A$, $I_B = \ker \pi_B$, $J_B := \pi_B(I_A) = \{b : (0, b) \in \mathbf{T}\}$. Then $(A \times B)/\mathbf{T} \cong B/J_B \cong \mathbf{T}/(I_A + I_B)$, and $\mathbf{T} = A \times_{B/J_B} B$.*

Proof. $\beta\colon A \to B/J_B$, $\pi_A(t) \mapsto \pi_B(t) + J_B$ is well defined ($\pi_A(t) = \pi_A(t')$ gives $t - t' \in I_A$) and a ring map; $\psi(a, b) := b - \beta(a)$ is surjective onto B/J_B with kernel $\mathbf{T}$: if $b - \beta(a) \in J_B$, pick t with $\pi_A(t) = a$, then $(0, b - \pi_B(t)) \in \mathbf{T}$ and $(a, b) = t + (0, b - \pi_B(t))$. Finally π_B induces $\mathbf{T}/I_B \cong B$, carrying $(I_A + I_B)/I_B$ to J_B. □

Proposition 3.5 (The Eisenstein–cuspidal module specialises)**.** *[proved here] from Theorem 2.2 and Lemma 3.4. $\mathcal{C}_{\mathrm{E}}(a) \cong \Lambda/J$ for the ideal $J = \{x \in \Lambda : (0, x) \in \mathbf{T}(a)\}$, a cyclic Λ-module; $\mathcal{C}_{\mathrm{E},k}(a) \cong \mathbb{Z}_\ell/\ell^{L_k}$ for an $L_k = L_k(a) \in \{0, 1, \dots\}$ ($L_k < \infty$ at $k \geq 3$ by Lemma 3.1), and for every $k \geq 2$ in the class*

$$\mathcal{C}_{\mathrm{E},k}(a) \;\cong\; \mathcal{C}_{\mathrm{E}}(a)/P_k\,\mathcal{C}_{\mathrm{E}}(a).$$

In particular the Eisenstein component meets the cuspidal locus ($\mathcal{C}_{\mathrm{E}}(a) \neq 0$) iff $L_k \geq 1$ for one, equivalently every, k in the class.

Proof. Lemma 3.4 for $\mathbf{T}(a) \subseteq \mathbf{h}(a) \times \Lambda$ gives $0 \to \mathbf{T}(a) \to \mathbf{h}(a) \times \Lambda \to \mathcal{C}_{\mathrm{E}}(a) \to 0$ with $\mathcal{C}_{\mathrm{E}}(a) = \Lambda/J$. Tensoring with Λ/P_k is right exact: $\mathbf{T}(a)/P_k \to \mathbf{h}_k \times \mathbb{Z}_\ell \to \mathcal{C}_{\mathrm{E}}(a)/P_k \to 0$ (Theorem 2.2), and the image of the first map is $\mathbf{T}_k(a)$. So $\mathcal{C}_{\mathrm{E}}(a)/P_k \cong (\mathbf{h}_k \times \mathbb{Z}_\ell)/\mathbf{T}_k(a)$, which is $\mathcal{C}_{\mathrm{E},k}(a)$ by Lemma 3.4 at weight k (both projections of $\mathbf{T}_k(a)$ are onto). It is a cyclic quotient of $\Lambda/P_k = \mathbb{Z}_\ell$. Last sentence: Nakayama ($P_k \subseteq \mathfrak{m}_\Lambda$). □

4 The finite-level reading: what one cell determines, and what it only bounds

Lemma 4.1 (The tower lemma)**.** *[proved here] For $k, k' \geq 2$ in one class and $m \geq 1$: $P_k\Lambda + \ell^m\Lambda = P_{k'}\Lambda + \ell^m\Lambda$ iff $k \equiv k' \pmod{(\ell-1)\ell^{m-1}}$, i.e. iff $n = k - 2$ and $n' = k' - 2$ lie in one Tw_m ([C, `def:hidatower`]). Consequently, for every finitely generated Λ-module M (for instance $\mathbf{h}(a)$, $\mathbf{T}(a)$, $\mathcal{C}_{\mathrm{E}}(a)$, any $\mathbf{T}(a)/\mathfrak{a}$), the module $M/(P_k, \ell^m)M$ — with its algebra structure and its named operators, when M carries them — depends only on the tower of level m through k.*

Proof. $P_k - P_{k'} = u^{k'} - u^k = u^k(u^{k'-k} - 1)$ and $v(u^x - 1) = 1 + v(x)$ for $u = 1 + \ell$, ℓ odd. If $\ell^{m-1} \mid k - k'$ the two ideals contain each other. Conversely, if they are equal then $u^k - u^{k'} \in (P_k, \ell^m) \cap \mathbb{Z}_\ell = \ell^m\mathbb{Z}_\ell$ (the quotient by (P_k, ℓ^m) is $\mathbb{Z}/\ell^m$ via $X \mapsto u^k - 1$). Within one class $k \equiv k' \pmod{\ell - 1}$ already. □

Proposition 4.2 (What one cell determines)**.** *[proved here] from Theorem 2.2, Lemma 2.3, Proposition 3.5 and Lemma 4.1. Let $k, k' \geq 3$ lie in one tower of level m.*

1. $\mathbf{h}_k/\ell^m \cong \mathbf{h}_{k'}/\ell^m$ *as $\mathbb{Z}/\ell^m$-algebras, $T_p \mapsto T_p$, $U \mapsto U$. Hence $c_{k,p} \equiv c_{k',p} \pmod{\ell^m}$ for every $p \neq \ell$ (Lemma 2.3(2): the characteristic polynomial of multiplication by T_p on the free module $\mathbf{h}_k/\ell^m$), and with Lemma 2.3(3) and $1+p^{k-1} \equiv 1+p^{k'-1} \pmod{\ell^m}$: $\chi^{\mathrm{ord}}_{n,p} \equiv \chi^{\mathrm{ord}}_{n',p} \pmod{\ell^m}$ and $h^1_{\mathrm{ord}}(n) = h^1_{\mathrm{ord}}(n')$ — the statements of [C, `Hi3`], [C, `Hi4`], [C, `Hi7`], [C, `Hi9`] of [C], derived here for $n \geq 1$ along the chain Hida (forms) → stabilisation → Eichler–Shimura, which runs through characteristic polynomials only and so never meets the lattice question of [C, `rmk:stabconvention`](3). ([C, `Hi8`], the lattice datum, is* not *derived: it is lattice-dependent.)*
2. *At $m = 1$: the decomposition of $\mathbf{h}_k/\ell$ into local factors, their residue fields, Hilbert functions and Loewy lengths depend only on the class.*
3. $\min(L_k, m) = \min(L_{k'}, m)$ *(Proposition 3.5: $\mathcal{C}_{\mathrm{E},k}/\ell^m = \mathcal{C}_{\mathrm{E}}/(P_k, \ell^m)$ is $\mathbb{Z}/\ell^{\min(L_k,m)}$).*

Proposition 4.3 (What one cell only bounds)**.** *[proved here] from Proposition 3.2. Let $k \geq 3$ lie in the class a.*

1. ***Components.*** *Distinct cuspidal components carry disjoint non-empty sets of weight-k eigenforms, so $\#\{\text{cuspidal components}\} \leq \#\{\mathrm{Gal}(\overline{\mathbb{Q}}_\ell/\mathbb{Q}_\ell)\text{-orbits on } \mathcal{E}_k\}$, and a component of Λ-rank r carries r eigenforms. If $c_{k,p}$ is squarefree for some p, the orbits are the $\mathbb{Q}_\ell$-irreducible factors of $c_{k,p}$, whose number bounds the number of components from above. From below, the number of maximal ideals of $\mathbf{h}(a)$ (= of $\mathbf{h}_k/\ell$) is at least the number of distinct irreducible factors of $\bar{c}_{k,p} \in \mathbb{F}_\ell[X]$, and each maximal ideal lies on at least one component.*
2. ***Intersections.*** *For components $P \neq P'$ meeting at weight k and $t \in \mathbf{T}_k(a)$, let $\chi_P(t)$, $\chi_{P'}(t)$ be the characteristic polynomials of t on the $\mathbb{Z}_\ell$-span of the eigenforms of P, P' at weight k, and $I_{P,k}$, $I_{P',k}$ the kernels of Definition 3.3. Then $\mathrm{length}_{\mathbb{Z}_\ell} \mathbf{T}_k/(I_{P,k} + I_{P',k}) \leq v\big(\mathrm{Res}(\chi_P(t), \chi_{P'}(t))\big)$ for every t.*

Proof. (1) Proposition 3.2(3); $\overline{\mathbb{Q}}_\ell$-points of a component above P_k come in Galois orbits, and a component of rank r has r of them counted with multiplicity, all simple there by the étaleness in the proof of (3). If $c_{k,p}$ is squarefree, T_p separates $\mathcal{E}_k$ and the orbits are the root sets of the irreducible factors. Distinct roots of $\bar{c}_{k,p}$ are distinct values of T_p modulo distinct maximal ideals. (2) Put $O' := \mathbf{T}_k/I_{P',k}$, a $\mathbb{Z}_\ell$-order in the étale algebra $\prod_{f \in P'} \overline{\mathbb{Q}}_\ell$ of rank r'. The element $\chi_P(t) \in \mathbf{T}_k$ lies in $I_{P,k}$ (Cayley–Hamilton on $\mathbf{T}_k/I_{P,k}$), so its image in O' lies in the image J' of $I_{P,k}$, and $\mathbf{T}_k/(I_{P,k} + I_{P',k}) = O'/J'$ is a quotient of $O'/\chi_P(t)O'$, of length $v(\det(\chi_P(t)|O')) = v\big(\prod_{f \in P'} \chi_P(t)(f)\big) = v(\mathrm{Res})$. □

Theorem 4.4 (The étale criterion)**.** *[proved here] from Theorem 2.2, Lemma 2.3, Lemma 4.1, Proposition 3.2 and (for (3)) Theorem 5.1. Suppose that for one $k \geq 3$ in the class a and one prime $p \neq \ell$ the reduction $\bar{c}_{k,p} \in \mathbb{F}_\ell[X]$ is squarefree, with irreducible factors of degrees $\delta_1, \ldots, \delta_r$. Then:*

1. $\mathbf{h}(a) \cong \prod_{i=1}^r W(\mathbb{F}_{\ell^{\delta_i}})[[X]]$ *is finite étale over Λ; its components are these r factors, pairwise disjoint, and it is generated over Λ by $T(p)$.*
2. *For every $k' \geq 3$ in the class, $\mathbf{h}_{k'} = \mathbb{Z}_\ell[T_p] \cong \mathbb{Z}_\ell[X]/(c_{k',p})$, étale over $\mathbb{Z}_\ell$, and $\bar{c}_{k',p} = \bar{c}_{k,p}$.*
3. *If moreover $\bar{c}_{k,p}(1+p^{k-1}) \neq 0$ in $\mathbb{F}_\ell$, then $\mathcal{C}_{\mathrm{E}}(a) = 0$: $\mathbf{T}(a) = \mathbf{h}(a) \times \Lambda$, and the Eisenstein component meets no cuspidal one. If instead $1+p^{k-1}$ is a root of $\bar{c}_{k,p}$* ***and*** *$\mathcal{C}_{\mathrm{E}}(a) \neq 0$ (equivalently $L_k \geq 1$; sufficient: $v(B_k/k) \geq 1$ with $k \geq 4$, $a \not\equiv 0$, by Theorem 5.1), it is a simple root and the Eisenstein component meets exactly one cuspidal component, the one of residue degree 1 carrying that root. (A root at one p alone does not suffice: it says $a_p(f) \equiv 1 + p^{k-1}$ for one f, and nothing about the $a_q(f)$, $q \neq p$.)*

Proof. Let $d = d_k = \deg c_{k,p}$. The subalgebra $\mathbb{F}_\ell[\bar{T}_p] \subseteq \mathbf{h}_k/\ell$ has dimension d: if $\mu(\bar{T}_p) = 0$ for a proper divisor μ of $\bar{c}_{k,p}$, lift μ to $\tilde{\mu} \in \mathbb{Z}_\ell[X]$; then $\tilde{\mu}(T_p) \in \ell\mathbf{h}_k$, so $v(\tilde{\mu}(a_p(f))) \geq 1$ for

all $f \in \mathcal{E}_k$; but a root $\bar{a}_p(f)$ of the (squarefree) cofactor is not a root of μ. As $\dim_{\mathbb{F}_\ell} \mathbf{h}_k/\ell = d$ (Lemma 2.3(2)), $\mathbf{h}_k/\ell = \mathbb{F}_\ell[\bar{T}_p] \cong \mathbb{F}_\ell[X]/(\bar{c}_{k,p})$, a product of the fields $\mathbb{F}_{\ell^{\delta_i}}$. By Theorem 2.2 this is $\mathbf{h}(a)/\mathfrak{m}_\Lambda \mathbf{h}(a)$. (1) By Nakayama $\mathbf{h}(a) = \Lambda[T(p)]$; $\Omega_{\mathbf{h}(a)/\Lambda} \otimes \mathbf{h}(a)/\mathfrak{m}_\Lambda = \Omega_{(\mathbf{h}/\mathfrak{m}_\Lambda)/\mathbb{F}_\ell} = 0$, so $\Omega = 0$ by Nakayama, and $\mathbf{h}(a)$ is free and unramified, i.e. finite étale, over the complete local Λ; the idempotents of the special fibre lift, and a local finite étale Λ-algebra with residue field $\mathbb{F}_q$ is $W(\mathbb{F}_q) \otimes_{\mathbb{Z}_\ell} \Lambda = W(\mathbb{F}_q)[[X]]$, a domain. (2) $\mathbf{h}_{k'} = \mathbf{h}(a)/P_{k'}$ is then étale over $\mathbb{Z}_\ell$ with special fibre $\mathbf{h}(a)/\mathfrak{m}_\Lambda$; the same Nakayama argument gives $\mathbf{h}_{k'} = \mathbb{Z}_\ell[T_p]$, and $\bar{c}_{k',p} = \bar{c}_{k,p}$ by Proposition 4.2(1). (3) If $\bar{e} := 1 + p^{k-1} \bmod \ell$ is not a root, $\bar{c}_{k,p}(\bar{e}) \neq 0$, so $L_k \leq v(c_{k,p}(1 + p^{k-1})) = 0$ by Theorem 5.1 below (its upper bound needs only Lemma 2.3), and $\mathcal{C}_{\mathrm{E}} = 0$ by Proposition 3.5. If it is a root it is simple (squarefreeness). Assume $\mathcal{C}_{\mathrm{E}}(a) \neq 0$. Then $P_{\mathrm{E}} + P_{\mathrm{S}} \neq \mathbf{T}(a)$, and the maximal ideal of $\mathbf{T}(a)$ containing P_{E} — $\mathfrak{m}_{\mathrm{E}} = (\mathfrak{m}_\Lambda, T(q) - \epsilon_q, U - 1)$, the only one, as $\mathbf{T}(a)/P_{\mathrm{E}} = \Lambda$ is local — contains $P_{\mathrm{E}} + P_{\mathrm{S}} = P_{\mathrm{E}} + \bigcap_i P_i$ (P_i the cuspidal minimal primes, Proposition 3.2(2)); being prime, it contains some P_i. On $\mathbf{T}(a)/P_i = W(\mathbb{F}_{\ell^{\delta_i}})[[X]]$ (part (1)) the image of $\mathfrak{m}_{\mathrm{E}}$ has residue field $\mathbb{F}_\ell$ and $T(p) \equiv \bar{e}$ there, so $\delta_i = 1$ and P_i is the factor of the simple root $\bar{e}$; it is unique, as distinct factors carry distinct roots of the squarefree $\bar{c}_{k,p}$. (Without $\mathcal{C}_{\mathrm{E}}(a) \neq 0$ this fails: then $J = \Lambda$, $(y, 0) \in \mathbf{T}(a)$ for every $y \in \mathbf{h}(a)$ by Lemma 3.4, and the image of $\mathfrak{m}_{\mathrm{E}}$ in $\mathbf{h}(a)$ is the unit ideal.) □

5 The Eisenstein component and its intersection with the cuspidal locus

Theorem 5.1 (The sandwich)**.** *[proved here] from Lemma 2.3 and [standard: the q-expansion formulas for T_n on $M_k(\mathrm{SL}_2(\mathbb{Z}))$, [Shi71, Ch. 3]; $E_4, E_6 \in 1 + q\mathbb{Z}[[q]]$; von Staudt–Clausen, [Was97, Thm. 5.10] (VERIFY)]. Let $k \geq 4$ lie in a class $a \not\equiv 0 \pmod{\ell - 1}$. Then*

$$v\left(\frac{B_k}{k}\right) \;\leq\; L_k \;\leq\; \min_{p \neq \ell} \; v\big(c_{k,p}(1 + p^{k-1})\big).$$

The upper bound holds for every $k \geq 3$ in every class.

Proof. Upper bound. $t := c_{k,p}(T_p) \in \mathbf{T}_k(a)$ has cuspidal component $c_{k,p}(T_p|_{\mathbf{h}_k}) = 0$ (Cayley–Hamilton for multiplication by T_p on $\mathbf{h}_k$, Lemma 2.3(2)) and Eisenstein component $c_{k,p}(1+p^{k-1})$. So $(0, c_{k,p}(1 + p^{k-1})) \in \mathbf{T}_k(a)$, and the ideal $J_k = \ell^{L_k}\mathbb{Z}_\ell$ of Lemma 3.4 contains $c_{k,p}(1 + p^{k-1})$.

Lower bound. Put $c := -B_k/2k$, so $E_k = c + \sum_{n \geq 1} \sigma_{k-1}(n) q^n \in M_k(\mathrm{SL}_2(\mathbb{Z}); \mathbb{Z}_\ell)$ (as $(\ell - 1) \nmid k$, B_k is ℓ-integral), and $\nu := v(c) = v(B_k/k)$; assume $\nu \geq 1$. With $k = 4\alpha + 6\beta$, $g := E_k - cE_4^\alpha E_6^\beta \in S_k(\mathrm{SL}_2(\mathbb{Z}); \mathbb{Z}_\ell)$ and $g \equiv E_k$ coefficientwise modulo ℓ^ν. The T_n preserve $M_k(\mathbb{Z}_\ell)$ and $T_n E_k = \sigma_{k-1}(n) E_k$, so the image $\bar{g}$ of g in $S_k(\mathbb{Z}/\ell^\nu)$ satisfies $T_n \bar{g} = \sigma_{k-1}(n)\bar{g}$; as $a_1(\bar{g}) = 1$, $\mathbb{Z}/\ell^\nu \bar{g}$ is free of rank one and stable, and the cuspidal level-one Hecke algebra $\mathbb{T}_k \otimes \mathbb{Z}_\ell$ (generated by all T_n on $S_k(\mathbb{Z}_\ell)$) acts on it through a ring map $\varphi\colon \mathbb{T}_k \otimes \mathbb{Z}_\ell \to \mathbb{Z}/\ell^\nu$, $\varphi(T_n) = \sigma_{k-1}(n)$. Since $\varphi(T_\ell) = 1 + \ell^{k-1}$ is a unit, $\varphi(e) = \lim \varphi(T_\ell)^{j!} = 1$, so φ factors through $e(\mathbb{T}_k \otimes \mathbb{Z}_\ell) = \mathbf{h}_k$ (Lemma 2.3(1)); $\varphi(U)$ is a unit root of $(Y - 1)(Y - \ell^{k-1})$ in $\mathbb{Z}/\ell^\nu$, so $\varphi(U) = 1$. The two ring maps $\mathbf{T}_k(a) \to \mathbb{Z}/\ell^\nu$, $t \mapsto \varphi(t_{\mathrm{S}})$ and $t \mapsto t_{\mathrm{E}} \bmod \ell^\nu$, agree on the generators $(T_p, 1 + p^{k-1})$, $(U, 1)$, hence everywhere; for $(0, x) \in \mathbf{T}_k(a)$ this gives $x \equiv 0 \pmod{\ell^\nu}$, i.e. $J_k \subseteq \ell^\nu \mathbb{Z}_\ell$. □

Corollary 5.2 (The Eisenstein locus is the irregular locus)**.** *[proved here] from Theorem 5.1, Proposition 3.5 and [standard: Kummer's congruence $B_k/k \equiv B_{k'}/k' \pmod{\ell}$ for $k \equiv k' \not\equiv 0 \pmod{\ell - 1}$, [Was82, Cor. 5.14, p. 61] (verified, L16-3); the weight congruence of mod-ℓ modular forms, over $\overline{\mathbb{F}}_\ell$: for $\ell \geq 5$ the algebra of mod-ℓ modular forms of level 1 is $\overline{\mathbb{F}}_\ell[E_4, E_6]/(E_{\ell-1} - 1)$, graded modulo $\ell - 1$, so if $f \in M_k(\overline{\mathbb{F}}_\ell)$, $f' \in M_{k'}(\overline{\mathbb{F}}_\ell)$ have the same non-zero q-expansion then $k \equiv k' \pmod{\ell - 1}$, [SD73, Thm. 2(iv), p. 22], [Ser73, n° 1.2, Th. 2, p. 416-04] (verified, slot-16 L16-3 / WP-8)]. For a class $a \not\equiv 0 \pmod{\ell - 1}$ the Eisenstein component meets the cuspidal*

locus iff $v(B_k/k) \geq 1$ *for one (equivalently every)* $k \geq 4$ *in the class, i.e. iff* (ℓ, a_0) *is an irregular pair* $(\ell \mid B_{a_0})$, $a_0 \in \{2, \ldots, \ell-3\}$ *the representative of* a.

Proof. If $v(B_k/k) \geq 1$ then $L_k \geq 1$ by Theorem 5.1. Conversely, if $\mathcal{C}_{\mathrm{E}}(a) \neq 0$ then $L_k \geq 1$ (Proposition 3.5), so a maximal ideal of $\mathbf{T}_k(a)$ contains $I_{\mathrm{E},k}+I_{\mathrm{S},k}$: some $f \in \mathcal{E}_k$ and a prime λ of $\overline{\mathbb{Q}}_\ell$ above ℓ with $a_p(f) \equiv 1+p^{k-1}$ for all $p \neq \ell$ and $\alpha_f \equiv 1$, hence $a_\ell(f) = \alpha_f+\ell^{k-1}\alpha_f^{-1} \equiv 1+\ell^{k-1}$, modulo λ. Both $(a_n(f))$ and $(\sigma_{k-1}(n))$ are determined by their values at primes through the same recursions, so $a_n(f) \equiv \sigma_{k-1}(n)$ for all $n \geq 1$ and $E_k - f \equiv c \pmod{\lambda}$ as q-series. If $v(c) = 0$, then reducing modulo λ, as forms over the residue field $\mathbb{F}_\lambda \subseteq \overline{\mathbb{F}}_\ell$ (the coefficients of f lie in the ring of integers $\mathcal{O}_\lambda$ of a finite extension of $\mathbb{Q}_\ell$), the constant $\bar{c} \neq 0$ would be a mod-λ modular form of weight k with the q-expansion of the weight-0 form $\bar{c}$, forcing $k \equiv 0 \pmod{\ell-1}$. So $v(B_k/k) = v(c) \geq 1$ at this k, hence at every k of the class by Kummer. □

Lemma 5.3 (Emerton's objects in the notation of [U])**.** *[proved here] from [literature: [Eme99], library copy (an 8-page preprint), read in place: §1, pp. 2–3 (the set-up), §3, pp. 5–6 (ordinary parts, Theorem 5), p. 7 (the shift against Mazur–Wiles); the identifications with Hida's objects are Emerton's own, p. 3 and p. 5, checked against [Hid86b, (1.12), (1.14)–(1.19), §3 pp. 249–250] (read-only PDF), and closed by [Hid88, (1.2), p. 9; (1.3), p. 10; Thm. 1.3, p. 17; Thm. 2.2, p. 20] (verified, read-only PDF; item (2))] and [standard: Kubota–Leopoldt interpolation, [Was82, Thm. 5.11, p. 57] (verified on the page images, first edition, slot-16 WP-8; re-pointed from the second edition's key at v0.2.1-pre.1)]. Let* a *be an even class,* $a \not\equiv 0 \pmod{\ell-1}$, *and write Emerton's* p *as* ℓ.[1]

1. ***(Components and*** Λ***.)*** *Emerton's* $\Lambda = \mathbb{Z}_\ell[[\mathbb{Z}_\ell^\times]] = \prod_{i \bmod \ell-1} \Lambda_{(i)}$, *with* $\mathbb{F}_\ell^\times \cong \mu_{\ell-1}$ *acting on* $\Lambda_{(i)}$ *through* ω^i, *and* $\langle\gamma\rangle$ *acting on* M_k *as* γ^k *(p. 3). So* $\mu_{\ell-1}$ *acts on* M_k *through* $\zeta \mapsto \zeta^k$ *and weight* k *lies in Emerton's component* i *iff* $k \equiv i$: ***his*** i ***is our class*** a, *with no shift. The ring isomorphism* $\iota_a\colon \Lambda_{(a)} \to \Lambda$, $\langle\gamma\rangle \mapsto (1+X)^{s(\gamma)}$ *for* $\gamma = u^{s(\gamma)} \in 1+\ell\mathbb{Z}_\ell$, *sends* $\langle x\rangle$, $x \in \mathbb{Z}_\ell^\times$, *to* $\omega(x)^a(1+X)^{s(x)}$ *with* $\langle x\rangle_\Gamma := x\omega(x)^{-1} = u^{s(x)}$, *and carries Emerton's weight-*k *specialisation* $(\langle\gamma\rangle \mapsto \gamma^k)$ *to* $X \mapsto u^k - 1$, *i.e. to our* P_k. *Hida's* Λ*-action is the same* $(f|z = z_p^k f$ *on weight* k, *[Hid86b, (1.12)];* $P_k = (1+X) - u^k$, *[Hid86b, p. 250]).*
2. ***(The cuspidal algebra.)*** *Emerton's* $\mathbf{T}_{\mathrm{par}}$ *is the quotient of his* $\mathbf{T}$ *(the closed* Λ*-algebra generated by the* T_q, $q \neq \ell$, *and* U_ℓ *on Katz's generalised* ℓ*-adic modular functions of tame level* 1*) acting faithfully on the cuspidal subspace; he identifies it with Hida's* h *(p. 3), and his superscript ord with Hida's superscript* 0 *(p. 5). The identification is checkable at three locators: (i) Hida's space of* ℓ*-adic modular forms of tame level* 1 *embeds in Katz's* V, *[Hid86b, (1.11a), p. 240], and is all of it by Katz's density theorem (VERIFY); Hida's algebra does not depend on the power of* ℓ *in the level, [Hid86b, (1.15a), p. 243]; (ii) Emerton's* $V_{\mathrm{par}} = \{f : a_0(f|\langle x\rangle) = 0 \ \forall x\}$ *is Hida's* $S(1;\mathbb{Z}_\ell)$: *[Hid88, Thm. 2.2, p. 20], the exact sequence* $0 \to S(N;\mathcal{O}) \to M(N;\mathcal{O}) \xrightarrow{\delta} C(S \times \mathbb{Z}_\ell^\times;\mathcal{O}) \to 0$ *with* $\delta(f)(s,z) = a(0, f|z)$, S *a point at* $N = 1$, *and [Hid88, (1.2), p. 9; (1.3), p. 10] for the independence of the* ℓ*-power level (Katz); (iii) the parabolic duality behind Emerton's Proposition 4 (Gouvêa's Cor. III.1.3, not in the library) is [Hid88, Thm. 1.3, p. 17]:* $h(N;\mathcal{O}) \cong \mathrm{Hom}(S(N;\mathcal{O}),\mathcal{O})$ *and* $S(N;\mathcal{O}) \cong \mathrm{Hom}(h(N;\mathcal{O}),\mathcal{O})$. *Hence, through* ι_a,

$$\mathbf{T}^{\mathrm{ord}}_{\mathrm{par},(a)} \;=\; \mathbf{h}(a) \qquad (T_q \mapsto T(q),\ U_\ell \mapsto U)$$

as Λ*-algebras (Definition 2.1:* $\mathbf{h}(a) = h^0(\ell, a)$, *the component on which* $\mu_{\ell-1}$ *acts through* $\zeta \mapsto \zeta^a$, *[Hid86b, p. 249]; Hida's algebra does not depend on the power of* ℓ *in the level,*

[1] Emerton's "elementary description" of V [Eme99, p. 2] (the ℓ-adic completion of $M \cap \mathbb{Z}_\ell[[q]]$, M the graded ring of level-one forms over $\mathbb{Q}_\ell$) is *not used* in [U], and is inaccurate as literally stated (the integral q-series $\sum_{n\geq 1}\sigma_{\ell-2}(n)q^n$ of $G_{\ell-1} - a_0(G_{\ell-1}) \in M \cap \mathbb{Z}_\ell[[q]]$ does not reduce to a mod-ℓ modular form of level one; slot-14 WP-U1 audit, §3). The objects used are Katz's geometric V and V_{par}, identified with Hida's spaces in item (2); the correct elementary description is the completion of $\bigoplus_k M_k(\mathbb{Z}_\ell)$.

[Hid86b, (1.19)]). "Parabolic" is Emerton's word for cuspidal; the ordinary algebra with the Eisenstein component is his $\mathbf{T}^{\mathrm{ord}}_{(a)}$*, whose image in* $\mathbf{h}(a)\times\Lambda$ *is our* $\mathbf{T}(a)$ *(Remark 2.7), and which is not needed below.*

3. ***(The Eisenstein ideal.)*** *Emerton's generator* $T_q - 1 - \langle q\rangle q^{-1}$ *(*$q\neq\ell$*) goes under* ι_a *to* $T(q) - \big(1+\omega(q)^a(1+X)^{s(q)}\,\omega(q)^{-1}\langle q\rangle_\Gamma^{-1}\big) = T(q)-\epsilon_q$ *(Definition 2.4), and* $U_\ell - 1$ *to* $U-1$*. So his* $I^0_{\mathrm{E},(a)}$*, the image in* $\mathbf{T}^{\mathrm{ord}}_{\mathrm{par},(a)}$ *of the ideal generated by these elements (p. 6), is*
$$I^0(a) := \big(T(q)-\epsilon_q\ (q\neq\ell),\ U-1\big) \subseteq \mathbf{h}(a).$$

4. ***(The L-function.)*** *Emerton's* $L_\ell(\omega^a, 1-s)\in\Lambda_{(a)}$ *is the element whose weight-*k *specialisation is the Kubota–Leopoldt value* $L_\ell(1-k,\omega^a) = -(1-\ell^{k-1})B_k/k$ *for* $k\equiv a$*,* $k\geq 1$ *(the character is written first and* s *is the weight). Its image under* ι_a *is therefore the* G_a *of Prediction 5.4 (which lies in* Λ *for* $a\not\equiv 0$*: [literature: [Was82, Thm. 7.10, p. 123, with the power series* $f(T,\theta)\in\mathcal{O}_\theta[[T]]$ *of p. 122 for* $\theta=\omega^a\neq 1$*], verified, first edition]): two elements of* Λ *with the same values at the infinitely many points* u^k-1 *are equal (Weierstrass preparation). Emerton's "the image of* $A_{0,(a)}$ *... is equal to* $L_\ell(\omega^a,1-s)$*" (p. 6) holds up to the unit* $\frac{1}{2}$*: the ordinary Eisenstein family normalised by* $a_1=1$ *has constant term* $\frac{1}{2}L_\ell(1-k,\omega^a)$ *at weight* k*; only the ideal is used.*

Proof. (1) and (3) are the computations shown: $\langle q\rangle = \omega(q)\langle q\rangle_\Gamma$ in $\mathbb{Z}_\ell^\times = \mu_{\ell-1}\times(1+\ell\mathbb{Z}_\ell)$, $\mu_{\ell-1}$ acts on $\Lambda_{(a)}$ through ω^a, so $\iota_a(\langle q\rangle q^{-1}) = \omega(q)^a(1+X)^{s(q)}\omega(q)^{-1}\langle q\rangle_\Gamma^{-1}$. (2) is the cited identification. (4): both sides specialise at $X=u^k-1$, $k\equiv a$, to $L_\ell(1-k,\omega^a)$, which for $k\equiv a$ is $-(1-\ell^{k-1})B_{k,\omega^{a-k}}/k = -(1-\ell^{k-1})B_k/k$; a non-zero element of Λ has finitely many zeros in $v(X)>0$. The constant term of $E_k - \ell^{k-1}E_k(\ell z)$ is $(1-\ell^{k-1})(-B_k/2k)$. □

What the tag of Prediction (U1) rests on. Two steps of [Eme99] are printed as sketches and are load-bearing; both are closed by standard published results (slot-14 WP-U1 audit, §3). *Corollary 2* (p. 4, $V_{(a)}\hat{\otimes}R\cong\mathrm{Hom}_{\mathrm{cont}}(\mathbf{T}_{(a)},R)$, "by the methods of" Gouvêa §III.1.2): its content is the finite-level duality $M_k(\mathbb{Z}_\ell)\cong\mathrm{Hom}(\mathbf{T}_k,\mathbb{Z}_\ell)$, $f\mapsto(T\mapsto a_1(f|T))$, in weights $k\not\equiv 0 \pmod{\ell-1}$ — injective as there are no constants in weight $k>0$, onto by the rank count $\operatorname{rank}\mathbf{T}_k = \dim M_k$ and because $a_0(f)\in\mathbb{Z}_\ell$ once $a_n(f)\in\mathbb{Z}_\ell$ for $n\geq 1$, by the weight congruence ([SD73, Thm. 2(iv)], [Ser73, Th. 2], verified; the argument of Corollary 5.2) — together with Katz's density theorem and q-expansion principle (VERIFY), which give $\mathbf{T}_{(a)}/\ell^m\cong\varprojlim_k\mathbf{T}_k/\ell^m$, Pontryagin dual to $V_{(a)}/\ell^m$, hence the statement for every ℓ-adically complete R, $R=\Lambda_{(a)}$ included. *Lemma 3*'s "easily seen" description of V_{par} is [Hid88, Thm. 2.2] (item (2)(ii)). So the tag rests on Emerton's Theorem 5 and these standard results; none of them is a Gorenstein property, a freeness hypothesis on a module over the Hecke algebra, or a main conjecture.

Prediction 5.4 ((U1) The congruence length is the Bernoulli valuation). [proved here] from [literature: Emerton [Eme99, Thm. 5, p. 6] (verified)] through Lemma 5.3, Lemma 3.4 and Proposition 3.5; **no Gorenstein, freeness, μ- or λ-invariant hypothesis**, and no main conjecture ([conjectural] through v0.1.1-pre.1; corroborated by [literature: Ohta [Oht03, Thm. (1.5.5)(II), p. 238] (verified)], whose proof in this case uses the Mazur–Wiles main conjecture, Remark 5.5). Let $\ell\geq 5$, let a be an even class with $a\not\equiv 0 \pmod{\ell-1}$, and let $G_a\in\Lambda$ be the Iwasawa power series of the Kubota–Leopoldt function, $G_a(u^k-1) = L_\ell(1-k,\omega^a) = -(1-\ell^{k-1})B_k/k$ for $k\equiv a$. Then:

(U1-Λ) $J = G_a\Lambda$ for the ideal J of Proposition 3.5; equivalently $\mathcal{C}_{\mathrm{E}}(a)\cong\Lambda/(G_a)$, as rings and as Λ-modules, and $\mathbf{h}(a)/I^0(a)\cong\Lambda/(G_a)$ with $I^0(a)$ of Lemma 5.3(3). In particular J is principal and $\mathbf{T}(a)$ is Λ-free of rank $d(a)+1$ (Remark 7.3).

(U1-k) For **every** $k\geq 2$ in the class, $\mathcal{C}_{\mathrm{E},k}(a)\cong\mathbb{Z}_\ell/\ell^{v(B_k/k)}$, i.e. $L_k = v(B_k/k)$; the lower bound of Theorem 5.1 is an equality. No condition on k beyond $k\equiv a \pmod{\ell-1}$ (at $k=2$, i.e. $a\equiv 2$: $L_2 = v(B_2/2) = 0$).

Checkable form (now a check of the implementation): at a computed cell, $\min_{p\in\mathcal{P}} v(c_{k,p}(1+p^{k-1})) \geq v(B_k/k) = L_k$ for every computed set $\mathcal{P}$ (Theorem 5.1), with equality once $\mathcal{P}$ is large enough; a value *below* $v(B_k/k)$ is a defect, a value above it at every computed p only says that $\mathcal{P}$ is too small. *Not covered*: the class $a \equiv 0$ (Emerton's Proposition 1 needs $i \not\equiv 0$, and G_0 has a pole; Remark 5.10), $\ell \in \{2,3\}$, tame level $N > 1$.

Proof. Step 1: P_{E} is generated by the Eisenstein differences, and $\pi_{\mathrm{S}}(P_{\mathrm{E}}) = I^0(a)$. Let $I \subseteq \mathbf{T}(a)$ be the ideal generated by $g_q := (T(q), \epsilon_q) - \epsilon_q \cdot 1$, $q \neq \ell$, and $g_\ell := (U,1) - 1$. Each generator of the Λ-algebra $\mathbf{T}(a)$ (Definition 2.6) is congruent modulo I to an element of $\Lambda \cdot 1$, so every monomial is, and $\mathbf{T}(a) = \Lambda \cdot 1 + I$. The composite $\Lambda \to \mathbf{T}(a)/I \to \mathbf{T}(a)/P_{\mathrm{E}} \cong \Lambda$ (the second map exists as $I \subseteq P_{\mathrm{E}}$, the isomorphism is π_{E}) is the identity, and the first map is onto; so $\mathbf{T}(a)/I \to \mathbf{T}(a)/P_{\mathrm{E}}$ is an isomorphism and $I = P_{\mathrm{E}}$. As $\pi_{\mathrm{S}}\colon \mathbf{T}(a) \to \mathbf{h}(a)$ is onto, $\pi_{\mathrm{S}}(P_{\mathrm{E}})$ is the ideal generated by the $\pi_{\mathrm{S}}(g_q) = T(q) - \epsilon_q$ and $\pi_{\mathrm{S}}(g_\ell) = U - 1$, i.e. $I^0(a)$.

Step 2: two readings of one kernel. Let $\phi\colon \Lambda \to \mathcal{C}_{\mathrm{E}}(a) = \mathbf{T}(a)/(P_{\mathrm{E}}+P_{\mathrm{S}})$ be the structure map (Λ embeds diagonally in $\mathbf{T}(a)$). Through $\mathbf{T}(a)/P_{\mathrm{E}} \cong \Lambda$ (π_{E}, the identity on Λ), $(P_{\mathrm{E}}+P_{\mathrm{S}})/P_{\mathrm{E}}$ is $\pi_{\mathrm{E}}(P_{\mathrm{S}}) = J$, so $\ker\phi = J$. Through $\mathbf{T}(a)/P_{\mathrm{S}} \cong \mathbf{h}(a)$ (π_{S}, which is Λ-linear), $(P_{\mathrm{E}}+P_{\mathrm{S}})/P_{\mathrm{S}}$ is $\pi_{\mathrm{S}}(P_{\mathrm{E}}) = I^0(a)$ (Step 1), so $\ker\phi = \{x \in \Lambda : x \cdot 1_{\mathbf{h}(a)} \in I^0(a)\}$, the pull-back of $I^0(a)$ along the structure map $\Lambda \to \mathbf{h}(a)$. (This is Lemma 3.4 read from both sides; $\mathbf{h}(a) = 0$ is allowed, and then $J = \Lambda$.)

Step 3: Emerton. By Lemma 5.3(1)–(3) the pull-back in Step 2 is, through ι_a, the pull-back of $I^0_{\mathrm{E},(a)}$ along $\Lambda_{(a)} \to \mathbf{T}^{\mathrm{ord}}_{\mathrm{par},(a)}$, which [Eme99, Thm. 5] computes as the principal ideal generated by $L_\ell(\omega^a, 1-s)$ (hypotheses: $\ell \geq 5$, $a \not\equiv 0 \pmod{\ell-1}$). By Lemma 5.3(4) that ideal is $G_a\Lambda$. So $J = G_a\Lambda$, $\mathcal{C}_{\mathrm{E}}(a) \cong \Lambda/J = \Lambda/(G_a)$ as rings (by ϕ, which is onto), and $\mathbf{h}(a)/I^0(a) \cong \mathcal{C}_{\mathrm{E}}(a)$ by Step 2. Freeness: $0 \to \{0\} \times J \to \mathbf{T}(a) \to \mathbf{h}(a) \to 0$ is exact ($P_{\mathrm{S}} = \{0\} \times J$) and splits, $\mathbf{h}(a)$ being Λ-free (Theorem 2.2); $J = G_a\Lambda \cong \Lambda$ as $G_a \neq 0$ ($J \neq 0$, Lemma 3.1).

Step 4: weight k. Proposition 3.5 (valid for every $k \geq 2$ in the class) gives $\mathcal{C}_{\mathrm{E},k}(a) \cong \mathcal{C}_{\mathrm{E}}(a)/P_k \cong \Lambda/(G_a, P_k) \cong \mathbb{Z}_\ell/G_a(u^k-1)\mathbb{Z}_\ell$, and $v(G_a(u^k-1)) = v\big((1-\ell^{k-1})B_k/k\big) = v(B_k/k)$, as $1-\ell^{k-1}$ is a unit for $k \geq 2$ (B_k/k is ℓ-integral, $(\ell-1) \nmid k$). □

Remark 5.5 (Ohta's theorem, and why no Gorenstein hypothesis enters). [literature: [Oht03, Thm. (1.5.5), p. 238; (1.1.4), (1.1.6), p. 229–230; (3.2.2)–(3.2.3), p. 253–254; (1.4.5), p. 235; Appendix A.1, p. 265], verified against the library copy.] At tame level $N = 1$ Ohta's pair is $(\theta, \psi) = (\omega^{a-2}, \mathbf{1})$: its weight-$k$ specialisations have nebentypus $\theta\omega^{2-k} = \omega^{a-k}$, trivial for $k \equiv a$, and $T(q)$-eigenvalue $\theta(q)q\langle q\rangle_\Gamma^{k-2} + 1 = 1 + q^{k-1}$. Hypothesis (1.5.2) holds for every even a ($\ell \nmid \varphi(1)$, $uv \in \{1, \ell\}$ — $\psi = \mathbf{1}$ has conductor $v = 1$, and $\theta = \omega^{a-2}$ has conductor $u = 1$ for $a \equiv 2$, so $uv = 1 = N$, and $u = \ell$ otherwise, so $uv = \ell = N\ell$ — and (θ, ψ) is not exceptional, $\theta|_{(\mathbb{Z}/\ell)^\times} = \omega^{a-2} \neq \omega^{-1}$). Case (II) of (1.5.5) is the sequence $0 \to I(\theta,\psi)_{\mathfrak{M}} \to e\mathbf{H}_{\mathfrak{M}} \to \Lambda \to 0$ of the full ordinary algebra, and its congruence module is, by (3.2.2)–(3.2.3), $(\mathbf{H}_1 \oplus \mathbf{H}_2)/\mathbf{H} = eh_{\mathfrak{M}}/I(\theta,\psi)_{\mathfrak{M}}$ — the ring $\mathbf{T}/(P_{\mathrm{E}} + P_{\mathrm{S}})$ of Definition 3.3, i.e. Lemma 3.4 in Ohta's language. The theorem gives $\Lambda/(A)$ with $A = G(T, \xi\omega^2) = G(T, \omega^a)$ ($N = 1$: no Euler factor in (1.5.4)); Ohta's variable is weight-2 normalised, $1 + T$ acting as u^{k-2} at weight k ((1.2.5), (1.4.5)), so $G(T, \omega^a) = G_a(u^2(1+T) - 1)$ — the shift $\langle\gamma\rangle \mapsto \gamma^{-2}\langle\gamma\rangle$ of [Eme99, p. 7]. This is (U1-Λ) again, for $a \not\equiv 0$, *provided* Ohta's $e\mathbf{H}(1;\mathbb{Z}_\ell)_{\mathfrak{M}}$ is identified with $\mathbf{T}(a)_{\mathfrak{m}_{\mathrm{E}}}$ (Remark 2.7: it acts faithfully on the ordinary Λ-adic forms, which span $eS \otimes \mathrm{Frac}(\Lambda) \oplus \mathrm{Frac}(\Lambda)E$ by (I), so it embeds in $\mathbf{h}(a) \times \Lambda$ with image generated by the Hecke pairs; the step is recorded, not audited). **Two routes, one statement**: in this case ($\psi = \mathbf{1}$) Ohta's proof of (II) runs through the appendix, i.e. Harder–Pink / Kurihara plus the Mazur–Wiles main conjecture (p. 254, p. 263); his elementary route (the duality (3.2.7)) needs $\psi \neq \mathbf{1}$, and he notes (p. 256) that a direct proof of the duality (3.3.1) ($em(N;\Lambda)_{\mathfrak{M}} = eM(N;\Lambda)_{\mathfrak{M}}$, p. 256) would give (II) directly — at $N = 1$ that is Emerton's Corollary 2 *at* $R = \Lambda_{(a)}$: a Λ-linear continuous functional on $\mathbf{T}_{(a)}$ is an element of $V_{(a)}\hat{\otimes}\Lambda_{(a)}$, i.e. a Λ-adic form with integral constant term. It is *not* Proposition 1 applied weight by weight: integrality of a Λ-adic constant term at every weight does not give

integrality in Λ ($(X^\ell - X)/\ell$ is $\mathbb{Z}_\ell$-valued on $\ell\mathbb{Z}_\ell$). *Why Gorenstein never enters*: [U]'s $\mathcal{C}_{\mathrm{E}}$ is the congruence module of the Hecke *algebra*, whose Eisenstein factor is Λ itself; Gorensteinness (or freeness of a module over the Hecke algebra) is what relates it to congruence modules of *modules* — Ohta's (III), (IV), Harder's denominator on $H^1_!$ (compare Harder–Pink's Satz 6–7 and §5, [EX] `s14:hp`) — and none of those is used here.

Remark 5.6 (The Iwasawa zero and the ladders). If $G_a = (\text{unit}) \cdot (X - \xi)$ (λ-invariant 1, $\mu = 0$), (U1) — a theorem since v0.1.1-pre.2 — reads $L_k = v(u^k - 1 - \xi) = 1 + v(k - s_a)$ with $u^{s_a} = 1 + \xi$, $s_a \in \mathbb{Z}_\ell$: $L_k = 1$ on the class except on the sub-tower $k \equiv s_a \pmod{\ell}$ (where $L_k \geq 2$), ≥ 3 on $k \equiv s_a \pmod{\ell^2}$, and so on — the congruence deepens exactly along the ℓ-adic approach of k to the zero s_a. Independently of (U1), Proposition 4.2(3) makes $\min(L_k, m)$ constant on every Tw_m. [verified computationally] Computed on cohomology at $k = 284$ ($\ell = 37$), $k = 228$ ($\ell = 103$) and, since v0.1.1-pre.6, $k = 914$ ($\ell = 59$, the cell $(912, 59)$), where $L_k = 2$: [EX] `s14:depthlaw` (slot-14 WP-C14 I-3b).

Proposition 5.7 (The two irregular classes at $\ell = 691$)**.** *[verified computationally] (Bernoulli numbers: `docs/implementation/slot14/wp8/bern.py`, exact B_k by `sympy` (slot 14, before the owner's no-CAS rule of 2026-10-03; every value used here since reproduced in-house, v0.5.0-pre.3: $B_k/k \bmod 691^8$ at $k = 12 + 690t$ and $200 + 690t$, $t = 0, \ldots, 5$, by tangent numbers in `HeckeLambdaValidation`'s read-only arm `HECKE_LAMBDA_ALL=1`, reproducing slot-19 R19-2's 691 data field by field ($\tilde{G}$, α, the rungs $342252 = 12 + 690 \cdot 496$, $342440 = 200 + 690 \cdot 496$), the tangent route itself gated against WP-NF's exact `BigInt` Bernoulli numbers for $k \leq 200$ in the default arm; and $v_{691}(B_k/k) = 2$ exactly at both rungs and $= 1$ at rung ± 690 in `HeckeLValueValidation`'s arm (B), by Voronoi and the Kummer–Mahler route; no value differs).* Primary route *for the class residues (slot-14 WP-9, `bern_check.py`, outside the tree): the three exact values B_{k_0+690t}, $t = 0, 1, 2$, fix the linear model of $(1 - \ell^{k-1})B_k/k \bmod \ell^2$ on each class, the third point checking linearity (slopes $80, 80$ on the class of 12, $99, 99$ on the class of 200), and its zero is at $t = 496$ on both classes; Kummer modulo ℓ^2, the one VERIFY this chain had through v0.2.0, is re-pointed at v0.2.1-pre.1 to verified first-edition locators (below).* Cross-check*: the scans by Voronoi's congruence [Was97, Prop. 9.5] (VERIFY), `voronoi_scan.py` with $a = 2$ (WP-8) and `voronoi_a3.py` with $a = 3$ (WP-9), agreeing at every test point, and against the exact $B_{12}/12$, $B_{702}/702$ modulo 691^2). 691 is irregular exactly for the classes of $k \equiv 12$ and $k \equiv 200 \pmod{690}$ [Was97] (VERIFY; cross-check, verified against the library copy: [CG13, Prop. 14 and the appendix table, row $p = 691$, 'E [12, 200]'], the weights $k \leq p + 1$ of the Eisenstein–cuspidal congruences mod 691; recomputed here: $v(B_k/k) = 1$ at $k = 12, 200, 702, 890, 1392, 1580$, and 0 at $k = 102, 202, 302, 402$). On the class of 12, $v_{691}(B_k/k) \geq 2$ iff $k \equiv 342252 \pmod{690 \cdot 691}$; on the class of 200, iff $k \equiv 342440 \pmod{690 \cdot 691}$; everywhere else on the two classes $v_{691}(B_k/k) = 1$. (By Kummer's congruences $(1 - \ell^{k-1})B_k/k \bmod \ell^2$ depends only on $k \bmod (\ell - 1)\ell$ for $k \not\equiv 0 \pmod{\ell - 1}$, so the 691 residues scanned cover each class. The inputs are [literature: [Was82, Thm. 5.11, p. 57; Thm. 5.12, p. 59; Cor. 5.14, p. 61], first edition, page images (slot-16 L16-3 and WP-8): Cor. 5.14 is the Kummer congruence, verified as a congruence modulo ℓ; whether it states the ℓ^{a+1} form itself was not recorded by the reader], and the modulus ℓ^2 follows from Thm. 5.12 by the argument of its proof (v0.2.1-pre.2: moved out of the [literature:] argument, the slot-17 referee F1's m-U1): for $k \equiv k' \pmod{(\ell - 1)\ell}$ put $\chi = \omega^k = \omega^{k'} \neq 1$; Thm. 5.11 gives $(1 - \ell^{k-1})B_k/k = -L_\ell(1 - k, \chi)$, and Thm. 5.12, $L_\ell(s, \chi) = \sum_i a_i(s-1)^i$ with $\ell \mid a_i$ for $i \geq 1$, gives $L_\ell(1 - k, \chi) - L_\ell(1 - k', \chi) = \sum_{i \geq 1} a_i\big((-k)^i - (-k')^i\big) \equiv 0 \pmod{\ell^2}$, as $\ell \mid k - k'$.)* A field-theoretic reading of the two rungs *(v0.3.0-pre.3; a cross-reference, nothing here rests on it): the rungs 342252 and 342440 are the unique lifts κ of ω^{12}, resp. ω^{200} ($\kappa \equiv 12$, resp. $200 \pmod{690}$) for which the cyclotomic unit $\eta_\kappa \in \mathbb{Q}(\zeta_{691^2})$ is a local 691^2-th power at $(1 - \zeta_{691^2})$* — *[verified computationally] by an equivariant group-ring computation that uses no Bernoulli number (slot-19 R19-3, `docs/implementation/slot19/r193/eqlayer2.py`; its method reproduced by the slot-19 referee F4 with a second implementation at 37 and 59) — so $X^{691^2} - \eta_{342252}$ and $X^{691^2} - \eta_{342440}$ define the ω^{-11}- and ω^{-199}-parts of the 691-Hilbert*

class field of $\mathbb{Q}(\zeta_{691^2})$ *([proved here] there given the computation and [literature:] inputs: [L37], `sec:fieldsribet`).*

Theorem 5.8 (L_k at the computed irregular cells)**.** *[proved here] from Theorem 5.1, Lemma 2.3(3), Proposition 4.2(3) and the [verified computationally] data named. At* $\ell = 691$*:* $L_{12} = L_{200} = L_{702} = 1$*; and* $L_k = 1$ *for every* $k \equiv 12 \pmod{690 \cdot 691}$ *and every* $k \equiv 200 \pmod{690 \cdot 691}$*. Data: * $c_{k,2}(1 + 2^{k-1})$ *has valuation exactly* 1 *at* $k = 12$ *(*$\tau(2) - (1 + 2^{11}) = -2073 = -3 \cdot 691$*), at* $k = 200$ *(the cell* $(198, 691)$ *over* $\mathbb{Z}/691^2$*, this package's run, report §3) and at* $k = 702$ *(the cell* $(700, 691)$ *over* $\mathbb{Z}/691^2$*, `docs/matrix-engine/results/hida_towers/`, slot-13 WP-10),* $c_{k,2}$ *read from* $\chi_{n,2}^{\mathrm{ord}} = (X - (1 + 2^{n+1}))c_{k,2}^2$ *modulo* 691^2 *(the monic square root modulo* ℓ^m *is unique: monic polynomials are non-zero-divisors);* $v(B_k/k) = 1$ *at the three weights (Proposition 5.7).* Independent cross-check *([verified computationally], slot-14 WP-9, `forms_side.py`, outside the tree, not from the project's cohomology data):* T_2 *on* $S_k(\mathrm{SL}_2(\mathbb{Z}))$ *from q-expansions (Miller basis, exact over* $\mathbb{Z}$*) gives* $v_{691}(c_{200,2}(1 + 2^{199})) = 1$ *(all* 16 *forms* 691*-ordinary) and* $v_{691}(\det(1 + 2^{701} - T_2 \mid S_{702})) = 1$*, an upper bound for the ordinary factor, so* $L_{200} = 1$ *and* $L_{702} \le 1$ *from q-expansions alone.*

Proof. The sandwich at each weight reads $1 \le L_k \le 1$. For the towers: $\min(L_k, 2) = \min(L_{12}, 2) = 1$ on Tw_2 through 12, and likewise through 200. □

Proposition 5.9 (Harder–Pink's sub-tower)**.** *[proved here] from [literature: Harder–Pink [HP92, Satz 5, p. 92; §4.2, pp. 92–93] (verified, slot-14 WP-Q2)] with the repair in the proof, Lemma 2.3(1), Theorem 5.1, Proposition 4.2(3), Kummer's congruences and the analytic class number formula [Was97, Thm. 4.17] (VERIFY); Harder–Pink's own inputs, their Satz 3 (iv), (v), are VERIFY. Let* $\ell \ge 5$*, let* $a \not\equiv 0 \pmod{\ell - 1}$ *be an even class, and put* $\delta_a := v(B_{1,\omega^{a-1}})$*. Then* $L_k = v(B_k/k) = \delta_a$ *for every* $k \ge 4$ *in the class with* $k \equiv 1 \pmod{\ell^{\delta_a}}$*. At* $\ell = 691$ *(*$\delta_a = 1$ *on both irregular classes, Proposition 5.7):* $L_k = 1$ *for* $k \equiv 7602$ *and for* $k \equiv 137510 \pmod{690 \cdot 691}$*.* Since v0.1.1-pre.2 a special case of Prediction 5.4 *([proved here] from Emerton's Theorem 5); recorded as a second route, independent of Emerton and Ohta. Its weights are disjoint from those of Theorem 5.8.*

Proof (slot-14 WP-Q2 §3). Harder–Pink's $\mathcal{T}_{\mathrm{E}}/I \cong \mathbb{Z}_\ell/\ell^{\delta'}$ is $\mathcal{C}_{\mathrm{E},k}(a)$, so $\delta' = L_k$: their algebra acts faithfully on the torsion-free $H^1_!$, which after $\otimes\mathbb{Q}$ is two copies of S_k, so its image is the level-one algebra; localised at $\mathfrak{m}_{\mathrm{E}}$ it is ordinary, $\mathbf{h}_k$ by Lemma 2.3(1); and their Eisenstein ideal contains $T_\ell - 1 - \ell^{k-1} = (\alpha - 1)(1 - \ell^{k-1}\alpha^{-1})$ with the second factor a unit (their Hilfssatz 3.1.5). Their §4.2 concludes $\delta' = \delta := v(B_k/k)$ at $k = 1 + j\ell^N$, $j = a - 1$, $N \gg 0$, under the hypothesis $\ell^{\delta'} \mid k - 1$, which the paper does not show. *Repair*: run their §3 with $\mathcal{T}_{\mathrm{E}}/(I, \ell^N) = \mathbb{Z}/\ell^{\min(\delta', N)}$; Satz 5's Fitting argument gives an unramified extension of degree $\ge \ell^{\min(L_k, N)}$ in the ω^{-j}-eigenspace of the (finite) class group of $\mathbb{Q}(\zeta_\ell)$, so $L_k < N$ for N large. Over all odd j at once: $\sum_j L_{k_j} \le \sum_j v\#A(\omega^{-j}) = v(h^-) = \sum_j v(B_{1,\omega^j})$ (the class number formula), $v(B_{1,\omega^j}) = v(B_{k_j}/k_j)$ for N large (the ℓ-adic limit $\zeta(-j\ell^N) \to -B_{1,\omega^j}$), and $\sum_j v(B_{k_j}/k_j) \le \sum_j L_{k_j}$ (Theorem 5.1); so every inequality is an equality, and $L_{k^*} = \delta_a$ at $k^* = 1 + (a - 1)\ell^N$. With $m = \delta_a + 1$, Proposition 4.2(3) gives $\min(L_k, m) = \min(L_{k^*}, m) = \delta_a$ for every k of the class with $k \equiv k^* \pmod{\ell^{m-1}}$, i.e. $k \equiv 1 \pmod{\ell^{\delta_a}}$, and Kummer's congruence modulo ℓ^m gives $v(B_k/k) = \delta_a$ there. At 691: $7602 \equiv 12 \pmod{690}$, $137510 \equiv 200 \pmod{690}$, both $\equiv 1 \pmod{691}$. □

Remark 5.10 (Class $a \equiv 0$). For $k \equiv 0 \pmod{\ell - 1}$ the Eisenstein series is $\equiv 1$ modulo ℓ after normalisation, Corollary 5.2's argument does not apply, and Emerton's theorem excludes the class ($i \not\equiv 0$; G_0 has a pole). [literature: [Oht03, Thm. (1.5.5), last clause, p. 238, proved in Appendix A.1, p. 265] (verified): for $(\theta, \psi) = (\omega^{-2}, \mathbf{1})$ — at $N = 1$ the class $a \equiv 0$ (Remark 5.5) — the Eisenstein ideal of the cuspidal ordinary algebra is the unit ideal, by Ohta's adaptation

[Oht03, App. A.1] of the construction of Harder–Pink and Kurihara [Kur93, §3] (whose own range $0 < k < \ell - 1$ excludes this class) and the triviality of the ω-part of the unramified Iwasawa module of $\mathbb{Q}(\zeta_{\ell^\infty})$ ("well known" there)]. The proof is by contradiction and *does not use the main conjecture*, which enters Ohta's Appendix A only for $(\theta, \psi) \neq (\omega^{-2}, \mathbf{1})$ (Thm. (A.1.13), p. 265): a proper I^* would give an unramified abelian pro-ℓ extension L of $\mathbb{Q}(\zeta_{\ell^\infty})$ on which $\mathrm{Gal}(\mathbb{Q}(\zeta_\ell)/\mathbb{Q})$ acts through ω, with $\mathrm{Gal}(L/\mathbb{Q}(\zeta_{\ell^\infty})) \otimes r \twoheadrightarrow B/I^*B$ for a faithful h^*-module B ((A.1.5), (A.1.10), (A.1.11), pp. 264–265). Its inputs are Ohta's (3.4.2) (p. 257: geometric Frobenius acts as $T^*(p)$ on the inertia invariants) and *Ohta's freeness theorem* — $X^+ = X^{I_\ell}$ free of rank one over h^*, $X/X^+ \otimes \mathcal{L}$ free of rank one (p. 258, citing his [16] (2.3.6) and [15] (5.1.2)) — a theorem of Ohta's about an étale cohomology module, not a hypothesis of [U]. The "well known" vanishing is [literature: [Was82, Prop. 6.16, p. 102] ($A_0 = A_1 = 0$: the ω-part of the ℓ-class group of $\mathbb{Q}(\zeta_\ell)$ vanishes) and [Was82, Prop. 13.22, p. 284] ($A_n \cong X/((1+T)^{\ell^n} - 1)X$ for a $\mathbb{Z}_\ell$-extension with exactly one ramified prime, totally ramified), verified, first edition]: $e_1 X/Te_1 X = e_1 A_0 = 0$, so $e_1 X = 0$ by Nakayama (X the unramified Iwasawa module of $\mathbb{Q}(\zeta_{\ell^\infty})$), and L, a quotient of $e_1 X$, is trivial. Ohta's ideal matches ours: $I^*(\omega^{-2}, \mathbf{1})$ is generated by $T(l) - (\theta(l)l\iota(l) + \psi(l))$, $T(\ell) - \psi(\ell)$ and $T(q,q) - (\theta\psi)(q)\iota(q)$ (p. 237, Definition (1.5.1) and the generators after it, read at primes); on the $\theta\psi = \omega^{-2}$-component the last vanishes and the others are $T(q) - \epsilon_q$, $U - 1$, so $I^*(\omega^{-2}, \mathbf{1}) \cdot (\text{that component}) = I^0(0) \otimes r$, and $I^0(0) = \mathbf{h}(0)$ follows over $\mathbb{Z}_\ell$ from the statement over r by faithfully flat descent (r is free over $\mathbb{Z}_\ell$). With Steps 1–2 of the proof of Prediction 5.4 (which do not use $a \not\equiv 0$): $\mathbf{h}(0)/I^0(0) = 0$, so $J = \Lambda$ and $\mathcal{C}_{\mathrm{E}}(0) = 0$ for every $\ell \geq 5$ — [proved here] from that [literature:] statement, through the identification of Ohta's cohomological cuspidal algebra e^*h^* with $\mathbf{h}(0)$ (his end of §1.2, via Hida; eigenvalues $1 + q^{k-1}$ as in Remark 5.5). ([conjectural] through v0.1.1-pre.1, with the heuristic: the residual representation would be an extension between $\mathbf{1}$ and ω^{-1}, whose $\omega^{\pm 1}$-parts of the class group of $\mathbb{Q}(\mu_\ell)$ vanish, [Was97, Ch. 6] (VERIFY).) Independently [proved here] at $\ell \in \{5, 7, 11, 13\}$ by Theorem 7.1 below.

6 No CM components at tame level one

Definition 6.1 (CM component)**.** A component P of $\mathbf{T}(a)$ has *CM by* an imaginary quadratic field K if the image of $T(p)$ in $\mathbf{T}(a)/P$ is 0 for every prime $p \neq \ell$ inert in K.

Theorem 6.2 (No CM at tame level one)**.** *[proved here] from Proposition 3.2(3), Lemma 2.3, and [standard: Deligne: for a normalised eigenform f of weight $k \geq 2$ and level 1 there is a continuous semisimple $\rho_f \colon G_{\mathbb{Q}} \to \mathrm{GL}_2(\overline{\mathbb{Q}}_\ell)$ unramified outside ℓ with $\det(1 - \rho_f(\mathrm{Frob}_p)Y) = 1 - a_p Y + p^{k-1}Y^2$, $\det \rho_f = \chi_{\mathrm{cyc}}^{k-1}$ [Del71]; as stated in [Hid86a, (2.1a,b), p. 556] (verified against the read-only PDF)], [standard: Chebotarev density and Brauer–Nesbitt] and [literature: Wiles [Wil88, Thm. 2.1.4, p. 561] (verified, page image, slot-16 L16-3, `docs/implementation/slot16/locators.tsv`; v0.2.0-pre.2): if moreover f is ℓ-ordinary then $\rho_f|_{G_{\mathbb{Q}_\ell}}$ is reducible with an unramified quotient character]. No component of $\mathbf{T}(a)$ has CM, for any $\ell \geq 5$ and any class a.*

Proof. The Eisenstein component has $\epsilon_p(u^k - 1) = 1 + p^{k-1} \neq 0$. Let P be cuspidal with CM by K. Take $k \geq 3$ in the class and a point of P above P_k (lying over, $\Lambda \subseteq \mathbf{T}/P$ integral); by Proposition 3.2(3) and Lemma 2.3 it is the eigencharacter of some $f \in \mathcal{E}_k$, so $a_p(f) = 0$ for every $p \neq \ell$ inert in K. Let χ_K be the quadratic character of K. For every prime $p \nmid \ell d_K$, $\chi_K(p) a_p(f) = a_p(f)$ (split: $\chi_K(p) = 1$; inert: $a_p = 0$), so $\mathrm{tr}\, \rho_f(\mathrm{Frob}_p) = \mathrm{tr}\,(\rho_f \otimes \chi_K)(\mathrm{Frob}_p)$ on a density-one set, and by Chebotarev and Brauer–Nesbitt $\rho_f \cong \rho_f \otimes \chi_K$ (both semisimple). *(a) Some prime $q \neq \ell$ divides d_K.* Then ρ_f is unramified at q, so the inertia group I_q acts trivially on ρ_f and through the non-trivial $\chi_K|_{I_q}$ on $\rho_f \otimes \chi_K$; isomorphic representations have isomorphic restrictions to I_q. Contradiction. *(b) Otherwise* ℓ is the only prime ramified in K, so $K = \mathbb{Q}(\sqrt{-\ell})$, $\ell \equiv 3 \pmod 4$, and $\chi_K|_{I_\ell} \neq 1$. By Wiles, $\rho_f|_{G_{\mathbb{Q}_\ell}}$ has semisimplification $\varepsilon_1 \oplus \varepsilon_2$

with ε_2 unramified; and $\{\varepsilon_1, \varepsilon_2\} = \{\varepsilon_1\chi_K, \varepsilon_2\chi_K\}$ on $G_{\mathbb{Q}_\ell}$. $\varepsilon_2 = \varepsilon_2\chi_K$ fails on I_ℓ, so $\varepsilon_2 = \varepsilon_1\chi_K$, whence $\varepsilon_1|_{I_\ell} = \chi_K|_{I_\ell}$ and $\det \rho_f|_{I_\ell} = \chi_K|_{I_\ell}$ has order 2 — but it is $\chi_{\mathrm{cyc}}^{k-1}|_{I_\ell}$, of infinite order for $k \geq 2$. Contradiction. □

Remark 6.3 (The standard route, and where CM lives). The plan's route [standard: a newform with CM by K has level divisible by $|d_K|$, [Rib77, §3] (VERIFY); an ordinary CM family by K needs ℓ split in K, [Hid86b, (7.1), p. 267] (verified: Hida's CM components are constructed under that hypothesis)] gives the same conclusion: ℓ unramified in K, so $|d_K| \geq 3$ divides the tame level. Hida's CM components [Hid86b, Thm. 7.1, pp. 268–269] (verified) have conductor $N_0 = d\,N(\mathfrak{c})$, divisible by $d = |d_K|$. The proof above avoids the level formula and case (b) needs ordinarity only at ℓ; with the crystallinity of $\rho_f|_{G_{\mathbb{Q}_\ell}}$ at level one (Faltings, Scholl; VERIFY) it would exclude CM for every level-one eigenform. CM ∩ GL(2) and CM ∩ Eisenstein therefore start at tame level $N \geq 3$ (§11).

Remark 6.4 (Dihedral residual systems at tame level one). (v0.1.1-pre.5; slot-14 WP-C16, `docs/implementation/slot14/wpc16/J-7_report.md` §2, §4 and its addendum, refereed in `wpc16/REFEREE_J8_WP-C16.md` §H.) Let $\ell \equiv 3 \pmod 4$, $K = \mathbb{Q}(\sqrt{-\ell})$, $h = h(-\ell)$; the ordinary dihedral blocks lie in the class $k \equiv (\ell+1)/2 \pmod{\ell-1}$. **(i)** [verified computationally] At every weight of that class run, the ordinary dihedral part of $H^1(\Gamma; M_n)$ has $\mathbb{F}_\ell$-dimension $h-1$, whether the Hecke eigenvalues are $\mathbb{F}_\ell$-rational or not ($\ell = 47, 103, 127, 151, 167, 191, 359, 367$; one block of dimension $2r_d$ per Galois orbit of non-trivial class-group characters χ up to $\chi \leftrightarrow \chi^{-1}$, so that $\sum 2r_d \cdot \#\{\text{orbits}\} = h-1$ is arithmetic, and $m = 1$ on these blocks is data, 20 / 20). The Λ-rank is not claimed. **(ii)** The non-ordinary dihedral ideals $\mathfrak{m}$ (the weights k_α, $\alpha \geq 1$) are non-reduced: the full local Hecke algebra of S_k at $\mathfrak{m}$, modulo ℓ, is $\mathbb{F}_q[\varepsilon]/(\varepsilon^2)$, $\mathbb{F}_q$ the residue field ([verified computationally]) — and their first-order direction is anticyclotomic ([verified computationally] on the 56 measured ideals). They are *not* CM in characteristic 0, but not by Theorem 6.2, which concerns the ordinary components: by the level, [standard: a CM newform has level divisible by $|d_K|$] (Remark 6.3's locator, VERIFY), and the level is 1. (That there is no level-one dihedral $\bar\rho$ at all when $\ell \equiv 1 \pmod 4$ — WP-C16's Proposition R — is recorded in [EX], not here: [U] is about the ordinary Λ-adic algebra.)

7 Cuspidal ∩ cuspidal, and the computed component structure

Theorem 7.1 (The ordinary algebra at tame level one for $\ell \leq 13$)**.** *[proved here] from Theorem 2.2, Lemma 2.3, Theorem 4.4, with the [verified computationally] ordinary ranks and polynomials of* `docs/matrix-engine/results/hida_towers/hida_towers.tsv` *(published 2026-09-27; every class of each ℓ occurs there with $n \geq 2$). For $\ell \in \{5, 7\}$, $\mathbf{T}(a) = \Lambda$ for every class (no ordinary cusp form: $h^1_{\mathrm{ord}} = 1$). For $\ell = 11$, $\mathbf{T}(a) = \Lambda$ except in the class $k \equiv 2 \pmod{10}$, where $\mathbf{T}(a) = \Lambda \times \Lambda$ (the Δ-family and the Eisenstein component, disjoint); for $\ell = 13$, likewise except in the class $k \equiv 0 \pmod{12}$. In particular every component is $\cong \Lambda$, no two components meet, and every ordinary local Hecke algebra of weight $k \geq 3$ at $\ell \leq 13$ is $\mathbb{Z}_\ell$ (Loewy length 1).*

Proof. Lemma 2.3(3) gives $d_k = (h^1_{\mathrm{ord}} - 1)/2$: 0 everywhere at 5, 7; at 11, 1 exactly for $n \equiv 0 \pmod{10}$, at 13, 1 exactly for $n \equiv 10 \pmod{12}$ (the published ranks, [C, `rmk:hidacheckable`] [C, `Hi3`]'s values). With $d(a) = d_k$ (Theorem 2.2), $\mathbf{h}(a) = 0$ or $\mathbf{h}(a)$ free of rank 1, i.e. Λ. For the Eisenstein intersection in the rank-one classes the catalogue gives $v(c_{k,2}(1 + 2^{k-1})) = 0$ on all 12 ($\ell = 11$) resp. 16 ($\ell = 13$) cells at $m = 1$ — including the class $a \equiv 0$ at 13 — so $L_k = 0$ (Theorem 5.1) and $\mathcal{C}_{\mathrm{E}}(a) = 0$ (Proposition 3.5); $\mathbf{T}(a) = \mathbf{h}(a) \times \Lambda$ by Lemma 3.4. □

Theorem 7.2 (Six classes at $\ell = 691$, one at $\ell = 2411$)**.** *[proved here] from Theorem 4.4 with the [verified computationally] polynomials of the catalogue and of this package's run of* (198, 691) *(report §3). Writing $\mathbf{T}(k_0)$ for the class of k_0 and δ-lists for the residue degrees of the cuspidal components:*

class of k	*cell used*	$d(a)$	*cuspidal components (residue degrees)*	*Eisenstein*
12	$(10, 691)$, $(700, 691)$	1	[1] *(Δ-family)*	*meets it,* $\mathcal{C}_{\mathrm{E}} \neq 0$, $L_{12} = L_{702} = 1$
200	$(198, 691)$	16	$[1, 1, 2, 2, 10]$	*meets one degree-1 component,* $L_{200} = 1$
102	$(100, 691)$	8	$[1, 7]$	*disjoint*
202	$(200, 691)$	16	$[1, 15]$	*disjoint*
302	$(300, 691)$	24	$[5, 19]$	*disjoint*
402	$(400, 691)$	33	$[6, 6, 21]$	*disjoint*
12 *at* $\ell = 2411$	$(10, 2411)$	0	*none*	$\mathbf{T} = \Lambda$

In every class listed, $\mathbf{h}(a)$ *is finite étale over* Λ*, its components are pairwise disjoint, and the only intersections of components are the Eisenstein ones of the two irregular classes; there* $\mathbf{T}(a)_{\mathfrak{m}_{\mathrm{E}}} \cong \Lambda \times_{\Lambda/J} \Lambda$ *with* J *the ideal of Proposition 3.5,* $\Lambda/(J, P_k) = \mathbb{Z}/\ell^{L_k}$ *for every* k *in the class; hence* $\mathbf{T}_k(a)_{\mathfrak{m}_{\mathrm{E}}} \cong \mathbb{Z}_\ell \times_{\mathbb{Z}/\ell^{L_k}} \mathbb{Z}_\ell$*. (That* $J = (G_a)$ *is principal comes from (U1), not from the data: Remark 7.3.)*

Proof. $\bar{c}_{k,2}$ (and $\bar{c}_{k,3}$) are squarefree at the cells named, with the listed factor degrees (report §2, §3); Theorem 4.4(1),(3). The Eisenstein value $1 + 2^{k-1}$ is not a root of $\bar{c}_{k,2}$ at $k = 102, 202, 302, 402$ and is a simple root at $k = 12, 200$, and $\mathcal{C}_{\mathrm{E}} \neq 0$ there by Theorem 5.1 with $v(B_k/k) = 1$ (Proposition 5.7), so Theorem 4.4(3) applies. In the irregular classes the cuspidal component through $\mathfrak{m}_{\mathrm{E}}$ has residue degree 1 and is étale, so it is Λ; Lemma 3.4 with $A = B = \Lambda$ gives $\mathbf{T}(a)_{\mathfrak{m}_{\mathrm{E}}} = \Lambda \times_{\Lambda/J} \Lambda$ ($J \neq 0$ by Lemma 3.1), and Proposition 3.5 gives $\Lambda/(J, P_k) = \mathcal{C}_{\mathrm{E},k} = \mathbb{Z}/\ell^{L_k}$; tensoring the fibre-product sequence with Λ/P_k (right exact) and comparing with Lemma 3.4 at weight k gives the finite-level statement. □

Remark 7.3 (Is J principal?). Every quotient of Λ is cyclic, but a non-zero ideal of the two-dimensional regular local ring Λ need not be principal ($\mathfrak{m}_\Lambda$ is not), and the data cannot decide it: $J = \mathfrak{m}_\Lambda$ also gives $L_k = 1$ for every k, exactly what is measured on the level-2 towers through 12 and 200. As a Λ-module $\mathbf{T}(a)_{\mathfrak{m}_{\mathrm{E}}} \cong \Lambda \oplus J$ (via $(x, y) \mapsto (y, x - y)$), so J **is principal iff** $\mathbf{T}(a)_{\mathfrak{m}_{\mathrm{E}}}$ **is Λ-free**. That would follow from Hida's freeness of the ordinary algebra of *modular* forms ([C, `thm:hidacontrol`](3), [Hid86b, Thm. 3.1], verified) *once* $\mathbf{T}(a)$ is identified with Hida's algebra (Remark 2.7); that identification needs the reducedness of Hida's algebra and that its only non-cuspidal component is the one of [Hid86b, Thm. 7.2], and is not carried out here. **Decided since v0.1.1-pre.2**: for $a \not\equiv 0$, $J = (G_a)$ is principal by (U1-Λ) (Prediction 5.4, [proved here] from Emerton's Theorem 5), so $\mathbf{T}(a)_{\mathfrak{m}_{\mathrm{E}}} \cong \Lambda \oplus \Lambda$ is Λ-free in the situation above (and $\mathbf{T}(a)$ is Λ-free in every class $a \not\equiv 0$); for $a \equiv 0$, $J = \Lambda$ (Remark 5.10). The data's ambiguity ($J = \mathfrak{m}_\Lambda$ versus $J = (G_a)$) is resolved by the literature, not by the data. The finite-level statements (Theorem 7.2, Proposition 8.2) do not depend on it.

Prediction 7.4 ((U2) Cuspidal congruence lengths along towers). For two cuspidal components $P \neq P'$ of one class that meet, let $L_k(P, P') := \operatorname{length} \mathbf{T}_k/(I_{P,k} + I_{P',k})$. [proved here] (as Proposition 3.5, with Lemma 3.4 for $\mathbf{T}/(P \cap P') \subseteq \mathbf{T}/P \times \mathbf{T}/P'$ when $\mathbf{T}/P$ and $\mathbf{T}/P'$ are Λ-free (e.g. of Λ-rank 1) — then $(\mathbf{T}/P)/P_k$ is torsion-free of rank $r = \operatorname{rank}_\Lambda \mathbf{T}/P$, hence equal to $\mathbf{T}_k/I_{P,k}$ (the kernel reading of Definition 3.3) by the étaleness at P_k shown in the proof of Proposition 3.2(3), and likewise for P': $L_k(P, P') = \operatorname{length} \mathcal{C}(P, P')/P_k$ and $\min(L_k(P, P'), m)$ is constant on Tw_m. [conjectural] in general: the same constancy without freeness. Upper bounds at one cell: Proposition 4.3(2). **No cuspidal ∩ cuspidal instance is known at tame level** 1: by Theorems 7.1, 7.2 no two cuspidal components meet in any computed class, and on [M1] §5.5's forms-side list ($k \leq 298$, $\ell \leq 23170$) every ordinary non-reduced block is a *ramified component*, not an intersection ([verified computationally], slot-14 WP-9 §3.4; Proposition 8.5(4): nine cells and $(68, 157)$; three further rows of the screened list are reduced). [verified computationally] (v0.1.1-pre.6; slot-14 WP-C14 I-8, in the wording of its referee

`docs/implementation/slot14/wpc14/REFEREE_I3b_I7b_I8_I9.md` §5): in the 320 weight-shift rungs of WP-C14 ($\ell \in \{7, 11, 13, 23, 31, 37, 59, 67, 101, 103, 131, 149, 157, 691\}$, the listed cells of [EX] `s14:wshida`), every ordinary cuspidal block has $g = 1$; the rank-2 ramified blocks of Proposition 8.5(4) ($(30, 67)$, $(26, 131)$, ...) are not among these cells and are the next targets of the intra-block reading.

8 Saturation, Gorensteinness and Loewy length

Let $\mathfrak{m}$ be a maximal ideal of $\mathbf{T}_k(a)$, $\mathbf{T}_{k,\mathfrak{m}}$ its localisation, and $\Lambda_{\mathfrak{m}}$ a $\mathbf{T}_k$-stable $\mathbb{Z}_\ell$-lattice on which $\mathbf{T}_{k,\mathfrak{m}}$ acts faithfully — for instance $e_{\mathrm{ord}}H(n)_{\mathfrak{m}}$ on the contract's lattice Sym^n or on Hida's lattice L_n ([C, `def:hidalattice`], [C, `rmk:stabconvention`](3)). As in [M1] Prop. 1, $\mathbf{T}_{\mathfrak{m}}^{\mathrm{sat}} := (\mathbf{T}_{k,\mathfrak{m}} \otimes \mathbb{Q}) \cap \mathrm{End}(\Lambda_{\mathfrak{m}})$, $\delta(\mathfrak{m}) := \mathrm{length}(\mathbf{T}_{\mathfrak{m}}^{\mathrm{sat}}/\mathbf{T}_{k,\mathfrak{m}})$, and the image A of $\mathbf{T}_{k,\mathfrak{m}}$ in $\mathrm{End}(\Lambda_{\mathfrak{m}}/\ell)$ is $\mathbf{T}_{k,\mathfrak{m}}/\ell$ iff $\delta(\mathfrak{m}) = 0$ ([M1] Prop. 1(2); [proved here] there).

Proposition 8.1 (Maximal orders are saturated)**.** *[proved here] If $\mathbf{T}_{k,\mathfrak{m}}$ is a maximal $\mathbb{Z}_\ell$-order in $\mathbf{T}_{k,\mathfrak{m}} \otimes \mathbb{Q}$ — e.g. étale over $\mathbb{Z}_\ell$ — then $\delta(\mathfrak{m}) = 0$ on* every *faithful stable lattice, and $A = \mathbf{T}_{k,\mathfrak{m}}/\ell$. This holds at every ordinary block of every class in Theorems 7.1, 7.2 except the Eisenstein blocks of the two irregular classes at* 691*.*

Proof. $\mathbf{T}_{\mathfrak{m}}^{\mathrm{sat}}$ is an order (a finitely generated $\mathbb{Z}_\ell$-algebra, being a submodule of $\mathrm{End}(\Lambda_{\mathfrak{m}})$) containing the maximal one. A block that is a local factor of an étale algebra ($\mathbf{h}_k$ by Theorem 4.4(2), or the factor $\mathbb{Z}_\ell$ of $\mathbf{h}_k \times \mathbb{Z}_\ell$) is étale. □

Proposition 8.2 (The Eisenstein blocks)**.** *[proved here] Let $\mathbf{T}_{k,\mathfrak{m}} = \mathbb{Z}_\ell \times_{\mathbb{Z}/\ell^L} \mathbb{Z}_\ell$ with $L \geq 1$ (Theorem 7.2). Then $\mathbf{T}_{k,\mathfrak{m}} = \mathbb{Z}_\ell[t]$ with $t = (\ell^L, 0)$, $t^2 = \ell^L t$, so $\mathbf{T}_{k,\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[t]/(t^2)$ (Loewy length 2, independent of L); $0 \leq \delta(\mathfrak{m}) \leq L$, with $\delta = L$ iff $\Lambda_{\mathfrak{m}}$ is the direct sum of its Eisenstein and cuspidal parts, and $\delta = 0$ iff $\Lambda_{\mathfrak{m}}/\ell$ is a faithful $\mathbb{F}_\ell[t]/(t^2)$-module. [verified computationally] ([M1] §5.1, the census cell $(10, 691)$ on Sym^{10}: $A_1 = A_3/691$ of Hilbert function $(1, 1)$, $\varepsilon_3 = 0$): $\delta = 0$ at $(10, 691)$.*

Proof. $\mathbf{T}_{k,\mathfrak{m}} = \{(x, y) : x \equiv y \ (\ell^L)\} = \mathbb{Z}_\ell(1, 1) \oplus \mathbb{Z}_\ell t$; $\mathbf{T}^{\mathrm{sat}} \subseteq \mathbb{Z}_\ell \times \mathbb{Z}_\ell$, of index ℓ^L over $\mathbf{T}$, and it is all of $\mathbb{Z}_\ell \times \mathbb{Z}_\ell$ iff the idempotent $(1, 0)$ preserves $\Lambda_{\mathfrak{m}}$. The last equivalence is [M1] Prop. 1(2). □

Remark 8.3 (Gorenstein, multiplicity one, and the lattice). [literature: Mazur–Tilouine [MT90], Wiles [Wil95, Ch. 2] (VERIFY): for a non-Eisenstein ordinary $\mathfrak{m}$ with ℓ-distinguished residual representation, the Λ-adic $\mathbf{T}_{\mathfrak{m}}$ is Gorenstein and the ordinary cohomology on Hida's lattice is free over it] — at tame level 1 the residual representation of an ordinary form of even weight k is automatically ℓ-distinguished ($\bar{\varepsilon}_1/\bar{\varepsilon}_2|_{I_\ell} = \omega^{k-1} \neq 1$, $k - 1$ odd, $\ell - 1$ even). Freeness on Hida's lattice gives $\delta = 0$ there ([M1] Prop. 1(2), sufficient condition); on Sym^n with $n \geq \ell$ the lattice differs ([C, `rmk:stabconvention`](3)), and nothing is transferred. In every computed class the ordinary non-Eisenstein blocks are étale, so Proposition 8.1 gives $\delta = 0$ on both lattices without the literature. *Since v0.5.0-pre.2* (slot-21 F4): on *ordinary* parts the two lattices give isomorphic Hecke modules, $e_1 H^1(\Gamma; L_n(\mathbb{Z}_\ell)) \cong e_1 H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2 \otimes \mathbb{Z}_\ell)$ (Lemma 14.3(3), part 4), so freeness on Hida's lattice *is* transferred to the contract's lattice on ordinary parts, and "nothing is transferred" above is superseded for them; the sentence is kept as the record of v0.1–0.4.

Prediction 8.4 ((U3) Ordinary blocks are saturated). [conjectural] On every ordinary block of $e_{\mathrm{ord}} H^1(\Gamma; \mathrm{Sym}^n \mathbb{Z}^2) \otimes \mathbb{Z}_\ell$, $\ell \geq 5$, $\delta(\mathfrak{m}) = 0$; equivalently the census-type algebra A of the block equals $\mathbf{T}_{k,\mathfrak{m}}/\ell$. Proved for the étale blocks (Proposition 8.1) and [verified computationally] at $(10, 691)$ (Proposition 8.2); open for non-étale ordinary blocks, of which the computed ones are the Eisenstein blocks of Theorem 7.2. Testable by slot-14 WP-6's route ($\varepsilon_K = 0$ at the stable K) or by [C, `Hi8`](c)'s comparison D_*.

Proposition 8.5 (Loewy length and components)**.** *[proved here] from Proposition 4.2(2). Let $\mathfrak{m}$ be an ordinary maximal ideal of $\mathbf{h}_k$, $k \geq 3$.*

1. $\mathrm{LL}(\mathbf{h}_{k,\mathfrak{m}}/\ell) \le \dim_{\mathbb{F}_\ell} \mathbf{h}_{k,\mathfrak{m}}/\ell = \mathrm{rank}_{\mathbb{Z}_\ell} \mathbf{h}_{k,\mathfrak{m}}$, *the number of eigenforms in the block; and it is a class invariant.*
2. $\mathrm{LL} = 1$ *iff* $\mathbf{h}(a)_{\mathfrak{m}}$ *is étale over* Λ *iff a single component* P *passes through* $\mathfrak{m}$ *and* $(\mathbf{T}/P)_{\mathfrak{m}}$ *is étale over* Λ. *So* $\mathrm{LL} \ge 2$ *means at least one of:* ***(i)*** *an intersection of components at* $\mathfrak{m}$*;* ***(ii)*** *a single component whose normalisation is ramified over* Λ *at* $\mathfrak{m}$*;* ***(iii)*** *a single non-normal component (normalisation étale over* Λ*, the component a proper suborder of it). If all components through* $\mathfrak{m}$ *are* $\cong \Lambda$ *(rank one),* LL *is at most their number.*
3. *Consequently an ordinary block with* $\mathrm{LL} \ge 3$ *needs at least three eigenforms in one residual system and, if its components are rank one, at least three components through* $\mathfrak{m}$. *At* $\ell \le 13$ *none exists (Theorem 7.1); this is the forms-side reading of [M1] Prop. 3 (non-reduced blocks at* $\ell \le 13$ *are non-ordinary) and agrees with [M1] §5.5 (no ordinary level-one block with* $\mathrm{LL} \ge 3$ *for* $\ell < 300$, *[verified computationally] there). (*Added at 0.7.0-pre.1, *the phenomenology proposal of* `docs/phenomenology/phenomena_catalogue.md` *§11 item 4; citation only, no statement moves.) The ordinary level-one Eisenstein block* $\mathbf{T}_{k,\mathfrak{m}}$ *at* $(n,\ell) = (484, 547)$, $k = 486$, *has* $\mathbf{T}_{k,\mathfrak{m}}/\ell \cong \mathbb{F}_\ell[y]/(y^3)$, *Loewy length* 3, *and* $\delta = 0$ *on* Sym^{484} *(part 6, Proposition 16.4(3), (6), [proved here] given the data (D1)–(D4) there, not from Proposition 4.2(2)); its cuspidal rank* 2, *and that* $(547, 486)$ *is the only irregular pair with* $\ell < 10^5$ *of cuspidal rank* ≥ 2, *are [literature: [BP19, §1.1, arXiv v1, p. 2] (page image verified)], the pair first in [literature: [Cal06, p. 68] (page image verified)]; this is a block* with *the Eisenstein component, so it is not an instance of the cuspidal statement above.*
4. *[verified computationally] (slot-14 WP-9 §3.4 on [M1] §5.5's screened list* `fl_search.txt` *— the rows with* $\ell \mid \mathrm{disc}\, c_{k,2}$ *—, corrected by slot-14 WP-14c §6 and its referee* `docs/implementation/slot14/wpc16/REFEREE_J8_WP-C16.md` *Part II; the scripts* `u2_candidates.py` *and, for the non-Eisenstein check,* `s6_eis_check.py`*, both outside the tree)* ***The first ordinary*** $\mathrm{LL} = 2$ ***cuspidal blocks at tame level*** 1 ***are all of kind (ii).*** *Of the* 12 *rows of that list with* $\ell \le 1000$, $k \le 60$, *nine are non-reduced — the cells* $(n,\ell) = (30, 67)$, $(26, 131)$, $(26, 139)$, $(32, 479)$, $(36, 181)$, $(36, 349)$, $(38, 73)$, $(44, 227)$, $(52, 71)$ *— and each of these carries one 2-dimensional block with a double root of* $\bar{c}_{k,2}$*, on which* T_ℓ *is invertible modulo* ℓ *(ordinary), which is not Eisenstein (the double root is not* $1 + 2^{k-1} \bmod \ell$*), and at which* $v_\ell(\mathrm{disc}\, c_{k,2}) = 1$ *with no root of* $c_{k,2}$ *modulo* ℓ^2 *lifting the double root: a ramified quadratic order, a single component,* $\mathbf{h}_{k,\mathfrak{m}}/\ell = \mathbb{F}_\ell[t]/(t^2)$. *Example* $(30, 67)$*:* $c_{32,2} = X^2 - 39960X - 2235350016$, $v_{67}(\mathrm{disc}) = 1$, $\det T_{67} \equiv 56 \pmod{67}$. *The other three rows,* $(46, 383)$, $(50, 677)$, $(54, 67)$, *are* reduced*: there* $v_\ell(\mathrm{disc}\, c_{k,2}) = 2$*, the double root of* $\bar{c}_{k,2}$ *lifts modulo* ℓ^2*, and* T_3 *separates two distinct residual systems on the* T_2*-generalised eigenspace (a conjugate pair over* $\mathbb{F}_{\ell^2}$ *at* 383 *and* 67*, two* $\mathbb{F}_\ell$*-rational systems at* 677*); the double root is a coincidence of* T_2*-eigenvalues, not a non-reduced block, and no statement of this proposition applies to them. (The earlier text listed all* 12 *as non-reduced: [M1]'s list is screened by* $\ell \mid \mathrm{disc}\, c_{k,2}$*, and the three reduced rows were carried over without the Loewy-length filter.) Among the* 215 *rows with* $\ell \le 23170$ *the only one with* $v_\ell(\mathrm{disc}\, c_{k,2}) \ge 2$ and *a non-reduced algebra* $\mathbf{h}_k/\ell$ *is* $(68, 157)$*: two double roots (*111, 96 *modulo* 157*), both blocks ordinary, both ramified (no lift modulo* 157^2*); every other row with* $v_\ell(\mathrm{disc}\, c_{k,2}) \ge 2$ *has reduced* $\mathbf{h}_k/\ell$. *So that list has* ***no*** *cuspidal* $\cap$ *cuspidal instance up to* $k \le 298$, $\ell \le 23170$. *At the nine cells with* $v_\ell(\mathrm{disc}) = 1$ *(and at the two blocks of* $(68, 157)$*, checked directly) the block's* $\mathbb{Z}_\ell[T_2]$ *is the maximal order of a ramified quadratic extension (its discriminant has odd valuation* 1*), so* $\mathbf{h}_{k,\mathfrak{m}}$ *is that maximal order and* $\delta(\mathfrak{m}) = 0$ *on every lattice (Proposition 8.1); slot-14 WP-14c §6 confirms* $\delta = 0$ *on both shipped routes at all eleven blocks. These are cheap tower cells for (U4), (U6) and (U3).*

Proof. (1) The chain $\mathbf{h}_{k,\mathfrak{m}}/\ell \supsetneq \mathfrak{m} \supsetneq \mathfrak{m}^2 \supsetneq \cdots$ strictly decreases in $\mathbb{F}_\ell$-dimension until it reaches 0 (Nakayama), so its length is at most $\dim_{\mathbb{F}_\ell}$; class invariance is Proposition 4.2(2). (2) $\mathrm{LL} = 1$

iff the special fibre $\mathbf{h}(a)_{\mathfrak{m}}/\mathfrak{m}_\Lambda$ is a field, iff (as in the proof of Theorem 4.4) $\mathbf{h}(a)_{\mathfrak{m}}$ is étale over Λ, which is a domain, so one component, étale. If one component passes through $\mathfrak{m}$ and is not étale there, either its normalisation is ramified at $\mathfrak{m}$, or the normalisation is étale and the component is a proper suborder of it; these and (i) exhaust the cases. For rank-one components $\mathbf{h}(a)_{\mathfrak{m}} \subseteq \Lambda^r$ has rank r, so (1) bounds LL by r. (3) follows. □

9 Predictions for the weight ladders

A *ladder* of level m through n_0 is $n_j = n_0 + j(\ell-1)\ell^{m-1}$, $j \geq 0$ — the cells of $\mathrm{Tw}_m(\ell; n_0)$ in order ([C, `def:hidatower`]); a sibling package proposes cluster runs along such ladders. The predictions below are numbered before any run (rule 20). A failure of a [proved here] one is a defect of the implementation or of the cited chain (Hida's control, stabilisation, Eichler–Shimura) and stops the run's analysis; a failure of a [conjectural] one is a finding.

(U1) [proved here] from [literature: Emerton, Thm. 5] (Prediction 5.4; [conjectural] through v0.1.1-pre.1) $L_k = v(B_k/k)$ at every rung of a class $a \not\equiv 0$. At $\ell = 691$: on the ladder $n_j = 10 + 690j$, $L = 1$ for every $j \not\equiv 496 \pmod{691}$ and $L \geq 2$ at $j \equiv 496$ ($n = 342250$); on $n_j = 198 + 690j$, likewise with the exceptional rungs $j \equiv 496$ ($n = 342438$) (Proposition 5.7). Checkable at a rung computed over $\mathbb{Z}/691^2$ as $\min_p v(c_{k,p}(1 + p^{k-1})) = 1$ — now a check of the implementation: a value 0 is a defect, a rung with all computed p giving ≥ 2 means too few p. *Already a theorem* (Theorem 5.8) on the level-2 ladders $n_j = 10 + 476790j$, $198 + 476790j$.

(U2) [conjectural] / [proved here] under freeness (Prediction 7.4) cuspidal congruence lengths: $\min(L_k(P, P'), m)$ constant along a level-m ladder. No cuspidal ∩ cuspidal instance is known at tame level 1: on [M1] §5.5's list ($k \leq 298$, $\ell \leq 23170$) every ordinary non-reduced block is a ramified component (slot-14 WP-9 §3.4, Proposition 8.5(4): nine cells and $(68, 157)$; three further rows of the screened list are reduced).

(U3) [conjectural] (Prediction 8.4) $\delta = 0$ on ordinary blocks; at 691 class 12 it says the census algebra of the Eisenstein block of every rung $n = 10 + 690j$ is $\mathbb{F}_\ell[t]/(t^2)$ (Hilbert function $(1, 1)$), as at $(10, 691)$.

(U4) [proved here] (Proposition 4.2, Theorem 4.4) **which components survive**: all of them, with the same residue degrees, at every rung of a level-1 ladder — h^1_{ord} is constant and $\bar{c}_{k,p}$ is the same polynomial in $\mathbb{F}_\ell[X]$ at every rung, for every $p \neq \ell$. Concretely at $\ell = 691$, with $\bar{c}_{k,2}$ factor degrees: ladder through $n = 10$: $h^1_{\mathrm{ord}} = 3$, [1] (predicted by [C, `Hi11`]; confirmed at $n = 700$ by `hida_towers/` and at $n = 1390, 2080, 2770$ by the slot-14 cluster runs, `docs/implementation/slot14/WP-C3_cluster_run1_report.md` §1–2 (`hi11_1390_691`, `hi11_2080_691`) and `WP-C4_cluster_run1_part2_report.md` §3 (`hi11_2770_691`), recorded in [C] as [D1]'s data remarks P2, P8; and at $n = 4150$ by the September second-cluster run, the WP-C10 report, §1 (`hi11_4150_691`), [C, `Hi11`]'s v1.14.1 sentence; [verified computationally], data signed off by the owner 2026-09-29); through 198: 33, $[1, 1, 2, 2, 10]$; through 100: 17, $[1, 7]$; through 200: 33, $[1, 15]$; through 300: 49, $[5, 19]$; through 400: 67, $[6, 6, 21]$. At $\ell = 2411$ through 10: 1, no cuspidal factor (predicted by [C, `Hi12`]; confirmed at $n = 2420$, `WP-C3_cluster_run1_report.md` §1–2, `hi12_2420_2411`; [D1] P1; and at $n = 4830$, `hi12_4830_2411`, the WP-C10 report, §1, [C, `Hi12`]'s v1.14.1 sentence). On level-m ladders the polynomials agree modulo ℓ^m.

(U5) [proved here] (Theorem 4.4(3), Lemma 4.1) **Eisenstein disjointness is a class invariant**: $v(c_{k,2}(1 + 2^{k-1})) = 0$ at every rung of the ladders through $100, 200, 300, 400$ at 691 and of every rank-one class at $\ell = 11, 13$; and ≥ 1 at every rung through 10 and 198 at 691.

(U6) [proved here] (Proposition 4.2(2), Proposition 8.2, Theorem 7.2) **which algebra**: $\mathbf{T}_{k,\mathfrak{m}}/\ell$ — the algebra slot-14 WP-6's $\mathbb{Z}/\ell^K$ route reads as A_K/ℓ at the stable K, *not* the census image A on $H^1(\mathbb{F}_\ell)$ (the two agree iff $\delta(\mathfrak{m}) = 0$, which is (U3), [conjectural]). It is constant

along a level-1 ladder on every non-Eisenstein block (there $\mathbf{T}_{k,\mathfrak{m}} = \mathbf{h}_{k,\mathfrak{m}'}$ and control at $m = 1$, Proposition 4.2(2)) and on the Eisenstein blocks of the classes through 10 and 198 at 691 (Proposition 8.2, $L_k \geq 1$ on the class, the cuspidal partner étale of residue degree 1): at 691 it is a field on every block of the classes through $100, 200, 300, 400$ and on the non-Eisenstein blocks through 198; $\mathbb{F}_\ell[t]/(t^2)$ on the Eisenstein blocks of the classes through 10 and 198. Constancy of $\mathbf{T}_{k,\mathfrak{m}}/\ell$ at a maximal ideal through which the Eisenstein component and a non-étale cuspidal component both pass is *not* claimed ($\mathbf{T}(a)/P_k \to \mathbf{T}_k(a)$ may have a torsion kernel, e.g. $\mathrm{Tor}_1(\Lambda/J, \Lambda/P_k) \neq 0$ for $J = \mathfrak{m}_\Lambda$). For the census A see (U3).

(U7) [proved here] (Theorem 7.1, [C, `cor:ordprecision`](1), Proposition 8.1) **the ordinary blocks of the ladders at** $\ell \in \{5, 7, 11, 13\}$, at every rung $n \geq 2$ of every shift that preserves the class ($s \equiv 0 \pmod{\ell - 1}$): the ordinary part of $H^1(\Gamma; M_n)$ is the Eisenstein line alone, except at $\ell = 11$, $n \equiv 0 \pmod{10}$ and at $\ell = 13$, $n \equiv 10 \pmod{12}$, where it is the Eisenstein line plus the 2-dimensional block of Δ's residual system ($(a_2, a_3, a_5) \equiv (9, 10, 1)$ at 11, $a_{11} \equiv 1$), whose algebra A is $\mathbb{F}_\ell$ (Loewy length 1) and which is disjoint from the Eisenstein system. Every other block met on these ladders (residual systems cut by every T_p, $p \leq p_{\max}$, **and by** U_ℓ, as the census does) — every block with Loewy length ≥ 2 in particular — is non-ordinary (U_ℓ nilpotent on it). (A block cut by the T_p, $p \neq \ell$, alone can mix the ordinary Δ-part with non-ordinary classes of the same prime-to-ℓ system, e.g. the mod-11 form $\theta^{10}\Delta$, filtration 132, $n = 130$ on the ladder, $a_{11} = 0$; at $\ell = 13$ the mod-13 form $\theta^{12}\Delta$, filtration 156, $n = 154$, $a_{13} = 0$, enters the Δ class the same way.)

(U4) and (U7) at every ℓ with weight-shift data. [verified computationally] (slot-14 WP-C14 I-8, `docs/implementation/slot14/wpc14/hida_types.py`, `I-8_report.md`; refereed; [EX] `s14:wshida`, `s14:depthlaw`) (U4) holds on 7412 ordinary pairs ($\ell = 7$, 11, 13, 23, 31, 37, 103, 131), all sharp; sharpness at the new ℓ (23, 31, 37, 103, 131) is at $m = 1$ only, since no two computed weights there differ by $(\ell - 1)\ell$. The ordinary rank is constant on 37 classes and equals $2 \dim M_{k_c}^{\mathrm{ord}}(\mathbb{F}_\ell) - 1$, $k_c \geq 4$ the least weight of the class. It exceeds the regular value by exactly 2 at the 11 computed irregular classes (Corollary 5.2): $g_E = 1$ there, read at one weight of each class ($D = 3$ on its Eisenstein block) and constant on the class by Proposition 4.2(2); whether $g_E = 1$ at every irregular class is not claimed. On (U7)'s pattern at the other ℓ: every ordinary non-Eisenstein block met (96, all of rank one) and every Eisenstein block with $g = 0$ (297) has $A = \mathbb{F}_\ell$, and the 17 Eisenstein blocks with $g = 1$ have $A = \mathbb{F}_\ell[t]/(t^2)$ (Proposition 8.2).

(U4) and (U7) on the weight-shift run at $\ell \in \{7, 11, 13\}$. [verified computationally] (v0.1.1-pre.5; second-cluster jobs `3465617` and `3467350`, all 287 rungs, $n \leq 1220$, data signed off by the owner 2026-09-29; `docs/implementation/slot14/wpl1/w1_287/` (`c9_stdout.txt`, `hida_pairs.tsv`); [EX] `s14:wsverdicts`, `s14:sept`, `s14:wshida`; [D1] P19, P21 with the counts re-derived on the 287 rungs) **(U7)**: the ordinary rank equals the prediction on 281 / 281 rungs (the other 6 are controls), the ordinary blocks are exactly those named on 353 / 353, and the 745 / 745 blocks of Loewy length ≥ 2 are non-ordinary. **(U4), sharp**: of the 7412 pairs above, the 7385 at $\ell \in \{7, 11, 13\}$ — 4540 at 7 (Eisenstein), $1122 + 946$ at 11 (Eisenstein + Δ), $399 + 378$ at 13, with $m = 1$, 2, 3 on 6575, 778, 32 — are every pair of rungs of one class with a common ordinary system, and each is congruent modulo ℓ^m and *not* modulo ℓ^{m+1} (the least level over $p \leq 23$ and ℓ): the level is exactly m, the extreme pair being $(10, 11) \to (1220, 11)$, exactly modulo 11^3 on both its Eisenstein and its Δ block. (The other 27 pairs are I-8's, at the five new ℓ, $m = 1$.) On the Eisenstein line this is lifting the exponent; on the Δ-family it says that $a_p(k)$ has a unit derivative in k at some $p \leq 23$.

The **non-ordinary** part is *not* governed by $\mathbf{T}$: nothing above predicts the census (non-ordinary) blocks or C-F9 ([E, `pr:latticesplit`]) along a ladder.

10 Testability of (U1)–(U3) with the shipped routes

The shipped tower route (`HeckeHidaHeavyCellProbe`, [C, `rmk:hidacheckable2`]) and slot-14 WP-6's $\mathbb{Z}/\ell^K$ Loewy route; from the referee's §4 (`docs/implementation/slot14/WP-9_referee.md`).

- **(U1): a theorem since v0.1.1-pre.2** (Prediction 5.4); the cells below, computed before the re-tag (confirmed, [EX] `s14:zt`), are now checks of the implementation. **Testable at small irregular primes.** The $L_k = 1$ rungs are theorems at $m = 2$ (Theorem 5.8); further rungs of the 691 classes over $\mathbb{Z}/691^2$ only repeat that. A genuine test of the Bernoulli-zero prediction $L_k \geq 2$ needs a rung with $v(B_k/k) = 2$, out of reach at 691 ($n = 342250$) but cheap at smaller irregular pairs. [verified computationally] (exact Bernoulli numbers: `u1_feasible.py`, each the first such rung of its class, linearity checked on a third point; the two cells re-checked in `u1_cells_check.py`; both outside the tree): $v_{103}(B_{228}/228) = 2$ and $v_{37}(B_{284}/284) = 2$ (also $v_{59}(B_{914}/914) = 2$, $v_{271}(B_{1434}/1434) = 2$). **Test cells**: $(n, \ell) = (226, 103)$ and $(282, 37)$ **over** $\mathbb{Z}/\ell^3$ ($103^3 = 1092727$ and $37^3 = 50653$ are $< 2^{32}$, the narrow band; the tower route at $m = 3$), reading $\min_p v(c_{k,p}(1+p^{k-1}))$ from $\chi^{\mathrm{ord}}_{n,p}$ as in Theorem 5.8. The value 2 confirms (U1) as the theorem $L_k = 2$ at that cell (Theorem 5.1); a value ≥ 3 at every computed p means too few p, a value ≤ 1 is a defect. Recommended for WP-14c / the ladder proposal.
- **(U2): not testable yet.** It needs two cuspidal components sharing a maximal ideal, and the natural candidate list has none (Proposition 8.5(4)). What the tower route *can* test on the ramified cells $(30, 67)$, $(26, 131)$, …, $(68, 157)$ (all tiny) is (U4)/(U6) on a ramified ordinary block — $\bar{c}_{k,2}$ with a double root and $\mathbf{h}_{k,\mathfrak{m}}/\ell = \mathbb{F}_\ell[t]/(t^2)$ at every rung — and, at $m = 2$, that the double root stays unlifted. $L_k(P, P')$ itself is readable neither from characteristic polynomials (Proposition 4.3(2) only bounds it) nor from WP-6's route as shipped (its idempotents split residual systems, not components inside one system); it needs the ℓ-adic factorisation of $c_{k,p}$ to precision K.
- **(U3): testable now** with WP-6's `HECKE_RING_LOEWY_CELLS`: on the Eisenstein blocks $(198, 691)$ ($K = 2$, ≈ 3 GB) and $(700, 691)$ ($K = 2$, behind the U_ℓ lever), prediction $\varepsilon_K = 0$, $A_1 = A_K/691$ of Hilbert function $(1, 1)$ as at $(10, 691)$; on the ramified ordinary blocks of Proposition 8.5(4) (e.g. $(30, 67)$, $K \leq 4$), $\varepsilon_K = 0$, $A_1 = A_K/\ell$ of Hilbert function $(1, 1)$ — there [proved here], not only predicted, at the nine cells with $v_\ell(\mathrm{disc}) = 1$ and the two blocks of $(68, 157)$ (maximal orders, Proposition 8.1; held on both routes at all eleven, slot-14 WP-14c §6), so a failure there is a defect.

11 Scope, and the code map

Out of scope. Tame level $N \geq 3$ (CM ∩ GL(2), CM ∩ Eisenstein, the Eisenstein ideal at level N), after composite N ([C, `sec:levelN`]; slot 15) — **since v0.2.0-pre.1 treated in part 2, §12**, except the Eisenstein ideal at level N; $\ell \in \{2, 3\}$; the Λ-adic algebra as a computed object — only its specialisations are computed, and every Λ-adic statement here is read from them through Theorem 2.2 (**amended at v0.3.0-pre.1**: its truncations $\mathbf{T}(a) \otimes \Lambda/(\ell^K, X^m)$ are in scope, part 3, §13; the whole Λ-adic algebra, every X-adic order at once, stays out of scope); the class $a \equiv 0$ in general (Remark 5.10); the non-ordinary part. **(U-1), a scope note** (v0.2.0-pre.1, owner ruling O4 of 2026-10-01; re-tagged at v0.2.0-pre.2 by the WP-14 referee's U-5): the hidden congruences of the Frobenius-layer kind — the ordinary block of an eigenform f meeting its Frobenius partner $V(f)$ after a θ-twist, [EX] `s15:hidden` — are *non-ordinary*, so they have no consequence for $\mathbf{T}$ and leave the control statements of this document untouched. [proved here] for every block in the image of θ^j, $j \geq 1$ ($U_\ell\theta = 0$, [C, `prop:thetatwist`]: such a block has $a_\ell = 0$); that the Frobenius-layer hidden blocks are such images is [EX] `s15:hidden` (CMPD). No statement of [U] changes with it. [verified computationally] (v0.2.1-pre.1; slot-16 WP-11, `docs/implementation/slot16/WP-11_report.md` §2.1, `HeckeNebentypusValidation`

arm (M2), gated) At the two first Frobenius-layer hidden blocks, $(n,\ell) = (220,17)$ and $(246,19)$ (the system of $\theta\Delta$, dim $= 4$, $a_\ell = 0$), every T_p, $p \le 97$, acts on $H^1(\Gamma;\mathrm{Sym}^n\mathbb{F}_\ell^2)_\mathfrak{m}$ as the scalar $p\,\tau(p) \bmod \ell$, while $\mathbb{T}_\mathfrak{m}/\ell \cong \mathbb{F}_\ell[x]/x^2$ (rank 2 by [C, `lem:heckerankone`]; for $\mathbb{Z}_\ell[T_2,T_3,T_5,T_7,T_\ell]$ certified by [C, `cor:stablecert`]): so the lattice $\Lambda_\mathfrak{m}$ needs 4 generators and is *not* free over $\mathbb{T}_\mathfrak{m}$, the full Hecke algebra (a generation certificate by the T_p, $p \le \dim S_k$), and the saturation defect is $\mathsf{A}_1 = \mathbb{F}_\ell \subsetneq \mathbb{T}_\mathfrak{m}/\ell$ ([C, `rmk:sattor`]: $\dim I = 1$). By the scope note above these blocks are non-ordinary, so this concerns neither **T** nor any statement of [U]; it is recorded here, beside (U-1), and not at Proposition 8.5, because the blocks are non-ordinary (v0.2.1-pre.2 wording, the slot-17 referee F1's m-U2); it bears on that proposition only by contrast: there the freeness of the ordinary blocks is what links saturation and Loewy length, and the Frobenius-layer blocks show that the link does not extend to them.

Code map. Not built. The readings of §4–§7 were made by out-of-tree scripts over the published catalogue (`docs/implementation/slot14/wp8/`); a component-reading validation is slot-14 WP-14c (stretch) or slot 15. The data came from the shipped `HeckeHidaHeavyCellProbe` ([C, `rmk:hidacheckable2`]) and the published `hida_towers/` catalogue. **Since v0.3.0-pre.1**: the truncated Λ-adic arm is planned as slot-19 WP-HL, gated on the referee of part 3 (its code map, §13); **built** at v0.3.0-pre.4.

12 Tame level N — CM components, CM ∩ GL(2), CM ∩ Eisenstein (part 2)

Status and scope of this part. Written as a pre-release (v0.2.0-pre.1), refereed by slot-16 WP-14 (Fable, ACCEPT WITH REPAIRS, repaired at pre.2 and confirmed), **released and owner-signed at v0.2.0** (ruling O2 of 2026-10-01). It extends §6 from tame level 1 to tame level $N \ge 1$, $\ell \nmid N$, $\ell \ge 5$, with a tame character. Its three questions are the ones §11 deferred: *which* components have CM (Theorem 12.3), *whether* a CM component meets a non-CM cuspidal one (CM ∩ GL(2), Remark 12.6) and *whether* it meets an Eisenstein one (CM ∩ Eisenstein, Proposition 12.5). Every statement here is about eigensystems and residual representations; the Λ-adic algebras are cited, not computed, as in Part 1. The finite-level reading is the ordinary part of the character model of [C, `def:nebmodel`] — [C, `sec:nebentypus`]'s $H^1(\Gamma_0(N);V_R(\chi))$ over $\mathbb{F}_q$ or $W_K(\mathbb{F}_q)$ — and its sanity cells are [C, `NB6`].

Conventions of this part. $N \ge 1$ with $\ell \nmid N$; $\chi\colon G_N = (\mathbb{Z}/N)^\times \to \mathbb{F}_q^\times$ a character ([C, `def:nebdata`](4); $\ell \nmid \mathrm{ord}\,\chi$), $\mathcal{O} := W(\mathbb{F}_q)$, $\tilde\chi := \mathfrak{t}\circ\chi$ its Teichmüller lift ([C, `lem:teichmuller`]), $\Lambda_\mathcal{O} := \mathcal{O}[[X]]$. K is an imaginary quadratic field of discriminant d_K, χ_K its quadratic character, c complex conjugation, $w_K := \left|\mathcal{O}_K^\times\right|$. A fixed embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}}_\ell$ determines, when ℓ splits in K, the prime $\mathfrak{l} \mid \ell$ of K it induces, and $\bar{\mathfrak{l}}$ its conjugate. $\omega_\mathfrak{l}$ is the Teichmüller character of $I_\mathfrak{l}$. Symbols new to this part: $\mathbf{h}(N,\chi;a)$, $\mathbf{h}_k(N,\chi)$, θ_λ, ε_λ, ρ_λ; they mean nothing else.

Definition 12.1 (The ordinary algebra at tame level N with character)**.** For a class $a \in \mathbb{Z}/(\ell-1)$ with $\chi(-1) = (-1)^a$, $\mathbf{h}(N,\chi;a)$ is Hida's cuspidal ordinary Hecke algebra of tame level N, tame character $\tilde\chi$ and class a: the $\Lambda_\mathcal{O}$-algebra generated by $T(p)$, $p \nmid N\ell$, and $U(q)$, $q \mid N\ell$, on the ordinary Λ-adic cusp forms of that tame level and character. For $k \ge 2$ with $k \equiv a \pmod{\ell-1}$, $\mathbf{h}_k(N,\chi)$ is the $\mathcal{O}$-algebra generated by the same operators on the ordinary part $e\,S_k(\Gamma_0(N\ell),\tilde\chi;\mathcal{O})$, $e = \lim U(\ell)^{j!}$. At $N = 1$ it is Definition 2.1. *Components* and *meeting* are as in Proposition 3.2(2) and Definition 3.3: two components meet iff they share a maximal ideal. A component has *CM by* K if the image of $T(p)$ vanishes for every $p \nmid N\ell$ inert in K (Definition 6.1).

Theorem 12.2 (Hida: freeness and control at tame level N)**.** *[literature: [Hid86b, Thm. 3.1 and Cor. 3.2, p. 250], verified (page image) by slot-16 L16-3, `docs/implementation/slot16/locators.tsv`: Cor. 3.2 reads "$\mathfrak{h}_0(N,a;\mathcal{O}_K)/P_k\mathfrak{h}_0(N,a;\mathcal{O}_K) \cong \mathfrak{h}_k^0(\Gamma_1(N_0p),a;\mathcal{O}_K)$ if $k\geq 2$", i.e. the source is at tame level N_0 on $\Gamma_1(N_0p)$. The character-χ algebra of Definition 12.1 is the direct factor cut out by the idempotent of χ on the prime-to-ℓ part of $(\mathbb{Z}/N)^\times$ ($\ell\nmid \operatorname{ord}\chi$), so freeness and control pass to it; VERIFY only for the identification of Hida's $\mathfrak{h}_0(N,a)$ with the $\Gamma_1(N_0p)$-algebra of all tame characters] $\mathbf{h}(N,\chi;a)$ is free of finite rank over $\Lambda_\mathcal{O}$, and for every $k\geq 2$ in the class, $\mathbf{h}(N,\chi;a)/P_k \cong \mathbf{h}_k(N,\chi)$ with $T(p)\mapsto T_p$, $U(q)\mapsto U_q$.*

Theorem 12.3 (Where CM can live)**.** *[proved here] from [standard: Deligne and Deligne–Serre/Ribet: for a normalised cuspidal eigenform f of weight $k\geq 2$, level $N\ell^r$ and character ψ there is a continuous irreducible $\rho_f\colon G_\mathbb{Q}\to \mathrm{GL}_2(\overline{\mathbb{Q}}_\ell)$, unramified outside $N\ell$, with $\operatorname{tr}\rho_f(\mathrm{Frob}_p)=a_p$ and $\det\rho_f=\psi\chi_{\mathrm{cyc}}^{k-1}$ ([Del71], [Rib77, §2]; VERIFY)], [standard: Chebotarev density, Brauer–Nesbitt, and Clifford theory: an irreducible $\rho\cong\rho\otimes\chi_K$ is induced from G_K] and [literature: Wiles [Wil88, Thm. 2.1.4, p. 561] (verified, page image, slot-16 L16-3, `docs/implementation/slot16/locators.tsv`): an ℓ-ordinary ρ_f is reducible on $G_{\mathbb{Q}_\ell}$ with an unramified quotient]. Let a component of $\mathbf{h}(N,\chi;a)$ have CM by K. Then:*

1. *ℓ **splits** in K;*
2. *every prime dividing d_K divides N;*
3. *[standard: a CM newform by K has level $|d_K|\,\mathrm{N}(\mathfrak{c})$, $\mathfrak{c}$ the conductor of its Hecke character, [Rib77, §3] (VERIFY)] hence $|d_K|\mid N$; in particular $N\geq 3$, and **tame levels** 1 **and** 2 **carry no CM component** (at $N=1$ this is Theorem 6.2).*

Proof. Take $k\geq 3$ in the class and a point of the component above P_k (Theorem 12.2): it is the eigensystem of an ℓ-ordinary eigenform f of weight k, level $N\ell^r$, character $\psi=\tilde{\chi}$ (as $k\equiv a$; Definition 12.1's $e\,S_k(\Gamma_0(N\ell),\tilde{\chi})$), with $a_p(f)=0$ for $p\nmid N\ell$ inert in K. As in the proof of Theorem 6.2, $\rho_f\cong\rho_f\otimes\chi_K$; ρ_f is irreducible, so $\rho_f\cong\mathrm{Ind}_{G_K}^{G_\mathbb{Q}}\lambda$ for a character λ of G_K. On I_ℓ, $\det\rho_f=\omega^{a-k}\chi_{\mathrm{cyc}}^{k-1}$ ($\tilde{\chi}$ is unramified at ℓ, as $\ell\nmid N$), of infinite order for $k\geq 2$; and by Wiles, $\rho_f|_{G_{\mathbb{Q}_\ell}}$ has a sub ε_1 and an unramified quotient ε_2. *(2)* A prime $q\neq\ell$ dividing d_K but not N: ρ_f is unramified at q while $\chi_K|_{I_q}\neq 1$ — the contradiction of Theorem 6.2(a). *(1), ℓ ramified in K:* $\{\varepsilon_1,\varepsilon_2\}=\{\varepsilon_1\chi_K,\varepsilon_2\chi_K\}$ on $G_{\mathbb{Q}_\ell}$ and $\chi_K|_{I_\ell}\neq 1$ force $\varepsilon_1|_{I_\ell}=\chi_K|_{I_\ell}$ (Theorem 6.2(b)), so $\det\rho_f|_{I_\ell}$ has order 2, against its infinite order. *(1), ℓ inert in K:* $K_\ell/\mathbb{Q}_\ell$ is unramified, so $I_\ell\subseteq G_K$ (after the embedding), $\rho_f|_{G_{K_\ell}}=\lambda\oplus\lambda^c$, and $\{\varepsilon_1,\varepsilon_2\}|_{I_\ell}=\{\lambda,\lambda^c\}|_{I_\ell}$; $\varepsilon_2|_{I_\ell}=1$ makes $\lambda|_{I_\ell}$ or $\lambda^c|_{I_\ell}$ trivial, hence both (choose the lift $\tilde{c}$ of c in D_ℓ — possible, $D_\ell\to\mathrm{Gal}(K_\ell/\mathbb{Q}_\ell)=\mathrm{Gal}(K/\mathbb{Q})$ being onto as ℓ is inert —, which normalises I_ℓ; WP-14 referee U-3), so $\rho_f|_{I_\ell}$ is unipotent and $\det\rho_f|_{I_\ell}=1$ — a contradiction. So ℓ splits. *(3)* By (1) $\ell\nmid d_K$; the newform underlying f has level $|d_K|\,\mathrm{N}(\mathfrak{c})\mid N\ell^r$, so $|d_K|\mid N$; $|d_K|\geq 3$. □

Proposition 12.4 (CM components exist where Theorem 12.3 allows)**.** *[literature: [Hid86b, Thm. 7.1, pp. 268–269], verified for the construction under ℓ split (Remark 6.3); the exact tame character and class of the family in the source's notation VERIFY] If ℓ splits in K and $\mathfrak{c}$ is an ideal of $\mathcal{O}_K$ prime to ℓ, there is a CM component by K of tame level $|d_K|\,\mathrm{N}(\mathfrak{c})$. [proved here] (classical, from [standard: Hecke's theta series: for a Hecke character λ of K of conductor $\mathfrak{c}$ and ∞-type $\alpha\mapsto\alpha^{k-1}$, $\theta_\lambda=\sum_\mathfrak{a}\lambda(\mathfrak{a})q^{\mathrm{N}\mathfrak{a}}\in S_k(\Gamma_0(|d_K|\,\mathrm{N}\mathfrak{c}),\chi_K\varepsilon_\lambda)$, ε_λ the restriction of λ's finite part to $\mathbb{Z}$; [Rib77, §3] (VERIFY)]) At a weight k, θ_λ is ℓ-ordinary iff ℓ splits in K: for $\ell=\pi\bar{\pi}$, $a_\ell=\lambda(\pi)+\lambda(\bar{\pi})$ is a unit at $\mathfrak{l}$ (exactly one summand is), and for ℓ inert $a_\ell=0$.*

Proposition 12.5 (CM ∩ Eisenstein)**.** *[proved here] from Theorem 12.3's inputs and Proposition 12.4. Let P be a CM component by K ($\ell=\mathfrak{l}\bar{\mathfrak{l}}$ split), whose weight-k points are θ_λ with λ of ∞-type $k-1$, $\rho_{\theta_\lambda}=\mathrm{Ind}_{G_K}^{G_\mathbb{Q}}\rho_\lambda$, where $\bar{\rho}_\lambda$ is unramified outside $\mathfrak{c}\ell$, $\bar{\rho}_\lambda|_{I_\mathfrak{l}}=\omega^{k-1}$ and $\bar{\rho}_\lambda|_{I_{\bar{\mathfrak{l}}}}=1$ (the ∞-type through the chosen embedding; $\mathfrak{c}$ prime to ℓ; WP-14 referee U-2).*

1. *If P meets an Eisenstein component (of the ordinary algebra with its Eisenstein components, Definition 2.6's analogue), then $\mathrm{Ind}\,\bar{\rho}_\lambda$ is reducible, which holds iff $\bar{\rho}_\lambda=\bar{\rho}_\lambda^c$; and*

that forces $a \equiv 1 \pmod{\ell-1}$ *(odd weights); then* $\bar\rho_\lambda$ *is unramified at* ℓ*, and* $\bar\varepsilon := \bar\rho_\lambda$ *is a* c*-invariant character of* G_K *unramified outside* $\mathfrak{c}$*. Then* $\bar\rho_P^{\mathrm{ss}} \cong \mu \oplus \mu\chi_K$ *for a character* μ *of* $G_{\mathbb{Q}}$ *with* $\mu|_{G_K} = \bar\varepsilon$*.*

2. ***The untwisted family meets Eisenstein.*** *Assume* $h_K = 1$ *(WP-14 referee U-1: for* $h_K > 1$ *the character below is defined on principal ideals only, its* h_K *extensions differ by class-group characters, and the congruence is the one of (1) with* $E_k(\mu, \mu\chi_K)$*,* μ *the character of (1),* $\mu = 1$ *when* $h_K = 1$*). Let* $\mathfrak{c} = 1$*,* $\lambda = \lambda_k$ *the character* $(\alpha) \mapsto \alpha^{k-1}$ *(defined iff* $w_K \mid k-1$*), so* $\theta_{\lambda_k} \in S_k(\Gamma_0(|d_K|), \chi_K)$*. If* $k \equiv 1 \pmod{\ell-1}$ *then, modulo* $\mathfrak{l}$*,* $a_p(\theta_{\lambda_k}) \equiv 1 + \chi_K(p)\, p^{k-1}$ *for every* $p \nmid d_K \ell$*: the eigensystem of* θ_{λ_k} *is congruent to that of the Eisenstein series* $E_k(\mathbf{1}, \chi_K)$ *of the same level and character.*

Proof. (1) Components meeting share a maximal ideal, hence a residual eigensystem; an Eisenstein one is $p \mapsto \psi_1(p) + \psi_2(p)p^{k-1}$, the trace of a reducible representation, so by Brauer–Nesbitt $\operatorname{Ind}\bar\rho_\lambda$ is reducible; by Mackey/Clifford an induced representation of a character of an index-2 subgroup is reducible iff the character is c-invariant. ρ_λ is unramified at $\bar{\mathfrak{l}}$ and equals $\chi_{\mathrm{cyc}}^{k-1}$ times an unramified character on $I_{\mathfrak{l}}$ (the ∞-type through the chosen embedding; $\mathfrak{c}$ prime to ℓ), so $\bar\rho_\lambda|_{I_{\mathfrak{l}}} = \omega_{\mathfrak{l}}^{k-1}$ and $\bar\rho_\lambda^c|_{I_{\mathfrak{l}}} = \bar\rho_\lambda|_{I_{\bar{\mathfrak{l}}}} \circ c = 1$. Equality forces $\omega^{k-1} = 1$, i.e. $(\ell-1) \mid (k-1)$; then $\bar\rho_\lambda$ is unramified at ℓ, $\bar\varepsilon := \det\bar\rho_\lambda$, and $\bar\varepsilon = \bar\varepsilon^c$. A c-invariant character of G_K extends to $G_{\mathbb{Q}}$ in two ways, μ and $\mu\chi_K$, and $\operatorname{Ind}\bar\varepsilon = \mu \oplus \mu\chi_K$. (2) For $p = \pi\bar\pi$ split ($\mathfrak{p} = (\pi)$ principal, as $h_K = 1$), $a_p = \pi^{k-1} + \bar\pi^{k-1}$ and $\pi, \bar\pi$ are units at $\mathfrak{l}$ with residue field $\mathbb{F}_\ell$, so both powers are $\equiv 1$ when $(\ell-1) \mid k-1$; $1 + \chi_K(p)p^{k-1} \equiv 2$. For p inert, $a_p = 0$ and $1 + \chi_K(p)p^{k-1} = 1 - p^{k-1} \equiv 0$. □

Remark 12.6 (CM ∩ GL(2)). [conjectural] with [literature: [HT93], [MT90]; VERIFY] (1) [proved here] A CM component P meets a non-CM cuspidal component only at a maximal ideal whose residual $\bar\rho$ is $\operatorname{Ind}_{G_K}\bar\rho_\lambda$ — dihedral if $\bar\rho_\lambda \neq \bar\rho_\lambda^c$, Eisenstein (Proposition 12.5) otherwise; a non-CM component through a dihedral maximal ideal is the phenomenon Remark 6.4(ii) met at tame level 1 (there with no CM component at all). (2) The congruence module of P against the rest of the algebra is governed by an anticyclotomic Katz ℓ-adic L-function of K — one divisibility by Hida–Tilouine, the converse the anticyclotomic main conjecture (Mazur–Tilouine), both VERIFY (WP-14 referee U-6) — so CM ∩ GL(2) at P is predicted exactly where ℓ divides that L-value; no numerical form of this is stated here. [verified computationally] reading for the code: at a cell of the character model on P's weight ladder, the ordinary generalised eigenspace at P's residual system has $\mathbb{F}_q$-dimension $2 \cdot \#\{\text{CM eigenforms of } P \text{ at that weight}\}$ plus the non-CM contribution; a positive non-CM contribution is a CM ∩ GL(2) instance.

Remark 12.7 ((U8)–(U10): the predictions of part 2, on [C, NB6]'s cells). Committed at v0.2.0-pre.1, before slot-16 WP-10's and WP-11's runs (rule 20); a failing [proved here] item is a defect, a failing [conjectural] item a finding. Cells are (N, χ, k, ℓ) of the character model, $\ell \geq 5$, $\ell \nmid N$.

(U8) [proved here] (Proposition 12.4, with [C, NB6]'s classical input) **Ordinarity of the CM cells.** At $(4, \chi_{-4}, 5)$, $(3, \chi_{-3}, 7)$, $(7, \chi_{-7}, 3)$ — the forms θ_{λ_k}, $K = \mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{-7})$ — the CM eigensystem lies in the ordinary part of H^1 iff ℓ splits in K ($\ell \equiv 1 \pmod 4$; $\ell \equiv 1 \pmod 3$; $\ell \equiv 1, 2, 4 \pmod 7$); for inert ℓ its T_ℓ-eigenvalue is 0.

(U9) [proved here] (Theorem 12.3) **No CM at tame level 1 or 2, and what the code must not read as CM.** No ordinary eigensystem in characteristic 0 at $N \in \{1, 2\}$ has CM. Modulo ℓ an ordinary block can still have T_p nilpotent at every p inert in some K: at $N = 1$ these are the residually dihedral blocks of Remark 6.4(i), $K = \mathbb{Q}(\sqrt{-\ell})$. A block with that property is *residually dihedral*, not CM; CM is a statement about the component (Definition 12.1), read on characteristic-0 or $\mathbb{Z}/\ell^K$ data at large K.

(U10) [proved here] (Proposition 12.5(2); $h_K = 1$ at [C, NB6]'s three fields $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, $\mathbb{Q}(\sqrt{-7})$) **CM ∩ Eisenstein.** At $(4, \chi_{-4}, 5, 5)$ and $(3, \chi_{-3}, 7, 7)$ — weight $\equiv 1 \pmod{\ell-1}$, ℓ split — and at $(7, \chi_{-7}, 11, 11)$ (on the CM ladder of $(7, \chi_{-7}, 3)$, $w_K = 2$), the CM eigensystem of $H^1(\Gamma_0(N); V_{\mathbb{F}_\ell}(\chi))$ coincides modulo ℓ with the Eisenstein system $p \mapsto 1 + \chi(p)p^{k-1}$: the joint generalised eigenspace of $(T_p)_{p \nmid N\ell}$ at that system contains the

Eisenstein classes and the two CM classes; both Eisenstein systems $1+\chi(p)p^{k-1}$ and $\chi(p)+p^{k-1}$ coincide modulo ℓ there ($p^{k-1}\equiv 1$ for $(\ell-1)\mid(k-1)$), so "the Eisenstein classes" means both and the eigenspace has dimension ≥ 4 (WP-14 referee U-7). [conjectural] The local algebra of the T_p ($p\nmid N\ell$) on that eigenspace is non-reduced (Loewy length ≥ 2), as at the Eisenstein blocks of tame level 1 with $g=1$ (Proposition 8.2).

Code map of part 2. Not built. The readings are WP-10's (`HeckeNebentypusValidation`, the [C, `NB6`] arm) and WP-11's runs; no validation of [U] is scheduled in slot 16.

13 The Λ-adic algebra, truncated (part 3)

Status and scope of this part. Written as a pre-release (v0.3.0-pre.1) for the slot-19 Fable referee F1 (WP-3), revised at v0.3.0-pre.2 on its ACCEPT WITH REPAIRS; **binding since v0.3.0** (the owner's release at the slot-19 close, ruling O3). Part 1 computes only specialisations of the Λ-adic algebras and reads every Λ-adic statement off them through Theorem 2.2 (§11). This part makes a *finite* piece of the Λ-adic algebra itself a computed object: its truncations modulo (ℓ^K, X^m) — the "$\mathbf{T}^{\mathrm{ord}}\otimes\Lambda/(\ell^k, X^m)$" of the slot-19 plan, written with K for the ℓ-power because k is a weight here. It states (i) what the truncation is and that finitely many weights determine it, with explicit precision ([proved here], elementary, from Part 1); (ii) which control statement is used and which literature it rests on; (iii) what is [proved here] at the slot-19 targets $\ell=691$, $(282,37)$, $(226,103)$, and what is [verified computationally] or [conjectural]. Tame level 1, $\ell\geq 5$, as in Part 1.

Definition 13.1 (Truncations)**.** For $K,m\geq 1$ put $\Lambda_{K,m}:=\Lambda/(\ell^K,X^m)=(\mathbb{Z}/\ell^K)[X]/(X^m)$, and for $x\in\ell\mathbb{Z}_\ell$, $\Lambda_{K,m}(x):=\Lambda/(\ell^K,(X-x)^m)$. For a class a and a finitely generated Λ-module M put $M_{K,m}:=M\otimes_\Lambda\Lambda_{K,m}$ and $M_{K,m}(x):=M\otimes_\Lambda\Lambda_{K,m}(x)$; in particular

$$\mathbf{T}_{K,m}(a):=\mathbf{T}(a)/(\ell^K,X^m)\mathbf{T}(a),\qquad \mathbf{h}_{K,m}(a),\qquad \mathcal{C}_{\mathrm{E};K,m}(a):=\mathcal{C}_{\mathrm{E}}(a)/(\ell^K,X^m),$$

finite $\mathbb{Z}/\ell^K$-algebras (Definitions 2.1, 2.6, 3.3), with the images of $T(p)$, U as named operators. For an integer k write $x_k:=u^k-1$, so that $P_k=X-x_k$ and $v(x_k)=1+v(k)$ (the proof of Lemma 4.1). The truncation *centred at the weight* k is $M_{K,m}(x_k)$. For the named algebras the argument is written after a semicolon: $\mathbf{T}_{K,m}(a;x):=\mathbf{T}(a)_{K,m}(x)$, $\mathbf{h}_{K,m}(a;x):=\mathbf{h}(a)_{K,m}(x)$, $\mathcal{C}_{\mathrm{E};K,m}(a;x):=\mathcal{C}_{\mathrm{E}}(a)_{K,m}(x)$ (v0.3.0-pre.2).

Proposition 13.2 (First properties)**.** *[proved here] from Theorem 2.2, Proposition 3.2, Lemma 3.1 and Prediction 5.4 (itself [proved here] from [literature: Emerton]). Let a be an even class.*

1. $\mathbf{h}_{K,m}(a)$ *is free over* $\Lambda_{K,m}$ *of rank* $d(a)$*, hence free over* $\mathbb{Z}/\ell^K$ *of rank* $m\,d(a)$*. If* $a\not\equiv 0$*,* $\mathbf{T}_{K,m}(a)$ *is free over* $\Lambda_{K,m}$ *of rank* $d(a)+1$*, and* $\mathcal{C}_{\mathrm{E};K,m}(a)\cong\Lambda_{K,m}/(G_a)$*.*
2. ***(Control for* $\mathbf{T}$*.)*** *If* $a\not\equiv 0$*, then* $\mathbf{T}(a)/P_k\mathbf{T}(a)\cong\mathbf{T}_k(a)$ *for every* $k\geq 3$ *in the class,* $T(p)\mapsto T_p$*,* $U\mapsto U$*.*
3. ***(At* $m=1$ *the truncation is a finite-level object.)*** $\mathbf{T}_{K,1}(a;x_k)\cong\mathbf{T}_k(a)/\ell^K$ *(*$a\not\equiv 0$*) and* $\mathbf{h}_{K,1}(a;x_k)\cong\mathbf{h}_k/\ell^K$ *for* $k\geq 3$ *in the class; and* $\mathbf{T}_{K,1}(a)=\mathbf{T}_{K,1}(a;x_k)$ *for any such* k *with* $\ell^{K-1}\mid k$ *(one exists by the Chinese remainder theorem).*
4. ***(Centres.)*** *If* $v(x-x')\geq K$*, then* $(\ell^K,(X-x)^m)=(\ell^K,(X-x')^m)$*. Hence* $M_{K,m}(x_k)$ *depends only on* k *modulo* $(\ell-1)\ell^{K-1}$*, i.e. only on the tower of level* K *through* k *([C,* `def:hidatower`*]); at* $m=1$ *this is Lemma 4.1.*

Proof. (1) $\mathbf{h}(a)$ is Λ-free of rank $d(a)$ (Theorem 2.2) and, for $a\not\equiv 0$, $\mathbf{T}(a)$ is Λ-free of rank $d(a)+1$ (Prediction 5.4, (U1-Λ)); base change preserves freeness and rank. $\mathcal{C}_{\mathrm{E}}(a)\cong\Lambda/(G_a)$ is (U1-Λ), and $\otimes\Lambda_{K,m}$ is right exact. (2) $\mathbf{T}(a)/P_k$ is $\mathbb{Z}_\ell$-free of rank $d(a)+1$ by (1) and maps onto $\mathbf{T}_k(a)$ (the generators go to the generators); $\mathbf{T}_k(a)$ is $\mathbb{Z}_\ell$-free of rank $d_k+1=d(a)+1$

(Lemma 3.1, Theorem 2.2). A surjection between free $\mathbb{Z}_\ell$-modules of the same finite rank is an isomorphism. (3) $(\ell^K, X - x_k)$ is the ideal (ℓ^K, P_k), and the quotients are $\mathbf{T}(a)/P_k$ and $\mathbf{h}(a)/P_k$ reduced modulo ℓ^K: (2) and Theorem 2.2. $(\ell^K, X) = (\ell^K, P_k)$ iff $v(x_k) \geq K$ iff $\ell^{K-1} \mid k$. (4) $(X-x)^m = \sum_j \binom{m}{j}(X-x')^{m-j}(x'-x)^j$, and every term with $j \geq 1$ lies in $\ell^K\Lambda$; symmetrically. $v(x_k - x_{k'}) = 1 + v(k - k')$, and $k \equiv k' \pmod{\ell - 1}$ inside a class. □

Lemma 13.3 (A product of weight primes). *[proved here] Let $k_1, \dots, k_N$ be integers, $x_j := x_{k_j}$. If the sum of the $N - m + 1$ smallest of the $v(x_j)$ is $\geq K$ — in particular if $N \geq K + m - 1$ — then $\prod_{j=1}^N (X - x_j) \in (\ell^K, X^m)$.*

Proof. The coefficient of X^t, $t < m$, in $\prod_j (X - x_j)$ is $\pm e_{N-t}(x_1, \dots, x_N)$, a sum of products of $N - t \geq N - m + 1$ distinct x_j; each $v(x_j) \geq 1$. □

Theorem 13.4 (Interpolation from finitely many weights). *[proved here] (elementary; Weierstrass division in Λ). Let $g \in \Lambda$, let $k_1, \dots, k_N$ be pairwise distinct integers with $N \geq K + m - 1$, $x_j := x_{k_j}$, and let $r \in \mathbb{Q}_\ell[X]$ be the interpolation polynomial of degree $< N$ with $r(x_j) = g(x_j)$.*

1. *$r \in \mathbb{Z}_\ell[X]$, and $g \equiv r \pmod{(\ell^K, X^m)}$: the truncation of g is determined by the N values $g(x_{k_j})$.*
2. ***(Precision.)** Suppose $k_j = k_0 + (j-1)(\ell - 1)$, $1 \leq j \leq N \leq \ell$ (consecutive weights of one class), and the values are known only modulo $\ell^{K'}$: $\tilde{g}_j \equiv g(x_j) \pmod{\ell^{K'}}$. Let $\tilde{r}$ be computed from the $\tilde{g}_j$ by Newton's divided differences (the $\tilde{D}_i$ formed in $\mathbb{Q}_\ell$: the $\tilde{g}_j$ lifted to $\mathbb{Z}$ before dividing by ℓ^i; the error analysis below covers it). If $K' \geq K + m - 1$, then $\tilde{r} \equiv g \pmod{(\ell^K, X^m)}$.*
3. ***(A further weight is a check.)** In the situation of (2), for every integer k' in the class, $\tilde{r}(x_{k'}) \equiv g(x_{k'}) \pmod{\ell^{\min(K', N)}}$.*

Proof. Write $r = \sum_{i=0}^{N-1} D_i B_i$ in Newton form, $B_i := \prod_{j \leq i}(X - x_j)$, $D_i := g[x_1, \dots, x_{i+1}]$ the divided difference. (1) For $g = X^n$, $D_i = h_{n-i}(x_1, \dots, x_{i+1})$ (complete homogeneous), integral; for $g = \sum c_n X^n$ the series $\sum c_n h_{n-i}(x)$ converges in $\mathbb{Z}_\ell$ ($v(h_{n-i}(x)) \geq n - i$), so $D_i \in \mathbb{Z}_\ell$ and $r \in \mathbb{Z}_\ell[X]$. $g - r \in \Lambda$ vanishes at the distinct points $x_j \in \ell\mathbb{Z}_\ell$; division by the distinguished polynomial $X - x_1$ gives $g - r = (X - x_1)h_1$ with $h_1 \in \Lambda$, and $h_1(x_j) = 0$ for $j \neq 1$ ($\mathbb{Z}_\ell$ is a domain); by induction $g - r \in \prod_j (X - x_j)\,\Lambda$, which lies in (ℓ^K, X^m) by Lemma 13.3. (2) Here $v(x_a - x_b) = 1 + v((a-b)(\ell-1)) = 1$ for $a \neq b$ ($|a - b| < \ell$), so $D_i = \sum_{a \leq i+1} g(x_a) / \prod_{b \leq i+1, b \neq a}(x_a - x_b)$ has denominators of valuation exactly i, and an error of valuation $\geq K'$ in the values changes D_i by an error of valuation $\geq K' - i$. Modulo X^m the coefficients of B_i are $\pm e_{i-t}(x_1, \dots, x_i)$, $t < m$, of valuation $\geq \max(0, i - m + 1)$. So the error of $D_i B_i$ modulo X^m has valuation $\geq K' - i + \max(0, i - m + 1) \geq K' - (m - 1) \geq K$; with (1). (3) $g(x_{k'}) - r(x_{k'}) = q(x_{k'}) \prod_j (x_{k'} - x_j)$ with $q \in \Lambda$ (the proof of (1)), of valuation $\geq N$; and the error of $D_i B_i(x_{k'})$ has valuation $\geq (K' - i) + i = K'$, as $v(x_{k'} - x_j) \geq 1$. □

For general distinct k_j the same argument gives (2) with $K' \geq K + \max_i \max_{a \leq i+1} \sum_{b \leq i+1,\, b \neq a} v(x_{k_a} - x_{k_b})$ (cruder; the X^m-gain is not used). Consecutive weights of one class are the economical choice, and the one WP-HL implements.

Proposition 13.5 (The truncated algebra from the weights). *[proved here] from Proposition 13.2 and Lemma 13.3. Let $a \not\equiv 0$, let $k_1, \dots, k_N \geq 3$ be pairwise distinct weights in the class, $N \geq K + m - 1$, and $\delta := \max_a \sum_{b \neq a} v(x_{k_a} - x_{k_b})$ ($= N - 1$ for consecutive weights, $N \leq \ell$).*

1. *$\mathbf{T}(a)/\prod_j P_{k_j} \to B := \prod_j \mathbf{T}_{k_j}(a)$ is injective; its image A is the Λ-subalgebra of B generated by the tuples $(T_p)_j$, $p \neq \ell$, and $(U)_j$, with X acting as $(x_{k_j})_j$; and $\ell^\delta B \subseteq A$.*
2. *$\mathbf{T}_{K,m}(a) \cong A/(\ell^K, X^m)A$.*

3. ***(Precision.)*** *If the $\mathbf{T}_{k_j}(a)$ are given modulo $\ell^{K'}$ with $K' \geq K+\delta$ (e.g. as the algebras of the Hecke matrices on a faithful $\mathbb{Z}_\ell$-free module such as $e\,M_{k_j}(\mathrm{SL}_2(\mathbb{Z});\mathbb{Z}_\ell)$, Remark 2.7), the Λ-subalgebra $\bar{A}'$ of $B/\ell^{K'}B$ generated by the images of the tuples $(T_p)_j$, $(U)_j$* and the weight tuple $(x_{k_j})_j$*, through which X acts (generated by the Hecke tuples alone over $\mathbb{Z}/\ell^{K'}$ it can be smaller; v0.3.0-pre.2, F1 m-U3), satisfies $\bar{A}'/\ell^{K'-\delta}\bar{A}' \cong A/\ell^{K'-\delta}A$, so it determines $\mathbf{T}_{K,m}(a)$.*
4. ***(The monogenic case.)*** *If a direct factor $\mathbf{T}(a)_\mathfrak{m}$ (a maximal ideal $\mathfrak{m}$) is generated over Λ by one element t — by Nakayama, iff $\mathbf{T}(a)_\mathfrak{m}/\mathfrak{m}_\Lambda$ is generated by $\bar{t}$ over $\mathbb{F}_\ell$ — then $\mathbf{T}(a)_\mathfrak{m} \cong \Lambda[Y]/(F)$, F the characteristic polynomial of multiplication by t on the free Λ-module $\mathbf{T}(a)_\mathfrak{m}$, $F(x_k,Y)$ is the characteristic polynomial of t on $\mathbf{T}_k(a)_\mathfrak{m}$ (the $\mathfrak{m}$-factor of $\mathbf{T}_k(a)$) for $k \geq 3$ in the class, and $\mathbf{T}_{K,m}(a)_\mathfrak{m} \cong \Lambda_{K,m}[Y]/(F)$. The coefficients of F modulo (ℓ^K, X^m) are determined by Theorem 13.4 from the characteristic polynomials at N consecutive weights modulo ℓ^{K+m-1}. The same holds for $\mathbf{h}(a)_\mathfrak{m}$, every a.*

Proof. (1) $\mathbf{T}(a)$ is Λ-free (Proposition 13.2(1)), so $\mathbf{T}(a)/\prod P_{k_j} \cong (\Lambda/\prod P_{k_j})^{d(a)+1}$ and $\mathbf{T}(a)/P_{k_j} \cong \mathbf{T}_{k_j}(a)$ (Proposition 13.2(2)). The P_{k_j} are pairwise non-associate primes of the factorial ring Λ, so $\bigcap_j P_{k_j}\Lambda = \prod_j P_{k_j}\Lambda$, and $\Lambda/\prod P_{k_j} \to \prod_j \Lambda/P_{k_j} = \mathbb{Z}_\ell^N$ is injective; by Weierstrass division its source is $\mathbb{Z}_\ell[X]_{<N}$, and the Lagrange polynomials $\prod_{b\neq a}(X-x_b)/\prod_{b\neq a}(x_a-x_b)$ show that ℓ^δ kills the cokernel. The image of $\mathbf{T}(a)$ is the image of the Λ-algebra generated by $T(p)$, U (Definition 2.6). (2) $\prod_j P_{k_j} \in (\ell^K, X^m)$ (Lemma 13.3). (3) If $\ell^{K'}b \in A$ with $b \in B$, then $\ell^{K'}b = \ell^{K'-\delta}(\ell^\delta b) \in \ell^{K'-\delta}A$; so $\ell^{K'}A \subseteq A \cap \ell^{K'}B \subseteq \ell^{K'-\delta}A$, and $\bar{A}' = A/(A \cap \ell^{K'}B)$. (4) $\mathbf{T}(a)$ is finite over the complete local ring Λ, hence the product of its localisations at maximal ideals; each factor is a direct summand of a free Λ-module, hence free. Nakayama gives $\mathbf{T}(a)_\mathfrak{m} = \Lambda[t]$, and a monogenic algebra free of rank r over Λ is $\Lambda[Y]/(F)$ with F the characteristic polynomial of t (Cayley–Hamilton and the rank count). Specialisation commutes with the characteristic polynomial, and $\mathbf{T}(a)_\mathfrak{m}/P_k$ is the $\mathfrak{m}$-factor of $\mathbf{T}_k(a)$ by (2) of Proposition 13.2. For $\mathbf{h}(a)$: Theorem 2.2. □

Lemma 13.6 (G_a at every integer, and the twisted Bernoulli numbers)**.** *[proved here] from [literature: [Was82, Thm. 5.11, p. 57; Thm. 7.10, p. 123] (both verified for [U], see Lemma 5.3(4); the variable of Thm. 7.10 read on the page by the slot-19 referee F1: "Let $\chi = \theta\psi$ be an* even *Dirichlet character (θ of the first kind, ψ of the second kind), $\zeta_\psi = \psi(1+q_0)^{-1}$. Then $L_p(s,\chi) = f(\zeta_\psi(1+q_0)^s - 1, \theta)$", $f(T,\theta) \in \mathcal{O}_\theta[[T]]$ for $\theta \neq 1$ (p. 122); and Thm. 5.11, p. 57: $L_p(1-n,\chi) = -(1-\chi\omega^{-n}(p)p^{n-1})B_{n,\chi\omega^{-n}}/n$, $n \geq 1$)]. Let $a \not\equiv 0 \pmod{\ell-1}$ be even. For* every *integer $k \geq 1$,*

$$G_a(x_k) = L_\ell(1-k,\omega^a) = -\big(1-\omega^{a-k}(\ell)\,\ell^{k-1}\big)\frac{B_{k,\omega^{a-k}}}{k},$$

which is $-(1-\ell^{k-1})B_k/k$ for $k \equiv a$ (Prediction 5.4) and $-B_{k,\omega^{a-k}}/k$ for $k \not\equiv a \pmod{\ell-1}$ (ω^{a-k} has conductor ℓ). So the generalised Bernoulli numbers B_{k,ω^j}, $j \not\equiv 0$, are values of the same G_{k+j} at points of other classes.

Proof. Thm. 7.10 is stated for *even* characters; $\chi = \omega^a$ with a even is even. With $\theta = \omega^a \neq 1$, $\psi = 1$ (so $\zeta_\psi = 1$) and $1+q_0 = 1+\ell = u$, it reads $L_\ell(s,\omega^a) = f(u^s-1)$ for $s \in \mathbb{Z}_\ell$, with $f \in \mathbb{Z}_\ell[[T]]$ — the u^s convention (verified on the page, slot-19 F1; at v0.3.0-pre.1 marked VERIFY). Put $G(X) := f\big(u(1+X)^{-1}-1\big) \in \Lambda$ ($u(1+X)^{-1}-1 = (\ell-X)(1+X)^{-1} \in \mathfrak{m}_\Lambda$). Then $G(x_k) = f(u^{1-k}-1) = L_\ell(1-k,\omega^a)$ for every integer k, and the value is Thm. 5.11 with $\chi = \omega^a$, $n = k$: $-(1-\chi\omega^{-k}(\ell)\ell^{k-1})B_{k,\chi\omega^{-k}}/k$. G and G_a agree at the infinitely many x_k, $k \equiv a$, so $G = G_a$ (Weierstrass preparation, as in Lemma 5.3(4)). □

Two remarks (v0.5.0-pre.5, slot 21; unnumbered, so that no number moves; no change of statement). (i) *The Eisenstein measure* (slot-21 P2, [E, `prop:lveismeas`]). The two factors $G_{a+1}(u(1+T)-1)$ and $G_{k-1-a}(u^{k-1}(1+T)^{-1}-1)$ of [E]'s Eisenstein congruence of measures

are, each with the sign -1, the branch series of the Bernoulli distribution E_1 and of $\iota_* E_{k-1}$, $\iota(x) = x^{-1}$ ($E_1 = -x\,\zeta_\ell$ on $\mathbb{Z}_\ell^\times$, ζ_ℓ the pseudo-measure with $\int \omega^b(x)\langle x\rangle^s d\zeta_\ell = G_b(u^s - 1)$; 0.5.0-pre.5 omitted the signs — F4m, 0.5.0-pre.6): the measure $\mu^{\mathrm{Eis}} = -E_1 \star \iota_* E_{k-1}$ on $\mathbb{Z}_\ell^\times$ has branch series exactly their product (with a minus sign) in $\Lambda \otimes \mathbb{Q}$ ([E]: [proved here] given Washington Thm. 12.2, p. 239, read on the page by F4m; [verified computationally] against the two group-ring routes of G_b). On the two pole classes one factor is G_0, and the polar term has residue $-\rho_\infty\, G_k(x_k)$, $\rho_\infty = (1-\ell)\log_\ell(u)/\ell$ — the value of this lemma at x_k times ζ_ℓ's residue (the constant [proved here] from the factorisation and $G_0 = \rho_\infty/T + (\Lambda)$, [E, `prop:lveismeas`](4) in [E] 1.21.0-pre.4; [verified computationally] as a calibration). (ii) *Wild points* (slot-21 P4, [E, `rmk:lvwild`](5)). For ψ_t the character of $(\mathbb{Z}/\ell^m)^\times$, $m \geq 2$, with $\psi_t(u) = \zeta^t$, ζ of order ℓ^{m-1}, $\ell \nmid t$, and G_b read as the group-ring element with $1 + X \leftrightarrow [u]$,

$$G_b(\zeta^t u^r - 1) = -\frac{B_{r,\omega^{b-r}\psi_t}}{r} \qquad (b \not\equiv 0 \text{ even},\ r \geq 1),$$

with no Euler factor (ℓ divides the conductor): [proved here] from Theorem 7.10 and Theorem 5.11 as quoted above (F4m, 0.5.0-pre.6; pre.5 called it "a reading"): $G_b(X) = f(u(1+X)^{-1} - 1, \omega^b)$ by the proof of this lemma, and at $1 + X = \zeta^t u^r$ the argument is $\zeta^{-t}u^{1-r} - 1 = \zeta_\psi(1+q_0)^s - 1$ with $s = 1 - r$ and $\zeta_\psi = \psi_t(1+q_0)^{-1} = \zeta^{-t}$ ($\chi = \omega^b\psi_t$ is even, ψ_t of the second kind), so $G_b(\zeta^t u^r - 1) = L_\ell(1-r, \omega^b\psi_t)$, and Theorem 5.11 gives $-(1-\omega^{b-r}\psi_t(\ell)\ell^{r-1})B_{r,\omega^{b-r}\psi_t}/r$ with $\omega^{b-r}\psi_t(\ell) = 0$. [verified computationally] (slot-21 P4, both group-ring routes, $r \leq 10$, $t \in \{1,2\}$, at $(\ell, m) = (5,2), (5,3), (7,2), (11,2), (37,2)$, against the definition of $B_{r,\chi}$; and independently by F4m at $(5,2), (7,2)$: 48/48, the convention ζ^{-t} 0/48); the convention ζ^{-t} fails at every non-zero value. On the pole class $b = 0$, $v_\pi(B_{r,\omega^{-r}\psi_t}/r) = -1$, $\pi = 1 - \zeta$ ([verified computationally]).

A third remark (v0.5.0-pre.7, slot 21 P5, [E, `rmk:lviwasawa`](6); unnumbered; no change of statement; refereed by F4 at v0.5.0-pre.8). *The Iwasawa invariants of G_b at the pairs used.* $\lambda(G_b) = 0$ iff $\ell \nmid B_b/b$ ([proved here] *given* $\mu(G_b) = 0$: the constant term is $\equiv G_b(x_b) = -(1-\ell^{b-1})B_b/b \pmod{\ell}$, by this lemma; a unit constant term gives $(\mu, \lambda) = (0,0)$ unconditionally, a divisible one gives $\lambda \geq 1$ only once $\mu = 0$ is known — F4, pre.8); $\lambda(G_b) = 1$ at the irregular pairs used in [U] and [E] — $(37,32)$, $(59,44)$, $(67,58)$, $(103,24)$, $(131,22)$, $(691,12)$, $(691,200)$ — [literature: [HHO17], abstract item (3), p. 1: "the Iwasawa invariants satisfy $\lambda_p = \nu_p = i_p$", read on the text layer for slot-21 P5 and again by F4, who also read §7 of the manuscript (λ_p, ν_p are the Iwasawa invariants of the p-class groups of $\mathbb{Q}(\zeta_{p^n})$; the test is four incongruences per irregular index)]; the step $\lambda_p = \sum_{b \text{ even}} \lambda(G_b)$ is [standard: Iwasawa's theorem for the minus part through the analytic class number formula, with the plus part trivial by Kummer–Vandiver, verified in [HHO17] item (2); recalled, locator unverified], and with $\lambda(G_b) \geq 1$ at each irregular index (Ferrero–Washington again) this forces $\lambda(G_b) = 1$; $\mu(G_b) = 0$ for every even b ([standard: Ferrero–Washington, recalled; not read for [U]]). Both re-derived ([verified computationally], slot-21 P5, `HeckeLValueIwasawaInvariantsValidation` arm I0; and independently by the referee F4 with own code, tangent-number Bernoulli numbers and divided differences at $s \leq 5$, `docs/implementation/slot21/rc/f4p5/f4p5_kl.py`: the seven pairs $(0,1)$, every other even class of the 16 primes $(0,0)$, $\mu = 0$ throughout): from exact Bernoulli numbers alone, by the divided differences of this lemma's values $G_b(x_r)$ at $r = b + s(\ell-1)$, $s = 0, \ldots, 3$ ($v(x_r - x_{r'}) = 1$; the reading of Lemma 13.7's proof), and by the Kubota–Leopoldt group ring at level 2; the two routes agree at every even class of $\ell \in \{11, \ldots, 47, 59, 67, 103, 131, 691\}$ (344/344 at 691), $\mu = 0$ everywhere, and $\lambda = 1$ exactly at the seven pairs listed. Proposition 13.8's four classes are among them; nothing there uses this remark.

Lemma 13.7 (The λ-criterion). *[proved here] (elementary). Let $a \not\equiv 0$ be an irregular class ($v(B_k/k) \geq 1$ for one, equivalently every, $k \equiv a$; Corollary 5.2), and $k, k' \equiv a$ with $\ell \nmid k - k'$. Then $\mu(G_a) = 0$ and $\lambda(G_a) = 1$ iff $v\big(G_a(x_k) - G_a(x_{k'})\big) = 1$. In that case $G_a = \varepsilon \cdot (X - \alpha_a)$, $\varepsilon \in \Lambda^\times$, $\alpha_a \in \ell\mathbb{Z}_\ell$, $v(G_a(x_{k''})) = v(x_{k''} - \alpha_a)$ for every integer k'' — which is $L_{k''}$ for $k'' \geq 2$ in*

the class, by (U1-k) — and, for the centred truncation, $k'' \geq 2$ in the class:

$$\mathcal{C}_{\mathrm{E};K,m}(a) \cong \Lambda_{K,m}/(X-\alpha_a) \cong \mathbb{Z}/\ell^{\min(K,\, m\, v(\alpha_a))}, \qquad \mathcal{C}_{\mathrm{E};K,m}(a; x_{k''}) \cong \mathbb{Z}/\ell^{\min(K,\, m\, L_{k''})}.$$

Proof. Write $G_a = \sum g_i X^i$. $g_0 \equiv G_a(x_k) \equiv 0 \pmod{\ell}$ ($v(x_k) \geq 1$). $G_a(x_k) - G_a(x_{k'}) = (x_k - x_{k'}) \sum_{i \geq 1} g_i h_{i-1}(x_k, x_{k'})$ with $v(x_k - x_{k'}) = 1$ and $v(h_{i-1}) \geq i-1$, so the valuation is 1 iff $g_1 \in \mathbb{Z}_\ell^\times$, which with $g_0 \in \ell\mathbb{Z}_\ell$ is $\mu = 0$, $\lambda = 1$. Weierstrass preparation gives ε, α_a. Then $\Lambda_{K,m}/(X-\alpha_a) = (\mathbb{Z}/\ell^K)/(\alpha_a^m)$ and $\Lambda_{K,m}(x)/(X-\alpha_a) = (\mathbb{Z}/\ell^K)/((\alpha_a - x)^m)$, with $v(\alpha_a - x_{k''}) = v(G_a(x_{k''}))$, which is $L_{k''}$ for $k'' \geq 2$ in the class (Prediction 5.4, (U1-k)); Proposition 13.2(1). □

Proposition 13.8 (The Eisenstein truncations at the slot-19 targets)**.** *[proved here] from Lemma 13.7, Prediction 5.4 and the [verified computationally] values below (exact Bernoulli numbers, in-house,* `docs/implementation/slot19/wp2/lambda_check.py`*, output* `lambda_check.out` *beside it, < 1 s). At $(\ell, a) \in \{(37,32), (103,24), (691,12), (691,200)\}$, with $k = a$, $k' = a + \ell - 1$: $v(G_a(x_k)) = v(G_a(x_{k'})) = 1$ and $v(G_a(x_k) - G_a(x_{k'})) = 1$. Hence at all four classes $\mu(G_a) = 0$, $\lambda(G_a) = 1$, $v(\alpha_a) = 1$, and for all $K, m \geq 1$*

$$\mathcal{C}_{\mathrm{E};K,m}(a) \cong \mathbb{Z}/\ell^{\min(K,m)}.$$

The root α_a modulo ℓ^2 predicts the first weight of the class with $L_k \geq 2$ ($x_k \equiv \alpha_a \pmod{\ell^2}$): $k = 284$ at 37 and $k = 228$ at 103 — the two depth-2 rungs of §10 and Remark 5.6, found there independently — and $k = 342252$ (class 12) and $k = 342440$ (class 200) at 691; the first is §10's "$n = 342250$". [verified computationally] (slot-19 referee F1, an independent route: $B_k \bmod \ell^3$ from the power sum $S_k(\ell^2)$, `docs/implementation/slot19/wp3/f1_lambda.py`*, output* `f1_lambda.out` *beside it, ≈ 25 s): the four readings reproduced; a third weight $k'' = a + 2(\ell-1)$ linear modulo ℓ^2, as $\lambda = 1$ demands; the two rungs at 691 computed, not only predicted: $v(B_k/k) = 2$ at $k = 342252$ and $k = 342440$, and 1 at $k \mp 690$ (where* `lambda_check.out` *reads "n/a"); and every other weight scanned ($k < 1500$ at 37 and 103, $k < 12000$ at 691) reads $L_k = 1$ (v0.3.0-pre.2).* Depth 3 *(v0.3.0-pre.3; slot-19 R19-2 and its referee F4, major M1). The root α_a modulo ℓ^3 gives the first weight of the class with $L_k \geq 3$:*

$k = 37580$ ($\ell = 37$), $k = 914250$ ($\ell = 103$), $k = 49928412$ *(class 12)*, $k = 159113510$ *(class 200)* ($\ell =$

[proved here] from Lemma 13.7, Theorem 13.4(1), Lemma 13.6, Prediction 5.4 and [verified computationally] values: five exact Bernoulli numbers per class, B_k at $k = a + j(\ell-1)$, $j = 0, \ldots, 4$ (all $k \leq 3650$; slot-19 R19-2, `docs/implementation/slot19/r192/lambda_trunc.py`*; reproduced by F4 with its own tangent-number Bernoulli numbers and Newton interpolation). At 37 and 103 Voronoi's congruence ([verified computationally],* `r192/rung_check.py`*, and F4) reads $v(B_k/k) = 3$ exactly at the rung and 1 at $k \mp (\ell-1)$; at 691 the Voronoi check (691^4 terms) is an optional cross-check, not what the statement rests on, which there gives $L_k \geq 3$.*

Proof. $v(\alpha_a) = v(g_0)$, and modulo ℓ^2, $G_a(x) \equiv g_0 + g_1 x$ for $x \in \ell\mathbb{Z}_\ell$; the two values determine $g_1 \bmod \ell$ and $g_0 \bmod \ell^2$, and the script finds $g_0 \not\equiv 0 \pmod{\ell^2}$ at all four. Lemma 13.7. That $\lambda = 1$ at these four is also the literature's ([literature: [HHO17]: the cyclotomic λ-invariants of all irregular pairs below 2^{31}, VERIFY]); nothing here uses it. *Depth* 3. $G_a = \varepsilon(X - \alpha_a)$ with $\varepsilon \in \Lambda^\times$, so $v(G_a(x_k)) = v(x_k - \alpha_a)$ exactly ($v(\varepsilon(x_k)) = 0$), and $L_k = v(G_a(x_k))$ for $k \geq 2$ in the class ((U1-k), Lemma 13.6: $1 - \ell^{k-1}$ is a unit); hence $L_k \geq 3$ iff $x_k \equiv \alpha_a \pmod{\ell^3}$. As $\alpha \in \ell\mathbb{Z}_\ell$ gives $\alpha^3 \in \ell^3\mathbb{Z}_\ell$, $\alpha_a \bmod \ell^3$ is the root in $\ell\mathbb{Z}/\ell^3$ of $G_a \bmod (\ell^3, X^3)$, unique by Hensel (g_1 a unit); and $G_a \bmod (\ell^3, X^3)$ is the interpolant through the $N = 5 = K + m - 1$ values $G_a(x_k) = -(1-\ell^{k-1})B_k/k$ (Theorem 13.4(1), $K = m = 3$; the values exact). As u has order ℓ^2 modulo ℓ^3, $k \mapsto x_k \bmod \ell^3$ is a bijection from the class modulo $(\ell-1)\ell^2$ onto $\ell\mathbb{Z}/\ell^3$, so exactly one $k \in [a, a + (\ell-1)\ell^2)$ in the class has $L_k \geq 3$; the script finds the four values displayed. □

(U12)'s checkable form, read ([verified computationally], v0.3.0-pre.3; slot-19 R19-2, `docs/implementation/slot19/R19-2_report.md` §§3–5, `r192/`, predictions committed first; refereed by F4). At the four classes, $K = 3$, $m = 1, 2, 3$, centred at x_{k_0} and at the depth-2 weight (24 truncations), from the forms side (q-expansions, ordinary part): the $\mathfrak{m}_E$-factor is $\Lambda_{3,m}[Y]/(Y(Y + F_1))$ with $Y = T_2 - \epsilon_2$ (Definition 2.4) and $F_1 = \epsilon_2 - a_2(f_X)$ (the sign fixed by F4, m10); $(F_1) = (\tilde{G}_a) = (X - \alpha_a)$ and the Hecke-side Eisenstein ideal $J = (\tilde{G}_a)$; the cokernel of (U12)'s exact sequence tensored with $\Lambda_{3,m}$ has order $\ell^{\min(3, m\,L_c)}$, L_c the depth at the centre, at 24/24; the truncation is Λ-free of rank 2, generated by Y. The $K + m - 1$ digits of Theorem 13.4(2) are sufficient and sharp (with $K + m - 2$ digits the truncation changes in 23 of the 24 cases); (U11)'s checkable form holds at every further weight; the forms-side characteristic polynomials reproduce all 753 comparable rows of the tower catalogue (`docs/matrix-engine/results/hida_towers/`). A one-line extra gate (F4, m10): $F_1(x_k)/G_{32}(x_k) \equiv 2 \pmod{37}$ is constant along the eight tower weights $k = 32, 68, \ldots, 284$ of $(37, 32)$ (v0.3.0-pre.4, slot-19 WP-HL, which gates it: F4's script formed $a_2(f_X) - \epsilon_2 = -F_1$, so its constant $35 \equiv -2$ is the ratio in the opposite sign; pre.3 paired 35 with this F_1 and counted six weights).

The rungs from the cyclotomic layer, Bernoulli-free ([verified computationally], v0.3.0-pre.4; slot-19 R19-7, `docs/implementation/slot19/R19-7_report.md` §§1–4, predictions committed first). The Coates–Wiles measure of the ω^a-part of the cyclotomic units of $\mathbb{Q}_\ell(\zeta_{\ell^m})$, built from one Iwasawa logarithm $\log_\ell(1 - \zeta_{\ell^m})$, has moments $\equiv (1 - \ell^{k-1})B_k/k = -G_a(x_k) \pmod{\ell^m}$ (the congruence proved in R19-7 §§1.1–1.2; the equality by Lemma 13.6) and so reads $G_a \bmod ((1 + X)^{\ell^{m-1}} - 1, \ell^m)$ without any Bernoulli number. Its unique zero modulo ℓ^m in the class is the depth-m rung: 284, 228, 342252, 342440 at $m = 2$ (the depth-2 rungs of Proposition 13.8), 37580 and 914250 at $m = 3$ (two of the depth-3 rungs proved above), and at $m = 4$, $(37, 32)$, the depth-4 rung $k = 1072544$; α_{32} read back is 925, 28305, 281570 modulo $37^2, 37^3, 37^4$ (those of R19-2's α; the depth-4 zero and α mod 37^4 were predicted from it before the run), $\alpha_{24} \equiv 2266 \pmod{103^2}$, and at 691 143037, 272945 modulo 691^2. The depth-3 rungs at 691 (49928412, 159113510) need the level-3 layer at 691 (a field of degree $\approx 3.3 \cdot 10^8$; R19-7 §8, a plan for slot 20). *Computed* (v0.4.0-pre.1; [verified computationally], the slot-19 run NF691 of `NumberFieldValidation` with `NUMBERFIELD_LAYER_CELLS="691:3:12/200:3:11"`, commit `adc71e8d`, `docs/implementation/slot19/nf691_run.md`, predictions committed first; 29.7 GiB, 1 h 15 min at 8 threads): the unique zero modulo 691^3 is $k = 49928412$ (class 12) and $k = 159113510$ (class 200), with $\alpha_{12} \equiv 139090008$, $\alpha_{200} \equiv 279599330 \pmod{691^3}$ — the depth-3 rungs proved above, now read on a second, Bernoulli-free route; every row modulo 691^2 equals the level-2 one. Nothing above rests on these readings.

Prediction 13.9 ((U14): the deeper rungs and α_a modulo ℓ^m at the slot-20 layer cells). (New at v0.4.0-pre.1, hecke-extensions slot 20, WP-2; stated before slot-20 WP-RC20's layer windows run (rule 20).) Let $(\ell, a, m) \in \{(37, 32, 5), (103, 24, 4), (59, 44, 4)\}$.

1. [proved here] from Lemma 13.7, Lemma 13.6, Prediction 5.4 and the [verified computationally] values below. At $(59, 44)$, $\mu(G_{44}) = 0$ and $\lambda(G_{44}) = 1$: $v(G_{44}(x_{44})) = v(G_{44}(x_{102})) = v(G_{44}(x_{44}) - G_{44}(x_{102})) = 1$ ([verified computationally], exact Bernoulli numbers, `docs/implementation/slot20/wp2/lambda_criterion.out`; at $(37, 32)$, $(103, 24)$ this is Proposition 13.8). Hence at all three classes $G_a = \varepsilon(X - \alpha_a)$, and exactly one weight $k \equiv a \pmod{\ell - 1}$ in $[a, a + (\ell - 1)\ell^{m-1})$ has $L_k \geq m$: the one with $x_k \equiv \alpha_a \pmod{\ell^m}$ (the depth-3 argument of Proposition 13.8, with m in place of 3).
2. [verified computationally] (`docs/implementation/slot20/wp2/rung_predict.py`, output `rung_predict.out` and `rung_predict_stability.out` beside it; Python `Fraction`, exact Bernoulli numbers by the tangent numbers, in-house, no CAS, ≈ 2 min). The values:

(ℓ, a)	m	$\alpha_a \bmod \ell^m$	the depth-m rung k ($n = k-2$)
$(37, 32)$	5	13400697	55777784
$(103, 24)$	4	3736634	60430740
$(59, 44)$	2, 3, 4	1711, 147913, 11649137	914, 86464, 8162384

How. P is the Newton interpolant of G_a through N points x_{k_j}, $k_j = a + j(\ell-1)$, $j < N$, from the exact values $G_a(x_{k_j}) = -(1-\ell^{k_j-1})B_{k_j}/k_j$. For $x \in \ell\mathbb{Z}_\ell$, $G_a(x) - P(x) = \prod_j (x - x_{k_j}) \cdot G_a[x_{k_0}, \ldots, x_{k_{N-1}}, x]$, and the divided differences of $G_a \in \Lambda$ (a power series with $\mathbb{Z}_\ell$-coefficients, part 1; v0.4.0-pre.2, F1 m-U1) are $\mathbb{Z}_\ell$-integral (complete homogeneous symmetric polynomials in the nodes), so $v(G_a(x)-P(x)) \geq N$ ([proved here], elementary). With $N = m+2$ the root of P in $\ell\mathbb{Z}/\ell^m$, found digit by digit, is $\alpha_a \bmod \ell^m$; $N = m+4$ gives the same values. The script reproduces every value already in this document or in [E]: 925, 28305, 281570 and the rungs 284, 37580, 1072544 at 37; 2266, 458453 and 228, 914250 at 103; 143037, 272945, 139090008, 279599330 and 342252, 342440, 49928412, 159113510 at 691; and at 59 the depth-2 rung 914 (slot-19 R19-3) and the depth-3 rung 86464 with $\alpha_{44} \equiv 147913 \pmod{59^3}$ (slot-19 WP-NF2's cell arm, Bernoulli-free). The new values are consistent with them: $13400697 \equiv 281570 \pmod{37^4}$, $3736634 \equiv 458453 \pmod{103^3}$, $11649137 \equiv 147913 \pmod{59^3}$.

3. *Checkable form* (slot-20 WP-RC20, the layer windows: `NumberFieldValidation` with `NUMBERFIELD_LAYER_CELLS`, its `EFFECTIVE` line read first; cost from `nf::cw_predict` before launch, at 37, $m=5$: 7.1 GB, ≈ 0.45 h at 8 threads; at 103, $m=4$: 11.5 GB, ≈ 0.47 h; at 59, $m=4$: predicted by WP-RC20 before launch). The level-m Coates–Wiles measure of the ω^a-part (the paragraph after Proposition 13.8) has, modulo ℓ^m, a *unique* zero in the class, equal to the rung of (2), and its α read back equals $\alpha_a \bmod \ell^m$ of (2); every row modulo ℓ^{m-1} equals the level-$(m-1)$ row. A mismatch is a finding against (1) or against the layer code, reported before any re-run. At 691, $m \geq 4$ is out of reach (the layer's $n_m \approx 2.3 \cdot 10^{11}$; slot-19 WP-NF2) and is recorded with its cost only.
4. *Outcome* (v0.4.0-pre.3; [verified computationally], slot-20 WP-RC20, `docs/implementation/slot20/RC20` the shared slot-20 binary, `ENGINE_THREADS=8`; each window `EFFECTIVE NumberFieldValidation arm=LAYER_CELLS cells=1, NUMBERFIELD_GATE checks=637 failing=0`). **Every prediction of (2) and (3) is met**: the level-m measure has a unique zero modulo ℓ^m in the class, and it and α read back equal (2) — 55777784, $\alpha_{32} \equiv 13400697 \pmod{37^5}$; 60430740, $\alpha_{24} \equiv 3736634 \pmod{103^4}$; 8162384, $\alpha_{44} \equiv 11649137 \pmod{59^4}$. The zero towers $[1874161, 50653, 1369, 37, 1]$, $[1092727, 10609, 103, 1]$, $[205379, 3481, 59, 1]$ are the depth-m rungs of (2) level by level, and every row agrees with the Bernoulli route and the control. Cost: 6.87, 11.20, 1.07 GB, 27, 34, 2 min (`nf::cw_predict`'s figures within 5 % of the `time -v` peaks (+4.4 %, +4.6 % at 37, 103), within 1 % of the high-water marks; F1′ m-U1). The prediction stays [proved here]; the readings are a second, Bernoulli-free route to it, as at 691 above. No finding.

Remark 13.10 (What the truncations rest on). **[literature:] (Part 1's, unchanged).** The Λ-freeness and control of $\mathbf{h}(a)$: Hida, Theorem 2.2. The Λ-freeness of $\mathbf{T}(a)$ and $\mathcal{C}_{\mathrm{E}}(a) \cong \Lambda/(G_a)$: Emerton's Theorem 5 through Lemma 5.3 (Prediction 5.4). The values of G_a: Washington, Lemma 13.6. **Not used.** Wiles's Λ-adic forms and the Λ-adic Galois representations [Wil88] (they would give the same algebra by duality); any Gorenstein property of $\mathbf{T}(a)$ or $\mathbf{h}(a)$; any main conjecture. **[proved here] here.** Proposition 13.2, Lemma 13.3, Theorem 13.4, Proposition 13.5, Lemma 13.7 (elementary, from the above). **[verified computationally].** The four λ-readings of Proposition 13.8 and the five exact Bernoulli numbers per class behind its depth-3 rungs; the one characteristic polynomial per class behind (U13) at 37 and 103 (v0.3.0-pre.3); the tower data of Part 1 (`docs/matrix-engine/results/hida_towers/`), which give $\chi_{n,p}^{\mathrm{ord}}$ modulo ℓ^m and hence $c_{k,p}$ modulo ℓ^m (Lemma 2.3(3); the monic square root is unique for ℓ odd). **What a tower row is not.** A row of level m is one point of $\Lambda/(\ell^m, P_k)$ (Lemma 4.1): it never sees

X^2; the X-direction comes only from *several* weights, through Theorem 13.4, at the cost of $m-1$ digits.

Prediction 13.11 ((U11)–(U13): the truncated algebra at the slot-19 targets). For slot-19 R19-2 (research) and WP-HL (code), stated before either runs (rule 20).

(U11) [proved here] (Proposition 13.5(4), Theorem 4.4(1)) **Étale classes.** In every class of Theorems 7.1, 7.2 — there $\bar{c}_{k,p}$ is squarefree for some $p \neq \ell$ — $\mathbf{h}_{K,m}(a) \cong \prod_i W_K(\mathbb{F}_{\ell^{\delta_i}})[X]/(X^m)$ and $\mathbf{h}_{K,m}(a) = \Lambda_{K,m}[Y]/(F_p)$, F_p the Λ-adic characteristic polynomial of $T(p)$. *Checkable form* (a failure is a defect of the code or of a tower datum): F_p mod (ℓ^K, X^m) interpolated from $N = K+m-1$ consecutive weights modulo ℓ^{K+m-1} predicts $c_{k',p}$ mod ℓ^{K+m-1} at every further weight k' of the class (Theorem 13.4(3)), the $\mathfrak{m}$-factors being matched across weights by their residual system (Lemma 4.1 at $m=1$). As Theorem 13.4(3) holds for *every* $g \in \Lambda$, the check tests the existence of the Λ-adic F_p (Hida's control) and the implementation, not the étale structure (v0.3.0-pre.2, F1 m-U6).

(U12) [proved here] (Proposition 13.8) **The Eisenstein truncations.** At $(37,32)$, $(103,24)$, $(691,12)$, $(691,200)$: $\mathcal{C}_{\mathrm{E};K,m}(a) \cong \mathbb{Z}/\ell^{\min(K,m)}$ and, centred at a weight, $\mathbb{Z}/\ell^{\min(K,mL_k)}$. *Checkable form*: on the Hecke side, $0 \to \mathbf{T}(a)/\prod_j P_{k_j} \to (\mathbf{h}(a) \times \Lambda)/\prod_j P_{k_j} \to \mathcal{C}_{\mathrm{E}}(a)/\prod_j P_{k_j} \to 0$ is exact ($\prod_j P_{k_j}$ is a non-zero-divisor on $\Lambda/(G_a)$, as $G_a(x_{k_j}) \neq 0$), both terms on the left embed into $\prod_j(\mathbf{h}_{k_j} \times \mathbb{Z}_\ell)$ as in Proposition 13.5(1), and the cokernel tensored with $\Lambda_{K,m}$ must have order $\ell^{\min(K,m)}$; on the Bernoulli side, $\Lambda_{K,m}/(\tilde{G})$ with $\tilde{G}$ interpolated (Theorem 13.4) from the values of Lemma 13.6 must agree. The cells $(282,37)$ ($k=284$, $L_k=2$) and $(226,103)$ ($k=228$, $L_k=2$) are the first weights of their classes at which the truncation centred at the weight reads $\mathbb{Z}/\ell^{\min(K,2m)}$ rather than $\mathbb{Z}/\ell^{\min(K,m)}$.

(U13) [proved here] (at all four classes since v0.3.0-pre.3; at pre.1, pre.2 [conjectural] at 37 and 103) **The Eisenstein-local algebra.** At the four classes of (U12), the cuspidal component through $\mathfrak{m}_{\mathrm{E}}$ is Λ (as [proved here] at 691, Theorem 7.2), so $\mathbf{T}(a)_{\mathfrak{m}_{\mathrm{E}}} = \Lambda \times_{\Lambda/(G_a)} \Lambda \cong \Lambda[Y]/(Y(Y-G_a))$ and

$$\mathbf{T}_{K,m}(a)_{\mathfrak{m}_{\mathrm{E}}} \cong \Lambda_{K,m}[Y]/\big(Y(Y-X+\alpha_a)\big),$$

free of rank 2 over $\Lambda_{K,m}$, local, with $\mathbf{T}_{K,m}(a)_{\mathfrak{m}_{\mathrm{E}}}/\ell \cong \mathbb{F}_\ell[X,Y]/(X^m, Y(Y-X))$. [proved here] at $\ell = 691$, classes 12 and 200 (the fibre product is Theorem 7.2 with $J=(G_a)$; the presentation: $(1,1)$ and $Y := (0, G_a)$ are a Λ-basis with $Y^2 = G_a Y$; then $G_a = \varepsilon(X-\alpha_a)$, $Y \mapsto \varepsilon Y$). [proved here] at $(37,32)$ and $(103,24)$ by the same argument (v0.3.0-pre.3; slot-19 R19-2 §3, the slot-19 referee F4, m8), from Theorem 4.4(1),(3), Lemma 3.4, (U1-Λ) ([literature: :] Hida's control, Emerton's Theorem 5) and a [verified computationally] input at one weight: $\mathrm{charpoly}(T_2 \mid S_{32}) = X^2 - 39960X - 2235350016$ has the roots 14, 23 modulo 37, with $23 \equiv 1 + 2^{31}$ simple, and $\mathrm{charpoly}(T_2 \mid S_{24}) = X^2 - 1080X - 20468736$ has the roots 70, 83 modulo 103, with $83 \equiv 1 + 2^{23}$ simple. So $\bar{c}_{k,2}$ (a divisor of these) is squarefree with the Eisenstein value a simple root, and $v(B_k/k) = 1$ at $k = 32$, resp. 24 (Proposition 13.8): the cuspidal component through $\mathfrak{m}_{\mathrm{E}}$ is étale of residue degree 1, i.e. Λ ($d_{\mathrm{Eis}} = 1$ at every weight of the class, as R19-2 reads at all 13, resp. 6, tower weights), and the fibre product follows as at 691. R19-2 also reads the presentation itself (the paragraph after Proposition 13.8).

Code (v0.3.0-pre.4, [verified computationally]): slot-19 WP-HL's `HeckeLambdaValidation` gates the checkable forms of (U11)–(U13) at all 12 truncations of $(37,32)$ and $(103,24)$, and with `HECKE_LAMBDA_ALL=1` also at $(691,12)$, $(691,200)$; all held (the code map below).

Code map of part 3. **Built** (v0.3.0-pre.4; slot-19 WP-HL, commit `e3a4be35`, after F1's ACCEPT of this part). `HeckeExt/HeckeHidaLambdaForms`: the input arm (Q) on q-expansions — by the owner's ruling of 2026-10-03 the primary input, the deviation from (i) below — the

Miller basis modulo ℓ^W, exact products by a multimodular NTT over 62-bit primes with Garner into $\mathbb{Z}/\ell^W$, T_p on q-expansions, the ordinary projector T_ℓ^N; `HeckeExt/HeckeHidaLambda`: the Λ stage — a certified Fitting split, the monogenic route through Theorem 13.4 with its precision bound *enforced* (refused below $K+m-1$ digits or weights, and on non-consecutive weights), the general route of Proposition 13.5(1) at $K+N-1$ (refused below), Bernoulli values by tangent numbers (= WP-NF's exact ones), α, the rungs, the Kubota–Leopoldt value; record `hecke_lambda/v1` (an FNV digest per line, not pinned; no golden line). The shipped tower route (`HeckeExt/HeckeHidaTower`) is a *cross-check* only, at 37 and 103 (arm (T): $(30,37)$, $(66,37)$, $(22,103)$, $m \le 2$). `HeckeLambdaValidation` (`standard`, ≈ 7 s, ≈ 70 MB, 138 checks; the planned name `HeckeHidaLambdaValidation` is superseded): R19-2's readings at $(37,32)$, $(103,24)$ reproduced field by field (the oracle is R19-2's data, never this code's output), (U11)–(U13) at all 12 truncations, the Kubota–Leopoldt value 47910634, the rungs $284\,/\,37580$, $228\,/\,914250$, the F4 m10 ratio, the tower catalogue at $\ell \le 13$ (744/744 rows), negative controls (the precision refusals, perturbed inputs caught); env `HECKE_LAMBDA_ALL=1` (read-only) adds $(691,12)$, $(691,200)$ (224 checks, ≈ 400 s, 0.15 GB). [verified computationally] where a value is read; no statement of this part rests on the code. *The plan, as written at v0.3.0-pre.1*: slot-19 WP-HL (upper core; Opus; only on F1's ACCEPT of this part), an arm of `HeckeExt/HeckeHidaTower` that (i) takes $N \ge K+m-1$ consecutive weights $k_0 + j(\ell-1)$ of one class, computed modulo ℓ^{K+m-1} on the shipped tower route ([C, `rmk:hidacheckable2`]); (ii) interpolates the block characteristic polynomials (Proposition 13.5(4)) and, where a block is not monogenic, the subalgebra of Proposition 13.5(1) at precision $K+\delta$; (iii) gates (U11)'s checkable form at the étale classes and (U12) against in-house Bernoulli numbers (WP-NF's `BigInt`, or modulo $\ell^{K'}$); `HeckeHidaLambdaValidation` (`standard`, ≤ 8 GB; heavy cells by an env interlock only; $(700,691)$ at $m=2$, 33.5 GiB, excluded). Research prototype: slot-19 R19-2 (from `results/hida_towers/` and new cells only by prediction). *Since v0.3.0-pre.3* (R19-2 §6, refereed by F4): the prototype ran on a q-expansion route; the shipped tower route cannot feed the Λ-adic arm at $\ell = 691$ ($(700,691)$ at $m=2$ already needs 33.5 GiB), so R19-2 recommends a q-expansion input arm — a deviation from (i) above that is the WP-HL planner's and the owner's decision. On either route the generator choice and the component split are made *on the ordinary part*: on the full S_k at $k = 140, 176, 284$ the Eisenstein eigenvalue $1+2^{k-1}$ of T_2 has multiplicity 2 modulo 37, with a non-ordinary companion, so on H^1 itself T_2 does not separate the $\mathfrak{m}_{\mathrm{E}}$-block there (F4, m9). No golden line, digest or record key is touched by this part.

14 The Λ-adic modular symbols, truncated (part 4)

Status and scope of this part. Written as a pre-release (v0.5.0-pre.1, **not signed**) for the slot-21 Fable referee, before any code of slot-21 P8 (rules 20, 21); v0.5.0-pre.2 applies that referee's repairs (F4, `docs/implementation/slot21/P8_F4_referee_report.md`: ACCEPT WITH REPAIRS; M1 the range of (BF), M2 the normalisation of Hida's P_n read on the page, M3 (U17)(a) as Harder's index on the ordinary block, and the minors m1–m7; no label added, renamed or removed; the tags of 0.4.0 unchanged). Parts 1 and 3 read the Λ-adic *algebra* from its specialisations; this part does the same for the Λ-adic *ordinary parabolic cohomology* of tame level 1 — Hida's Λ-adic modular symbols — and fixes what slot-21 P9 (the two-variable measure) needs from it: (i) the finite-weight objects and their structure per sign (Definition 14.1, Lemma 14.2); (ii) that the ordinary part does not see the difference between the contract's lattice Sym^n and Hida's lattice L_n (Lemma 14.3, [proved here], new); (iii) multiplicity one — freeness of rank one over the component — at every *étale* component, at finite weight (Proposition 14.4, [proved here], elementary) and Λ-adically and in every truncation (Theorem 14.7, [proved here] from [literature:] Hida); (iv) a normalisation of the Λ-adic generator by one linear functional, its unit check modulo (ℓ, X), and the period-free invariants it produces (Proposition 14.9, [proved here]; the functional itself: Lemma 14.8,

[proved here] from [E]); (v) the predictions (U15)–(U17) (Prediction 14.10): (U15) [proved here] — the match with P7's Λ-adic q-expansions at eigenvector level; (U16) [conjectural] — the Λ-adic coherence of the normalised measures; (U17) [conjectural] — the Eisenstein gluing on H^1. Tame level 1, $\ell \geq 5$, as in part 1. *Not in this part*: $\Gamma_1(\ell^r)$, wild characters and the ω-type truncations (slot-21/22 P10), non-étale ordinary components (the ramified blocks of Proposition 8.5(4)), whose modular symbols are module-valued and are refused by name (Remark 14.11). (The LL = 4 cells of [L37] are not among them: they are *non-ordinary*, U_{37} nilpotent by [L37] `prop:nonord` (indeed $U_{37} \in \mathfrak{m}_A^2$, [L37] `prop:Uposition`), hence not objects of [U] at all.)

Notation of this part. $\ell \geq 5$, a an even class, $k \geq 3$ in the class, $n = k-2$. $V_\ell := \mathrm{Sym}^n \mathbb{Z}^2 \otimes \mathbb{Z}_\ell$ with ρ_n of [C, `def:rhon`] on all of $\mathrm{Mat}_2(\mathbb{Z})$ (multiplicative, [C, `prop:symconv`](2)), $u_t = x^{n-t}y^t$, and $V_n' := \bigoplus_t \binom{n}{t}\mathbb{Z}_\ell u_t \subseteq V_\ell$, the image of Hida's lattice $L_n(\mathbb{Z}_\ell)$ under $f \mapsto Df$ ([C, `rmk:stabconvention`](3)), a Δ-stable lattice ([E, `prop:lvlattice`]); $m_n := \max_t v\binom{n}{t}$, so $\ell^{m_n} V_\ell \subseteq V_n'$ and $m_n = 0$ iff $V_n' = V_\ell$. ∂ is the boundary of [C]'s pair ([C, `def:pair`]), H_c^1 the cohomology of the pair, and $H_!^1 := \mathrm{im}(H_c^1 \to H^1) = \ker(H^1 \to H^1(\partial))$ (exactness of the sequence of the pair; for $\ell \geq 5$ this is Hida's parabolic cohomology H_P^1). $e_1 := e_{\mathrm{ord}}(T_\ell)$ at level 1 and $e_\ell := e_{\mathrm{ord}}(U_\ell^{(\ell)})$ at level ℓ, the ℓ-adic projectors of [C, `rmk:ordtorsionfree`] and [C, `lem:levelelltorsion`](3). ι^* is [C]'s involution ([C, `def:iota`]), normalised so that the Eisenstein class lies in H_+^1 ([C, `rmk:iotasign`]); for $2 \in R^\times$ every module below splits Hecke-stably into its $\pm$-parts ([C, `prop:iotasplit`], [E, `rmk:lviotak`]). For a maximal ideal $\mathfrak{m}$ and a module M, $M_\mathfrak{m}$ is the $\mathfrak{m}$-part (Fitting / generalised eigenspace).

Definition 14.1 (The finite-weight objects)**.** For $R \in \{\mathbb{Z}_\ell, \mathbb{Z}/\ell^K\}$ put

$$\mathsf{S}_k^\pm(\mathbb{Z}_\ell) := e_1 H_!^1(\Gamma; V_\ell)^\pm, \qquad \mathsf{H}_k^\pm(\mathbb{Z}_\ell) := e_1 H^1(\Gamma; V_\ell)^\pm, \qquad \mathsf{S}_k^\pm(\mathbb{Z}/\ell^K) := \mathsf{S}_k^\pm(\mathbb{Z}_\ell)/\ell^K,$$

with the operators T_p ($p \neq \ell$) and $U := u_{\mathrm{st}}$, the unit root of $Y^2 - T_\ell Y + \ell^{k-1}$ in the closure of $\mathbb{Z}_\ell[T_\ell]$ ([C, `prop:stabilisation`](2),(3)), so that $T_\ell = U + \ell^{k-1}U^{-1}$ on $e_1 H$. The *level-ℓ incarnation* is $e_\ell H^1(\Gamma_0(\ell); V_\ell)^\pm$, identified with $\mathsf{H}_k^\pm(\mathbb{Z}_\ell)$ by $e_\ell \circ \mathrm{dg}_\ell$, with $U_\ell^{(\ell)} \leftrightarrow U$ ([C, `prop:stabilisation`](3),(4)); here ι^* at level ℓ is defined by the recipe of [C, `def:iota`] (ι normalises $\Gamma_0(\ell)$ and commutes with $g = \mathrm{diag}(1,\ell)$, so $\mathrm{dg}_\ell = [\Gamma\, g\, \Gamma_0(\ell)]$ commutes with it and the identification respects the signs). $\mathsf{S}_k^\pm(\mathbb{Z}_\ell)$ is the **ordinary modular-symbol module of weight k and sign $\pm$**; for a maximal ideal $\mathfrak{m}$ of $\mathbf{h}_k$, an **ordinary eigensymbol** of $(\mathfrak{m}, \pm)$ is a generator of $\mathsf{S}_k^\pm(\mathbb{Z}_\ell)_\mathfrak{m}$ over $\mathbf{h}_{k,\mathfrak{m}}$ when that module is free of rank one.

Lemma 14.2 (Structure over a field, per sign)**.** *[proved here] from Lemma 2.3, [C, `rmk:iotasign`], [C, `Hi5`] and [literature: [Hid86a, Lemma 6.4, pp. 587–588], read on the page for this text (and by the slot-21 referee F4, pp. 587–588 on the extracted text: "For any subfield K of $\mathbb{C}$ or Ω, $H_P^1(\Gamma_1(M), L_n(K))$ is free of rank 2 over the Hecke algebra $h_{n+2}(\Gamma_1(M); K)$ for each positive integer M", quoted from Shimura [25, Th. 3.51 and Th. 8.4]; in the proof $V^-(R) \cong V^+(R) \cong h_{n+2}(\Gamma_1(M); R)$): for a field K and $M \geq 1$ — $M = 1$ included — $H_P^1(\Gamma_1(M), L_n(K))$ is free of rank 2 over the Hecke algebra $h_{n+2}(\Gamma_1(M); K)$, and each of Hida's eigenspaces $V^\pm(K)$ of the involution $[\varepsilon]$ is free of rank 1 over it; the matrix ε is garbled in the extracted text and is not transcribed (VERIFY): it normalises $\Gamma_1(M)$ and has determinant -1, and for the two candidates $\begin{pmatrix}-1&0\\0&1\end{pmatrix}$, $\begin{pmatrix}1&0\\0&-1\end{pmatrix}$ the operators agree for n even ($-I$ acts by $(-1)^n$)].* *At $M = 1$ the cited statement is classical (Eichler–Shimura and multiplicity one at level one, [C, `obl:oracle`]), so nothing here depends on the hypotheses of Hida's §4 (v0.5.0-pre.2, F4 m7). Let $k \geq 3$ lie in the class a. Then $\mathbf{h}_k$ acts on $\mathsf{S}_k^\pm(\mathbb{Z}_\ell)$ (through $T_p \mapsto T_p$, $U \mapsto U$), $\mathbf{T}_k(a)$ acts on $\mathsf{H}_k^+(\mathbb{Z}_\ell)$ and $\mathbf{h}_k$ on $\mathsf{H}_k^-(\mathbb{Z}_\ell)$, and over $\overline{\mathbb{Q}}_\ell$:*

$$\mathsf{S}_k^\pm \otimes \overline{\mathbb{Q}}_\ell \cong \bigoplus_{f \in \mathcal{E}_k} \overline{\mathbb{Q}}_\ell\, \varphi_f^\pm, \qquad \mathsf{H}_k^+ \otimes \overline{\mathbb{Q}}_\ell = (\mathsf{S}_k^+ \otimes \overline{\mathbb{Q}}_\ell) \oplus \overline{\mathbb{Q}}_\ell\, \mathcal{E}_k, \qquad \mathsf{H}_k^- \otimes \overline{\mathbb{Q}}_\ell = \mathsf{S}_k^- \otimes \overline{\mathbb{Q}}_\ell,$$

$\varphi_f^{\pm}$ *an eigenvector with the eigensystem of* f *(each* $f \in \mathcal{E}_k$ *exactly once in each sign),* $\mathcal{E}_k$ *the Eisenstein class (*$T_p \mapsto 1 + p^{k-1}$, $U \mapsto 1$*). So* $\operatorname{rank} \mathsf{S}_k^{\pm} = d_k$, $\operatorname{rank} \mathsf{H}_k^{+} = d_k + 1$, $\operatorname{rank} \mathsf{H}_k^{-} = d_k$, $\sum = h^1_{\mathrm{ord}}(n)$ *(Lemma 2.3(3)).*

Proof. Over $\mathbb{Q}_\ell$, $L_n \otimes \mathbb{Q} = V \otimes \mathbb{Q}$ through D ([C, `rmk:stabconvention`](3)), Hecke- and ι-equivariantly. Hida's Lemma 6.4 at $M = 1$, $K = \overline{\mathbb{Q}}_\ell$: each sign of $H^1_!$ is free of rank one over $h_k(\mathrm{SL}_2(\mathbb{Z}); \overline{\mathbb{Q}}_\ell) = \overline{\mathbb{Q}}_\ell^{\{\text{eigenforms}\}}$ (multiplicity one at level 1), so every level-one eigenform occurs once in each sign; e_1 keeps the f with $v(a_\ell(f)) = 0$, i.e. $\mathcal{E}_k$, with $U \mapsto \alpha_f$ (Lemma 2.3(1)). For either candidate ε, $[\varepsilon] = \pm\iota^*$ ([C]'s $\iota = \mathrm{diag}(-1, 1)$ and the overall sign of [C, `rmk:iotasign`]); renaming the two eigenspaces does not change "once in each sign". $H^1/H^1_! \subseteq H^1(\partial)$ is, after e_1 and $\otimes\mathbb{Q}$, the Eisenstein line ([C, `Hi5`]; E_k is ordinary), which lies in H^1_+ ([C, `rmk:iotasign`]). The algebra statements: $\mathsf{S}_k^{\pm}$ and $\mathsf{H}_k^{\pm}$ are $\mathbb{Z}_\ell$-free ([C, `rmk:ordtorsionfree`]), so the image of $\mathbb{Z}_\ell[T_p, U]$ in their endomorphisms embeds into the product of the eigencharacters that occur, which is $\mathbf{h}_k$ resp. $\mathbf{T}_k(a)$ by their definition (Definition 2.6, Lemma 2.3(1)). □

Lemma 14.3 (The ordinary part does not see the lattice)**.** *[proved here] from [C,* `def:hecke`*] ([C,* `eq:heckechain`*]), [C,* `prop:symconv`*](2), [C,* `prop:levelcorrespondence`*], [C,* `def:bdryop`*], [C,* `prop:ordinary`*](3) and [E,* `prop:lvlattice`*] (the* Δ*-stability of* V'_n*). Put* $Q_n := V_\ell / V'_n \cong \bigoplus_t \big(\mathbb{Z}_\ell/\binom{n}{t}\big) u_t$*, a finite* Δ*-module.*

1. *For every* $A \in \mathrm{Mat}_2(\mathbb{Z})$ *with* $|\det A| = \ell^N$, $N \geq m_n$*:* $\rho_n(A) V_\ell \subseteq V'_n$*, i.e.* $\rho_{Q_n}(A) = 0$*.*
2. $T_\ell^{m_n} = 0$ *on* $H^q(\Gamma; Q_n)$ *and on* $H^q(\partial; Q_n)$ *for every* q*, and* $(U_\ell^{(\ell)})^{m_n} = 0$ *on* $H^q(\Gamma_0(\ell); Q_n)$*. Hence* e_1 *and* e_ℓ *kill all of them.*
3. *The inclusion* $V'_n \subseteq V_\ell$ *induces isomorphisms* $e_1 H^q(\Gamma; V'_n) \cong e_1 H^q(\Gamma; V_\ell)$*,* $e_1 H^q(\partial; V'_n) \cong e_1 H^q(\partial; V_\ell)$*, hence* $e_1 H^1_!(\Gamma; V'_n) \cong \mathsf{S}_k(\mathbb{Z}_\ell)$*, and the same at level* ℓ *with* e_ℓ*; all of them equivariant for every* T_p*, for* U *and for* ι^**. With* D*:* $e_1 H^1_!(\Gamma; L_n(\mathbb{Z}_\ell)) \cong \mathsf{S}_k(\mathbb{Z}_\ell)$*, Hida's lattice against the contract's.*
4. *Consequently, for every* $f \in \mathcal{E}_k$ *the change-of-lattice index of [E,* `prop:lvlattice`*](2) vanishes:* $d_f^{\pm} = 0$*, and* $c_f^{\pm} = e\, v_\ell(n!)$*.*

Proof. (1) Write $A = \gamma\, \mathrm{diag}(d_1, d_2)\, \gamma'$ with $\gamma, \gamma' \in \mathrm{GL}_2(\mathbb{Z})$ and $d_1 \mid d_2$, $|d_1 d_2| = \ell^N$ (Smith). V'_n is stable under $\rho_n(B)$ for *every* $B \in \mathrm{Mat}_2(\mathbb{Z})$, determinant < 0 included: it is the image of the divided powers $\Gamma^n\mathbb{Z}^2 \to \mathrm{Sym}^n\mathbb{Z}^2$ ([C, `rmk:stabconvention`](3)), a natural transformation of functors of $\mathbb{Z}^2$, so every endomorphism of $\mathbb{Z}^2$ preserves the image (this is more than the Δ-stability of [E, `prop:lvlattice`], which covers $\det > 0$ only; v0.5.0-pre.2, F4 m1). So $\rho_n(\gamma)$, $\rho_n(\gamma')$ are automorphisms of V_ℓ preserving V'_n, $\rho_n(A) V_\ell = \rho_n(\gamma)\rho_n(\mathrm{diag}(d_1, d_2)) V_\ell$, and it suffices to treat $\mathrm{diag}(d_1, d_2)$. If $\ell \mid d_1$, $\rho_n(\mathrm{diag}(d_1, d_2))$ is divisible by ℓ^n and $n \geq m_n$ ($\binom{n}{t} < \ell^n$). Otherwise $d_1 = \pm 1$, $d_2 = \pm\ell^N$, and $u_t \mapsto \pm\ell^{Nt} u_t$; for $t = 0$ the component of Q_n is 0 ($\binom{n}{0} = 1$), for $t \geq 1$, $Nt \geq m_n \geq v\binom{n}{t}$. (2) By [C, `eq:heckechain`] the N-fold composite of the cochain map $\widetilde{T}$ of $T_{\Gamma g \Gamma}$, $g = \mathrm{diag}(1, \ell)$, sends ϕ to a $\mathbb{Z}$-linear combination of terms $\rho(\delta_{j_1}) \cdots \rho(\delta_{j_N})\, \phi(c) = \rho(\delta_{j_1} \cdots \delta_{j_N})\, \phi(c)$ with c a chain and $\det(\delta_{j_1} \cdots \delta_{j_N}) = \ell^N$; with coefficients in Q_n and $N = m_n$ every term vanishes by (1). At level ℓ the blocks of [C, `prop:levelcorrespondence`] are ρ_n of matrices of determinant ℓ, on the boundary those of [C, `def:bdryop`]; the same argument applies. $e_{\mathrm{ord}} = \lim U^{j!}$ is then 0. (3) The long exact sequences of $0 \to V'_n \to V_\ell \to Q_n \to 0$ on Γ and on ∂ are sequences of finitely generated $\mathbb{Z}_\ell$-modules, T_ℓ-equivariant ($\widetilde{T}$ is natural in the coefficient Δ-module, so it commutes with the maps of the short exact sequence of cochain complexes and with the connecting maps). On a finite exact complex of finitely generated $\mathbb{Z}_\ell$-modules with a compatible endomorphism T, the image B of $\mathbb{Z}_\ell[X]$ ($X \mapsto T$) in the product of the endomorphism rings is a finite commutative $\mathbb{Z}_\ell$-algebra, a product of local rings, and $e := \lim T^{j!} \in B$ is the idempotent of the factors in which T is a unit; multiplication by an idempotent of B is exact on complexes of B-modules, and $eM = e_{\mathrm{ord}} M$ in the sense of [C, `rmk:ordtorsionfree`] (the finite levels, [C, `prop:ordinary`](3), and the limit; v0.5.0-pre.2, F4 m2). So e_1 is exact on both long exact sequences; the Q_n-terms vanish

by (2). The two isomorphisms commute with restriction to ∂, so they identify the kernels $H^1_!$. Every operator here is a formula in the $\rho_n(\delta)$ and integral chain data, so it commutes with the inclusion of Δ-modules; ι acts diagonally on the u_t. For the last sentence: D is an isomorphism of $\mathrm{Mat}_2(\mathbb{Z})$-modules $L_n(\mathbb{Z}_\ell) \cong V_n'$, so it identifies $H^1_!(\Gamma; L_n(\mathbb{Z}_\ell))$ with $H^1_!(\Gamma; V_n')$ compatibly with [C]'s operators; Hida's $T(p)$ and $T(\ell)$ at level 1 (the action through the main involution, [C, `rmk:stabconvention`](2)) agree with [C]'s T_p, T_ℓ on $H^1(\Gamma; L_n(\mathbb{Z}_\ell)) \otimes \mathbb{Q}$ (both act on the f-line by $a_p(f)$: Eichler–Shimura on either side), hence on the torsion-free $e_1 H^1_!(\Gamma; L_n(\mathbb{Z}_\ell))$ and on every Hecke-module isomorphism class read on it (v0.5.0-pre.2, F4 m3). (4) $d_f^{\pm}$ is the index of $H^1(\Gamma; V_n')[f]^{\pm}$ in $H^1(\Gamma; V_\ell)[f]^{\pm}$; for f ordinary both f-lines lie in the e_1-parts (an f-line of the free quotient lies in the direct summand $e_1(H/\mathrm{tors})$, the complement being torsion-free), which coincide by (3); and $e_1 H^1(\Gamma; V_n')$ is torsion-free by (3), so it is the ordinary part of its free quotient as in [C, `rmk:ordtorsionfree`](2). Then [E, `prop:lvlattice`](2). □

Remarks on Lemma 14.3. (a) It upgrades slot-13 WP-5's finding (Hi8)(c) — $D_*\colon M_m^{\mathrm{H}}(n) \to M_m(n)$ an isomorphism at 45/45 cells including the 31 non-Lucas ones (slot-13 WP-5 report, (Hi8)(c) row) — to a theorem, and it resolves the caution of [C, `rmk:stabconvention`](3) ("lattice-dependent: the isomorphism class of a finite-level module") and of Remark 8.3 ("nothing is transferred") *for ordinary parts*; neither text is edited here, and whether [C] records it is the owner's call (a [C] erratum candidate, for the referee). (b) Checkable consequence: the five eigenlines at $(40, 37)$ with $d_f = 1$ ([E, `rmk:lvlatticevalues`]) are non-ordinary. (c) The lemma is about ordinary parts only; non-ordinary blocks do see the lattice ([E, `rmk:lvlatticevalues`]: $d_f = 1$ on ω^{b_j}-twists). (d) (v0.5.0-pre.2, F4.) Two consistency checks already in the record: $d_f = 0$ on both signs at the $\mathfrak{m}_{\mathrm{E}}$-blocks of $(282, 37)$, $(226, 103)$, $(258, 37)$ ([E, `rmk:lvlatticevalues`]; the cusp forms there are ordinary), and the slot-13 (Hi8)(c) count of (a). Not a contradiction of a statement of [C]: [C, `rmk:stabconvention`](3) classifies the finite-level datum as *lattice-dependent* in the sense "not shown independent"; item (3) shows it independent on ordinary parts. The literature transfer of Remark 8.3 now applies to the contract's lattice on ordinary parts (the sentence added there).

Proposition 14.4 (Multiplicity one at étale components, finite weight)**.** *[proved here] from Lemmas 14.2, 14.3, 3.4, Theorem 4.4, [C, `prop:bdrychar`], [C, `cl:chi`], [C, `cor:ordprecision`](1) and [C, `prop:stabilisation`]. Let $k \geq 3$ lie in the class a and let $\mathfrak{m}$ be a maximal ideal of $\mathbf{h}_k$ with $\mathbf{h}_{k,\mathfrak{m}}$ étale over $\mathbb{Z}_\ell$, so $\mathbf{h}_{k,\mathfrak{m}} \cong W(\mathbb{F}_q)$, $q = \ell^r$, and r eigenforms of $\mathcal{E}_k$ lie in $\mathfrak{m}$ (Theorem 4.4(2)).*

1. *$\mathsf{S}_k^{\pm}(\mathbb{Z}_\ell)_{\mathfrak{m}}$ is free of rank one over $\mathbf{h}_{k,\mathfrak{m}}$, for both signs. Hence $\mathsf{S}_k^{\pm}(\mathbb{Z}/\ell^K)_{\mathfrak{m}}$ is free of rank one over $W_K(\mathbb{F}_q)$, and for every $t \in \mathbf{h}_{k,\mathfrak{m}}$ with $\mathbf{h}_{k,\mathfrak{m}} = \mathbb{Z}_\ell[t]$ (e.g. PΥ's T_{p_0}) it is cyclic over $(\mathbb{Z}/\ell^K)[t]$, with annihilator the characteristic polynomial of t on $\mathbf{h}_{k,\mathfrak{m}}$. The same holds on Hida's lattice and at level ℓ (Lemma 14.3(3), [C, `prop:stabilisation`](3),(4)).*
2. ***(The boundary away from the Eisenstein ideals.)** Let $\mathfrak{m}_b := (\ell,\ T_p - p^b - p^{n+1-b} : p \neq \ell)$ for $b \in \mathbb{Z}/(\ell-1)\mathbb{Z}$, the ideals of the boundary characters χ_b of [C, `cl:chi`] ($\mathfrak{m}_0$ is the Eisenstein ideal; $\mathfrak{m}_b = \mathfrak{m}_{b'}$ iff $b' \equiv b$ or $b' \equiv n+1-b$, [C, `cl:chi`](3)). If $\mathfrak{m} \neq \mathfrak{m}_b$ for* every *$b \in \mathbb{Z}/(\ell-1)\mathbb{Z}$ — condition **(BF)** — then $H^q(\partial; V_\ell \otimes R)_{\mathfrak{m}} = 0$, and $\mathsf{S}_k^{\pm}(R)_{\mathfrak{m}} = \mathsf{H}_k^{\pm}(R)_{\mathfrak{m}} = e_1 H^1_c(\Gamma; V_\ell \otimes R)^{\pm}_{\mathfrak{m}} = e_1 H^1(\Gamma; V_\ell \otimes \mathbb{Z}/\ell^K)^{\pm}_{\mathfrak{m}}$; on H^1_c it is read on the modular symbols $W(R)$ of [E, `rmk:lvshapes`] (B).*
3. ***(The cuspidal component through $\mathfrak{m}_{\mathrm{E}}$.)** Let $a \not\equiv 0$ and suppose the cuspidal component through $\mathfrak{m}_{\mathrm{E}}$ is étale of residue degree one (as at the four classes of (U13), Prediction 13.11), so $\mathbf{T}_{k,\mathfrak{m}_{\mathrm{E}}} = \mathbb{Z}_\ell \times_{\mathbb{Z}/\ell^{L_k}} \mathbb{Z}_\ell$. Then (1) holds for $\mathsf{S}_k^{\pm}(\mathbb{Z}_\ell)_{\mathfrak{m}}$, both signs. On H^1, $\mathsf{H}_k^{+}(\mathbb{Z}_\ell)_{\mathfrak{m}_{\mathrm{E}}} \cong \mathbb{Z}_\ell \times_{\mathbb{Z}/\ell^j} \mathbb{Z}_\ell$ for a unique $j = j_k \in [0, L_k]$, and it is free of rank one over $\mathbf{T}_{k,\mathfrak{m}_{\mathrm{E}}}$ iff $j_k = L_k$. On H^1_c, the boundary symbol $p_0 = u_0 - u_n$ spans the kernel of $e_1 H^1_c(\mathbb{Z}_\ell) \to e_1 H^1(\mathbb{Z}_\ell)$ (it lies in the minus part of [C], geometric-even, [E, `rmk:lvshapes`] (B)); if the H^1_c-primitive cuspidal symbol r of that sign satisfies $r \equiv u\, p_0 \pmod{\ell^d}$ exactly, u a unit*

([E, `rmk:lvlegc`] (L1)), then the class of $w := (r - u\,p_0)/\ell^d$ generates $\mathsf{S}_k^-(\mathbb{Z}_\ell)_\mathfrak{m}$ and the image of r is ℓ^d times a generator.

Proof. (1) $M := \mathsf{S}_k^\pm(\mathbb{Z}_\ell)_\mathfrak{m}$ is $\mathbb{Z}_\ell$-free and $\mathbf{h}_k$ acts on it through $\mathbf{h}_{k,\mathfrak{m}}$ (Lemma 14.2); $\mathbf{h}_{k,\mathfrak{m}} = W(\mathbb{F}_q)$ is a discrete valuation ring and M is finitely generated and torsion-free over it, hence free, of rank $\mathrm{rank}_{\mathbb{Z}_\ell} M/r = r/r = 1$ (the r eigenforms of $\mathfrak{m}$, once each in each sign). Modulo ℓ^K by base change; cyclicity is the case $\mathbf{h}_{k,\mathfrak{m}} = \mathbb{Z}_\ell[t] \cong \mathbb{Z}_\ell[Y]/(\text{charpoly})$. (2) $H^q(\partial; V_\ell \otimes R)_\mathfrak{m} = 0$ for both q iff $H^q(\partial; V_\ell \otimes \mathbb{F}_\ell)_\mathfrak{m} = 0$ for both q (dévissage along the sequences $0 \to V/\ell^i \to V/\ell^{i+1} \to V/\ell \to 0$, all q at once, and Nakayama for $R = \mathbb{Z}_\ell$). On $H^q(\partial; V_\ell \otimes \mathbb{F}_\ell)$, $q = 0, 1$, the T_p^∂, $p \neq \ell$, are simultaneously diagonalisable and every joint eigencharacter is some χ_b, $b \in \mathbb{Z}/(\ell-1)\mathbb{Z}$ — [C, `lem:bdrytorus`](2),(3): the eigenvalue on a torus weight vector of weight $a^{n-j}d^j$ is $\chi_j(p)$ in degree 0 and $\chi_{n-j}(p)$ in degree 1 — with [C, `prop:bdrychar`](1) for the convention. Which b occur is [C, `cl:chi`](1) (degree 0: exactly $0 \le b \le \lfloor n/\ell \rfloor$, [proved here]) and [C, `cl:chi`](2) (degree 1: the same multiset, [conjectural]), which is why (BF) quantifies over all of $\mathbb{Z}/(\ell-1)\mathbb{Z}$ and not over $0 \le b \le \lfloor n/\ell \rfloor$ (v0.5.0-pre.2, F4 M1: through pre.1 the range was $0 \le b \le \lfloor n/\ell \rfloor$, which rests on the [conjectural] half of [C, `cl:chi`](2)); so under (BF) the $\mathfrak{m}$-parts of the boundary terms of the sequence of the pair vanish, and H_c^1, $H_!^1$, H^1 agree after localisation. $e_1 H^1(\Gamma; V_\ell \otimes \mathbb{Z}/\ell^K) = e_1 H^1(\Gamma; V_\ell)/\ell^K$ is [C, `cor:ordprecision`](1). (3) $\mathsf{S}_k^\pm$ carries only cuspidal eigencharacters (Lemma 14.2), so the argument of (1) applies to its $\mathfrak{m}_\mathrm{E}$-part. On $\mathsf{H}^+_{k,\mathfrak{m}}$ the two eigenlines are the cuspidal and the Eisenstein one; with A_1, A_2 the intersections of the lattice with them, the lattice modulo $A_1 \oplus A_2$ is cyclic (a subgroup of $(\mathbb{Q}_\ell/\mathbb{Z}_\ell)$ through either projection), $\cong \mathbb{Z}/\ell^j$, and after rescaling one coordinate by a unit the lattice is $\{(x, y) : x \equiv y\ (\ell^j)\}$; it is stable under $(\ell^L, 0) \in \mathbf{T}_{k,\mathfrak{m}}$ iff $j \le L$, and j is an invariant of the $\mathbf{T}_{k,\mathfrak{m}}$-module (the eigenlines are intrinsic); $j = L$ is $\mathbf{T}_{k,\mathfrak{m}}$ itself (Lemma 3.4). For H_c^1: over $\mathbb{Z}_\ell$, $H^0(\Gamma; V_\ell) = 0$ ($n \ge 1$) and $H^0(\partial; V_\ell) = \mathbb{Z}_\ell u_0$ with T_p-eigenvalue $1 + p^{k-1}$ and U-eigenvalue 1 ([C, `prop:bdrychar`](3)), whose image is $\mathbb{Z}_\ell p_0$ ([E, `rmk:lvlegc`]); if $r - u\,p_0 = \ell^d w$, then $r \mapsto \ell^d[w]$, and $[w]$ is not divisible by ℓ since r is not $\equiv$ a multiple of p_0 modulo ℓ^{d+1}. □

Definition 14.5 (Hida's Λ-adic modular symbols)**.** Let $V^0(a)$ be the a-part of Hida's Λ-module V^0 of [C, `thm:hidacontrol`](1) (the Pontryagin dual of the ordinary part of $\varinjlim_r H^1(X_r, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$, i.e. of the parabolic cohomology; [Hid86a, Thm. 3.1, p. 562], Λ-free of rank $2d(a)$ by [Hid86a, p. 589, Step II]), with the Hecke operators of Hida's universal algebra and his involution $[\varepsilon]$. Here "the a-part" is the component of V^0 on which $\mathbf{h}(a)$ acts — in Hida's natural indexing by the tame diamond character this is $V^0(a-2)$ ("$h_2^0(a)$ acts on $\mathcal{V}^0(a-2)$", [Hid86a, p. 593], read on the page for part 5): the weight-2 forms of the class a have nebentypus ω^{a-2}. The Λ-structure on it is Hida's twisted one ($1 + X \mapsto u^2\langle u\rangle$ on the tower, Definition 15.1), so that P_k is the weight-k point. Labelling only: every statement of this part uses $\mathbf{M}(a)$ as the module $\mathbf{h}(a)$ acts on (text v0.2 / [U] 0.6.0-pre.1, F-U5). Put

$$\mathbf{M}(a) := \mathrm{Hom}_\Lambda\big(V^0(a), \Lambda\big), \qquad \mathbf{M}(a) = \mathbf{M}(a)^+ \oplus \mathbf{M}(a)^- \ (2 \in \Lambda^\times),$$

with the transposed operators; for a maximal ideal $\mathfrak{m}$ of $\mathbf{h}(a)$, $\mathbf{M}(a)^\pm_\mathfrak{m}$; and $\mathbf{M}_{K,m}(a)^\pm := \mathbf{M}(a)^\pm \otimes_\Lambda \Lambda_{K,m}$, $\mathbf{M}_{K,m}(a; x)^\pm$ likewise (Definition 13.1).

Proposition 14.6 (Control for $\mathbf{M}(a)$)**.** *[proved here] from [literature: [Hid86a, Thm. 3.1, p. 562; Lemma 4.6, p. 571; Prop. 4.7, p. 572; Thm. 6.6, p. 590], transcribed with verified locators in [C, `thm:hidacontrol`], [C, `thm:hidaweight`]; Prop. 4.7 (including its sentence "the isomorphism in each direction is explicitly given by $e \circ \mathrm{res}$ and $e_0 \circ \mathrm{Tr}$, which preserve parabolic cohomology classes") and Thm. 6.6 read on the page for this text;* **and the normalisation of Hida's weight prime, read on the page by the slot-21 referee F4 (v0.5.0-pre.2; VERIFY lifted)***: [Hid86a, p. 545, Introduction] — "the evaluation of power series in Λ at the point $\varepsilon(u)u^k - 1$ ($u = 1+p \in \Gamma$) gives an algebra homomorphism of Λ into Ω for each finite order character ε of Γ and each integer k. Let P_{k-2} be the prime ideal of Λ which is the kernel of this morphism" — so Hida's P_n with*

$\varepsilon = 1$ *is the kernel of* $X \mapsto u^{n+2} - 1$, *i.e.* $P_n = P_{n+2} = P_k$ *in the notation of this document, the index shift being built into Hida's notation; [C,* `thm:hidacontrol`*](2) does not restate it] and Lemma 14.3. For* $k \geq 3$ *in the class there are isomorphisms, compatible with* $T(p) \mapsto T_p$ $(p \neq \ell)$ *and* $U \mapsto U$, *carrying the two* $[\varepsilon]$*-eigenspaces onto the two* ι^**-eigenspaces (in some order, fixed for the class):*

$$\mathbf{M}(a)/P_k\mathbf{M}(a) \;\cong\; eH^1_P(\Phi_1; L_n(\mathbb{Z}_\ell)) \;\cong\; e_0H^1_P(\mathrm{SL}_2(\mathbb{Z}); L_n(\mathbb{Z}_\ell)) \;\cong\; \mathsf{S}_k(\mathbb{Z}_\ell).$$

Proof. Thm. 6.6 gives $\mathcal{V}^0(a)[P_n] \cong eH^1_P(\Phi_1; L_n(\mathbb{Z}_\ell)) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell$ as Hecke modules; Pontryagin duality turns it into $V^0(a)/P_nV^0(a) \cong \mathrm{Hom}_{\mathbb{Z}_\ell}(eH^1_P(\Phi_1; L_n(\mathbb{Z}_\ell)), \mathbb{Z}_\ell)$ (the latter module is $\mathbb{Z}_\ell$-free, Lemma 4.6). As $V^0(a)$ is Λ-free, $\mathbf{M}(a)/P_n = \mathrm{Hom}_{\mathbb{Z}_\ell}(V^0(a)/P_n, \mathbb{Z}_\ell)$, the double dual, i.e. $eH^1_P(\Phi_1; L_n(\mathbb{Z}_\ell))$ with its own operators. Prop. 4.7 ($r = 1$, $N = 1$, parabolic version) gives the second isomorphism; at level 1 Hida's $T(p)$ is this document's T_ℓ ([C, `def:hidalattice`], variant (H): $\Gamma \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) \Gamma \;=\; \Gamma \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma$). The third is D and Lemma 14.3(3). The U-compatibility: on the image of level one, Hida's U and [C]'s $U_\ell^{(\ell)}$ both act as the unit root ([C, `rmk:stabconvention`](4), [C, `prop:stabilisation`](3)). Restriction, trace and D commute with $\varepsilon = \iota$. □

Theorem 14.7 (Λ-adic multiplicity one at étale components)**.** *[proved here] from Proposition 14.6, Proposition 14.4(1), Theorem 4.4(1) and Nakayama. Let* $\mathfrak{m}$ *be a maximal ideal of* $\mathbf{h}(a)$ *with* $\mathbf{h}(a)_\mathfrak{m}$ *étale over* Λ, *so* $\mathbf{h}(a)_\mathfrak{m} \cong W(\mathbb{F}_q)[[X]]$, $q = \ell^r$.

1. $\mathbf{M}(a)^\pm_\mathfrak{m}$ *is free of rank one over* $\mathbf{h}(a)_\mathfrak{m}$, *for both signs.*
2. *For all* $K, m \geq 1$ *and* $x \in \ell\mathbb{Z}_\ell$: $\mathbf{M}_{K,m}(a)^\pm_\mathfrak{m} \cong \mathbf{h}_{K,m}(a)_\mathfrak{m} \cong W_K(\mathbb{F}_q)[X]/(X^m)$ *and* $\mathbf{M}_{K,m}(a; x)^\pm_\mathfrak{m} \cong \mathbf{h}_{K,m}(a; x)_\mathfrak{m}$.
3. *A generator* $\Phi^\pm$ *of (1) — a* Λ***-adic ordinary eigensymbol*** *of* $(\mathfrak{m}, \pm)$ — *specialises at every* P_k, $k \geq 3$ *in the class, to an ordinary eigensymbol* $\varphi_k^\pm$ *of* $(\mathfrak{m}_k, \pm)$ *in the sense of Definition 14.1,* $\mathfrak{m}_k$ *the image of* $\mathfrak{m}$; *two generators differ by* $\mathbf{h}(a)^\times_\mathfrak{m}$.

Proof. First, $\mathbf{h}(a)$ acts on $\mathbf{M}(a)$: the Λ-subalgebra of $\mathrm{End}_\Lambda(\mathbf{M}(a))$ generated by the $T(p)$, $p \neq \ell$, and U embeds into $\prod_k \mathrm{End}(\mathbf{M}(a)/P_k) = \prod_k \mathrm{End}(\mathsf{S}_k(\mathbb{Z}_\ell))$ over the $k \geq 3$ of the class ($\mathbf{M}(a)$ is Λ-free and $\bigcap_k P_k\Lambda = 0$), hence into $\prod_k \mathbf{h}_k$ (Lemma 14.2), as the Λ-subalgebra generated by the tuples $(T_p)_k$, $(U)_k$ with X acting as $(x_k)_k$; so does $\mathbf{h}(a)$, with the same image (Theorem 2.2, the argument of Proposition 13.5(1)); so the two are identified, $T(p) \mapsto T(p)$, $U \mapsto U$, Λ-linearly (v0.5.0-pre.2, F4 m4). $\mathbf{h}(a)$ is finite over the complete local Λ, so $\mathbf{M}(a)^\pm = \bigoplus_\mathfrak{m} \mathbf{M}(a)^\pm_\mathfrak{m}$ and each summand is Λ-free (a direct summand of the Λ-free $\mathbf{M}(a)$, over a local ring). Its Λ-rank is the $\mathbb{Z}_\ell$-rank of its specialisation at one P_k, which is $\mathsf{S}_k^\pm(\mathbb{Z}_\ell)_{\mathfrak{m}_k}$ (Proposition 14.6) of rank r (Proposition 14.4(1)). $\mathbf{M}^\pm_\mathfrak{m}/\mathfrak{m}_\Lambda$ is a module over $\mathbf{h}(a)_\mathfrak{m}/\mathfrak{m}_\Lambda = \mathbb{F}_q$ of $\mathbb{F}_\ell$-dimension r, hence one-dimensional; by Nakayama over the local ring $\mathbf{h}(a)_\mathfrak{m}$ it is cyclic, $\mathbf{h}(a)_\mathfrak{m} \twoheadrightarrow \mathbf{M}^\pm_\mathfrak{m}$, a surjection of free Λ-modules of the same rank r, hence an isomorphism. (2) is base change; (3) Proposition 14.6 and (1). □

Lemma 14.8 (The ball masses on S_k, and their $\mathbf{h}$-valued form)**.** *[proved here] from [E,* `def:lvmtt`*] and [E,* `prop:lvmtt`*](1),(2),(6) (their proofs are linear in the symbol). For* $x \in e_1H^1_c(\Gamma; V_\ell)$ *with symbol* Φ_x *([E,* `rmk:lvshapes`*] (B)),* $c_x(a, m)$ *as in [E,* `def:lvmtt`*], put*

$$\mu_x\big(a + \ell^m\mathbb{Z}_\ell\big) := c_{U^{-m}x}(a, m) \;-\; \ell^{k-2}\, c_{U^{-m-1}x}(a, m-1) \qquad (m \geq 0),$$

the formula of [E, `def:lvmtt`*] with the unit root* α *replaced by the operator* U *of Definition 14.1 — on* $e_1H^1_c$ *itself* U *is the unique unit root of* $Y^2 - T_\ell Y + \ell^{k-1}$ *in the closure of* $\mathbb{Z}_\ell[T_\ell] \subseteq \mathrm{End}(e_1H^1_c)$, *which exists by Hensel's lemma exactly as in Lemma 2.3(1) (*T_ℓ *is a unit on the ordinary part, the root* $Y = T_\ell$ *of* $Y(Y - T_\ell)$ *is simple modulo the radical of this finite* $\mathbb{Z}_\ell$*-algebra), whether or not* $e_1H^1_c$ *is torsion-free; it induces the* U *of Definition 14.1 on* $e_1H^1_!$ *and acts as* 1 *on* p_0 *(v0.5.0-pre.2, F4 m5).*

1. *μ_x is a $\mathbb{Z}_\ell$-valued distribution on $\mathbb{Z}_\ell$, $\mathbb{Z}_\ell$-linear in x, with μ_{tx} for $t \in \mathbf{h}_k$ given by t acting on x; on a T_ℓ-eigenclass it is [E, `def:lvmtt`]'s μ_x.*
2. *On compact open sets inside $\mathbb{Z}_\ell^\times$ it vanishes on $\mathbb{Z}_\ell p_0$, so it descends to $\mathsf{S}_k^\pm(\mathbb{Z}_\ell)$ there.*
3. *For $\mathfrak{m}$ as in Proposition 14.4 and a compact open $B \subseteq \mathbb{Z}_\ell^\times$ there is a unique $\mathbf{h}_{k,\mathfrak{m}}$-linear $\mu^{\mathbf{h}}(B)\colon \mathsf{S}_k^\pm(\mathbb{Z}_\ell)_\mathfrak{m} \to \mathbf{h}_{k,\mathfrak{m}}$ with $\mathrm{Tr}_{\mathbf{h}_{k,\mathfrak{m}}/\mathbb{Z}_\ell}\big(c\,\mu_x^{\mathbf{h}}(B)\big) = \mu_{cx}(B)$ for all c; it is the **h-valued mass** ($= \mu_x(B)$ when $r = 1$).*

Proof. (1) Linearity of [E, `prop:lvmtt`](1) in the symbol gives $\sum_b c_x(a + b\ell^m, m+1) = c_{T_\ell x}(a,m) - \ell^{k-2}c_x(a, m-1)$ for every x, and $c_y(a,m)$ depends on $a \bmod \ell^m$; so $\sum_b \mu_x(a + b\ell^m, m+1) = c_{(T_\ell U^{-1} - \ell^{k-1}U^{-2})U^{-m}x}(a,m) - \ell^{k-2}c_{U^{-m-1}x}(a, m-1) = \mu_x(a + \ell^m\mathbb{Z}_\ell)$, as $T_\ell = U + \ell^{k-1}U^{-1}$. U commutes with $\mathbf{h}_k$; on an eigenclass $U = \alpha$. (2) U acts as 1 on p_0 (its T_ℓ-eigenvalue is $1 + \ell^{k-1}$), and [E, `prop:lvmtt`](6) with $\alpha = 1$ gives $D_{p_0}^{(1)}(a,m) = 0$ for $m \geq 1$; the kernel of $e_1H_c^1 \to e_1H_!^1$ is $\mathbb{Z}_\ell p_0$ (proof of Proposition 14.4(3)). (3) $\mathbf{h}_{k,\mathfrak{m}} = W(\mathbb{F}_q)$ is étale over $\mathbb{Z}_\ell$, so its trace form is perfect and $M \cong \mathbf{h}_{k,\mathfrak{m}}$ (Proposition 14.4(1)); $\mu^{\mathbf{h}}$ is the image of the $\mathbb{Z}_\ell$-linear $c \mapsto \mu_{cx}(B)$ under $\mathrm{Hom}_{\mathbb{Z}_\ell}(W, \mathbb{Z}_\ell) \cong W$. □

Proposition 14.9 (Normalisation of the Λ-adic eigensymbol)**.** *[proved here] (elementary). In the situation of Theorem 14.7, let $\lambda\colon \mathbf{M}(a)_\mathfrak{m}^\pm \to \Lambda$ be Λ-linear and $\lambda^{\mathbf{h}}$ its $\mathbf{h}(a)_\mathfrak{m}$-linear form ($\mathrm{Tr}_{\mathbf{h}/\Lambda}(c\lambda^{\mathbf{h}}(x)) = \lambda(cx)$; $\mathbf{h}(a)_\mathfrak{m}$ is finite étale over Λ).*

1. *λ is **admissible** if $\lambda^{\mathbf{h}}(\Phi) \in \mathbf{h}(a)_\mathfrak{m}^\times$ for one, equivalently every, generator Φ; equivalently $\lambda^{\mathbf{h}} \not\equiv 0$ on the one-dimensional $\mathbb{F}_q$-space $\mathbf{M}_\mathfrak{m}^\pm/\mathfrak{m}\mathbf{M}_\mathfrak{m}^\pm$ — **the unit check modulo** (ℓ, X). Then $\Phi^\lambda := \lambda^{\mathbf{h}}(\Phi)^{-1}\Phi$ is the unique generator with $\lambda^{\mathbf{h}}(\Phi^\lambda) = 1$.*
2. *If λ specialises at P_k to $\lambda_k\colon \mathsf{S}_k^\pm(\mathbb{Z}_\ell)_{\mathfrak{m}_k} \to \mathbb{Z}_\ell$ (through Proposition 14.6), then Φ^λ specialises to $\Phi_k^\lambda := \lambda_k^{\mathbf{h}}(\varphi_k)^{-1}\varphi_k$ for any eigensymbol φ_k — the normalisation "$\lambda_0(P_k) = 1$" — and admissibility is decided at one weight: $\lambda_k^{\mathbf{h}}(\varphi_k) \in W(\mathbb{F}_q)^\times$ at one k iff at every k.*
3. *For any second such λ', $\lambda'^{\mathbf{h}}(\Phi^\lambda) \in \mathbf{h}(a)_\mathfrak{m}$ does not depend on Φ, and its specialisation at P_k is the* period-free *ratio $\lambda_k'^{\mathbf{h}}(\varphi_k)/\lambda_k^{\mathbf{h}}(\varphi_k) \in W(\mathbb{F}_q)$, for any φ_k.*

Proof. $\mathbf{M}_\mathfrak{m}^\pm = \mathbf{h}_\mathfrak{m}\Phi$ (Theorem 14.7), $\lambda^{\mathbf{h}}(t\Phi) = t\lambda^{\mathbf{h}}(\Phi)$, and an element of the local ring $\mathbf{h}_\mathfrak{m}$ is a unit iff it is non-zero modulo $\mathfrak{m}$; $\mathfrak{m}$ contains $\mathfrak{m}_\Lambda$ and the specialisation $\mathbf{h}_\mathfrak{m} \to \mathbf{h}_{k,\mathfrak{m}_k}$ is local, so "unit" is read at any one weight. The trace form commutes with base change along $\Lambda \to \Lambda/P_k$. (3): the ratio is $\mathbf{h}_\mathfrak{m}^\times$-invariant. □

The normaliser of record (the convention of P8; Prediction 14.10). $\lambda_0 := \mu^{\mathbf{h}}(B_0)$ of Lemma 14.8(3) for the first ball $B_0 = b_0 + \ell^{M_0}\mathbb{Z}_\ell \subseteq \mathbb{Z}_\ell^\times$ in the order $M_0 = 1, 2, \ldots,$ $b_0 = 1, \ldots, \ell^{M_0} - 1$ with $\ell \nmid b_0$, at which $\mu_{\varphi_c}^{\mathbf{h}}(B_0) \not\equiv 0 \pmod{\mathfrak{m}}$ at the centre weight c; refused by name (`no_unit_ball`) if none exists up to $M_0 = M_{\max}$ (default 2). That λ_0 is the specialisation of a Λ-adic functional — what Proposition 14.9(2) needs — is (U16), not proved here. On the sign of S_k that carries p_0 on H_c^1 at $\mathfrak{m}_\mathrm{E}$ the symbol to use is w (Proposition 14.4(3)); the H_c^1-primitive r has all masses $\equiv 0 \pmod{\ell^d}$ on $\mathbb{Z}_\ell^\times$ ([E, `rmk:lvmeasplus`](ii)) and would fail the unit check.

Prediction 14.10 ((U15)–(U17): Λ-adic modular symbols at the slot-21 cells). (New at v0.5.0-pre.1; stated before slot-21 P8 runs (rule 20).) Cells: the étale cuspidal components of P7 — $(\ell, a) = (11, 2)$ (the Δ family), $(13, 0)$, $(23, 2)$ (residue degree 2), and the components through $\mathfrak{m}_\mathrm{E}$ at $(37, 32)$, $(103, 24)$ — with $K = 3$, $m = 1, 2, 3$, $N = K + m - 1$ consecutive weights $k_0 + j(\ell - 1)$ of the class computed modulo ℓ^{K+m-1} (Theorem 13.4(2)), and at least three further weights; at $\ell \in \{5, 7\}$ every class has $\mathsf{S}_k = 0$ (Theorem 7.1).

(U15) [proved here] (Lemma 14.2, Proposition 14.4(1),(2), Proposition 13.5(4)) **The match with P7 at eigenvector level.** For P7's component of residual factor g of T_{p_0} ($\mathfrak{m} = (\ell, g(T_{p_0}))$, (BF) checked), at every computed weight k and both signs: $\mathsf{S}_k^\pm(\mathbb{Z}/\ell^{K'})_\mathfrak{m}$ is free of rank r over $\mathbb{Z}/\ell^{K'}$ and cyclic over T_{p_0}; its characteristic polynomial is P7's $F(x_k, Y)$; and $T_p = P_p(x_k, T_{p_0})$, $U = \alpha(x_k, T_{p_0})$ as operators on it, for every prime $p \leq B$, $p \neq \ell$ — for

$r = 1$: $T_p\varphi_k = a_p(x_k)\varphi_k$, $U\varphi_k = \alpha(x_k)\varphi_k$ — modulo $\ell^{K'}$ against P7's forms data at the weight itself, and modulo $\ell^{\min(K,\, m\, v(x_k - x_c))}$ against P7's truncation centred at x_c. At level ℓ (cells with $\ell \le 13$): $U_\ell^{(\ell)} = \alpha(x_k, T_{p_0})$ on $e_\ell H^1(\Gamma_0(\ell); V_\ell \otimes \mathbb{Z}/\ell^{K'})_{\mathfrak{m}}$, which has rank $2r$. A failure is a defect of the code or of P7.

(U16) [conjectural] **Λ-adic coherence of the normalised measures.** For every compact open $B \subseteq \mathbb{Z}_\ell^\times$ there is $\nu_B^\pm \in \mathbf{h}(a)_{\mathfrak{m}}$ with $\nu_B^\pm(x_k) = \mu^{\mathbf{h}}_{\Phi_k^\pm}(B)$ for every $k \ge 3$ in the class, $\Phi_k^\pm$ the eigensymbol normalised by λ_0 — i.e. the normalised measures of the weights are the specialisations of one $\mathbf{h}(a)_{\mathfrak{m}}$-valued measure, and λ_0 is admissible in the sense of Proposition 14.9. *Checkable form*: (a) the unit check holds at every computed weight (Proposition 14.9(2)); (b) for each ball of level $M \le 2$, the coordinates of $\mu^{\mathbf{h}}_{\Phi_k}(B)$ in the basis $1, T_{p_0}, \dots, T_{p_0}^{r-1}$, interpolated from the N weights (Theorem 13.4), predict every further weight modulo $\ell^{\min(K', N)}$ (Theorem 13.4(3)); (c) the distribution relation holds for the normalised family at every weight (Lemma 14.8(1), a code check). *Expected literature*: the Λ-adic modular-symbol measure of Greenberg–Stevens [GS93] and Kitagawa's construction (not held; `docs/implementation/slot21/missing_literature.md` LIT-08). v0.5.0-pre.3: Greenberg–Stevens **read** (rendered from the image-only scan, `docs/implementation/slot21/gs_reading_notes.md`): Prop. 6.1 (W^0 Λ-free, $\Phi_\kappa = 0 \iff \Phi \in P_\kappa W^0$), Thm. 5.13 (control after localising at an arithmetic point, in characteristic 0) and (5.5), (5.10), Thm. 5.15 give a *weaker* form of (U16): a local, characteristic-0, meromorphic coherence of the period-free ratios $\mu(B)/\mu(B_0)$; the integral $\mathbf{h}(a)_{\mathfrak{m}}$-form with a unit normaliser is *not* in Greenberg–Stevens. The tag stays [conjectural]. *Partial reason, not a proof*: for balls of level 1 and modulo ℓ, $c_x(a, 1)$ is the value at $\{a/\ell, \infty\}$ of the image of x under $P \mapsto P(0,1)$, which is $\Gamma_0(\ell)$-equivariant modulo ℓ into $\mathbb{F}_\ell$ with the character $d \mapsto d^n$, and [literature: [Hid86a, Cor. 4.5, p. 570], read on the page: $l_n \circ \mathrm{res}\colon eH^1(\Phi_1, L_n(\mathbb{Z}/p^r\mathbb{Z})) \cong eH^1(\Phi_r, \mathbb{Z}/p^r\mathbb{Z}(n))$, also for parabolic cohomology] makes the ordinary part at $r = 1$ depend on n only modulo $\ell - 1$; transporting Hida's $T(p) = U^*$ ([C, `rmk:stabconvention`](2)) to this document's U is not carried out, so the tag stays [conjectural]. At level M the same mechanism needs $\Phi_M = \Gamma_0(\ell^M)$ — where the Λ-action lives (P10).

(U17) [conjectural] (for (a): [proved here] from [standard:] at the weights with $n < \ell - 1$, see below) **The Eisenstein gluing on H^1, and the reduced symbol.** At $(37, 32)$ and $(103, 24)$, at every computed weight: (a) $j_k = L_k$ in Proposition 14.4(3), i.e. $\mathsf{H}_k^+(\mathbb{Z}_\ell)_{\mathfrak{m}_{\mathrm{E}}}$ is free of rank one over $\mathbf{T}_{k,\mathfrak{m}_{\mathrm{E}}}$ (evidence: the cuspidal symbol of the other geometric sign is $\equiv s\, P^{\mathrm{E}} \pmod{\ell^d}$ to depth exactly d, [E, `cl:lvmeascong`](2), and $\delta = 0$ at $(10, 691)$, Proposition 8.2; expected from Harder's denominator of the Eisenstein class, [Har87], not read: to be confirmed); (b) on S_k^- at $\mathfrak{m}_{\mathrm{E}}$ the reduced symbol w passes the unit check (evidence: the reduced plus measure has $\mu = 0$ at $(30, 37)$, $(282, 37)$, $(226, 103)$, [E, `rmk:lvmeasplus`](iii), an observation there). *What (a) is* (v0.5.0-pre.2, F4 M3). With $M := \mathsf{H}_k^+(\mathbb{Z}_\ell)_{\mathfrak{m}_{\mathrm{E}}}$, $A_1 = M \cap (\text{cusp line}) = \mathsf{S}_k^+(\mathbb{Z}_\ell)_{\mathfrak{m}_{\mathrm{E}}}$ ($H^1_! = \ker \mathrm{res}$ is saturated in H^1) and $A_2 = \mathbb{Z}_\ell E$, E the primitive Eisenstein class, the proof of Proposition 14.4(3) gives $\ell^{j_k} = [M : A_1 \oplus A_2]$. Since e_1 is a direct-summand projector of the free module $H^+ := H^1(\Gamma; V_\ell)^+/\mathrm{tors}$ with $E \in e_1 H^+$ and $(1 - e_1)H^+ \subseteq H^+_!$ ($H^+/H^+_! \otimes \mathbb{Q}$ is the Eisenstein line), and the other $\mathfrak{m}$-blocks of $e_1 H^+$ lie in $H^+_!$ whenever $H^1(\partial; V_\ell)$ is torsion-free — true for $n < \ell$ ([C, `prop:bdrychar`](1): V_R^T has rank 1) — one has $\ell^{j_k} = [H^+ : H^+_! \oplus \mathbb{Z}_\ell E]$ for $n < \ell$: **(a) is the index statement of [E, `cl:lveis`](2) (Harder's denominator of the Eisenstein class) restricted to the ordinary $\mathfrak{m}_{\mathrm{E}}$-block**, with $v(B_k/k) = L_k$ by (U1). So (a) is [proved here] from [standard:] ([E, `cl:lveis`](2), [Har22], VERIFY there) for $n < \ell - 1$ — at the cells $(30, 37)$ ($k = 32$), $(22, 103)$ ($k = 24$) and $(10, 691)$ — and stays [conjectural] for $n \ge \ell - 1$ exactly as [E, `cl:lveis`](2) does (the P8 weights $k \ge 68$ at 37, $k \ge 126$ at 103); and $\delta(\mathfrak{m}_{\mathrm{E}}) = 0$ of Proposition 8.2 is *equivalent* to $j_k = L_k$ (on M the element $t = (\ell^L, 0)$ maps into ℓM iff $j_k < L$, and on the rank-one minus block t

acts as ℓ^L), so (a) is [verified computationally] at $(10, 691)$ as well. For the code, G11 of `P8_spec.md` is a *defect* at $k = 32$, $k = 24$ and a *finding* at the other weights.

Evidence (v0.5.0-pre.3; [verified computationally], no tag moves). **P8** (`HeckeLambdaModSymValidation`, commit `96cdaaac`): at the four default classes 1126 checks and 179 finding gates, 0 failing, 0 findings fired — (U15) at every computed weight; (U16)(a) the unit check and (U16)(b) the prediction of every further weight; (U17)(a) $j_k = L_k$ at all 8 weights of 37 ($j_k = 2$ at $k = 284$); env arm `HECKE_LAMBDA_MSYM_ALL=1` (adds $(103, 24)$, weights to $k = 738$; $j_k = L_k$ there too, $L = 2$ at $k = 228$, as read in [RC] 0.1.0-pre.5's gluing table) 1830 checks and 290 finding gates, 0 failing (v0.5.0-pre.4). The referee F4's recommended $(691, 12)$ (`P8_spec.md` §2, S5) is *not* in the committed env arm. **P9** (`HeckeLambdaTwoVarValidation`, [E, `rmk:lvtwovar`]): the interpolated masses predict three further weights modulo ℓ^5 at all four classes; at $(11, 2)$ the masses predicted at x_2 equal 11a's own normalised MTT masses on every unit ball to level 3 modulo 11^5 ((U16) extended to $k = 2$, with P8's normaliser of record $2 + 11\mathbb{Z}_{11}$, resp. $3 + 11\mathbb{Z}_{11}$); at $(37, 32)$ the two-variable Eisenstein cross products vanish at $X = \alpha_{32}$ modulo 37^3 (the full Kubota–Leopoldt precision), and at $(103, 24)$ modulo 103^2 at $X = \alpha_{24}$ (env arm). **P3** (`HeckeLValueCoefficientRingValidation`, [E, `rmk:lvcoeffring`]): at $\ell = 29$ and the residually non-split component $\bar{g} = X^2 + 22X + 15$ the level-2 unit-ball measures of $k = 52, 80, 108$ are proportional modulo $\mathfrak{m}$ to that of $k = 24$ by a scalar in $\mathbb{F}_{29}^\times$ (0/812 balls mismatching) — (U16) modulo $\mathfrak{m}$ at $r = 2$ outside P8's cells. None of this proves (U16) or (U17)(b). **P2, P4** (v0.5.0-pre.5; [E, `prop:lveismeas`], [E, `rmk:lvwild`]): the exact Eisenstein measure and wild characters are computed at *single* weights only — no evidence for (U16) or (U17); their bearing on part 3 is the two remarks after Lemma 13.6. Wild characters along a family (Remark 14.11, P10) remain not built.

Remark 14.11 (What part 4 rests on, and what it does not do). **[literature: .]** Hida's Λ-freeness and control of the ordinary parabolic cohomology (Invent. Thm. 3.1, Thm. 6.6), the level comparison (Prop. 4.7), multiplicity one over fields (Lemma 6.4): Proposition 14.6, Theorem 14.7, Lemma 14.2. **VERIFY.** Only the matrix ε of Lemma 6.4 (garbled; both candidates give the same operator); the normalisation of Hida's P_n is read on the page since v0.5.0-pre.2 (p. 545: $P_{k-2} = \ker(X \mapsto u^k - 1) = P_k$). **[proved here] here, elementary.** Lemma 14.3, Proposition 14.4, Lemma 14.8, Proposition 14.9, (U15). **[conjectural].** (U16); (U17)(b); (U17)(a) for $n \geq \ell - 1$ ((U17)(a) is [proved here] from [standard:] for $n < \ell - 1$: [E, `cl:lveis`](2) on the ordinary block). **Not used.** Any Gorenstein property beyond étaleness; any Galois representation; any main conjecture; Greenberg–Stevens and Kitagawa (cited only as the expected source of (U16)). **What a weight is not.** $\mathsf{S}_k^{\pm}(\mathbb{Z}/\ell^K)$ is one point of $\mathbf{M}(a)/(\ell^K, P_k)$: at level $\Gamma_0(\ell)$ the cohomology carries no Λ-action, so the X-direction comes only from several weights, through Theorem 13.4 and a normalisation — the generators φ_k are each defined up to $W(\mathbb{F}_q)^\times$, and the Λ-adic datum is exactly which tuple of them is the image of one Φ; Proposition 14.9 fixes it by λ_0, provided (U16). **Refused by name.** A non-étale component (residually non-squarefree T_{p_0} on $e\,S_k$ at every $p_0 \leq B$: P7's refusal), where $\mathbf{M}_{\mathfrak{m}}$ need not be free and a "two-variable function" is a Fitting ideal, not a function; a component failing (BF) on H^1_c (other than $\mathfrak{m}_{\mathrm{E}}$, handled by Proposition 14.4(3)); $\Gamma_1(\ell^r)$ and wild characters (P10) — see part 5, §15.

Code map of part 4. Built (v0.5.0-pre.3; slot-21 P8, commit `96cdaaac`, after F4's ACCEPT WITH REPAIRS of this part: `HeckeLambdaModSymValidation`, `standard`, ≈ 24 s, 0.29 GB, 1126 checks and 179 finding gates, record `hecke_lambda_msym/v1`; and slot-21 P9, `HeckeExt/HeckeLambdaTwoVar`, `HeckeLambdaTwoVarValidation`, `standard`, ≈ 7.5 s, 71 checks and 41 finding gates, record `hecke_lambda_twovar/v1`, [E, `rmk:lvtwovar`]). *The plan, as written at v0.5.0-pre.1*: **Planned** (slot-21 P8; Opus; only after the Fable referee's ACCEPT of this part, rule 22; implementation spec `docs/implementation/slot21/P8_spec.md`). A new `HeckeExt/HeckeLambdaModSym` on top of P7's `HeckeExt/HeckeLambdaQExp` (`BuildLambdaQExpansion`, `SpecialiseAt`, `FactorSquarefreeModEll`) and WP-HL's

`ComputeFormsWeight`, with the symbols, T_p, ι and the measure of slot-20 WP-MEAS / WP-LIN (`HeckeExt/LValue/BirchModularSymbol`, `ModularSymbolMeasure`); the level-ℓ arm on `Hecke/HeckeLevelEll`; record `hecke_lambda_msym/v1` (FNV per line, not pinned, no golden line); validation `HeckeLambdaModSymValidation` (`standard`) gating (U15) and the checkable forms of (U16), (U17), with the lattice check of Lemma 14.3 at small cells and its consequence at $(40, 37)$. No statement of this part rests on the code.

15 The tower $\Gamma_1(\ell^r)$, wild characters, and the Ω-type truncations (part 5)

Status and scope of this part. Written as a pre-release (v0.6.0-pre.1, **not signed**) for the slot-22 Fable referee F-U5, before any code of slot-22 P10 (rules 20–22). Parts 3 and 4 obtain the Λ-adic algebra and the Λ-adic modular symbols of a class from finitely many weights, by interpolation (Theorem 13.4): "at level $\Gamma_0(\ell)$ the cohomology carries no Λ-action, so the X-direction comes only from several weights" (Remark 14.11). This part reads the Λ-action where Hida defines it — on the ordinary cohomology of the tower $\Gamma_1(\ell^r)$, through the diamond operators — at *one* level and weight 2: (i) the Λ-structure on the tower, Hida's $\omega_{2,r}$ and the truncations $\Lambda/(\ell^K, \Omega_r) \cong (\mathbb{Z}/\ell^K)[G_r]$ (Definition 15.1, Lemma 15.2); (ii) the computed objects, on the tame-character groups Γ_r^{t} (Definition 15.3); (iii) the dictionary between the diamonds and the classes (Proposition 15.4, [verified computationally] at $r = 1$ and gated); (iv) Hida's control read on these objects: freeness over the group ring, the tower, the Hecke algebra modulo (ℓ^K, Ω_r), the weight fibres, the wild points (Theorem 15.5, [proved here] from [literature:] Hida); (v) the Eisenstein family in closed form (Proposition 15.6); (vi) multiplicity one and the computed structure constants $a_p^{(r)}$ at étale rank-one components, and the X-adic truncations they contain *without interpolation* (Proposition 15.7, Corollary 15.8); (vii) the predictions (U18)–(U22) (Prediction 15.9). Tame level 1, $\ell \geq 5$, as in part 1; $n = 0$ (weight 2) throughout — Hida's theorem makes weight 2 sufficient. *Not in this part*: tame level $N > 1$ (slot-23 P11); components glued to the Eisenstein component ($L_k > 0$, e.g. $(37, 32)$) and non-étale components (refused by name, Remark 15.10); Kitagawa's measure is a stretch with its own [conjectural] (Remark 15.11).

Notation of this part. $u = 1 + \ell$, $x_k = u^k - 1$, $P_k = X - x_k$ (part 3). $\mathsf{H}_r := \mu_{\ell-1}(\mathbb{Z}/\ell^r) \leq G_{\ell^r} = (\mathbb{Z}/\ell^r)^\times$, the Teichmüller subgroup, and $G_r := G_{\ell^r}/\mathsf{H}_r \cong (1 + \ell\mathbb{Z})/(1 + \ell^r\mathbb{Z})$, cyclic of order ℓ^{r-1}, generated by the class g of u. $\Gamma_r^{\mathrm{t}} := \Gamma_{\mathsf{H}_r}(\ell^r)$ of [C, `def:nebgammaH`] ($\Gamma_1^{\mathrm{t}} = \Gamma_0(\ell)$, as $\mathsf{H}_1 = G_\ell$). For an even class a, $\chi_a := \omega^{a-2}$, viewed as a character of G_{ℓ^r} through $(\mathbb{Z}/\ell^r)^\times \to \mathbb{F}_\ell^\times$ and the Teichmüller lift. For $p \neq \ell$, $s(p) \in \mathbb{Z}_\ell$ with $\langle p \rangle_\ell = u^{s(p)}$ (as in Definition 2.4). O_j, $\mathcal{O}_{j,e}$, π, ψ_t, $\chi_{i,t}$ are those of [C, `def:nebwild`].

Definition 15.1 (The Λ-structure on the tower)**.** For $r \geq 1$ put $\Omega_r(X) := (1+X)^{\ell^{r-1}} - u^{2\ell^{r-1}} \in \Lambda$ and $\Lambda_K^{(r)} := \Lambda/(\ell^K, \Omega_r)$. On a $\mathbb{Z}_\ell[G_r]$-module M (here: G_r acting by the diamonds of [C, `prop:nebgammaH`](4)), the **tower Λ-structure** is

$$\iota_r \colon \Lambda \to \mathrm{End}(M), \qquad 1 + X \mapsto u^2\langle u \rangle.$$

Lemma 15.2 (Ω_r and the group ring)**.** *[proved here] (elementary; Hida's notation [literature: [Hid86a, p. 552–553], read on the page for this text: $P_\chi = \iota(u) - \chi(u)$, $P_{k,\varepsilon}$ for $\chi(\gamma) = \gamma^k\varepsilon(\gamma)$, $\omega_{k,r} = \prod_\varepsilon P_{k,\varepsilon}$ over the characters ε of Γ/Γ_r])*

1. $\Omega_r = \prod_\varepsilon \big((1 + X) - \varepsilon(u)u^2\big)$ *over the* ℓ^{r-1} *characters* ε *of* G_r — *Hida's* $\omega_{2,r}$ — *and* $\Omega_r \equiv X^{\ell^{r-1}} \pmod{\ell}$*: a distinguished polynomial of degree* ℓ^{r-1}*.*
2. ι_r *induces* $\Lambda/\Omega_r \cong \mathbb{Z}_\ell[G_r]$ *and* $\Lambda_K^{(r)} \cong (\mathbb{Z}/\ell^K)[G_r]$*, local with residue field* $\mathbb{F}_\ell$*;* $1, X, \ldots, X^{\ell^{r-1}-1}$ *is a* $\mathbb{Z}/\ell^K$*-basis (Weierstrass division by* Ω_r*). The* ***Weierstrass representative*** *of an element is its expression in this basis.*

3. *For $r' \le r$, $\Omega_{r'} \mid \Omega_r$, and $\Lambda/\Omega_{r'} = \mathbb{Z}_\ell[G_r]/(g^{\ell^{r'-1}} - 1)$.*
4. *For an integer k: $\Omega_r(x_2) = 0$ and $v(\Omega_r(x_k)) = r + v(k-2)$ for $k \neq 2$. Hence $\Lambda/(\ell^K, \Omega_r, P_k) = \mathbb{Z}/\ell^{K_r(k)}$, $K_r(k) := \min(K, r + v(k-2))$ ($K_r(2) = K$).*
5. *For $c \in \mathbb{Z}$ with $v(c-2) = 0$ and $m \ge 1$: $\Omega_r \in (\ell^K, (X - x_c)^m)$ iff $K \le r$ when $m = 1$, and iff $m \le \ell^{r-1}$ and $K \le r - 1 - \lfloor \log_\ell(m-1) \rfloor$ when $m \ge 2$. The same holds for (ℓ^K, X^m) ($c = 0$). In that case $\Lambda_{K,m}(x_c)$ (resp. $\Lambda_{K,m}$) of Definition 13.1 is a quotient of $\Lambda_K^{(r)}$.*

Proof. (1) $\prod_{\zeta^{\ell^{r-1}}=1}(Y - \zeta u^2) = Y^{\ell^{r-1}} - u^{2\ell^{r-1}}$ at $Y = 1 + X$; modulo ℓ, $(1+X)^{\ell^{r-1}} \equiv 1 + X^{\ell^{r-1}}$ and $u \equiv 1$. (2) $g \mapsto u^{-2}(1+X)$ sends $g^{\ell^{r-1}} - 1$ to $u^{-2\ell^{r-1}}\Omega_r$; Weierstrass preparation for the distinguished Ω_r. (3) The roots of $\Omega_{r'}$ are among those of Ω_r; $g^{\ell^{r'-1}} - 1 \mapsto$ a unit times $\Omega_{r'}$. (4) $\Omega_r(x_k) = u^{2\ell^{r-1}}(u^{(k-2)\ell^{r-1}} - 1)$ and $v(u^m - 1) = 1 + v(m)$. (5) The Taylor coefficients of Ω_r at x_c are $\binom{\ell^{r-1}}{i}(1+x_c)^{\ell^{r-1}-i}$ ($1 \le i \le \ell^{r-1}$, of valuation $r - 1 - v(i)$ for $i < \ell^{r-1}$ and 0 at $i = \ell^{r-1}$) and $\Omega_r(x_c)$, of valuation $r + v(c-2) = r$; Ω_r lies in $(\ell^K, (X - x_c)^m)$ iff the coefficients of order $< m$ have valuation $\ge K$. [verified computationally] check: `docs/implementation/slot22/p10c/p10c_predict.py` §4, the criterion against the definition at $\ell \in \{5, 7, 11, 13\}$, $r \le 3$, $K \le 4$, $m \le \ell^{r-1} + 2$: 0 mismatches. □

Definition 15.3 (The computed objects)**.** Let a be an even class, $r, K \ge 1$, $R \in \{\mathbb{Z}_\ell, \mathbb{Z}/\ell^K\}$. On the Γ_r^{t}-model of [C, `def:nebgammaH`] with character χ_a, $n = 0$, let T_p ($p \neq \ell$), $U := U_\ell$ and $\langle d \rangle$ ($d \in G_r$) be the operators of [C, `prop:nebgammaH`](4),(5) (on H^1 by [C, `prop:nebgammaHheilbronn`]), $e := \lim_j U^{j!}$, and

$$\mathsf{M}_r(a; R) := e\, H^1(\Gamma_r^{\mathrm{t}}; R(\chi_a)),$$

a Λ-module through ι_r and a module over the Hecke operators. For $b \in \mathbb{Z}/(\ell - 1)$ let $\mathfrak{m}_b$ be the residual **boundary systems** $T_p \mapsto \omega^b(p) + \omega^{a-2-b}(p)\, p$ ($p \neq \ell$), $U \mapsto 1$; $\mathsf{M}_r^{\mathrm{E}}$ is the sum of the generalised eigenspaces of the $\mathfrak{m}_b$ in M_r and $\mathsf{M}_r^{\mathrm{S}}$ the sum of the others (condition (BF′) of a component $\mathfrak{m}$: $\mathfrak{m} \neq \mathfrak{m}_b$ for every b). $\mathbf{h}_r(a; K)$ is the Λ-subalgebra of $\mathrm{End}(\mathsf{M}_r^{\mathrm{S}}(a; \mathbb{Z}/\ell^K))$ generated by the T_p and U (with Λ acting through ι_r), and $a_p^{(r)}$, $\alpha^{(r)}$ denote the elements of $\Lambda_K^{(r)}$ by which T_p, U act when they act by scalars (Proposition 15.7).

Proposition 15.4 (The dictionary: diamonds, classes, and the orientation of U)**.** *Let a be an even class, $r \ge 1$, $\psi\colon G_r \to \mathcal{O}^\times$ a character. On $eH^1(\Gamma_0(\ell^r); \mathcal{O}(\chi_a\psi))$ — by [C, `lem:nebshapiro`](a) the ψ-specialisation of $\mathsf{M}_r(a; \cdot)$ — the Hecke systems (T_p, $p \neq \ell$; U) are those of the U-ordinary weight-2 eigenforms of level ℓ^r and character $\chi_a\psi$ in the convention $f|_2\sigma_d = \chi_a\psi(d)f$, $U \mapsto a_\ell$ — the normalisation in which $E_2(1, \chi)$ has $a_p = 1 + \chi(p)p$, $a_\ell = 1$, and $E_2(\chi, 1)$ has $a_p = \chi(p) + p$, $a_\ell = \ell$, so that "the ordinary Eisenstein line carries $1 + \chi(p)p$" is a statement about [C]'s model alone (v0.6.0-pre.2, F-U5 m3); their residual systems are those of the class-a ordinary forms of level one (for $k \ge 3$ in the class: $T_p \equiv a_p(f_k)$), together with the Eisenstein system $1 + \chi_a\psi(p)\, p$, $U = 1$.* ***Status.*** *[proved here] for the T_p-systems at $r = 1$ from [C, `NB4`](c) (the untwisted constituent Sym^{a-2} lies in the ω^{a-2}-part); [verified computationally] for the U-orientation at $r = 1$ (`docs/implementation/slot22/p10c/p10c_orient.py`, a transcription of [C, `def:cosetrepsN`], [C, `eq:MHchi`], [C, `def:kappaN`] and [C, `rmk:nebheilbronn`], no project code: at $(\ell, \omega^i) \in \{(5, \omega^2), (7, \omega^2), (7, \omega^4), (11, 1), (11, \omega^2), (13, \omega^{10}), (13, \omega^2), (13, \omega^4)\}$ the U-ordinary part of the ω^i-part has rank 1, or 3 at $(11, 1)$ and $(13, \omega^{10})$, its Eisenstein line carries $T_p = 1 + \omega^i(p)p$ and, at the seven cells with $\omega^i \neq 1$ where the two differ, never $\omega^i(p) + p$, and at $(13, \omega^{10})$ the cusp system $T_2 \equiv \tau(2)$, $T_3 \equiv \tau(3) \pmod{13}$ is ordinary while at $(13, \omega^2)$ the cusp form is not);* ***gated*** *on the project code at every default cell and every r by (U18)(o) — a failure is a defect of* this text *and stops P10 before any Λ-adic reading.*

Remark on the orientation (why it has to be checked, not assumed; reworded at v0.6.0-pre.2, referee F-U5 m1–m3). [C, `rmk:stabconvention`](2): Hida's $T(p)$ at level p^r is this document's

U^*, not U_ℓ; and Hida lets $\sigma \equiv \left(\begin{smallmatrix} * & * \\ 0 & z \end{smallmatrix}\right)$ act on cohomology through the main involution. On the $\Gamma_1(\ell^r)$ cohomology three conventions move *together* under either the passage $\alpha \mapsto \operatorname{adj}\alpha$ or conjugation by the Atkin–Lehner element: $U \leftrightarrow U^*$, $\langle d\rangle \leftrightarrow \langle d\rangle^{-1}$, and $T_p \leftrightarrow T_p\langle p\rangle^{\mp 1}$ for $p \neq \ell$ ($\Gamma_1 \left(\begin{smallmatrix} 1 & 0 \\ 0 & p \end{smallmatrix}\right) \Gamma_1$ and $\Gamma_1 \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) \Gamma_1$ are distinct double cosets, differing by a diamond; on forms, $a_p(f|W) = \bar\chi(p)a_p(f)$). So no formal cancellation argument fixes the dictionary; under the *swapped* dictionary the ordinary cusp system would sit on the χ_a^{-1}-part twisted by $\chi_a^{-1}(p)$ (at $\ell = 13$: on ω^2 instead of ω^{10}, with $T_p \equiv p^2\tau(p)$), and the ordinary Eisenstein line would read $\chi(p) + p$ or $1 + \bar\chi(p)p$. The computation decides: in [C]'s ω^i-character model the U_ℓ-ordinary Eisenstein line carries $1 + \omega^i(p)p$ — the system of $E_2(1, \omega^i)$, whose $a_\ell = 1$ is a unit, while $E_2(\omega^i, 1)$ has $a_\ell = \ell$ — and the untwisted $\tau(p)$ is ordinary on ω^{10} at 13 (on ω^0 at 11); i.e. *[C]'s χ-model is the cohomology of the forms of nebentypus χ in the classical normalisation $f|_2\sigma_d = \chi(d)f$, and [C]'s U_ℓ is the classical U*: the class a sits on the $\chi_a = \omega^{a-2}$-part of [C]'s U (Serre's untwisted congruence $f_k \leftrightarrow$ weight 2, nebentypus ω^{k-2}), matching Hida's "$h_2^0(a)$ acts on $\mathcal{V}^0(a-2)$" [literature: [Hid86a, p. 593], read on the page]. Only the Eisenstein half of this is provable from the boundary; the cusp half (which constituent of [C, NB4](c) is U_ℓ-ordinary) is outside (NB4)(c)'s monoid (determinant prime to ℓ) and stays [verified computationally] + gated.

Theorem 15.5 (Hida's control, read on the tower)**.** *Let a be an even class, $d := d(a) = \operatorname{rank}_\Lambda \mathbf{h}(a)$ (Theorem 2.2), $r \geq 1$, $K \geq 1$.*

1. ***(Freeness.)*** *$\mathsf{M}_r(a; \mathbb{Z}_\ell)$ is free over $\Lambda/\Omega_r \cong \mathbb{Z}_\ell[G_r]$ of rank $2d+1$, and $\mathsf{M}_r(a; \mathbb{Z}/\ell^K) = \mathsf{M}_r(a; \mathbb{Z}_\ell)/\ell^K$ is free over $\Lambda_K^{(r)}$ of rank $2d+1$. If the cuspidal and Eisenstein components of $\mathbf{T}(a)$ are disjoint ($L_k = 0$; every class at $\ell \leq 13$, Theorem 7.1) then $\mathsf{M}_r = \mathsf{M}_r^{\mathrm{S}} \oplus \mathsf{M}_r^{\mathrm{E}}$ with $\mathsf{M}_r^{\mathrm{S}}$ free of rank $2d$ and $\mathsf{M}_r^{\mathrm{E}}$ free of rank 1.*
2. ***(The tower.)*** *For $1 \leq r' \leq r$: $\mathsf{M}_r(a; R)/(g^{\ell^{r'-1}} - 1) \cong \mathsf{M}_{r'}(a; R)$ as modules over the T_p, U and the diamonds (ranks, characteristic polynomials, and the structure constants $a_p^{(r)} \mapsto a_p^{(r')}$ of Proposition 15.7: what (U18)(b) tests). The left side is $eH^1(\Gamma_{\mathsf{H}'}(\ell^r); R(\chi_a))$ by [C, `lem:nebshapiro`](b); the cohomological map realising the isomorphism in [C]'s U-convention is $e \circ \mathrm{dg}_{\ell^{r-r'}}$, not $e \circ \mathrm{res}$ (Hida's Cor. 4.5 is stated for $e(T(p)) = e(U^*)$ and transports under W_{ℓ^r}, which normalises $\Gamma_1(\ell^{r'}) \cap \Gamma_0(\ell^r)$ but not $\Gamma_1(\ell^{r'})$, to the other degeneracy, exactly as in [C, `rmk:stabconvention`](2) at $r' = 0$); that map is not written in [C] for Γ_{H} with a character and is* not *a statement of this text (v0.6.0-pre.2, F-U5 M2).*
3. ***(The Hecke algebra.)*** *$\mathbf{h}(a) \to \mathrm{End}(\mathsf{M}_r^{\mathrm{S}}(a; \mathbb{Z}/\ell^K))$ ($T(p) \mapsto T_p$, $U \mapsto U$, Λ through ι_r) factors through $\mathbf{h}(a)/(\ell^K, \Omega_r)$ and is onto $\mathbf{h}_r(a; K)$; at every component $\mathfrak{m}$ with $\mathbf{h}(a)_\mathfrak{m}$ étale, $\mathbf{h}_r(a; K)_\mathfrak{m} \cong \mathbf{h}(a)_\mathfrak{m}/(\ell^K, \Omega_r)$.*
4. ***(Weight fibres.)*** *At such a component, for every integer k in the class: the characteristic polynomial of T_p on $\mathbf{h}_r(a; K)_\mathfrak{m} \otimes_\Lambda \Lambda/P_k$ is $c_{k,p}$ restricted to $\mathfrak{m}$ (Lemma 2.3(2); for $k = 2$ the level-ℓ weight-2 ordinary forms) modulo $\ell^{K_r(k)}$ (Lemma 15.2(4)); likewise U against the unit roots.*
5. ***(Wild points.)*** *For $\psi = \psi_t \colon G_r \to \mu_{\ell^j}$ ($j \leq r-1$) and $e \leq K\varphi_j$: $\mathsf{M}_r(a; \mathbb{Z}/\ell^K) \otimes_{\Lambda_K^{(r)}} \mathcal{O}_{j,e}$, $1 + X \mapsto u^2\zeta^t$, is $eH^1(\Gamma_0(\ell^r); \mathcal{O}_{j,e}(\chi_a\psi_t))$, Hecke-equivariantly, free of rank $2d+1$ over $\mathcal{O}_{j,e}$.*

Status. *(1)–(3) [proved here] from [literature: [Hid86a, Thm. 3.1(i)–(iii), p. 562; Cor. 4.5, p. 570; Thm. 1.2 and the display after it, p. 553; Lemma 4.6, p. 571; the twist, p. 593], all read on the page for this text and re-read on the page images by the referee F-U5], [standard: the ordinary weight-2 forms of level ℓ^r whose character has conductor $\leq \ell^{r'}$ are old from level $\ell^{r'}$: a newform of level ℓ^r, $r \geq 2$, with character of conductor dividing ℓ^{r-1} has $a_\ell = 0$ [Miy89, Thm. 4.6.17] VERIFY], [C, `lem:nebshapiro`], [C, `lem:nebwildtors`] and Theorem 2.2 — **given Proposition 15.4** (the forms-side dictionary "class $a \leftrightarrow$ nebentypus ω^{a-2}, untwisted, U-ordinary" on [C]'s model:*

[verified computationally] at $r=1$, gated by (U18)(o); every Hida convention is the W_{ℓ^r}-conjugate of [C]'s, so the rank $2d+1$ of the χ_a-part rests on it; v0.6.0-pre.2, F-U5 M3); (4) [proved here] from (3) and Lemma 15.2(4); (5) [proved here] from (1) and [C, `lem:nebshapiro`](a).

Proof. (1) Let Y_r be the open modular curve of $\Gamma_1(\ell^r)$ and $\mathcal{W}^0_r := eH^1(Y_r;\mathbb{Q}_\ell/\mathbb{Z}_\ell)$, W^0 the Pontryagin dual of $\varinjlim_r \mathcal{W}^0_r$. By Hida's Thm. 3.1(ii) W^0 is Λ-free, the group $Z \supseteq \Gamma = 1+\ell\mathbb{Z}_\ell$ acting through $\Gamma_0(\ell^r)\to(\mathbb{Z}/\ell^r)^\times$, $\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\mapsto d$ (p. 562); by Thm. 3.1(i) $\mathcal{W}^0_r = (\varinjlim \mathcal{W}^0)^{\Gamma_r}$, so $W^0/(\gamma_r-1)W^0$, $\gamma_r = u^{\ell^{r-1}}$, is the dual of $\mathcal{W}^0_r$ and is free over $\mathbb{Z}_\ell[\Gamma/\Gamma_r]=\mathbb{Z}_\ell[G_r]$. The dual of $\mathcal{W}^0_r = eH^1(Y_r;\mathbb{Z}_\ell)\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell$ ($eH^1(Y_r;\mathbb{Z}_\ell)$ is $\mathbb{Z}_\ell$-free: Lemma 4.6, p. 571, and $eH^0=0$, U acting on H^0 by ℓ) is $\mathrm{Hom}_{\mathbb{Z}_\ell}(eH^1(Y_r;\mathbb{Z}_\ell),\mathbb{Z}_\ell)$; a $\mathbb{Z}_\ell$-dual of a free $\mathbb{Z}_\ell[G]$-module is free ($\mathrm{Hom}_{\mathbb{Z}_\ell}(\mathbb{Z}_\ell[G],\mathbb{Z}_\ell)\cong\mathbb{Z}_\ell[G]$). So $eH^1(\Gamma_1(\ell^r);\mathbb{Z}_\ell)$ is $\mathbb{Z}_\ell[G_r]$-free. Its χ_a-part for the tame diamonds ($|\mathsf{H}_r|=\ell-1$ is a unit; [C, `prop:nebgammaH`](6)) is $\mathsf{M}_r(a;\mathbb{Z}_\ell)$, free over $\mathbb{Z}_\ell[G_r]$. Its rank is the Λ-rank of the matching component of W^0: by the twist of p. 593 the class a of the Hecke algebra corresponds to the natural tame character $a-2$, which on [C]'s side is the χ_a-part (Proposition 15.4); there $\mathrm{rank}_\Lambda V^0 = 2\,\mathrm{rank}_\Lambda h^0 = 2d$ (Thm. 3.1(iii), component by component through Thm. 6.6 and Lemma 6.4, as in Proposition 14.6), and $\mathrm{rank}\,W^0 - \mathrm{rank}\,V^0 = \frac12\varphi(\ell) = \frac{\ell-1}{2}$ in total at $N=1$ (Thm. 3.1(iii)), one in each of the $\frac{\ell-1}{2}$ even classes, which each carry the Λ-adic Eisenstein series (Remark 2.7, [literature: [Hid86b, Thm. 7.2, p. 272]]). Modulo ℓ^K: $H^2=0$ and $eH^0=0$, so $eH^1(\mathbb{Z}/\ell^K)=eH^1(\mathbb{Z}_\ell)/\ell^K$. The splitting: under disjointness the Hecke algebra of M_r is a product, and the residual systems of the cuspidal factor are not boundary systems (the boundary of $\Gamma^{\mathrm t}_r$ carries the Eisenstein systems $\varphi(p)+\psi(p)p$ of $E_2(\varphi,\psi)$, $\varphi\psi=\chi_a\cdot(\text{wild})$, residually $\omega^b(p)+\omega^{a-2-b}(p)p$ — [standard: [DS05, Thm. 4.5.2, 4.6.2] VERIFY]; [C, `lem:bdrytorus`] is level 1 with trivial character and does *not* cover this, F-U5 m8; (BF′) is checked, (U18)(c)); the ranks are those of V^0 and W^0/V^0. (2) Hida's Cor. 4.5 (p. 570, first display, $A=\mathbb{Z}_\ell$, $n=0$): res: $eH^1(\Gamma_1(\ell^{r'});\mathbb{Z}_\ell)\cong eH^1(\Gamma_1(\ell^{r'})\cap\Gamma_0(\ell^r);\mathbb{Z}_\ell)$ for $r>r'>0$; take χ_a-parts; and [C, `lem:nebshapiro`](b) for $\Gamma^{\mathrm t}_r\trianglelefteq\Gamma_{\mathsf H'}(\ell^r)$, $\mathsf H'$ the preimage of $\mathsf H_{r'}$, quotient $\langle g^{\ell^{r'-1}}\rangle$; then reduce modulo ℓ^K as in (1). (3) Hida p. 553: $h^0(N;\mathcal O)/\omega_{k,r}h^0\cong h^0_k(\Gamma_1(Np^r);\mathcal O)$ for $k\ge2$, with the Λ-structure of p. 552 ($l\equiv1 \bmod Np$ acts by $l^2T(l,l)$). At $k=2$ and on cohomology $T(l,l)=\langle l\rangle$ ([C, `prop:nebgammaH`](5) at $n=0$ with $\chi_a(l)=1$ for $l\equiv1 \bmod \ell$), so $l^2T(l,l)=\iota_r((1+X)^{s(l)})$: the Λ-structures agree, and $\omega_{2,r}=\Omega_r$ (Lemma 15.2(1)). The weight-2 ordinary Hecke algebra of $\Gamma_1(\ell^r)$ acts faithfully on $eH^1_!(\Gamma_1(\ell^r);\mathbb{Z}_\ell)$ (Eichler–Shimura and multiplicity one over a field, [literature: [Hid86a, Lemma 6.4, pp. 587–588]], as in Lemma 14.2), hence $\mathbf{h}(a)/\Omega_r\to\mathrm{End}(\mathsf M^{\mathrm S}_r(a;\mathbb{Z}_\ell))$ is injective with image the $\mathbb{Z}_\ell$-algebra of the operators; reduce modulo ℓ^K (onto). At an étale component $\mathsf M^{\mathrm S}_{r,\mathfrak m}$ is free over $\mathbf{h}(a)_{\mathfrak m}/\Omega_r$ (Proposition 15.7), so the reduction stays faithful. (4) Specialisation commutes with characteristic polynomials; $\mathbf{h}(a)/P_k=\mathbf{h}_k$ (Theorem 2.2) and $(\ell^K,\Omega_r,P_k)=(\ell^{K_r(k)},P_k)$. (5) [C, `lem:nebshapiro`](a) with $\Gamma''=\Gamma^{\mathrm t}_r$, $\Gamma'=\Gamma_0(\ell^r)$, $P=\mathcal O_{j,e}$ through $g\mapsto\zeta^t$, and (1); e commutes ([C, `lem:nebshapiro`]); the torsion of [C, `lem:nebwildtors`](1) is not ordinary. □

The route of record for (1), without Thm. 3.1 and without any convention transport (promoted from "a second route" by the referee F-U5, v0.6.0-pre.2; the Thm. 3.1 argument above is kept as the second route): $\mathrm{rank}_{\mathbb{Z}_\ell}\mathsf M_r(a;\mathbb{Z}_\ell)=\sum_\psi \mathrm{rank}_{\mathcal O}\,eH^1(\Gamma_0(\ell^r);\mathcal O(\chi_a\psi))=\ell^{r-1}(2d+1)$ (Shapiro over $\overline{\mathbb{Q}}_\ell$) by Hida's Thm. 1.2 ($h^0/P_{2,\varepsilon}\cong h^0_2(\Phi_r,\varepsilon)$, rank d on the class-a part), Lemma 6.4 (rank two over the Hecke algebra over a field) and one ordinary Eisenstein line per character (for ψ of conductor $<\ell^r$ the old Eisenstein space $E(\ell^j z)$ has U-ordinary part of rank one, as for cusp forms); $\mathsf M_r(a;\mathbb{Z}_\ell)$ is $\mathbb{Z}_\ell$-free (the applicability paragraph of [C, `lem:nebchainrref`]); $\mathsf M_r/(g-1)=eH^1(\Gamma_0(\ell^r);\mathbb{Z}_\ell(\chi_a))$ by [C, `lem:nebshapiro`](b), of rank $2d+1$ by the [standard:] input of the Status line (ordinary = old from level ℓ, one U-ordinary line per form, [C, `prop:stabilisation`]'s argument) and torsion-free ([C, `lem:nebwildtors`]'s argument), so $\dim_{\mathbb{F}_\ell}\mathsf M_r/(\ell,g-1)=2d+1$; Nakayama over the local ring $\mathbb{Z}_\ell[G_r]$ gives a surjection $\mathbb{Z}_\ell[G_r]^{2d+1}\to$

M_r of $\mathbb{Z}_\ell$-free modules of equal rank, hence an isomorphism. The orientation enters only through "class $a \leftrightarrow$ nebentypus ω^{a-2}" on the forms side (Hida p. 589: $r(a) = \operatorname{rank} h_2^0(\Phi_1, \omega^a)$), which is what (U18)(o) gates.

Proposition 15.6 (The Eisenstein family on the tower)**.** *[proved here] from Theorem 15.5(1),(5), Definition 2.4 and [standard: the ordinary weight-2 Eisenstein series $E_2(1,\chi)$ of character χ: $a_p = 1 + \chi(p)p$ $(p \nmid \ell)$, $a_\ell = 1$; [DS05, Thm. 4.6.2] VERIFY]. Under disjointness, $\mathsf{M}_r^{\mathrm{E}}(a; \mathbb{Z}/\ell^K) \cong \Lambda_K^{(r)}$ with*

$$T_p = \epsilon_p(X) = 1 + \omega(p)^{a-1}\langle p\rangle_\ell^{-1}(1+X)^{s(p)}, \qquad U = 1, \qquad \langle p\rangle = \omega^{a-2}(p)\, g^{s(p)} \quad \text{in } \Lambda_K^{(r)},$$

ϵ_p the Λ-adic Eisenstein eigenvalue of part 1 with the same a. At $X = \zeta^t u^2 - 1$ it is $1 + p\,\chi_a\psi_t(p)$; at $X = x_k$ it is $1 + p^{k-1}$ (Lemma 2.5).

Proof. $\mathsf{M}_r^{\mathrm{E}}$ is free of rank one over the local ring $\Lambda_K^{(r)}$ (Theorem 15.5(1)), so T_p acts by an element $\epsilon \in \Lambda_K^{(r)}$; at the $\mathbb{Z}_\ell$-level ($\mathsf{M}_r^{\mathrm{E}}(a;\mathbb{Z}_\ell)$, rank one over $\mathbb{Z}_\ell[G_r] \subseteq \prod_\psi \mathcal{O}_\psi$, injective) its value at every ψ is the T_p-eigenvalue of the ordinary Eisenstein line of character $\chi_a\psi$ (Theorem 15.5(5), Proposition 15.4), i.e. $1 + p\chi_a\psi(p)$, which is $\epsilon_p(\psi(u)u^2 - 1)$: $\omega(p)^{a-1}\langle p\rangle_\ell^{-1}(\psi(u)u^2)^{s(p)} = \omega(p)^{a-1}\langle p\rangle_\ell^{-1}\psi(p)\langle p\rangle_\ell^2 = p\,\omega^{a-2}(p)\psi(p)$. So $\epsilon - \epsilon_p$ vanishes at every point of $\mathbb{Z}_\ell[G_r]$, hence is 0; reduce. The diamond: $\langle p\rangle = \langle\omega(p)\rangle\langle\langle p\rangle_\ell\rangle$ and $\langle\zeta\rangle = \chi_a(\zeta)$ on the χ_a-model for $\zeta \in \mathsf{H}_r$ ([C, `prop:nebgammaH`](4) with the extension χ_a). [verified computationally] check of the identity: `p10c_predict.py` §6, 144/144 cells (ℓ, a, p), $\ell \le 13$, $p \le 29$. □

Proposition 15.7 (Multiplicity one along the tower, and the computed structure constants)**.** *[proved here] from Theorem 15.5(1),(3), Theorem 14.7(1) and [literature: Hida, Lemma 6.4] (as in Lemma 14.2). Let $\mathfrak{m}$ be a component of $\mathbf{h}(a)$ with $\mathbf{h}(a)_\mathfrak{m}$ étale, $\cong W(\mathbb{F}_q)[[X]]$, satisfying (BF′).*

1. *$\mathsf{M}_r^{\mathrm{S}}(a;\mathbb{Z}/\ell^K)_\mathfrak{m}$ is free of rank 2 over $\mathbf{h}_r(a;K)_\mathfrak{m} \cong W_K(\mathbb{F}_q)[X]/(\Omega_r)$; with ι^* ([C, `def:iota`]; ι normalises Γ_r^{t} and fixes a_γ) each sign is free of rank one.*
2. *If $q = \ell$ ($\mathbf{h}(a)_\mathfrak{m} = \Lambda$; e.g. $(11,2)$, $(13,0)$, Theorem 7.1): every T_p acts on $\mathsf{M}_{r,\mathfrak{m}}^{\mathrm{S}}$ by the scalar $a_p^{(r)} \in \Lambda_K^{(r)}$, the image of $a_p(X) := T(p) \in \mathbf{h}(a)_\mathfrak{m} = \Lambda$, and U by $\alpha^{(r)}$, the image of $\alpha(X) := U$; in particular T_p lies in the image of the diamond algebra, $T_p = a_p^{(r)}(u^2\langle u\rangle - 1)$. The $a_p^{(r)}$ are the **structure constants of the tower**: their Weierstrass representatives are polynomials in X of degree $< \ell^{r-1}$ over $\mathbb{Z}/\ell^K$; under $\Lambda_K^{(r)} \to \Lambda_K^{(r')}$ they map to $a_p^{(r')}$.*
3. *The Λ-adic q-expansion along the tower: $a_{p^{s+1}} = a_p a_{p^s} - p\,\omega^{a-2}(p)g^{s(p)}a_{p^{s-1}}$ and multiplicativity, in $\Lambda_K^{(r)}$ (the operator identity $T(p,p) = \langle p\rangle$ of [C, `prop:nebgammaH`](5) at $n = 0$, and $p\langle p\rangle = \omega(p)^{a-1}\langle p\rangle_\ell^{-1}(1+X)^{s(p)}$ under ι_r).*

Proof. (1) Directly (F-U5 m10, v0.6.0-pre.2; the base-change route through Theorem 14.7(1) and the identification $\mathsf{M}_{r,\mathfrak{m}}^{\mathrm{S}}(\mathbb{Z}_\ell) = \mathsf{M}_\mathfrak{m} \otimes \Lambda/\Omega_r$, $\mathsf{M}_\mathfrak{m}$ the $\mathfrak{m}$-part of the Λ-dual of $V^0(a-2)$, needs Hida's duality transported by W_{ℓ^r} and is kept only as a second route): by Theorem 15.5(1),(2) and [C, `lem:nebshapiro`](b), $\mathsf{M}_{r,\mathfrak{m}}^{\mathrm{S}}/(\ell, g-1)$ is the $\mathfrak{m}$-part of the cuspidal weight-2 cohomology of $\Gamma_1^{\mathrm{t}} = \Gamma_0(\ell)$ modulo ℓ, free of rank 2 over $\mathbf{h}_{2,\mathfrak{m}}/\ell = \mathbb{F}_q$ (Lemma 6.4 at level ℓ over $\mathbb{F}_q$, as in Lemma 14.2); so $\mathsf{M}_{r,\mathfrak{m}}^{\mathrm{S}}/\mathfrak{m}_h$ has $\mathbb{F}_q$-dimension 2, Nakayama over the local $\mathbf{h}_r(a)_\mathfrak{m} \cong W(\mathbb{F}_q)[X]/(\Omega_r)$ gives a surjection $\mathbf{h}_{r,\mathfrak{m}}^2 \to \mathsf{M}_{r,\mathfrak{m}}^{\mathrm{S}}$ of $\mathbb{Z}_\ell$-free modules of equal rank $2[\mathbb{F}_q : \mathbb{F}_\ell]\ell^{r-1}$, hence an isomorphism; the signs by ι^* ($2 \in \mathbb{Z}_\ell^\times$), each of rank one by the same count at $r = 1$. (2) A scalar endomorphism of a free module of rank two over a commutative ring is determined by the ring element; $\mathbf{h}(a)_\mathfrak{m} = \Lambda$ acts through $\Lambda \to \Lambda_K^{(r)}$. (3) The Hecke relations of [C, `thm:nebrelations`](2) on each specialisation, transported (the $\mathbb{Z}_\ell[G_r]$-level module is torsion-free, so a relation holding at every ψ holds), and $p\langle p\rangle$ computed as in Proposition 15.6. □

Corollary 15.8 (Truncations from one level, without interpolation)**.** *[proved here] from Proposition 15.7(2) and Lemma 15.2(5). In the situation of Proposition 15.7(2):*

1. *For (K,m,c) with $\Omega_r \in (\ell^K,(X-x_c)^m)$ the reduction of $a_p^{(r)}$ modulo $(X-x_c)^m$ is the truncation $a_p \bmod (\ell^K,(X-x_c)^m)$ of Proposition 13.5(4) — read from one level ℓ^r at weight 2; no interpolation, no consecutive weights, no loss of precision. For X-adic truncations ($c=0$): $\Lambda_{1,m}$ for $m \le \ell^{r-1}$, $\Lambda_{2,m}$ for $m \le \ell^{r-2}$, …; at $r=3$: Λ_{1,ℓ^2}, $\Lambda_{2,\ell}$, $\Lambda_{3,1}$, of which Λ_{1,ℓ^2} and $\Lambda_{2,\ell}$ exceed the reach $N = K+m-1 \le \ell$ of Theorem 13.4(2).*
2. *In general $a_p^{(r)}$ and the interpolated truncation of part 3 have equal images in $Q := \Lambda/(\ell^K,\Omega_r,(X-x_c)^m)$ (both are images of $a_p(X)$). The lengths of Q at the P7 centres are in `p10c_predict.py` §5.*
3. *For k in the class, $a_p^{(r)}(x_k) \equiv a_p(f_k) \pmod{\ell^{K_r(k)}}$; and if $\Omega_r \in (\ell,(X-x_k)^2)$ ($r \ge 2$, $v(k-2)=0$), the first-order coefficient of $a_p^{(r)}$ at x_k modulo ℓ is that of $a_p(X)$.*

Prediction 15.9 ((U18)–(U22): the tower at $\ell \le 13$, $r \le 3$). (New at v0.6.0-pre.1; stated before slot-22 P10 runs (rule 20).) Cells: every even class at $\ell \in \{5,7\}$ ($d=0$: the $S_k=0$ controls); at $\ell=11$ the class $a=2$ (the Δ family, $11a$ at weight 2) and the control $a=4$; at $\ell=13$ the class $a=0$ (the Δ family) and the control $a=4$; $r \in \{1,2,3\}$ ($r=3$ at $\ell \in \{11,13\}$ in the env arm of `P10_spec.md`), $K=3$. Numbers: `P10_spec.md` §7, from `p10c_predict.py` (output of record `out_predict.txt`).

(U18) [proved here] (Theorem 15.5(1),(2), Proposition 15.6; (a)–(c) *given* (o), which is [verified computationally] at $r=1$: a failure of (o) is a defect of this text and stops P10, F-U5 M3) **Ranks, freeness, the tower.** (o) *Orientation* (Proposition 15.4): the ordinary cusp system lies in the χ_a-part and is untwisted; the ordinary Eisenstein line carries $1+\chi_a\psi(p)p$. (a) $\operatorname{rank}_{\mathbb{Z}/\ell^K} \mathsf{M}_r(a) = \ell^{r-1}(2d+1)$: at $\ell \in \{5,7\}$ and the controls ℓ^{r-1}; at $(11,2)$ $3,33,363$; at $(13,0)$ $3,39,507$ for $r=1,2,3$; free over $\Lambda_K^{(r)}$ (the F_ℓ-dimension of $\mathsf{M}_r/(\ell,g-1)$ is $2d+1$). (b) The tower: $\mathsf{M}_r/(g^{\ell^{r'-1}}-1) \cong \mathsf{M}_{r'}$ and $a_p^{(r)} \mapsto a_p^{(r')}$. (c) (BF′) at $(11,2)$, $(13,0)$; the Eisenstein closed form of Proposition 15.6 on $\mathsf{M}_r^{\mathrm{E}}$, every cell.

(U19) [proved here] (Theorem 15.5(3),(4), Proposition 15.7, Corollary 15.8) **The structure constants against the forms side.** At $(11,2)$, $(13,0)$: T_p acts by scalars $a_p^{(r)}$ (off-diagonal coordinates 0 in a $\Lambda_K^{(r)}$-basis); $a_p^{(r)}(x_k) \equiv a_p(f_k) \pmod{\ell^{K_r(k)}}$ at $k=2$ ($11a$; the ω^{10}-form of level 13), 12 (Δ), 22 (ΔE_{10}), 24 (the ordinary line of S_{24}) and 112 resp. 158; the first-order coefficients at x_{12} modulo ℓ at $r \ge 2$; and the images of $a_p^{(r)}$ and of P7's truncations $(K,m)=(3,1),(3,2),(3,3)$ centred at x_{12} agree in Q.

(U20) [proved here] (Theorem 15.5(5), [C, `NB8`], [C, `NB9`]) **Wild points.** At ψ_1 of order ℓ ($r=2$, $e=3(\ell-1)$) and of order ℓ^2 ($r=3$, at $\ell=5$ with $e=40$): the character model over $\mathcal{O}_{j,e}$ has the lengths of [C, `NB8`], its ordinary part is free of rank $2d+1$, and T_p, U on it are $a_p^{(r)}$, $\alpha^{(r)}$ evaluated at $g \mapsto \zeta$.

(U21) [proved here] from [verified computationally] (`p10c_predict.py` §3: $a_2'(x_{12}) \equiv 2 \pmod{11}$, $a_3'(x_{12}) \equiv 12 \pmod{13}$, both non-zero) **The Hecke algebra without diamonds.** At $(11,2)$, $(13,0)$ and every r,K, the $\mathbb{Z}/\ell^K$-algebra generated by the T_p ($p \le 30$, $p \ne \ell$) on $\mathsf{M}_r^{\mathrm{S}}$ is all of $\Lambda_K^{(r)}$: $t := a_{p_0} - a_{p_0}(x_{12})$ lies in $\mathfrak{m}_\Lambda \setminus (\mathfrak{m}_\Lambda^2 + \ell\Lambda)$, so $\Lambda = \mathbb{Z}_\ell[[t]]$ and t is nilpotent in the Artinian $\Lambda_K^{(r)}$ — the ordinary Hecke algebra of the 11^{r-1} resp. 13^{r-1} weight-2 forms of the family is the group ring $(\mathbb{Z}/\ell^K)[C_{\ell^{r-1}}]$, generated by one T_p.

(U22) [conjectural] (stretch, P10(c); Remark 15.11) **Kitagawa's measure on the tower** equals P9's interpolated two-variable measure in the common quotient.

Remark 15.10 (What part 5 rests on, and what it does not do). **[literature: .]** Hida, Invent. 85: Thm. 3.1 (freeness, invariants, ranks), Cor. 4.5 (raising the Γ_0-level does not change the ordinary part), Thm. 1.2 with the $\omega_{k,r}$ display (control at $k=2$ for $\Gamma_1(\ell^r)$), the Λ-structure of p. 552 and its twist against the natural action, p. 593 — all read on the page for this text —

and Lemma 6.4 (read for part 4). **[proved here] here.** Lemma 15.2, Corollary 15.8, Proposition 15.6 (given [standard:] for $E_2(1,\chi)$), Proposition 15.7; and [C]'s new lemmas. **[verified computationally] and gated.** The orientation (Proposition 15.4): [verified computationally] at $r = 1$ (re-derived independently by the referee F-U5 at all eight cells), gated at every cell and every r by (U18)(o); Theorem 15.5(1),(5), Proposition 15.6, [C, `NB8`]'s rank clause and (U18)(a)–(c), (U20) are [proved here] *given* it (F-U5 M3). **[conjectural].** (U22). **Hypotheses, each named where used**: ordinarity (only e-parts are claimed); étaleness of the component (Proposition 15.7; non-étale components are *refused by name*, `tower_nonetale`: there $\mathsf{M}_{\mathfrak{m}}$ need not be free and the structure constants are a Fitting ideal); disjointness of the cuspidal and Eisenstein components (Theorem 15.5(1); a glued component as at $(37, 32)$ is *refused by name*, `tower_eisenstein_glued` — it needs the gluing of (U17) along the tower, not stated here); multiplicity one over fields (Lemma 14.2); the freeness of the Λ-adic symbols (Theorem 14.7). **Not used.** Any Gorenstein property beyond étaleness; Galois representations; Kitagawa (only in (U22)). **What a level is.** $\mathsf{M}_r(a)$ is $\mathbf{M}(a)/(\Omega_r)$ together with the Eisenstein part: one level of the tower determines the Λ-adic object modulo Ω_r, which is *not* a truncation of the form (ℓ^K, X^m) but contains them as Corollary 15.8 says.

Remark 15.11 (Kitagawa's measure on the tower: stretch, recalled). [conjectural] (the construction is *recalled*: Kitagawa (LIT-08) is not held; Greenberg–Stevens [GS93] is held image-only, read for (U16) in a weaker, local, characteristic-0 form). For $\Phi \in \mathsf{M}_r^{\mathrm{S}}(a; \mathbb{Z}/\ell^K)^{\pm}_{\mathfrak{m}}$ and a ball $b + \ell^M\mathbb{Z}_\ell \subseteq \mathbb{Z}_\ell^\times$, $M \le r$, put $\mu_\Phi(b + \ell^M\mathbb{Z}_\ell) :=$ the value of $U^{-M}\Phi$ on the modular symbol $\{\infty\} - \{b/\ell^M\}$ of Γ_r^{t} (the $n = 0$ case of [E, `def:lvmtt`]'s functional, read on the Γ_r^{t}-model through the boundary-symbol identification of [E, `rmk:lvshapes`] (B) at level ℓ^r, not yet written). *Expected* (Kitagawa, recalled): a $\Lambda_K^{(r)}$-valued distribution, whose specialisation at x_k is the weight-k measure of part 4 modulo $\ell^{K_r(k)}$ up to the normalisation of Proposition 14.9. (U22) compares it with P9's interpolated two-variable measure ([E, `rmk:lvtwovar`]) in Q. Nothing else in this part depends on it.

Code map of part 5. Planned (slot-22 P10; Opus; only after F-U5's ACCEPT, rule 22; implementation spec `docs/implementation/slot22/P10_spec.md`): the Γ_r^{t}-model on WAKE-1's `GammaHLevelN` with the tame scalar ([C, `rmk:nebcosetlevel`]), the chain-ring cokernel ([C, `lem:nebchainrref`]) and the Heilbronn U_ℓ, T_p ([C, `prop:nebgammaHheilbronn`]), the ordinary part by a certified block-Krylov iteration of U; `HeckeExt/HeckeLambdaTower` (the reading over $\Lambda_K^{(r)}$: the free basis, the $a_p^{(r)}$, $\alpha^{(r)}$, their Weierstrass representatives, the comparisons with P7 (`HeckeExt/HeckeLambdaQExp`) and with WP-HL's `ComputeFormsWeight`); record `hecke_lambda_tower/v1` (FNV per line, not pinned, no golden line); validation `HeckeLambdaTowerValidation` (`standard`; env arm `HECKE_LAMBDA_TOWER_R3=1` for $r = 3$ at $\ell \in \{11, 13\}$) gating (U18)–(U21). No statement of this part rests on the code.

16 The Eisenstein component over Λ at irregular pairs (part 6)

Status and scope of this part. Written as a pre-release (v0.7.0-pre.1, **not signed**; refereed ACCEPT WITH MINORS, repaired at v0.7.0-pre.2) for an adversarial referee, before the code of the slot-25/26 package H1 (rules 20–22). It answers, for the Eisenstein component of a class $a \not\equiv 0$, the questions "is $\mathbf{T}(a)_{\mathfrak{m}}$ Gorenstein, a complete intersection, regular; is the Eisenstein ideal principal, and by which operator" over Λ, from finitely many finite-weight data: (i) **one weight decides the ring-theoretic type** (Lemma 16.1); (ii) at cuspidal rank $g_a = 1$ the structure is forced (Proposition 16.3); (iii) at the one known pair of cuspidal rank 2, $(\ell, a) = (547, 486)$, the cuspidal family is a regular local ring $\mathbb{Z}_\ell[[Y]]$ of degree 2 over Λ (Proposition 16.4), with multiplicity one at weight 486 at a non-étale component, outside Theorem 14.7; (iv) the predictions (U23)–(U24). Tame level 1, $\ell \ge 5$, as in

part 1. **Literature and novelty.** No novelty is claimed for the finite-level facts at $(547, 486)$: the pair, its cuspidal rank 2 and its uniqueness for $\ell < 10^5$ are [Cal06, p. 68] and [BP19, §1.1]; the finite-level Gorenstein and DVR properties are known consequences of [Kur93] (Remark 16.5). The Λ-adic statements of Proposition 16.4 were not found in the sources searched (slot-24 LIT, `docs/implementation/slot24/heckez/LIT-report.md` §3, §3a: Calegari 2006, Bellaïche–Pollack 2019, Kurihara 1993, Wake 2015 and arXiv:1401.3764, Wake–Wang-Erickson 2018 and arXiv:1707.01894, Pollack–Wake arXiv:2401.14518); that is a statement about a search, not a novelty claim. *Not in this part*: the class $a \equiv 0$ (Remark 5.10: no Eisenstein congruence there); the cuspidal congruence modules of the ramified families of Proposition 8.5(4) (slot-26 H2); tame level $N > 1$.

Notation of this part. $a \not\equiv 0$ is an even class, $\mathfrak{m}$ the Eisenstein maximal ideal of $\mathbf{T}(a)$ (the one over the Eisenstein component; it always exists, as the quotient of $\mathbf{T}(a)$ by the Eisenstein component is Λ, which is local; its image in $\mathbf{h}(a)$ is a proper ideal, i.e. $\mathfrak{m}$ lies on a cuspidal component, iff the pair is irregular, Corollary 5.2), and also its images in $\mathbf{h}(a)$ and in $\mathbf{T}_k(a)$, $\mathbf{h}_k$. $x_k := u^k - 1$, so $P_k = X - x_k$. $I^0(a) \subseteq \mathbf{h}(a)$ is the Eisenstein ideal of Lemma 5.3(3) and G_a the series of Prediction 5.4; ϵ_q is Definition 2.4. The *cuspidal rank* is $g_a := \operatorname{rank}_\Lambda \mathbf{h}(a)_\mathfrak{m}$; by Theorem 2.2 it is $\operatorname{rank}_{\mathbb{Z}_\ell} \mathbf{h}_{k,\mathfrak{m}}$ at every $k \geq 3$ in the class, the number of ordinary normalised eigenforms of weight k congruent to E_k, counted over $\overline{\mathbb{Q}}_\ell$. By Prediction 5.4, $\operatorname{rank}_\Lambda \mathbf{T}(a)_\mathfrak{m} = 1 + g_a$.

Lemma 16.1 (One weight decides the ring-theoretic type)**.** *[proved here] from Theorem 2.2, Proposition 13.2(2) and [standard: a local ring that is finite and free over the regular local ring Λ is Gorenstein, resp. a complete intersection, iff its quotient by a regular sequence is: [BH93, Prop. 3.1.19(b), Thm. 2.3.4(a)]VERIFY]. Let $a \not\equiv 0$, $k \geq 3$ in the class, and $\mathfrak{m}$ a maximal ideal of $\mathbf{T}(a)$ (not necessarily the Eisenstein one). Write R for $\mathbf{T}(a)_\mathfrak{m}$ and R_k for $\mathbf{T}_k(a)_\mathfrak{m}$, or $R = \mathbf{h}(a)_\mathfrak{m}$ and $R_k = \mathbf{h}_{k,\mathfrak{m}}$ when $\mathfrak{m}$ comes from $\mathbf{h}(a)$.*

1. *$R/\mathfrak{m}_\Lambda R \cong R_k/\ell R_k$.*
2. *R is Gorenstein, resp. a complete intersection, iff the Artinian ring $R_k/\ell R_k$ is (Gorenstein: a socle of dimension 1 over the residue field).*
3. *If $R_k/\ell R_k$ is generated, as an algebra over its residue field, by the image of one $t \in R$, and the residue field is $\mathbb{F}_\ell$, then $R = \Lambda[t] \cong \Lambda[Y]/(F)$ with F the characteristic polynomial of t over Λ, monic of degree $\operatorname{rank}_\Lambda R$; R is a complete intersection.*

Proof. (1) $\mathfrak{m}_\Lambda = (\ell, X) = (\ell, P_k)$ as $x_k \in \ell\mathbb{Z}_\ell$. $\mathbf{T}(a)/P_k\mathbf{T}(a) \cong \mathbf{T}_k(a)$ (Proposition 13.2(2)) and $\mathbf{h}(a)/P_k \cong \mathbf{h}_k$ (Theorem 2.2); a finite Λ-algebra is the product of its localisations at its maximal ideals, each of which contains $\mathfrak{m}_\Lambda \supseteq (P_k)$, so the quotients commute with localisation. (2) R is a direct factor of a free Λ-module, hence Λ-free, and ℓ, P_k is a regular sequence on Λ, hence on R (all in the maximal ideal of R). The cited facts apply to $R \to R/(\ell, P_k)R = R_k/\ell R_k$. (3) By (1), $\Lambda[t] + \mathfrak{m}_\Lambda R = R$, so $\Lambda[t] = R$ by Nakayama (R is finite over Λ). Cayley–Hamilton for t acting on the free module R gives $F(t) = 0$, so $\Lambda[Y]/(F) \to R$ is a surjection of free Λ-modules of the same rank $r = \deg F$, hence an isomorphism. For the complete-intersection property, replace t by $t - c$, $c \in \mathbb{Z}_\ell$ a lift of the residue of t (the residue field is $\mathbb{F}_\ell$; this changes neither $\Lambda[t]$ nor $\Lambda[Y]/(F)$ up to $Y \mapsto Y + c$): then $t \in \mathfrak{m}_R$, F is distinguished, and, R being complete and local, $R \cong \mathbb{Z}_\ell[[X, Y]]/(F)$, a regular local ring modulo one non-zero-divisor. □

Remark 16.2 (The census reading). [proved here] from Lemma 16.1, given the certificate named. The census column `gorenstein` at an Eisenstein row (n, ℓ), $k = n+2$, is a statement about the Λ-adic $\mathbf{T}(a)_\mathfrak{m}$ whenever the row's algebra A is certified to be $\mathbf{T}_k(a)_\mathfrak{m}/\ell$. The certificate is generation by saturation (the slot-24 HeckeZ plan's Prop. 2.2, [C] 1.25.0-pre.3 [C, `prop:heckeZgen`](1), a pre-release): $\dim_{\mathbb{F}_\ell} A = 1 + g_a$, with $1 + g_a$ known a priori, **provided** $k \leq \ell + 1$ **or** T_ℓ **is among the operators of the row, and the row's block has dimension** $1 + 2g_a$

(the row's `module_D`), so that it is the $\mathfrak{m}$-part. A census block is cut out by finitely many operators, so it may also contain a form that is Eisenstein only at those; then the lattice is larger than $H_{\mathfrak{m}}$, the a-priori rank of [C, `prop:heckeZgen`](1) is not $1+g_a$, and $\dim A = 1+g_a$ certifies nothing (at $(484,547)$, (D4) has `module_D=5`). Without T_ℓ the block cut out by the T_p, $p \neq \ell$, at the residual Eisenstein system contains every eigenform f with $\bar\rho_f^{\mathrm{ss}} \cong \mathbf{1} \oplus \omega^{k-1}$, ordinary or not, while g_a counts the ordinary ones. For $2 \le k \le \ell+1$ a non-ordinary level-one eigenform has $\bar\rho_f|_{I_\ell} \cong \omega_2^{k-1} \oplus \omega_2^{\ell(k-1)}$, irreducible ([standard: Fontaine's theorem, as stated in [Edi92, Thm. 2.6]]VERIFY), so the two blocks coincide; this covers $(484,547)$ ($k = 486 < 548$) and every irregular-pair row at $k = a < \ell$. For $k > \ell+1$ a non-ordinary congruent form is possible and the row certifies nothing about $\mathbf{T}(a)_{\mathfrak{m}}$ unless T_ℓ is read ($T_\ell \equiv 1+\ell^{k-1} \equiv 1$ is then a unit on the block, which makes it ordinary).

Proposition 16.3 (Cuspidal rank one: the structure is forced)**.** *[proved here] from Prediction 5.4 ([proved here] from [literature: Emerton]), Lemma 3.4 and Theorem 2.2. Let (ℓ, a) be irregular with $g_a = 1$. Then $\mathbf{h}(a)_{\mathfrak{m}} = \Lambda$, $I^0(a)_{\mathfrak{m}} = G_a\Lambda$, and*

$$\mathbf{T}(a)_{\mathfrak{m}} = \Lambda \times_{\Lambda/(G_a)} \Lambda \;\cong\; \Lambda[t]/(t^2 - G_a t), \qquad t = (G_a, 0),$$

a complete intersection, hence Gorenstein. For a prime $q \neq \ell$: $I^0(a)_{\mathfrak{m}} = (T(q) - \epsilon_q)$ iff $v\big(a_q(f_k) - (1+q^{k-1})\big) = v(B_k/k)$ for one $k \ge 3$ in the class, f_k the ordinary eigenform of weight k congruent to E_k (and then for every such k).

Proof. A local Λ-algebra, free of rank 1, is Λ. Localised at $\mathfrak{m}$, (U1-Λ) gives $\mathbf{h}(a)_{\mathfrak{m}}/I^0(a)_{\mathfrak{m}} \cong \Lambda/(G_a)$, so $I^0_{\mathfrak{m}} = (G_a)$, and Lemma 3.4 with $J = G_a\Lambda$ gives the fibre product $(\mathbf{T}(a) \subseteq \mathbf{h}(a) \times \Lambda$, localised at $\mathfrak{m}$; the factor Λ is local and lies over $\mathfrak{m}$). $\{(x,y) : x \equiv y \ (G_a)\} = \Lambda(1,1) \oplus \Lambda t$ and $t^2 = G_a t$; $\Lambda[t]/(t^2 - G_a t)$ is a regular ring modulo a monic polynomial. For the generator, $T(q) - \epsilon_q \in I^0_{\mathfrak{m}}$, so $T(q) - \epsilon_q = wG_a$ in $\mathbf{h}(a)_{\mathfrak{m}} = \Lambda$; it generates iff $w \in \Lambda^\times$, iff $w(x_k) \in \mathbb{Z}_\ell^\times$ for one (every) k, as $w(x_k) \equiv w(0) \pmod{\ell}$. At $X = x_k$, $T(q) - \epsilon_q \mapsto a_q(f_k) - (1+q^{k-1})$ (Theorem 2.2, Lemma 2.5) and $v(G_a(x_k)) = v(B_k/k)$ ($1 - \ell^{k-1}$ is a unit). □

Where $g_a = 1$ ([verified computationally], uncertified forms-side scripts: slot-17 R17-1 `gw_scan.py` and its independent re-derivation in the slot-24 HeckeZ audit P2a). At the weight $k = a < \ell+1$ every residually Eisenstein form is ordinary (Remark 16.2), so g_a is at most the multiplicity of the root $1 + 2^{a-1}$ of $\chi(T_2 \mid S_a) \bmod \ell$, and $g_a \ge 1$ at an irregular pair (Corollary 5.2); a multiplicity 1 therefore proves $g_a = 1$. Of the 155 irregular pairs with $\ell < 2000$, every one has multiplicity 1 except $(547, 486)$ (multiplicity 2). [literature: [BP19, §1.1, arXiv v1, p. 2] (page image verified, slot-24 LIT §3)]: cuspidal rank one holds at every irregular pair with $p < 10^5$ except $(547, 486)$, where the cuspidal algebra has $\mathbb{Z}_\ell$-rank 2; the scan is a check of the project's code against that published result, and Proposition 16.3 applies at every other pair below 10^5 (their $\mathbb{T}^{\mathrm{c}}_{\mathfrak{m}^{\mathrm{c}}}$ read as $\mathbf{h}(a)_{\mathfrak{m}}$ from their notation, not re-derived). Example: at $(n,\ell) = (10, 691)$, $k = 12$, $\tau(2) - (1 + 2^{11}) = -2073 = -3 \cdot 691$ and $v_{691}(B_{12}/12) = 1$, so $I^0(12)_{\mathfrak{m}} = (T(2) - \epsilon_2)$, which re-derives the Eisenstein block of Theorem 7.2. The generator test at the other pairs is one integer per pair (slot-25/26 H1, on the C2 chains of the HeckeZ plan).

Proposition 16.4 (Cuspidal rank two at $(547, 486)$)**.** *[proved here] given the data (D1)–(D4) below ([verified computationally]), from Lemma 16.1, Prediction 5.4, Lemma 3.4, Theorem 2.2, Corollary 5.2, Lemma 2.5, Lemma 5.3(3), Lemma 14.2, Remark 16.2 with [M1] Prop. 1(2), [C, `rmk:ordtorsionfree`], and [standard: Weierstrass preparation in $\mathbb{Z}_\ell[[X,Y]]$; Krull–Schmidt]. Let $\ell = 547$, $a = 486$, $\mathfrak{m}$ the Eisenstein maximal ideal, and $Y := T(2) - \epsilon_2 \in \mathbf{h}(a)_{\mathfrak{m}}$.*

1. *$\mathbf{h}(a)_{\mathfrak{m}} = \Lambda[Y] \cong \Lambda[Y]/(F)$, $F = Y^2 + bY + c$, with $b \in \mathfrak{m}_\Lambda$, $c = w\,G_a$, $w \in \Lambda^\times$, and $G_a = w_1(X - \xi)$ with $w_1 \in \Lambda^\times$, $\xi \in \ell\mathbb{Z}_\ell$ ($\lambda(G_a) = 1$, $\mu(G_a) = 0$). Hence $\mathbf{h}(a)_{\mathfrak{m}} \cong \mathbb{Z}_\ell[[Y]]$ is a **regular local domain of dimension** 2: a single cuspidal Hida family of Λ-degree $g_a = 2$, on which $X - \xi$ has Weierstrass degree 2 in Y.*

2. *$I^0(a)_{\mathfrak{m}} = (Y)$: the Eisenstein ideal is principal, generated by $T(2) - \epsilon_2$, and $\mathbf{h}(a)_{\mathfrak{m}}/(Y) \cong \Lambda/(G_a) \cong \mathbb{Z}_\ell$.*
3. *$\mathbf{T}(a)_{\mathfrak{m}} = \Lambda[Y'] \cong \Lambda[Y]/\big(Y\,F(Y)\big)$, $Y' = (Y, 0) = (T(2), \epsilon_2) - \epsilon_2$: a complete intersection, hence Gorenstein, of Λ-rank 3, with special fibre $\mathbb{F}_\ell[y]/(y^3)$.*
4. *At every $k \geq 3$ in the class, $\mathbf{h}_{k,\mathfrak{m}} = \mathbb{Z}_\ell[T_2] \cong \mathbb{Z}_\ell[x]/(f_k)$ with $f_k(x + 1 + 2^{k-1}) = x^2 + b(x_k)x + c(x_k)$, and $v\big(c(x_k)\big) = v(B_k/k)$. Where $v(B_k/k) = 1$, $v(\operatorname{disc} f_k) = 1$: $\mathbf{h}_{k,\mathfrak{m}}$ is the ring of integers of a **ramified** quadratic extension of $\mathbb{Q}_\ell$, and the two ordinary eigenforms congruent to E_k are Galois-conjugate.*
5. ***(The rung.)** Where $v(B_k/k) \geq 2$ — in particular at $k^* = 55086$, where $v_{547}(B_{k^*}/k^*) = 2$ ([verified computationally], below) — the order $\mathbf{h}_{k,\mathfrak{m}}$ is **not maximal**: its index in the maximal order of $\mathbf{h}_{k,\mathfrak{m}} \otimes \mathbb{Q}$ is ≥ 547. Whether the two forms there are conjugate over a ramified or an unramified extension, or rational, depends on $b(x_k)$, which (D1)–(D4) do not determine.*
6. ***(Multiplicity one at weight** 486**.)** Let $H := e_{\mathrm{ord}} H^1(\Gamma; \mathrm{Sym}^{484}\mathbb{Z}^2) \otimes \mathbb{Z}_\ell$ localised at $\mathfrak{m}$, $H = H^\varepsilon \oplus H^{-\varepsilon}$ by the involution, ε the sign that carries the Eisenstein class ($\varepsilon = +$ in [C]'s convention: Lemma 14.2, [C, `rmk:iotasign`]). Then $H^\varepsilon \cong \mathbf{T}_{486}(a)_{\mathfrak{m}}$ and $H^{-\varepsilon} \cong \mathbf{h}_{486,\mathfrak{m}}$, free of rank one, with $H^\varepsilon/\ell \cong \mathbb{F}_\ell[y]/(y^3)$ and $H^{-\varepsilon}/\ell \cong \mathbb{F}_\ell[y]/(y^2)$; and $\delta(\mathfrak{m}) = 0$ on this lattice. This is multiplicity one at a* non-étale *component, outside the hypothesis of Theorem 14.7, at the one weight* 486*; the Λ-adic statement is (U24) below.*

The data *(slot-24 HeckeZ stage 05 §2.4; scratch data outside the tree):* ***(D1)*** *T_2 on $S_k(\mathrm{SL}_2(\mathbb{Z}))$ modulo 547^4 at $k = 486, 1032, 1578, 2124$: the root $1 + 2^{k-1}$ has multiplicity 2 modulo 547, and its Hensel factor f_k has $v(\operatorname{disc} f_k) = 1$ and $v(f_k(1 + 2^{k-1})) = 1$; at $k = 486$, $f = x^2 + 18353242915\,x + 40544750686 \bmod 547^4$ (`blk547.py`, pure Python, uncertified; controls $g = 1$ at six pairs and $\tau(2)$ at $(691, 12)$).* ***(D2)*** *On cohomology, `HeckeHidaComponentsValidation` with `HECKE_HIDA_COMPONENTS_CELLS="484:547:3"`: the Eisenstein block has dimension $5 = 1 + 2g_a$, $L_{\mathrm{coh}} = 1$.* ***(D3)*** *$v_{547}(B_k/k) = 1$ at $k = 486, 1032$, and $v\big(G_a(x_{486}) - G_a(x_{1032})\big) = 1$ (`bern547.py`, power sums modulo ℓ^4; controls: the rungs* 284 *at* 37*, Remark 5.6, and* 342252 *at* 691*, Proposition 5.7); and, evaluated directly at precision 547^3 by the slot-24 audit P2a, $v_{547}(B_{55086}/55086) = 2$, $= 1$ at 55086 ± 546.* ***(D4)*** *The census-4.0 row $(484, 547)$, `eis(0,485): length_A=3, hilbert=[1,1,1], embdim=1, gorenstein=true, module_D=5, module_loewy=[2,2,1], min_generators=2.` (D1) and (D3) are uncertified scripts; (D2) is a shipped validation and (D4) a signed census row. The tag becomes "[proved here] given certified data" when H1 reproduces (D1), (D3) in C++ and the Lean checker K9 of the HeckeZ plan checks them.*

Proof. Step 1 (weight 486*).* By (D1) the root $\bar{e}$, $e := 1 + 2^{485}$, of $\chi(T_2 \mid S_{486})$ modulo ℓ has multiplicity 2; let E be the T_2-generalised eigenspace of $\bar{e}$ in $S_{486} \otimes \mathbb{Z}_\ell$, of rank 2, and $f := f_{486}$ the characteristic polynomial of T_2 on it (its Hensel factor). $f(x + e) = x^2 + \beta x + \gamma$ with $\beta \equiv \gamma \equiv 0$ (e is a double root modulo ℓ) and $v(\gamma) = 1$: f is Eisenstein at e, so irreducible over $\mathbb{Q}_\ell$, and $\mathbb{Z}_\ell[x]/(f)$ is a DVR, the maximal order of $\mathbb{Q}_\ell[x]/(f)$. Every eigenform congruent to E_{486} has $a_2 \equiv e$, so the block $\mathbf{h}_{486,\mathfrak{m}}$ lies in E. One such eigenform f_1 exists ($g_a \geq 1$, Corollary 5.2); the other eigenform in E is its $\mathbb{Q}_\ell$-conjugate (the two roots of the irreducible f are conjugate), which is ordinary and congruent to E_{486} at every prime too (the conjugation fixes the unique prime above ℓ of the ramified quadratic field). Hence $g_a = 2$, E is the block, and $\mathbb{Z}_\ell[T_2] \subseteq \mathbf{h}_{486,\mathfrak{m}}$ forces equality. (Neither (D2) nor Remark 16.2 is used here; (D2), the cohomological block of dimension $5 = 1 + 2g_a$, is a check.) So $\mathbf{h}_{486,\mathfrak{m}}/\ell = \mathbb{F}_\ell[\bar{y}]/(\bar{y}^2)$, $\bar{y}$ the image of $T_2 - e$. *Step 2 ((1), (2)).* $\epsilon_2(x_{486}) = e$ (Lemma 2.5), so Y maps to $T_2 - e$, and Lemma 16.1(1), (3) give $\mathbf{h}(a)_{\mathfrak{m}} = \Lambda[Y]/(F)$, $F = Y^2 + bY + c$; $F \equiv Y^2$ modulo $\mathfrak{m}_\Lambda$ gives $b, c \in \mathfrak{m}_\Lambda$. $Y \in I^0(a)$ (Lemma 5.3(3)), so $\Lambda/(c) = \mathbf{h}(a)_{\mathfrak{m}}/(Y) \twoheadrightarrow \mathbf{h}(a)_{\mathfrak{m}}/I^0(a)_{\mathfrak{m}} \cong \Lambda/(G_a)$ ((U1-Λ), localised), a map of Λ-algebras; hence $c \in (G_a)$, $c = wG_a$. At $X = x_{486}$: $c(x_{486}) = f(e)$ has valuation 1 (D1) and $v(G_a(x_{486})) = v(B_{486}/486) = 1$ (D3), so $w(x_{486})$, hence w, is a unit; then $\Lambda/(c) \to \Lambda/(G_a)$ is a surjection between isomorphic Noetherian rings, so injective, and $I^0_{\mathfrak{m}} = (Y)$. By (D3),

$G_a(0) \equiv G_a(x_{486}) \equiv 0 \pmod{\ell}$ and the difference quotient $\big(G_a(x_{486}) - G_a(x_{1032})\big)/(x_{486} - x_{1032})$ is a unit ($v(x_{486} - x_{1032}) = v(u^{546} - 1) = 1$), and it is $\equiv$ the X-coefficient of G_a modulo ℓ; so $G_a = w_1(X - \xi)$ by Weierstrass preparation, $v(\xi) \geq 1$. Then $c = w_2(X - \xi)$, $w_2 := ww_1 \in \Lambda^\times$. F is distinguished in Y, so $\Lambda[Y]/(F) = \mathbb{Z}_\ell[[X, Y]]/(F)$; modulo (ℓ, Y), $F \equiv \bar{w}_2(X)\, X$ (as $\xi \equiv 0$), so F is distinguished of degree 1 in X, $F = (\text{unit})(X - \varphi(Y))$, and $\mathbb{Z}_\ell[[X, Y]]/(F) \cong \mathbb{Z}_\ell[[Y]]$. In it $X - \xi = -w_2^{-1}(Y^2 + bY)$ with $b \equiv 0$ modulo (ℓ, Y), so $X - \xi$ has Weierstrass degree 2 in Y. *Step 3 ((3)).* $\mathbf{T}(a)_\mathfrak{m} = \mathbf{h}(a)_\mathfrak{m} \times_{\Lambda/(G_a)} \Lambda$ (Lemma 3.4 with (U1-Λ), localised as in Proposition 16.3), the gluing map sending $Y \mapsto 0$. $Y' \in \mathbf{T}(a)$, and for $h = h_0 + Yh_1(Y) \in \mathbf{h}(a)_\mathfrak{m} = \Lambda \oplus \Lambda Y$ and $(h, \lambda) \in \mathbf{T}(a)_\mathfrak{m}$: $\lambda - h_0 \in G_a\Lambda$, and $(h, \lambda) = h_0 + Y'h_1(Y') + (\lambda - h_0)\, G_a^{-1} w^{-1} F(Y')$, because $F(Y') = (F(Y), F(0)) = (0, c) = (0, wG_a)$. So $\mathbf{T}(a)_\mathfrak{m} = \Lambda[Y']$. $Y'F(Y') = (YF(Y), 0) = 0$, and $\Lambda[Y]/(YF) \to \mathbf{T}(a)_\mathfrak{m}$ is a surjection of free Λ-modules of rank 3 (Prediction 5.4: rank $1 + g_a$), hence an isomorphism; $YF \equiv Y^3$ modulo $\mathfrak{m}_\Lambda$. (The fibre $\mathbb{F}_\ell[y]/(y^3)$ is the census row (D4), `hilbert=[1,1,1]`: a consistency check, not an input.) *Step 4 ((4), (5)).* Theorem 2.2 and Lemma 16.1(1): $\mathbf{h}_{k,\mathfrak{m}} = \mathbf{h}(a)_\mathfrak{m}/P_k = \mathbb{Z}_\ell[Y]/(F(x_k; Y))$, $Y \mapsto T_2 - (1 + 2^{k-1})$. $v(c(x_k)) = v(G_a(x_k)) = v(B_k/k)$. disc $= b(x_k)^2 - 4c(x_k)$ with $v(b(x_k)^2) \geq 2$: at $v(B_k/k) = 1$ it has valuation 1, $F(x_k; \cdot)$ is Eisenstein, and the ring is a ramified DVR, of rank 2 with the two roots conjugate. At $v(B_k/k) \geq 2$ both roots y_1, y_2 of $F(x_k; \cdot)$ have $v \geq 1$ (Newton polygon: $v(b(x_k)) \geq 1$, $v(c(x_k)) \geq 2$), and $F(x_k; \cdot)$ is squarefree ($\mathbf{h}_k$ is reduced). If it is irreducible, y_1/ℓ is integral and $\mathbb{Z}_\ell[y_1] = \mathbb{Z}_\ell \oplus \mathbb{Z}_\ell y_1 \not\ni y_1/\ell$; if it splits, $\mathbb{Z}_\ell[Y]/(F) = \{(p(y_1), p(y_2))\}$ has index $\ell^{v(y_1 - y_2)} \geq \ell$ in $\mathbb{Z}_\ell^2$. Either way the index is $\geq \ell$. *Step 5 ((6)).* H is $\mathbb{Z}_\ell$-free ([C, `rmk:ordtorsionfree`]), and $\mathbf{T}_{486}(a)_\mathfrak{m}$ acts on it faithfully. By (D4), $\dim_{\mathbb{F}_\ell} A = 3 = 1 + g_a = \operatorname{rank} \mathbf{T}_{486}(a)_\mathfrak{m}$ for the image A of $\mathbf{T}_{486}(a)_\mathfrak{m}$ in $\operatorname{End}(H/\ell)$, so $A = \mathbf{T}_{486}(a)_\mathfrak{m}/\ell \cong \mathbb{F}_\ell[y]/(y^3)$ and $\delta(\mathfrak{m}) = 0$ (generation by saturation, Remark 16.2; [M1] Prop. 1(2)). H/ℓ is an $\mathbb{F}_\ell[y]/(y^3)$-module of dimension 5, with 2 generators and Loewy layers $(2, 2, 1)$, so $H/\ell \cong \mathbb{F}_\ell[y]/(y^3) \oplus \mathbb{F}_\ell[y]/(y^2)$. The involution commutes with the Hecke operators (ℓ odd), and $H^\pm \otimes \mathbb{Q}$ has ranks $1 + g_a = 3$ and $g_a = 2$ (each cuspidal system occurs once in each sign, the Eisenstein system once, in the sign ε; Lemma 14.2). By Krull–Schmidt, $H^\varepsilon/\ell \cong \mathbb{F}_\ell[y]/(y^3)$ and $H^{-\varepsilon}/\ell \cong \mathbb{F}_\ell[y]/(y^2)$. Both are cyclic, so $H^\pm$ are cyclic by Nakayama; H^ε is faithful over $\mathbf{T}_{486}(a)_\mathfrak{m}$ (every system of the block occurs in it) and $H^{-\varepsilon}$ over $\mathbf{h}_{486,\mathfrak{m}}$, through which $\mathbf{T}_{486}(a)_\mathfrak{m}$ acts on it. A cyclic faithful module is free of rank one. □

Remark 16.5 (The literature at $(547, 486)$: known consequences, not results of [U]). (The sentence in the [literature:] tag is the owner-signed P-K2 of 2026-11-05; the rest of this remark is new at 0.7.0-pre.1.) Let T^E be Kurihara's algebra: the finite-level cuspidal Hecke algebra of weight k (all T_l, T_p included) completed at the Eisenstein maximal ideal, [Kur93, pp. 282–283]; at $(547, 486)$ it is $\mathbf{h}_{486,\mathfrak{m}}$ (Lemma 2.3). The pair is in Kurihara's range ($0 < 486 < 546$), and [literature: T^E is Gorenstein by [Kur93, Rmk. 4.5(3)], as $g = 2$; a DVR by [Kur93, Thm. 0.4] given Vandiver at 547 (page images verified, slot-24 LIT §3a, [GT] `lit:hz-kurihara-gor`)]. In detail: Rmk. 4.5(3), "at most two liftings of G_k mod p", makes T^E monogenic over $\mathbb{Z}_\ell$, hence Gorenstein; $n = v(B_{486}/486) = 1$ gives $\mathfrak{I}T^E = \mathfrak{m}$ (his Lemma 3.1); and under $A^{\omega^{62}} = 0$ (Vandiver at 547, an input from outside that paper, not read here) his Thm. 0.4(ii) makes $\mathfrak{m}$ principal. This is the case $k = 486$ of Proposition 16.4(4), which Step 1 of its proof obtains from (D1) without Vandiver. Kurihara does not name 547; the pair is named first by [Cal06, p. 68] ("$\mathbf{T}_\mathfrak{J}/p \simeq \mathbf{F}_p[x]/x^2$" at one finite level), and its cuspidal rank 2 and uniqueness for $p < 10^5$ are [BP19, §1.1, §3.9]. Kurihara proves unconditionally only the implication "T^E Gorenstein $\Rightarrow$ the ω^{1-k}-part of the class group is cyclic" (Thm. 0.3) and gives *no* non-Gorenstein example. None of these sources treats the Λ-adic ring $\mathbf{h}(a)_\mathfrak{m}$, the principality of $I^0(a)$ over Λ, or the rung.

Prediction 16.6 ((U23), (U24) The 547 family).(U23) **(The rung order.)** Stated here as a test of the implementation, before the computation (slot-25/26 H1-R, the forms side on all of S_{k^*}, $\dim S_{k^*} = 4590$, modulo 547^3, with T_2, T_3 only). At $k^* = 55086$: [proved here] given (D1)–(D4), from Proposition 16.4(4), (5): the characteristic polynomial f_{k^*} of T_2 on the ordinary block $\mathbf{h}_{k^*,\mathfrak{m}}$ has degree 2 and $v(\operatorname{disc} f_{k^*}) \geq 2$, $\mathbb{Z}_\ell[T_2|_{\text{block}}]$ has index ≥ 547 in its

normalisation, and f_{k^*} divides the Hensel factor of $\chi(T_2 \mid S_{k^*})$ at $1+2^{k^*-1}$. [verified computationally] (`docs/implementation/slot25/U6_referee.md` §1, `rung.py`, an uncertified referee script, modulo 547; H1-R reproduces it): that Hensel factor has degree exactly 2, so it *is* f_{k^*} (at $k^* > \ell+1$ a non-ordinary form with $a_2 \equiv 1+2^{k^*-1}$ is not excluded a priori, Remark 16.2; none occurs). H1 reads b, c modulo (ℓ^3, X^3), so it predicts f_{k^*} modulo 547^3; the sharp form of the test is that the predicted polynomial equals the Hensel factor. **The anchor** (v0.7.0-pre.3, written before H1-R runs, rule 22; [verified computationally]: slot-25 H1, `HECKE_LAMBDA_CELLS="547:486:3:3"`, `docs/implementation/slot25/H1_report.md` §2, reproduced independently by Newton interpolation in the audit `H1_audit.md` §1):

$$f_{k^*} \equiv x^2+56702273\,x+83587893 \pmod{547^3}, \qquad v_{547}(\operatorname{disc} f_{k^*}) = 2, \quad v_{547}\big(f_{k^*}(1+2^{k^*-1})\big) = 2.$$

H1-R's Hensel factor of $\chi(T_2 \mid S_{k^*})$ at $1+2^{k^*-1}$, reduced modulo 547^3, must be this polynomial. A Hensel factor of degree $\neq 2$, or $v(\text{disc}) \leq 1$, at k^* is a defect of the implementation, of (D1)–(D4) or of that computation.

(U24) **(Λ-adic multiplicity one at a non-étale component.)** [conjectural] At $(547, 486)$, $\mathbf{M}(a)^{\pm}_{\mathfrak{m}}$ (Definition 14.5, the ordinary parabolic Λ-adic modular symbols of part 4) is free of rank one over $\mathbf{h}(a)_{\mathfrak{m}}$, for both signs. Expected from Proposition 16.4(6) by Nakayama and control (Proposition 14.6, [C, `thm:hidacontrol`](1), (2), Lemma 14.3, Lemma 2.3); the chain of identifications is not written out, so it is not tagged [proved here]. ($\mathbf{T}(a)_{\mathfrak{m}}$ acts on $\mathbf{M}(a)^{\pm}_{\mathfrak{m}}$ through $\mathbf{h}(a)_{\mathfrak{m}}$, as the specialisations are cuspidal; a Λ-adic analogue of $H^{\varepsilon} \cong \mathbf{T}_{486}(a)_{\mathfrak{m}}$ would need a Λ-adic full H^1, which [U] does not define.)

Remark 16.7 (Comparison: Mazur's Eisenstein ideal at prime level (slot-25 H3)). [verified computationally] (`docs/implementation/slot25/H3_report.md`, `H3_cells.tsv`; refereed ACCEPT_WITH_MINORS with independent spot checks, `H3_referee.md`). Outside the scope of [U] (prime level N, weight 2; [U] cites no statement of it), recorded because it is the finite-level model of Proposition 16.4(4)–(5). On the 180 cells (N, ℓ), $\ell \mid \nu = \text{num}((N-1)/12)$, of the H3 list, the cuspidal Eisenstein-local ring $\mathbb{T}^0_{\mathfrak{m}} = \mathbb{Z}_\ell[\eta_q] \cong \mathbb{Z}_\ell[y]/(F)$ (q a good prime that passes the valuation check $v_\ell F(0) = v_\ell(\nu)$) has rank $e = 1, 2, 3$ at 148, 29, 3 cells, $e \geq 2$ exactly at the Merel cells, and $\deg F = (\dim W - 1)/2$ on every cell. Where $\ell \| \nu$ the ring is a totally ramified DVR (the analogue of (4)); where $\ell^2 \mid \nu$ and $e = 2$ it is **not maximal**, of two of the three kinds that (5) leaves open (ramified, unramified, rational; no ramified non-maximal cell occurs): $(401, 5)$ and $(701, 5)$ give the index-5 order $\mathbb{Z}_5 + 5\mathbb{Z}_{25}$ (unramified splitting field), $(751, 5)$ the split fibre product $\mathbb{Z}_5 \times_{\mathbb{F}_5} \mathbb{Z}_5$. So the trichotomy of (5) is not vacuous in two of its branches. The agreement of $\deg F$ with $\dim W$ is evidence for [C]'s prediction [C, `LN4`] ("equality is part of the prediction") on these cells, not a proof: [C, `LN4`] stays a prediction, and [C]'s [C, `LN6`] reading is unchanged.

Remark 16.8 (Steps 1–2 of Proposition 16.4 at any cuspidal rank $g_a \geq 2$). [proved here] from Lemma 16.1(1), (3), Prediction 5.4, Lemma 5.3(3), Theorem 2.2, Lemma 2.5 and [standard: Weierstrass preparation in $\mathbb{Z}_\ell[[X, Y]]$; q-expansion duality $S_k(\mathbb{Z}_\ell)_{\mathfrak{m}} \cong \text{Hom}_{\mathbb{Z}_\ell}(\mathbf{h}_{k,\mathfrak{m}}, \mathbb{Z}_\ell)$]. Let (ℓ, a) be irregular, $\mathfrak{m}$ the Eisenstein maximal ideal, $k \geq 3$ in the class, $g := g_a \geq 2$, $q \neq \ell$ a prime, $Y := T(q) - \epsilon_q \in \mathbf{h}(a)_{\mathfrak{m}}$, and B the $\mathfrak{m}$-part of $e_{\text{ord}} S_k(\mathbb{Z}_\ell)$ (rank g), cut out by the full residual system, T_ℓ included. Suppose (a) the image $\bar{y}$ of $T_q - (1+q^{k-1})$ in $\text{End}(B/\ell)$ satisfies $\dim_{\mathbb{F}_\ell} \mathbb{F}_\ell[\bar{y}] = g$; (b) $v\big(\det(T_q - (1+q^{k-1}) \mid B)\big) = v(B_k/k)$; (c) $v\big(G_a(x_k) - G_a(x_{k+\ell-1})\big) = 1$. Then the conclusions (1), (2) of Proposition 16.4 hold with $T(2)$ replaced by $T(q)$ and 2 by g: $\mathbf{h}(a)_{\mathfrak{m}} = \Lambda[Y] \cong \Lambda[Y]/(F) \cong \mathbb{Z}_\ell[[Y]]$, F monic of degree g, a regular local domain of dimension 2, $I^0(a)_{\mathfrak{m}} = (Y)$, and $X - \xi$ has Weierstrass degree g in Y. *Proof.* By duality B is a faithful $\mathbf{h}_{k,\mathfrak{m}}$-module of rank $g = \text{rank}_{\mathbb{Z}_\ell} \mathbf{h}_{k,\mathfrak{m}}$, and $B/\ell \cong \text{Hom}(\mathbf{h}_{k,\mathfrak{m}}/\ell, \mathbb{F}_\ell)$ is faithful over $\mathbf{h}_{k,\mathfrak{m}}/\ell$, so (a) says that $\bar{y}$ generates $\mathbf{h}_{k,\mathfrak{m}}/\ell$ (dimension g); the residue field is $\mathbb{F}_\ell$. This replaces Step 1. Y maps to $T_q - (1+q^{k-1})$ at x_k (Lemma 2.5), so Lemma 16.1(1), (3) give $\mathbf{h}(a)_{\mathfrak{m}} = \Lambda[Y] \cong \Lambda[Y]/(F)$, $F = Y^g + b_{g-1}Y^{g-1} + \cdots + b_1 Y + c$, and $F \equiv Y^g$ modulo $\mathfrak{m}_\Lambda$ ($\bar{y}$ is nilpotent in the local Artinian ring), so $b_i, c \in \mathfrak{m}_\Lambda$. Step 2 then runs verbatim: $Y \in I^0(a)$ (Lemma 5.3(3)) gives

$\Lambda/(c) \twoheadrightarrow \Lambda/(G_a)$, so $c = wG_a$; $F(x_k;\cdot)$ is the characteristic polynomial of $T_q - (1+q^{k-1})$ on $\mathbf{h}_{k,\mathfrak{m}}$ (Theorem 2.2), equal to that on B (duality), so $v(c(x_k)) = v(\det) = v(B_k/k) = v(G_a(x_k))$ by (b), w is a unit and $I^0_{\mathfrak{m}} = (Y)$; (c) and $v(x_k - x_{k+\ell-1}) = 1$ make the X-coefficient of G_a a unit, $G_a = w_1(X-\xi)$, $c = w_2(X-\xi)$, and modulo (ℓ, Y), $F \equiv \bar{w}_2(X)X$, so $\mathbb{Z}_\ell[[X,Y]]/(F) \cong \mathbb{Z}_\ell[[Y]]$, in which $X - \xi = -w_2^{-1}(Y^g + \cdots + b_1 Y)$ has Weierstrass degree g in Y. □

Steps 3–5 (the full algebra, the specialisations, multiplicity one) use data of $(547, 486)$ and are not claimed here. At $g = 1$ the hypotheses (a) and (c) are not needed and the statement is Proposition 16.3's generator test. Where (a), (b) or (c) fails no conclusion is drawn. This is the rule of the record's G8 word `regular_deg_g` ([C, `sec:recordschema`] item 8, the G8 bullet: (a) and (b) for the first q that passes, (c) by the λ test; otherwise `open`). Below $\ell < 10^5$ the only pair with $g_a \geq 2$ is $(547, 486)$ [BP19, §1.1], where the remark is Proposition 16.4(1), (2).

Code map of part 6. Built (v0.7.0-pre.3; slot-25 H1, commit `006f60a9`, audited PASS_WITH_MINORS in `docs/implementation/slot25/H1_audit.md`): `HeckeExt/HeckeHidaEisLambda.{h` (the ordinary Eisenstein block of $e_{\mathrm{ord}}S_k$ at one weight, the generator test, K9's integers, the G8 source), `HeckeHidaEisLambdaValidation` (Proposition 16.3 on the 11 C2 chain pairs, (U1) on the forms side and on cohomology, the record's G8), and the record's G8 under the non-default settings value `heckez.G8 = hida` ([C, `sec:recordschema`] item 8; never golden). As planned (slot-25/26 H1, after this part's referee ACCEPT, rule 22): the read-only env arm `HECKE_LAMBDA_CELLS="ell:a:K:m,..."` on `HeckeLambdaValidation` (`EFFECTIVE` line) for the truncation of $\mathbf{T}(486)_{\mathfrak{m}}$ at $\ell = 547$, $K = m = 3$ (Theorem 13.4: 5 weights at precision 5), reproducing (D1) and (D3) and reading b modulo (ℓ^3, X^3); Proposition 16.3's generator test on the C2 chains (one integer per pair); the Lean checker K9 (a few integers per pair, `decide +kernel`). H1-R (a second workstation, owner-launched) is the test of (U23). Slot-25 HZ-LOC (`HeckeZLocalValidation`) supplies the per-maximal-ideal certificate of Remark 16.2 for census rows. No statement of this part rests on code.

References

[BH93] Winfried Bruns and Jürgen Herzog. *Cohen–Macaulay Rings*, volume 39 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 1993. Prop. 3.1.19(b): a Noetherian local ring is Gorenstein iff its quotient by a regular sequence is. Not checked against the page (VERIFY); added at [C] v1.25.0-pre.0 (sec:heckeZ: prop:heckeZmono).

[BP19] Joël Bellaïche and Robert Pollack. Congruences with Eisenstein series and μ-invariants. *Compositio Math.*, 155(5):863–901, 2019. arXiv:1806.04240v1. §1.1, p. 2, page image verified (slot-24 LIT §3): (Cuspidal rank one) holds for every irregular pair (p,k) with $p < 10^5$ except $(547, 486)$, where $\dim_{\mathbf{Z}_p} \mathbb{T}^c_{\mathfrak{m}^c} = 2$; §3.9, p. 21.

[Cal06] Frank Calegari. Eisenstein deformation rings. *Compositio Math.*, 142(1):63–83, 2006. p. 68, page image verified (slot-24 LIT §3): at $p = 547$, $k = 486$, "$\mathbf{T}_{\mathfrak{I}}/p \simeq \mathbf{F}_p[x]/x^2$". The first printed source naming the pair.

[CG13] Craig Citro and Alexandru Ghitza. Computing level one Hecke eigensystems (mod p). *LMS J. Comput. Math.*, 16:246–270, 2013. arXiv:1102.3321v2 (the library copy, read). Cor. 13: $(p-1)^2/4$ Eisenstein eigensystems mod p; Prop. 14; the appendix table: the number of non-Eisenstein level-one eigensystems mod p for $p < 2000$, with the weights of the Eisenstein–cuspidal congruences. Journal data NOT checked against the copy (an arXiv version).

[Del71] Pierre Deligne. Formes modulaires et représentations ℓ-adiques. In *Séminaire Bourbaki, vol. 1968/69, exposé 355*, volume 179 of *Lecture Notes in Math.*, pages 139–172.

Springer, 1971. The Galois representations of eigenforms of weight $k \geq 2$; used as recalled in [Hid86a, (2.1a,b), p. 556] (that page verified).

[DS05] Fred Diamond and Jerry Shurman. *A First Course in Modular Forms*, volume 228 of *Graduate Texts in Mathematics*. Springer-Verlag, 2005. §5.5 (the Petersson inner product, the adjoints of the Hecke operators, a basis of simultaneous eigenforms). Cited by lem:heckerankone (v1.17.0-pre.2) for the eigenbasis; a secondary pointer since v1.17.0-pre.5 (primary: diamond_im). NOT verified against the text: the book is not in the project's library (slot-17 L17-5).

[Edi92] Bas Edixhoven. The weight in Serre's conjectures on modular forms. *Invent. Math.*, 109:563–594, 1992. §3: Tate's θ-cycles. NOT verified against the text (rule A2).

[Eme99] Matthew Emerton. The Eisenstein ideal in Hida's ordinary Hecke algebra. *Internat. Math. Res. Notices*, (15):793–802, 1999. Thm. 5, p. 6 of the library copy (= the Theorem of the introduction, p. 2), VERIFIED: for $p \geq 5$ and $i \not\equiv 0 \pmod{p-1}$, $\Lambda_{(i)}/L_p(\omega^i, 1-s) \to \mathbf{T}^{\mathrm{ord}}_{\mathrm{par},(i)}/I^0_{\mathrm{Eis}}$ is an isomorphism, at tame level 1. The library copy is an 8-page preprint without journal header, so the journal data (IMRN 1999, no. 15) are NOT checked against it. Also used at [U] v0.1.1-pre.2 (lem:u-emerton): pp. 2–3 (the set-up, $\langle\gamma\rangle$ acting as γ^k on M_k, $\mathbf{T}_{\mathrm{par}}$ = Hida's h), p. 5 (ord = Hida's superscript 0), p. 7 (the shift by 2 against Mazur–Wiles).

[GS93] Ralph Greenberg and Glenn Stevens. p-adic L-functions and p-adic periods of modular forms. *Invent. Math.*, 111(2):407–447, 1993. Added at [U] 0.5.0-pre.1 (slot-21 P8) as the expected source of the Lambda-adic modular-symbol measure behind (U16). The library copy (papers/g/Greenberg/, 43 pp.) is image-only; NO statement or locator of it was read for [U] (rule A2): every use is marked VERIFY / to be confirmed. Kitagawa's construction (missing_literature.md LIT-08) is not held at all.

[Har87] Günter Harder. Eisenstein cohomology of arithmetic groups. The case GL_2. *Invent. Math.*, 89:37–118, 1987. STANDARD input to the cone chapter's identification of T_p on $H^\bullet_c$ with the correspondence on the Borel–Serre pair. Title, journal, volume, year and pages verified 2026-09-16 against the published record; the entry previously carried the WRONG title ("Eisenstein cohomology for $\mathrm{SL}_2(\mathbb{Z})$ and special values of L-functions"), which is not a paper of this author in this volume. The SECTION pointer §§1–2 used in the text is NOT verified against the article (rule A2), exactly as for borel_serre.

[Har22] Günter Harder. Cohomology of arithmetic groups, vol. iii. manuscript; `docs/literature/Volume-III-07.-08.-2022.pdf`, 2022.

[HHO17] William Hart, David Harvey, and Wilson Ong. Irregular primes to two billion. *Math. Comp.*, 86(308):3031–3049, 2017. Added at [U] 0.3.0-pre.1 (slot-19 WP-2) for the cyclotomic lambda-invariants of the irregular pairs; bibliographic data as in loewy_and_galois_refs.bib; the statement about lambda NOT read in the text (rule A2, VERIFY). [U] does not use it: the four lambda-readings of prop:u-eistrunc are computed in-house.

[Hid86a] Haruzo Hida. Galois representations into $\mathrm{GL}_2(\mathbf{Z}_p[[x]])$ attached to ordinary cusp forms. *Invent. Math.*, 85:545–613, 1986. The Λ-adic ordinary Hecke algebra and control (added at [C] v1.12.0-pre.1). Locators Thm. 3.1 (p. 562), (4.6a) and pp. 563–566, Lemma 4.6 (p. 571), Prop. 4.7 (p. 572), Thm. 6.6 (p. 590) VERIFIED against the PDF at [C] v1.13.0-pre.1, slot-13 WP-2; the source is read-only and is not copied into this repository.

[Hid86b] Haruzo Hida. Iwasawa modules attached to congruences of cusp forms. *Ann. Sci. École Norm. Sup. (4)*, 19:231–273, 1986. The ordinary projector; the rank of the ordinary part depends only on the weight modulo $p-1$ (added at [C] v1.12.0-pre.1, slot-12 WP-2). Locators Thm. 3.1 and Cor. 3.2, p. 250, VERIFIED against the PDF (numdam scan) at [C] v1.13.0-pre.1, slot-13 WP-2; the source is read-only and is not copied into this repository.

[Hid88] Haruzo Hida. A p-adic measure attached to the zeta functions associated with two elliptic modular forms. II. *Ann. Inst. Fourier (Grenoble)*, 38(3):1–83, 1988. Added at [U] v0.1.1-pre.3 (slot-14 WP-U1 audit, R1). VERIFIED against the owner's library copy (numdam scan, papers/h/Hida/, read in place, not copied): (1.2), p. 9 and (1.3), p. 10 (Katz: the p-power level does not matter); Thm. 1.3, p. 17 (the duality $h(N;\mathcal{O}) \cong \mathrm{Hom}(S(N;\mathcal{O}),\mathcal{O})$ and back); Thm. 2.2, p. 20 (the exact sequence $0 \to S(N;\mathcal{O}) \to M(N;\mathcal{O}) \to C(S \times \mathbf{Z}_p^{\times};\mathcal{O}) \to 0$, $\delta(f)(s,z) = a(0, f|z)$). Ohta's reference [7]; Ohta cites its Thm. 2.2 as his (3.3.2). The scan's running head reads "38, 1"; numdam lists no. 3.

[HP92] Günter Harder and Richard Pink. Modular konstruierte unverzweigte abelsche p-Erweiterungen von $\mathbf{Q}(\zeta_p)$ und die Struktur ihrer Galoisgruppen. *Math. Nachr.*, 159:83–99, 1992. Added at [U] v0.1.1-pre.5 (proposal P23 of [D1], slot-14 WP-Q2). Read in place in the owner's library (papers/h/Harder/Harder_Pink.pdf, 17 pages, not copied) by slot-14 WP-Q2: Satz 5, p. 92, and §4.2, pp. 92–93, VERIFIED; its Satz 3 (iv), (v) (cited there to Faltings, Fontaine–Messing and Ribet) are inputs NOT verified here (rule A2).

[HT93] Haruzo Hida and Jacques Tilouine. Anti-cyclotomic Katz p-adic L-functions and congruence modules. *Ann. Sci. École Norm. Sup. (4)*, 26(2):189–259, 1993. Cited for the congruence module of a CM component (added at [U] 0.2.0-pre.1, slot-16 WP-12). Bibliographic data and content NOT verified (rule A2).

[Kur93] Masato Kurihara. Ideal class groups of cyclotomic fields and modular forms of level 1. *J. Number Theory*, 45(3):281–294, 1993. Read on the page images (image-only scan, printed = PDF + 280), slot-24 LIT §3a, 2026-11-04; [GT] row lit:hz-kurihara-gor. §3 (Lemma 3.3, Thm. 3.4): the ideals I_B, I_C, the construction credited to Harder–Pink in footnote 1; Thm. 0.3 (p. 283): T^E Gorenstein $\Rightarrow$ Conj. 0.1; Thm. 0.4 (p. 283): given $A^{\omega^{2-k}} = 0$, $\mathfrak{I}$ principal $\Leftrightarrow$ T^E Gorenstein $\Leftrightarrow$ Conj. 0.1; Rmk. 4.5(3) (p. 294): at most two lifts of G_k mod $p \Rightarrow T^E$ monogenic. Range: level one, $p \geq 5$, $0 < k < p-1$; T^E the finite-level cuspidal algebra.

[Miy89] Toshitsune Miyake. *Modular Forms*. Springer-Verlag, Berlin, 1989. Reprinted in Springer Monographs in Mathematics, 2006. Added at [U] 0.6.0-pre.1 (slot-22 P10-C; the key added by the Fable referee F-U5 at text v0.2) for Thm. 4.6.17: a newform of level ℓ^r, $r \geq 2$, whose character has conductor dividing ℓ^{r-1} has $a_\ell = 0$. Locator NOT verified (rule A2, VERIFY).

[MT90] Barry Mazur and Jacques Tilouine. Représentations galoisiennes, différentielles de Kähler et "conjectures principales". *Publ. Math. IHÉS*, 71:65–103, 1990. Gorenstein property of ordinary Hecke algebras; also cited ([U] v0.2.0-pre.2, Remark rmk:u-cmgl2) for the anticyclotomic main conjecture, the converse of Hida–Tilouine's divisibility for the congruence module of a CM component. NOT verified (rule A2).

[Oht03] Masami Ohta. Congruence modules related to Eisenstein series. *Ann. Sci. École Norm. Sup. (4)*, 36(2):225–269, 2003. Thm. (1.5.5), p. 238, VERIFIED (journal data from the library copy): under (1.5.2) the congruence modules of the four Λ-adic

Eisenstein sequences (I)–(IV) of tame level N are all $\Lambda_{\mathfrak{r}}/(A(T;\theta,\psi))$, A of (1.5.4), $G(T,\theta\omega^2)$ normalised by (1.4.5). Also used at [U] v0.1.1-pre.2 (rmk:u-ohta, rmk:u-classzero): (1.1.4), (1.1.6) pp. 229–230; (3.2.2)–(3.2.3) pp. 253–254; the $\psi = \mathbf{1}$ case of (II) via Appendix A (from p. 263, the vanishing case p. 265).

[Rib77] Kenneth A. Ribet. Galois representations attached to eigenforms with Nebentypus. In *Modular Functions of One Variable V (Bonn 1976)*, volume 601 of *Lecture Notes in Math.*, pages 17–51. Springer, 1977. §3: forms with complex multiplication. Locator NOT verified (rule A2).

[SD73] H. P. F. Swinnerton-Dyer. On ℓ-adic representations and congruences for coefficients of modular forms. In *Modular Functions of One Variable III (Antwerp 1972)*, volume 350 of *Lecture Notes in Math.*, pages 1–55. Springer, 1973. Mod-ℓ level-1 forms, θ and the filtration, the Ramanujan congruences. NOT verified against the text (rule A2).

[Ser73] Jean-Pierre Serre. Congruences et formes modulaires [d'après H. P. F. Swinnerton-Dyer]. In *Séminaire Bourbaki, vol. 1971/72, exposé 416*, volume 317 of *Lecture Notes in Math.*, pages 319–338. Springer, 1973. The weight congruence of mod-ℓ modular forms. NOT verified (rule A2).

[Shi71] Goro Shimura. *Introduction to the Arithmetic Theory of Automorphic Functions*. Princeton University Press, 1971. Ch. 3 (pp. 51–95) and Ch. 8 (pp. 223–241) read on the owner's scan, slot-21 R20-L: Prop. 3.8 (p. 54), Prop. 3.16, 3.17 (pp. 57–58), Thm. 3.20 (p. 59), Thm. 3.24 (p. 63), Lemma 3.29 (p. 66), Prop. 3.30–3.33, Thm. 3.34, 3.35 (pp. 67–71), Prop. 3.36 (p. 72), Thm. 3.41, 3.42 (p. 77), Thm. 8.4 (p. 234), Prop. 8.5 (pp. 237–238).

[Was82] Lawrence C. Washington. *Introduction to Cyclotomic Fields*, volume 83 of *Graduate Texts in Mathematics*. Springer, 1 edition, 1982. The first edition (gl2_refs.bib's `washington` is the second). Added at [U] v0.1.1-pre.3 (slot-14 WP-U1 audit, R3, R5). VERIFIED on rendered pages of the owner's library scan (books/Zahlentheorie/, read in place, not copied; PDF page = book page + 10): Prop. 6.16, p. 102 ($A_0 = A_1 = 0$); Prop. 13.22, p. 284 ($A_n \cong X/((1+T)^{p^n} - 1)X$ for a $\mathbf{Z}_p$-extension with exactly one ramified prime, totally ramified); Thm. 7.10, p. 123, with the power series $f(T,\theta) \in \mathcal{O}_\theta[[T]]$ of p. 122 ($L_p(s,\chi) = f(\zeta_\psi(1+q_0)^s - 1, \theta)$).

[Was97] Lawrence C. Washington. *Introduction to Cyclotomic Fields*, volume 83 of *Graduate Texts in Mathematics*. Springer, 2 edition, 1997.

[Wil88] Andrew Wiles. On ordinary λ-adic representations associated to modular forms. *Invent. Math.*, 94:529–573, 1988. Thm. 2.1.4, p. 561: the local shape at p of the representation of an ordinary form. VERIFIED (page image) by slot-16 L16-3, docs/implementation/slot16/locators.tsv ([U] v0.2.0-pre.2, WP-14 referee U-4).

[Wil95] Andrew Wiles. Modular elliptic curves and Fermat's Last Theorem. *Ann. of Math. (2)*, 141:443–551, 1995. Ch. 2: Gorenstein and freeness of ordinary Hecke modules. NOT verified (rule A2).

A Revision history

[U] keeps [C]'s practice: a published version's text never changes under its own number. This appendix is the record of the versions. Its first subsection is the release record, one paragraph per version, newest first (up to 0.7.0-pre.3 it was the date field of the title page); the second holds the changelogs of parts 1–6, which up to 0.7.0-pre.3 stood at the end of each part; the third quotes the earlier title page.

A.1 Release record

Version 0.7.3 — 2026-11-10 (hecke-extensions slot 26, **signed by the owner 2026-11-10**: the owner-signed release of 0.7.3-pre.1; no change of mathematical content — header, date and this record only; no statement, label, number or tag moves. Companion pointers: [C] v1.26.4, [E] 1.25.3.)

Version 0.7.3-pre.1 — 2026-11-10 (slot 26, a **pre-release, not signed**: title page only, no change of mathematical content: the date line of the title page names only the version, the date and this appendix; the words "signed by the owner 2026-11-09" leave the title page, and the sign-off record of every version stays in this record; no statement, label, number or tag moves. Companion pointers: [C] v1.26.4, [E] 1.25.3 (the same title-page change; signed with 0.7.3).)

Version 0.7.2 — 2026-11-09 (hecke-extensions slot 26, **signed by the owner 2026-11-09**: the owner-signed release of 0.7.2-pre.1; no change of mathematical content — header, date and this record only; no statement, label, number or tag moves. Companion pointers: [C] v1.26.3, [E] 1.25.2.)

Version 0.7.2-pre.1 — 2026-11-09 (slot 26, a **pre-release, not signed**: text-only: private path removed, no change of mathematical content (release audit of 2026-11-09, finding S1): the seven paths into the author's scratch directory were reduced to the scripts' file names, or to "outside the tree"; the two citations of the September cluster run's report name it "the WP-C10 report" instead of a file name that carried the cluster's name; no statement, label, number or tag moves. Companion pointers: [C] v1.26.3 (the same scrub; signed with 0.7.2), [E] 1.25.2.)

Version 0.7.1 — 2026-11-09 (hecke-extensions slot 26, **signed by the owner 2026-11-09**: the owner-signed release of 0.7.1-pre.1; no change of content — header, date and this record only; no statement, label, number or tag moves. Companion pointers: [C] v1.26.1, [E] 1.25.1.)

Version 0.7.1-pre.1 — 2026-11-09 (slot 26, a **pre-release, not signed**: no change of mathematical content: machine names and private paths removed for the public release (owner ruling 2026-11-09).)

Version 0.7.0 — 2026-11-07 (hecke-extensions slot 25, **signed by the owner 2026-11-07**: the owner-signed release of 0.7.0-pre.4; no change of content — header, date and this record only; no statement, label, number or tag moves. Companion pointers: [C] v1.25.0, [E] 1.25.0.)

Version 0.7.0-pre.4 — 2026-11-07 (**a pre-release, not signed, not yet seen by the owner**: front matter and layout only. The title page has a one-line date, the uniform author block and a generic abstract that describes all six parts; the date field of 0.7.0-pre.3 (the release record below) and the changelogs of parts 1–6, which stood at the end of each part (in part 5 with the paragraphs "Wording repair at 0.6.0-pre.2 (part 4)" and "Release candidate 0.6.0"), are moved here verbatim; the earlier author line and abstract are quoted in the last subsection. The sections of parts 2–6 are retitled by subject; their earlier titles were "Part 2: tame level N — CM components, CM $\cap$ GL(2), CM $\cap$ Eisenstein", "Part 3: the Λ-adic algebra, truncated", "Part 4: Λ-adic modular symbols, truncated", "Part 5: the tower $\Gamma_1(\ell^r)$, wild characters, and the Ω-type truncations" and "Part 6: the Eisenstein component over Λ at irregular pairs". No statement, label, number or status tag changes; the status paragraph of §1 and the code maps are unchanged).

Version 0.7.0-pre.3 — 2026-11-06 (hecke-extensions slot 25, H1 merge, **a pre-release, not signed, not yet seen by the owner**: in part 6 the explicit (U23) anchor polynomial, Remark 16.8 (Steps 1–2 of Proposition 16.4 at any $g_a \geq 2$) and the code map *Built*; no result changes, no statement number moves; see the changelog of part 6);

Version 0.7.0-pre.2 — 2026-11-06 (hecke-extensions slot 25, U6 fix round, **a pre-release, not signed**: the referee's F1–F13 in part 6 and one status sentence of §1; no result changes; see the changelog of part 6);

Version 0.7.0-pre.1 — 2026-11-06 (hecke-extensions slot 25, U6, **a pre-release, not signed**: new part 6, §16 — the Eisenstein component over Λ at irregular pairs: one weight decides the ring-theoretic type, cuspidal rank one, the regular family at $(547, 486)$, the rung, multiplicity one

at weight 486, the predictions (U23)–(U24); Remark 5.10 credits Ohta's adaptation of Harder–Pink and Kurihara (owner-signed P-K1); a Bellaïche–Pollack citation in Proposition 8.5(3); no statement, label, number or tag of 0.6.0 moves; see the changelog of part 6);

Version 0.6.0 — 2026-10-16 signed by the owner (text of 2026-10-12, slot 22 CLOSE-T);

Version 0.6.0 (release candidate) — 2026-10-12 (hecke-extensions slot 22, CLOSE-T: the **release candidate** from 0.6.0-pre.3, **owner sign-off pending**, not yet signed; header, date and changelog only — no statement, label, number or tag moves);

Version 0.6.0-pre.3 — 2026-10-12 (hecke-extensions slot 22, **a pre-release, not signed**: CLOSE-T T1 repairs — two wording points of the F-T1 release referee (a paragraph title, a version label in Definition 14.5); no statement, label, number or tag moves);

Version 0.6.0-pre.2 — 2026-10-08 (hecke-extensions slot 22, **a pre-release, not signed**: one wording repair in part 4's "Not in this part" sentence — the Loewy-4 cells of [L37] are non-ordinary, not non-étale ordinary components; no statement, label, number or tag moves);

Version 0.6.0-pre.1 — 2026-10-06 (hecke-extensions slot 22, **a pre-release, not signed**: new part 5, §15 — the tower $\Gamma_1(\ell^r)$, wild characters and the Ω-type truncations, as refereed by F-U5; an addendum in Definition 14.5 (labelling only); no statement, label, number or status tag of 0.5.0 moved; see the changelog of part 5);

Version 0.5.0 — 2026-10-06 (hecke-extensions slot 21, the close: **release** from v0.5.0-pre.8, released with the slot-21 close under the owner's decision D16 (accepted as recommended, 2026-10-06); part 4 as refereed by F4, F4m and the F4 of P5; pointers to [C] v1.21.0 and [E] 1.21.0; no statement, label, number or status tag moved; see the changelog of part 4);

Version 0.5.0-pre.8 — 2026-10-06 (hecke-extensions slot 21, **a pre-release, not signed**: the Fable referee F4's repairs to the third remark after the proof of Lemma 13.6 — "$\lambda(G_b) = 0$ iff $\ell \nmid B_b/b$" is [proved here] given $\mu(G_b) = 0$, and the step $\lambda_p = \sum_b \lambda(G_b)$ behind [HHO17] is tagged [standard: ;] no statement, label, tag or number changed; see the changelog of part 4);

Version 0.5.0-pre.7 — 2026-10-06 (hecke-extensions slot 21, **a pre-release, not signed**: the slot-21 P5 result as a third unnumbered remark after the proof of Lemma 13.6 — the Iwasawa invariants of G_b at the irregular pairs used, re-derived from exact Bernoulli numbers; no statement, label, tag or number changed; see the changelog of part 4);

Version 0.5.0-pre.6 — 2026-10-06 (hecke-extensions slot 21, **a pre-release, not signed**: the measure-side referee F4m's repairs to the two remarks after the proof of Lemma 13.6 — the sign of the two factors, the residue's constant and the wild extension proved; no statement, label or number changed; see the changelog of part 4);

Version 0.5.0-pre.5 — 2026-10-06 (hecke-extensions slot 21, **a pre-release, not signed**: the slot-21 P2 and P4 results as two remarks after the proof of Lemma 13.6 (no number moves) and one evidence sentence after Prediction 14.10; no statement, label or tag changed; see the changelog of part 4);

Version 0.5.0-pre.4 — 2026-10-05 (hecke-extensions slot 21, **a pre-release, not signed**: one evidence sentence after Prediction 14.10, the $(103, 24)$ env arm and the absent $(691, 12)$; see the changelog of part 4);

Version 0.5.0-pre.3 — 2026-10-05 (hecke-extensions slot 21, **a pre-release, not signed**: status text only — Proposition 5.7's Bernoulli numbers reproduced in-house (the validations named), Greenberg–Stevens read for (U16) (a weaker form; the tag stays), the slot-21 evidence of P8, P9 and P3 after Prediction 14.10, the code map of part 4 built; no statement, label or tag changed; see the changelog of part 4);

Version 0.5.0-pre.2 — 2026-10-05 (hecke-extensions slot 21, the F4 repairs to part 4, **a pre-release, not signed**: the slot-21 Fable referee of part 4 (`docs/implementation/slot21/P8_F4_referee_report.md`, ACCEPT WITH REPAIRS) applied — (BF) over all $b \in \mathbb{Z}/(\ell-1)\mathbb{Z}$, Hida's $P_n = P_{n+2}$ read on p. 545, (U17)(a) as Harder's index on the ordinary block ([proved here] from [standard:] for $n < \ell - 1$), seven minors; no label added, renamed or removed; see the changelog of part 4);

Version 0.5.0-pre.1 — 2026-10-05 (hecke-extensions slot 21, P8, **a pre-release, not signed**, for the slot-21 Fable referee: **part 4**, the Λ-adic modular symbols truncated, §14 — the ordinary modular-symbol modules per sign, that their ordinary parts do not see the lattice, multiplicity one at étale components at finite weight and Λ-adically, the normalisation by one functional with its unit check modulo (ℓ, X), the predictions (U15)–(U17); no statement of 0.4.0 changed; see the changelog of part 4);

Version 0.4.0 — 2026-10-05 (hecke-extensions slot 20, the close: **release** from v0.4.0-pre.3 under the owner's ruling O6 of 2026-10-04, on the slot-20 referees F1 (ACCEPT) and F1′ (ACCEPT; its optional m-U1 applied); pointers to [C] v1.20.0 and [E] 1.20.0; no statement, label or status tag moved; see the changelog of part 3);

Version 0.4.0-pre.3 — 2026-10-04 (hecke-extensions slot 20, the close delta for the Fable referee F1′, **a pre-release, not signed**: Prediction 13.9 gains item (4), the outcome — (U14) **met** at all three layer cells by slot-20 WP-RC20 ([verified computationally]); no statement, label or tag moved; pointers [C] v1.20.0-pre.2, [E] 1.20.0-pre.6);

Version 0.4.0-pre.2 — 2026-10-04 (hecke-extensions slot 20, WP-3 repairs after the Fable referee F1 (`docs/implementation/slot20/WP-3_F1_review.md`: [U] ACCEPT, the optional m-U1 applied), **a pre-release, not signed**: Prediction 13.9(2) names $G_a \in \Lambda$ as the series whose divided differences are integral; no statement, label or tag moved; pointers [C] v1.20.0-pre.1, [E] 1.20.0-pre.2);

Version 0.4.0-pre.1 — 2026-10-04 (hecke-extensions slot 20, WP-2, **a pre-release, not signed**, for the slot-20 Fable referee F1 (WP-3): part 3 gains Prediction 13.9 ((U14): the depth-5 rung at 37, the depth-4 rungs at 103 and 59, and α_a modulo ℓ^m, [proved here] from exact Bernoulli numbers, stated before slot-20 WP-RC20's layer windows), and the 691 depth-3 rungs read on the Bernoulli-free layer route (slot-19 NF691, [verified computationally]); one label added (`pr:u-rungs`), none renamed or removed, no tag moved; pointers [C] v1.20.0-pre.1, [E] 1.20.0-pre.1; see the changelog of part 3);

Version 0.3.0 — 2026-10-04 (hecke-extensions slot 19, the close: **release** from v0.3.0-pre.4 under the owner's ruling O3, on the slot-19 referees F1 (part 3, ACCEPT WITH REPAIRS), F4 (ACCEPT WITH REPAIRS) and F1-delta (ACCEPT); part 3 binding with the rest; pointers to [C] v1.19.0 and [E] 1.19.0; no statement, label or status tag moved; see the changelog of part 3);

Version 0.3.0-pre.4 — 2026-10-04 (hecke-extensions slot 19, WP-2 continued, **a pre-release, not signed**: the code map of part 3 **built** (slot-19 WP-HL, `HeckeLambdaValidation`); the F4 m10 ratio gate corrected in sign and count (WP-HL); the depth-3 and depth-4 rungs read from the Coates–Wiles measure of the layer $\mathbb{Q}_\ell(\zeta_{\ell^m})$ (slot-19 R19-7, [verified computationally]); no label added, renamed or removed, no tag moved; see the changelog of part 3);

Version 0.3.0-pre.3 — 2026-10-04 (hecke-extensions slot 19, WP-2 continued, **a pre-release, not signed**: the slot-19 research results R19-2, R19-3 folded into part 3 with the tags of the Fable referee F4 (ACCEPT WITH REPAIRS) — Proposition 13.8's depth-3 rungs [proved here] from five exact Bernoulli numbers per class (F4's major M1) and R19-2's [verified computationally] reading of (U12); (U13) [proved here] at $(37, 32)$ and $(103, 24)$ (F4 m8); Proposition 5.7's rungs gain a cross-reference to the $\mathbb{Z}/691^2$ Kummer fields; no label added, renamed or removed; see the changelog of part 3);

Version 0.3.0-pre.2 — 2026-10-03 (hecke-extensions slot 19, WP-3 repairs, **a pre-release, not signed**: the slot-19 Fable referee F1's six minors to part 3 (ACCEPT WITH REPAIRS, textual only) — Lemma 13.6's Washington locator verified on the page (the u^s convention) and its VERIFY lifted; the semicolon notation of Definition 13.1; Proposition 13.5(3)'s generating set includes the weight tuple; Lemma 13.7's reading as $L_{k''}$ for $k'' \geq 2$ in the class; Proposition 13.8's 691 rungs computed (F1); (U11)'s checkable form; no statement changed in substance, no label added, renamed or removed, no tag moved; see the changelog of part 3);

Version 0.3.0-pre.1 — 2026-10-03 (hecke-extensions slot 19, WP-2, **a pre-release, not**

signed, for the slot-19 Fable referee F1 (WP-3); owner ruling O5 of 2026-10-03: **part 3**, the Λ-adic algebra truncated modulo (ℓ^K, X^m), §13 — what the truncation is, that finitely many weights determine it and at which precision, the Eisenstein truncations at 37, 103, 691, the predictions (U11)–(U13); the scope sentence of §1 and §11 amended; no statement of 0.2.1 changed; see the changelog of part 3);

Version 0.2.1 — 2026-10-02 (hecke-extensions slot 17, the close: **release** from v0.2.1-pre.2 under the owner's ruling O3 of 2026-10-01; the referee F1 (slot-17 WP-3) was ACCEPT WITH REPAIRS, 0 blocking, its three minors applied at pre.2; over pre.2 the [C] / [E] pointers only, now [C] v1.17.0 and [E] 1.17.0; no statement, label or status tag moved; see the changelog of part 2);

Version 0.2.1-pre.2 — 2026-10-02 (hecke-extensions slot 17, WP-2 resumed, **a pre-release, not signed**: the slot-17 referee F1's three minors — the ℓ^2 argument of Proposition 5.7 moved out of its [literature:] citation, the placement sentence of §11's [verified computationally] reworded, the [C] pointer note; no statement, label or status tag moved; see the changelog of part 2);

Version 0.2.1-pre.1 — 2026-10-01 (hecke-extensions slot 17, WP-2, drafted during the slot-16 close, **a pre-release, not signed**: locators and one [verified computationally] sentence — the last two Washington VERIFY sites re-pointed to verified first-edition locators, and the slot-16 WP-11 sentence on the Frobenius-layer hidden blocks beside §11's (U-1); no statement, label or status tag of 0.2.0 moved; referee slot-17 WP-3; see the changelog of part 2);

Version 0.2.0 — 2026-10-01 (hecke-extensions slot 16, the close: **owner-signed release** from v0.2.0-pre.2, owner ruling O2 of 2026-10-01; the referee (slot-16 WP-14) was ACCEPT WITH REPAIRS, every repair applied at pre.2 and confirmed; over pre.2 locators only, verified by slot-16 WP-8; no statement, label or status tag moved; part 2 is binding with the rest; see the changelog of part 2);

v0.1.1 — 2026-10-01 (**owner-signed**, ruling O2 of 2026-10-01: the text of v0.1.1-pre.6 signed as 0.1.1, no change of content; recorded at the slot-16 close with 0.2.0; see the changelog);

v0.2.0-pre.2 — 2026-10-01 (hecke-extensions slot 16, WP-12 resumed, **a pre-release, not signed**: the WP-14 referee's repairs (ACCEPT WITH REPAIRS; U-1 the hypothesis $h_K = 1$, U-2–U-7); see the changelog of part 2);

v0.2.0-pre.1 — 2026-10-01 (hecke-extensions slot 16, WP-12, **a pre-release, not signed**: **part 2**, tame level N — where CM components can live, CM $\cap$ Eisenstein, CM $\cap$ GL(2), the predictions (U8)–(U10), §12; proposal (U-1) in §11 for the referee; no statement of v0.1.x changed; referee slot-16 WP-14);

v0.1.1-pre.6 — 2026-09-30 (slot-14 inventory patch list, **a pre-release, not signed**: $k = 914$ in Remark 5.6, and the referee's rewritten [verified computationally] sentence after Prediction 7.4; no statement, label or status tag moved; see the changelog);

v0.1.1-pre.5 — 2026-09-29 (slot-14, owner rulings of 2026-09-29 on the contract-currency audit, **a pre-release, not signed**: a dated **erratum** to Proposition 8.5(4) — nine non-reduced cells, not twelve; the (U4) ladder at $n = 4150$ and $(4830, 2411)$; (U4) exactly sharp and (U7) on all 287 weight-shift rungs; new Proposition 5.9 (Harder–Pink's sub-tower, a second route) and Remark 6.4; no existing tag moved; see the changelog);

v0.1.1-pre.4 — 2026-09-29 (slot-14 WP-C14, **a pre-release, not signed**: two [verified computationally] additions as the Fable referee of WP-C14 accepted them — the depth-2 cells in Remark 5.6, and (U4)/(U7) at every ℓ with weight-shift data after §9's list; no statement, label or status tag moved; see the changelog);

v0.1.1-pre.3 — 2026-09-28 (slot-14 WP-U1 audit repairs, **a pre-release, not signed**: the Fable audit's repairs R1–R6 — locators and wording only, every new locator verified on the page; no statement, label or status tag moved; see the changelog);

v0.1.1-pre.2 — 2026-09-28 (slot-14 WP-U1, **a pre-release, not signed**: (U1) proved from Emerton's Theorem 5 — both halves re-tagged [proved here] from [literature:]; the class $a \equiv 0$

from Ohta's Theorem (1.5.5); a Fable audit follows; see the changelog);

v0.1.1-pre.1 — 2026-09-28 (slot-14 references edit, **a pre-release, not signed**: citations only — Emerton, Ohta and Citro–Ghitza verified against the owner's library; no statement, label or status tag moved; see the changelog);

v0.1.0 — 2026-09-28 (hecke-extensions slot 14, package WP-13/Q5, **owner-signed release**: from v0.1.0-pre.2, owner decision 2026-09-28 "Go for publication and release" ([S14] Q5 at the recommended option). [U] becomes **binding**, like [C] and [E]: owner-released, cited by label, code gated on it. Scope tame level 1 only; tame $N \geq 3$ is slot 15. No statement changed from pre.2; every VERIFY locator stays marked VERIFY, including Theorem 6.2(b)'s dependence on Wiles. Release note: `docs/implementation/slot14/WP-13_release_note.md` §6);

v0.1.0-pre.2: 2026-09-28 (hecke-extensions slot 14, packages WP-8 and WP-9, **a pre-release, non-binding** (owner question Q5 at its default): the first version with the referee's repairs. Reports: `docs/implementation/slot14/WP-8_report.md`, `docs/implementation/slot14/WP-9_referee.md`)

A.2 Changelogs of the parts

Part 1 is Sections 1–11; parts 2–6 are Sections 12–16. The changelogs are quoted verbatim in their original wording.

Changelog. v0.1.0-pre.1 (2026-09-28, slot-14 WP-8): first version. v0.1.0-pre.2 (2026-09-28, slot-14 WP-9, repairs half): the referee's findings applied. **B1**: Theorem 4.4(3)'s second half needs $\mathcal{C}_{\mathrm{E}}(a) \neq 0$ (supplied at $k = 12,200$ by $v(B_k/k) = 1$). **B2**: Theorem 7.2 no longer claims J principal; the fibre product is over Λ/J, Remark 7.3 (principal iff $\mathbf{T}(a)_{\mathfrak{m}_{\mathrm{E}}}$ is Λ-free). **S1** Lemma 3.1 (Deligne's bound), cited in Propositions 3.2, 3.5 and Theorem 7.2; **S2** $I_{P,k}$ defined (Definition 3.3); **S3** (U6) names $\mathbf{T}_{k,\mathfrak{m}}/\ell$ and is restricted to what is proved; **S4** (U7)'s blocks are cut by U_ℓ too; **S5** the weight congruence over $\mathbb{F}_\lambda$; **S6** Proposition 8.5(2)'s trichotomy and (4), the ramified ordinary blocks of [M1] §5.5, and (U2)'s wording. **m1**–**m7**: the d_k warning and the $k \geq 3$ convention, (U1)'s direction, the three-point Bernoulli route, the forms-side cross-check, the cluster-report pointers in (U4), Ribet's construction. New §10. **v0.1.0** (2026-09-28, slot-14 WP-13/Q5, **owner-signed release**): [U] becomes **binding**, like [C] and [E] — owner-released, cited by label, code gated on it (owner decision 2026-09-28 "Go for publication and release", [S14] Q5 at the recommended option); **scope** recorded in §1 and §11: tame level 1 only, tame $N \geq 3$ deferred to slot 15. Released **unbuilt** (§11's code map, unchanged). No statement changed from pre.2: every finding of the WP-9 referee stands as applied, and every VERIFY locator (rule A2, [S14] Q6 at its default) stays marked VERIFY, **including Theorem 6.2's no-CM proof, whose case (b) depends on Wiles [Wil88, Thm. 2.1.4]** (case (a) and the rest of the theorem do not need it; the crystalline alternative of Remark 6.3 remains a remark, itself VERIFY). [C] and [E] citations repointed to v1.14.0 and v1.15.0 (§1), the versions released alongside this one. Report: `docs/implementation/slot14/WP-13_release_note.md` §6. **v0.1.1-pre.1** (2026-09-28, slot-14 references edit, **not signed**): citations only, after the owner added Emerton's and Ohta's papers to the library. Prediction 5.4's source line: the locators [Eme99, Thm. 5] and [Oht03, Thm. (1.5.5)] verified and quoted, "none is in the project's library" corrected (Mazur–Wiles is there too, locator still VERIFY); Proposition 5.7: the verified cross-check [CG13] for the two irregular classes of 691; bibliography entries completed (`hida_hecke_algebra_refs.bib`). No statement, label or status tag moved: (U1) stays [conjectural]. *For the owner*: Emerton's Theorem 5 is, up to the dictionary between his $\mathbf{T}^{\mathrm{ord}}_{\mathrm{par},(i)}$, $L_\ell(\omega^i, 1-s)$ and our $\mathbf{h}(a)$, G_a (not written out here), the Λ-adic half of (U1), with no Gorenstein hypothesis; whether (U1) is re-tagged is an owner decision (compare proposal P24 of [D1]). **v0.1.1-pre.2** (2026-09-28, slot-14 WP-U1, **not signed**; owner decision: write the matching out, re-tag if it holds; a Fable audit follows). New Lemma 5.3 (the dictionary: Emerton's component

i is our class a with no shift, $\mathbf{T}^{\mathrm{ord}}_{\mathrm{par},(a)} = \mathbf{h}(a)$, $I^0_{\mathrm{E},(a)} = I^0(a)$, $L_\ell(\omega^a, 1-s) \mapsto G_a$), a proof of Prediction 5.4 (Steps 1–4: P_{E} is generated by the Eisenstein differences; the kernel of $\Lambda \to \mathcal{C}_{\mathrm{E}}(a)$ is both J and Emerton's pull-back; his Theorem 5; Proposition 3.5), and Remark 5.5 (Ohta's theorem at $N = 1$, its weight-2 shift, its main-conjecture route for $\psi = \mathbf{1}$, why no Gorenstein hypothesis enters). **Tags moved**: Prediction 5.4 [conjectural] → [proved here] from [literature:], both halves, (U1-Λ) and (U1-k) for every $k \geq 2$ in a class $a \not\equiv 0$, $\ell \geq 5$ ((U1) of §9 likewise); Remark 5.10 [conjectural] → [proved here] from [literature:] (Ohta's vanishing clause). Remark 7.3 decided ($J = (G_a)$; $\mathbf{T}(a)$ is Λ-free); Theorem 7.2's parenthetical, Remark 5.6 and §10's (U1) item reworded accordingly. No other statement moved. Supersedes [D1]'s P24 split (its (U1-Λ) "stays [conjectural]") and WP-Q2 §4's "Gorensteinness or freeness gives the other" (dated addenda there). Standalone matching: `docs/implementation/slot14/WP-U1_emerton_matching.md`. **v0.1.1-pre.3** (2026-09-28, slot-14 WP-U1 audit repairs, **not signed**): the Fable audit `docs/implementation/slot14/WP-U1_fable_audit.md` (verdict: the re-tag stands) applied; every new locator read on the page. **R1** Lemma 5.3(2): the identification $\mathbf{T}^{\mathrm{ord}}_{\mathrm{par},(a)} = \mathbf{h}(a)$ made checkable — [Hid86b, (1.11a), (1.15a)], [Hid88, (1.2), (1.3), Thm. 1.3, Thm. 2.2] (new bib entry). **R2** a paragraph after the lemma recording what the tag rests on: Emerton's Corollary 2 filled in (finite-level duality in weights $k \not\equiv 0 \pmod{\ell - 1}$, Katz's density and q-expansion principle, VERIFY) and Lemma 3 by [Hid88, Thm. 2.2]. **R3** Remark 5.10: the "well known" vanishing is [Was82, Prop. 6.16, Prop. 13.22] plus Nakayama (first edition, new bib entry; its VERIFY removed); Ohta's inputs named ((3.4.2), the freeness of X^+, [15] (5.1.2), [16] (2.3.6)); the main conjecture enters his Appendix A only for $(\theta, \psi) \neq (\omega^{-2}, \mathbf{1})$; the identification of $I^*(\omega^{-2}, \mathbf{1})$ with $I^0(0) \otimes r$ and faithfully flat descent. **R4** Remark 5.5: the conductor $u = 1$ for $a \equiv 2$; "Corollary 2 at $R = \Lambda_{(a)}$" in place of "Proposition 1 and Corollary 2". **R5** Lemma 5.3(4): the existence of $G_a \in \Lambda$ is [Was82, Thm. 7.10, p. 123] (verified; [Was82, Thm. 5.11, p. 57] for the values (verified, slot-16 WP-8; this locator updated at v0.2.0)). **R6** a footnote: Emerton's p. 2 description of V is not used. No statement, label or status tag moved. **v0.1.1-pre.4** (2026-09-29, slot-14 WP-C14, **not signed**): two [verified computationally] additions, as the Fable referee `docs/implementation/slot14/wpc14/REFEREE_I3b_I7b_I8_I9.md` §5 accepted them. Remark 5.6: one sentence pointing to the depth-2 cells $k = 284$ ($\ell = 37$) and $k = 228$ ($\ell = 103$), computed on cohomology ([EX] `s14:depthlaw`; the law itself stays where it was, no new proposition; $k = 914$ at $\ell = 59$ is not added, its rung not having finished). §9: a paragraph on (U4) and (U7) at every ℓ with weight-shift data (WP-C14 I-8), with the referee's changes (sharpness at the new ℓ at $m = 1$ only; $g_E = 1$ at the 11 classes as data). I-8's proposed addition to Prediction 7.4 was rejected as worded by the referee and is not applied. No statement, label or status tag moved. **v0.1.1-pre.5** (2026-09-29, owner rulings of that day on the Fable audit `docs/implementation/slot14/AUDIT_contract_currency_2026-09-29.md`, **not signed**). **Q-A**: [U] is binding, as recorded at v0.1.0 (§1; the [C] pointer moves to v1.14.1). **Erratum** (dated 2026-09-29; slot-14 WP-14c §6, refereed in `docs/implementation/slot14/wpc16/REFEREE_J8_WP-C16.md` Part II, whose corrected text II.3 is applied): Proposition 8.5(4) listed 12 non-reduced rows of [M1]'s screened list; three of them, $(46, 383)$, $(50, 677)$, $(54, 67)$, are reduced ($v_\ell(\operatorname{disc} c_{k,2}) = 2$, the double root lifts modulo ℓ^2, two residual systems), carried over from WP-9's filter without the Loewy-length condition. The item now names the nine non-reduced cells and the three reduced rows; the sentence on $(68, 157)$ is kept (it was true as written) and sharpened; the item's conclusion (kind (ii) only, no cuspidal ∩ cuspidal instance) and Prediction 7.4's stand. §10's (U3) item: "the 12 cells" → the nine cells and the two blocks of $(68, 157)$; the (U2) item and Prediction 7.4 name the nine cells and $(68, 157)$. **Q-B** (the second cluster run signed off): L5 of `WP-L1_binding_doc_proposals.md` — (U4)'s ladder confirmed at $n = 4150$ ($\ell = 691$) and $n = 4830$ ($\ell = 2411$). [D1] **P19**: (U7)'s parenthetical gains $\theta^{12}\Delta$ at $\ell = 13$, and a [verified computationally] paragraph after §9's list gives (U7) on all 287 rungs (281 / 353 / 745, the counts of [EX] `s14:sept` in place of the 266-rung 260 / 321 / 607). **P21**: in the same paragraph, (U4) exactly sharp on the 7385 pairs at $\ell \in \{7, 11, 13\}$ ($m = 1, 2, 3$ on 6575, 778, 32).

The count: [D1]'s 6662 was the 266-rung run; the audit asked to re-derive 7385 (WP-L1's L6) against I-8's 7412. Both are right, on different sets: 7385 is the weight-shift run's pairs at 7, 11, 13 (`wpl1/w1_287/hida_pairs.tsv`, reproduced row for row by I-8), and $7412 = 7385 + 27$ adds I-8's pairs at 23, 31, 37, 103, 131 (sharp at $m = 1$ only), as pre.4's paragraph states. **P23**: new Proposition 5.9 ([proved here] from [literature: Harder–Pink], with the repair of WP-Q2 §3.2), after Theorem 5.8, as a second route to a special case of Prediction 5.4; bibliography entry `harder_pink`. **J-8's [U] remark**: new Remark 6.4 after Remark 6.3 — (i) [verified computationally] (the ordinary dihedral part has $\mathbb{F}_\ell$-dimension $h(-\ell)-1$ on 8 primes; the Λ-rank not claimed); (ii) with the referee's change: the non-ordinary dihedral blocks are non-reduced, $\mathbb{F}_q[\varepsilon]/(\varepsilon^2)$, anticyclotomic ([verified computationally]), and not CM by the *level* ([standard:]), not by Theorem 6.2; Proposition R stays in [EX]. **Labels added**: `prop:u-HP`, `rmk:u-dihedral`; none renamed or removed; no existing status tag moved. **v0.1.1-pre.6** (2026-09-30, slot-14 inventory patch list, `docs/implementation/slot14/INVENTORY_findings_vs_latex_2026-09-29.md` §3 items 6–7, **not signed**): two referee-accepted items that had not been written down. Remark 5.6: $k = 914$ ($\ell = 59$, the cell $(912, 59)$) joins the depth-2 cells computed on cohomology, as the referee's proviso asked once its rung had finished (it has, [EX] `s14:depthlaw`). After Prediction 7.4: the [verified computationally] sentence in the referee's rewritten wording (`wpc14/REFEREE_I3b_I7b_I8_I9.md` §5, "reject as worded, rewrite accepted"): in the 320 weight-shift rungs every ordinary cuspidal block has $g = 1$, and the ramified rank-2 blocks of Proposition 8.5(4) are the next targets. No statement, label or status tag moved. **v0.1.1** (signed 2026-10-01, **owner-signed**; owner ruling O2 of 2026-10-01, "sign [U] 0.1.1"): the text of v0.1.1-pre.6 is signed as 0.1.1 with **no change of content** — pre.1–pre.6 above, among them (U1) proved from Emerton's Theorem 5 (pre.2, re-tag audited, pre.3), the erratum to Proposition 8.5(4) (pre.5), Proposition 5.9 and Remark 6.4 (pre.5). The signature was approved on 2026-10-01 together with the decision to build part 2 on it and is recorded at the slot-16 close, with the release of 0.2.0 (§12's changelog); 0.2.0 is built on the 0.1.1 text, and every change it makes to it is listed in the changelog of part 2. Report: `docs/implementation/slot16/contract_release_report.md`.

Changelog of part 2. **v0.2.0-pre.1** (2026-10-01, slot-16 WP-12, **a pre-release, not signed**; referee WP-14): new §12 — Definition 12.1, Theorem 12.2 ([literature: ,] VERIFY), Theorem 12.3 ([proved here] from [standard: /][literature:)], Proposition 12.4, Proposition 12.5, Remark 12.6, Remark 12.7 ((U8)–(U10)); in §11 the out-of-scope sentence points here, and proposal (U-1) of `docs/implementation/slot15/WP-12_U_E_proposals.md` is added as a [conjectural] sentence **for the referee** (owner ruling O4 of 2026-10-01). Bibliography: `hida_tilouine`. No statement of v0.1.x changed, no label renamed or removed, no tag moved. **v0.2.0-pre.2** (2026-10-01, slot-16 WP-12 resumed, **a pre-release, not signed**): the repairs answering the Fable referee `docs/implementation/slot16/WP-14_referee.md` ([U] ACCEPT WITH REPAIRS; 0 blocking, 1 major, 6 minor), every one as worded. **U-1** (major) Proposition 12.5(2) assumes $h_K = 1$ (else the congruence is with $E_k(\mu, \mu\chi_K)$), and (U10) records $h_K = 1$ at its three fields. **U-2** Proposition 12.5's set-up stated as three clauses on $\bar\rho_\lambda$, and $\bar\varepsilon := \bar\rho_\lambda$ after unramifiedness. **U-3** Theorem 12.3's proof: the lift of c chosen in D_ℓ. **U-4** Wiles Thm. 2.1.4, p. 561, verified (L16-3) at Theorems 12.3, 6.2 and in the bib note; Theorem 12.2 cites Hida Cor. 3.2 (p. 250) verbatim with the direct-factor sentence. **U-5** (U-1) in §11 re-tagged [proved here] for θ^j-images ([C, `prop:thetatwist`]), citing [EX] `s15:hidden` (CMPD). **U-6** Remark 12.6(2): one divisibility Hida–Tilouine, the converse Mazur–Tilouine; the `mazur_tilouine` bib note amended. **U-7** (U10): both Eisenstein systems, dimension ≥ 4; Theorem 12.3's proof writes $\psi = \tilde\chi$. No label added, renamed or removed. **v0.2.0** (2026-10-01, the slot-16 close, **owner-signed release**; owner ruling O2 of 2026-10-01, "release [C] v1.16.0 and [U] 0.2.0 since WP-14 accepted and the repairs are confirmed; sign [U] 0.1.1"): released from v0.2.0-pre.2. The referee's repairs U-1–U-7 are confirmed, 16 of 16 findings with

[C]'s, by `docs/implementation/slot16/WP-14_repair_confirmation.md`. Over pre.2, **locators only** (slot-16 WP-8, `docs/implementation/slot16/WP-8_citations.md` §§1–2, on L16-3's readings): Corollary 5.2 cites Kummer's congruence as [Was82, Cor. 5.14, p. 61] and the weight congruence as [SD73, Thm. 2(iv), p. 22], [Ser73, n° 1.2, Th. 2, p. 416-04], all verified (VERIFY removed there); the paragraph after Lemma 5.3 cites the same weight congruence, verified; v0.1.1-pre.3's R5 entry records Washington's Theorem 5.11 (the values) at p. 57 of the first edition, verified. The status paragraph of §1 and this part's record the signatures; [C] and [E] pointers move to v1.16.0 and v1.16.2. No statement, label or status tag moved. **Not in this release** (new mathematics, for a referee): slot-16 WP-11's proposed sentence on the F1 hidden blocks beside Proposition 8.5, and any statement beyond the readings of (U8)–(U10). **v0.2.1-pre.1** (2026-10-01, slot-17 WP-2, drafted during the slot-16 close, **a pre-release, not signed**; referee slot-17 WP-3, release as 0.2.1 at the slot-17 close under the owner's ruling O3): **locators and one [verified computationally] sentence**. (1) The two Washington VERIFY sites the 0.2.0 release left (`docs/implementation/slot16/contract_release_report.md` §2): Lemma 5.3's Kubota–Leopoldt interpolation now cites [Was82, Thm. 5.11, p. 57] (verified, slot-16 WP-8), and Proposition 5.7's Kummer congruence modulo ℓ^2 cites [Was82, Thm. 5.11, Thm. 5.12, Cor. 5.14] with the two-line argument from Thm. 5.12 written out (Cor. 5.14 verified as a congruence modulo ℓ by L16-3; the ℓ^2 form is the argument, not a quoted locator); the chain sentence of Proposition 5.7 follows. (2) In §11, beside (U-1): slot-16 WP-11's sentence on the Frobenius-layer hidden blocks $(220, 17)$, $(246, 19)$ — not free over the full Hecke algebra, $\mathsf{A}_1 = \mathbb{F}_\ell$ against $\mathbb{T}_{\mathfrak{m}}/\ell = \mathbb{F}_\ell[x]/x^2$ — as [verified computationally] (gated by `HeckeNebentypusValidation`), placed there rather than inside Proposition 8.5 because the blocks are non-ordinary. No statement, label or status tag moved; [C] and [E] pointers stay at v1.16.0 and v1.16.2 (their 1.17.0-pre.1 drafts, with [C, `rmk:sattor`], are in the tree for the same referee). **v0.2.1-pre.2** (2026-10-02, slot-17 WP-2 resumed, **a pre-release, not signed**): the repairs of the slot-17 Fable referee F1 (`docs/implementation/slot17/WP-3_referee_F1.md`; [U] ACCEPT WITH REPAIRS, 0 blocking, 3 minor), each as worded. m-U1: in Proposition 5.7 the two-line ℓ^2 argument from Thm. 5.12 stands in the text, the [literature:] citation names only the theorems and what was verified (whether Washington's Cor. 5.14 states the ℓ^{a+1} form remains unread). m-U2: the [verified computationally] sentence of §11 says it is recorded beside (U-1), not at Proposition 8.5, and why. m-U3: the citation paragraph notes that at the release the pointers read [C] v1.17.0 and the then-current [E] ([C, `rmk:sattor`] exists only from [C] 1.17.0-pre.1 on; 1.17.0-pre.3 is in the tree). No statement, label or status tag moved. **v0.2.1** (2026-10-02, the slot-17 close, WP-18): **the release** from v0.2.1-pre.2 under the owner's ruling O3 of 2026-10-01, as v0.2.0 was released under O2. Over pre.2 only the citation paragraph's pointers, now [C] v1.17.0 and [E] 1.17.0 (m-U3). No statement, label or status tag moved.

Changelog of part 3. v0.3.0-pre.1 (2026-10-03, slot-19 WP-2, **a pre-release, not signed**; referee slot-19 WP-3 (Fable F1); owner ruling O5 of 2026-10-03): new §13 — Definition 13.1, Proposition 13.2, Lemma 13.3, Theorem 13.4, Proposition 13.5, Lemma 13.6, Lemma 13.7, Proposition 13.8 ([verified computationally]: `docs/implementation/slot19/wp2/lambda_check.py`), Remark 13.10, Prediction 13.11 ((U11), (U12) [proved here], (U13) [proved here] at 691 and [conjectural] at 37, 103). In §1 and §11 the out-of-scope sentence on the Λ-adic algebra is amended (its truncations are in scope; the whole algebra is not), and the code map points here. Bibliography: `hho`. No statement of v0.2.1 changed, no label renamed or removed, no tag moved. **v0.3.0-pre.2** (2026-10-03, slot-19 WP-3 repairs, **a pre-release, not signed**; report `docs/implementation/slot19/WP-3_F1_referee_report.md`, repair table `docs/implementation/slot19/WP-3_repairs.md`): m-U1 Lemma 13.6, Thm. 7.10 (1st ed., p. 123) quoted, even χ, $1 + q_0 = u$, both VERIFY removed (Proposition 13.8's [HHO17] stays VERIFY, unused); m-U2 Definition 13.1, $\mathbf{T}_{K,m}(a; x)$ etc.; m-U3 Proposition 13.5(3), the Λ-

subalgebra with the weight tuple; m-U4 Lemma 13.7, $k'' \geq 2$ in the class; m-U5 Proposition 13.8, F1's power-sum route and scans, `docs/implementation/slot19/wp3/f1_lambda.py` + `.out` tracked; m-U6 (U11)'s checkable form, and the $\tilde{D}_i$ in $\mathbb{Q}_\ell$ in Theorem 13.4(2). Tags unchanged. **v0.3.0-pre.3** (2026-10-04, slot-19 WP-2 continued, **a pre-release, not signed**; the slot-19 research results R19-2, R19-3, refereed by the Fable referee F4, `docs/implementation/slot19/R19-6_F4_referee_report.md`, ACCEPT WITH REPAIRS; change table `docs/implementation/slot19/WP-2_pre3_changes.md`): Proposition 13.8 gains the depth-3 rungs 37580, 914250, 49928412, 159113510, [proved here] from five exact Bernoulli numbers per class with their proof (F4's major M1: R19-2 had the two 691 rungs as "predicted"), and the [verified computationally] reading of (U12)'s checkable form by R19-2 (F4 m10's sign of F_1 and its ratio gate); (U13) [conjectural] → [proved here] at $(37, 32)$, $(103, 24)$ (F4 m8); Proposition 5.7 a cross-reference to R19-3's $\mathbb{Z}/691^2$ Kummer fields; Remark 13.10's [verified computationally] list; the code map of part 3: R19-2's route recommendation and F4 m9 (the ordinary part). No label added, renamed or removed; one tag moved ((U13) at 37, 103). **v0.3.0-pre.4** (2026-10-04, slot-19 WP-2 continued, **a pre-release, not signed**; change table `docs/implementation/slot19/WP-2_pre5_changes.md`): the code map of part 3 **built** (slot-19 WP-HL; arm (Q) primary by the owner's ruling of 2026-10-03; `HeckeLambdaValidation`, the planned name superseded) and a *Code* line in Prediction 13.11; the F4 m10 ratio corrected to $F_1/G_{32} \equiv 2$ at eight weights (WP-HL); a [verified computationally] paragraph after Proposition 13.8: the rungs and α modulo ℓ^m read from the level-m Coates–Wiles measure (slot-19 R19-7), the depth-4 rung 1072544 at $(37, 32)$. No label added, renamed or removed; no tag moved. **v0.3.0** (2026-10-04, slot-19 close, WP-18; **the release** from v0.3.0-pre.4 under the owner's ruling O3): the slot-19 F1-delta read the pre.3 → pre.4 delta of [U] as ACCEPT (`docs/implementation/slot19/F1delta_referee_report.md`; its optional m-U1 not applied, `docs/implementation/slot19/F1delta_repairs.md`). Release lines only: part 3 binding with the rest; the citation paragraph's pointers read [C] v1.19.0 and [E] 1.19.0. No statement, label or status tag moved. **v0.4.0-pre.1** (2026-10-04, slot-20 WP-2, **a pre-release, not signed**; referee slot-20 WP-3 (Fable F1); list of new and changed labels `docs/implementation/slot20/WP-2_drafts.md`): a new Prediction 13.9 ((U14)) — at $(37, 32)$, $m = 5$, the rung 55777784 and $\alpha_{32} \equiv 13400697 \pmod{37^5}$; at $(103, 24)$, $m = 4$, the rung 60430740 and $\alpha_{24} \equiv 3736634 \pmod{103^4}$; at $(59, 44)$ $\lambda = 1$ ([verified computationally]) and, $m = 2, 3, 4$, the rungs 914, 86464, 8162384 with $\alpha_{44} \equiv 11649137 \pmod{59^4}$ — [proved here] from exact Bernoulli numbers (`docs/implementation/slot20/wp2/rung_predict.py`), with the checkable form that slot-20 WP-RC20's layer windows gate; the paragraph after Proposition 13.8: the 691 depth-3 rungs and α modulo 691^3 read on the level-3 layer (slot-19 NF691, [verified computationally]). One label added; none renamed or removed; no tag moved; no golden line, digest or record key touched. **v0.4.0-pre.2** (2026-10-04, slot-20 WP-3 repairs, **a pre-release, not signed**): the slot-20 Fable referee F1 (`docs/implementation/slot20/WP-3_F1_review.md`) read [U] as ACCEPT and reproduced (U14) by an independent route; its optional m-U1 is applied (Prediction 13.9(2): "the divided differences of $G_a \in \Lambda$"). No statement, label or status tag moved (`docs/implementation/slot20/WP-3_repairs.md`). **v0.4.0-pre.3** (2026-10-04, slot-20 close delta, **a pre-release, not signed**; referee F1′; list `docs/implementation/slot20/close_delta.md`): Prediction 13.9 gains item (4), the outcome of slot-20 WP-RC20's layer windows — (U14) **met** at $(37, 32, 5)$, $(103, 24, 4)$, $(59, 44, 4)$ ([verified computationally]). No statement changed; no label added, renamed or removed; no tag moved. **v0.4.0** (2026-10-05, slot-20 close, WP-18; **the release** from v0.4.0-pre.3 under the owner's ruling O6 of 2026-10-04): the slot-20 F1′ read the pre.2 → pre.3 delta of [U] as ACCEPT (`docs/implementation/slot20/F1prime_review.md`); its optional m-U1 (the cost figures of Prediction 13.9(4) against the `time -v` peaks) was applied by slot-20 WP-17 (`docs/implementation/slot20/WP-17_repairs.md`). Release lines only: the citation paragraph's pointers read [C] v1.20.0 and [E] 1.20.0. No statement, label or status tag moved.

Changelog of part 4. **v0.5.0** (2026-10-06, slot-21 release, the close): released with the slot-21 close from v0.5.0-pre.8 under the owner's decision D16 (accepted as recommended); release lines only — the status paragraph's sentence on part 4, the pointers [C] v1.21.0 and [E] 1.21.0, the date line. No statement, label, number or status tag moved. **v0.5.0-pre.8** (2026-10-06, slot 21, **a pre-release, not signed**; the Fable referee F4 of P5, `docs/implementation/slot21/rc/F4_referee_p5_report.md`): the third remark after Lemma 13.6 repaired — "$\lambda(G_b) = 0$ iff $\ell \nmid B_b/b$" [proved here] given $\mu(G_b) = 0$; the step $\lambda_p = \sum_b \lambda(G_b)$ tagged [standard:] (Iwasawa, Vandiver from [HHO17] item (2)); [HHO17] §7 read by the referee; the seven pairs re-derived with the referee's own code. No statement, label, tag or number changed. **v0.5.0-pre.7** (2026-10-06, slot 21, **a pre-release, not signed**; slot-21 P5, `docs/implementation/slot21/P5_spec.md` v0.2): a third unnumbered remark after Lemma 13.6 (part 3) — $\lambda(G_b) = 1$ at the irregular pairs used ([literature: [HHO17]]), $\mu(G_b) = 0$ ([standard: Ferrero–Washington, recalled]), both re-derived from exact Bernoulli numbers ([verified computationally]). No statement, label, tag or number changed. **v0.5.0-pre.6** (2026-10-06, slot 21, **a pre-release, not signed**; the measure-side Fable referee F4m, `docs/implementation/slot21/rc/F4_referee_measure_report.md`): the two remarks after Lemma 13.6 (part 3) repaired — (i) each G-factor is *minus* the branch series ($E_1 = -x\,\zeta_\ell$; the product unchanged; Washington Thm. 12.2 read on the page), the residue's constant [proved here] from the factorisation; (ii) the wild extension written out as a proof from Thm. 7.10 / 5.11 ([verified computationally] → [proved here], with [verified computationally]). No statement, label or number changed. **v0.5.0-pre.5** (2026-10-06, slot 21, **a pre-release, not signed**): the slot-21 P2 and P4 results — two unnumbered remarks after the proof of Lemma 13.6 (part 3; the G-factors of the Eisenstein measure, and the wild extension $G_b(\zeta^t u^r - 1) = -B_{r,\omega^{b-r}\psi_t}/r$, [verified computationally]) and one sentence in the Evidence paragraph (single weights: no evidence for (U16), (U17)). No statement, label or tag changed; no number moves. **v0.5.0-pre.4** (2026-10-05, slot 21, **a pre-release, not signed**): the Evidence paragraph gains the $(103, 24)$ env arm ($j_k = L_k$ to $k = 738$, $L = 2$ at 228) and the note that F4's $(691, 12)$ is not in the committed env arm. No statement, label or tag changed. **v0.5.0-pre.3** (2026-10-05, slot 21, **a pre-release, not signed**): the code map **built** (P8, P9); an *Evidence* paragraph after Prediction 14.10 (P8, P9, P3; [verified computationally]; no tag moves); (U16)'s literature sentence — Greenberg–Stevens read, a weaker (local, characteristic-0, meromorphic) form, not the integral one. No statement, label or tag changed. **v0.5.0-pre.2** (2026-10-05, slot-21 F4 repairs after the Fable referee of part 4, **a pre-release, not signed**; `docs/implementation/slot21/P8_F4_referee_report.md`, ACCEPT WITH REPAIRS, 0 blocking, 3 major, 7 minor, all applied): **M1** Proposition 14.4(2): (BF) quantifies over all $b \in \mathbb{Z}/(\ell-1)\mathbb{Z}$ (the pre.1 range $0 \le b \le \lfloor n/\ell \rfloor$ rested on the [conjectural] half of [C, `cl:chi`](2); the proof now cites [C, `lem:bdrytorus`](2),(3)); **M2** Proposition 14.6: Hida's P_n read on p. 545, $P_{k-2} = \ker(X \mapsto u^k - 1) = P_k$, VERIFY lifted (the only remaining VERIFY of part 4 is the matrix ε of Lemma 6.4); **M3** (U17)(a) identified with [E, `cl:lveis`](2) on the ordinary $\mathfrak{m}_{\mathrm{E}}$-block, hence [proved here] from [standard:] for $n < \ell - 1$ ($k = 32$ at 37, $k = 24$ at 103, $(10, 691)$) and [conjectural] beyond, with $\delta(\mathfrak{m}_{\mathrm{E}}) = 0 \iff j_k = L_k$; **m1** Lemma 14.3(1): V_n' is $\mathrm{Mat}_2(\mathbb{Z})$-stable (divided powers), not only Δ-stable; **m2** its (3): the exactness of e_1 on finitely generated $\mathbb{Z}_\ell$-modules written out; **m3** its (3): Hida's level-one operators agree with [C]'s on the torsion-free ordinary part (Eichler–Shimura on both sides); **m4** Theorem 14.7: the identification of the algebra acting on $\mathbf{M}(a)$ with $\mathbf{h}(a)$ written out; **m5** Lemma 14.8: U on $e_1 H_c^1$ by Hensel; **m6** Remark 8.3: a pointer sentence (the literature transfer now applies on ordinary parts); **m7** Lemma 14.2: Lemma 6.4's statement quoted ("for each positive integer M"), classical at $M = 1$. Remark (d) after Lemma 14.3; the status paragraph; Remark 14.11's VERIFY line. No statement of 0.4.0 changed except the pointer sentence of Remark 8.3; no label added, renamed or removed; no tag of 0.4.0 moved. **v0.5.0-pre.1** (2026-10-05, slot-21 P8, **a pre-release, not signed**; for the slot-21 Fable referee): new §14 — Definition 14.1,

Lemma 14.2, Lemma 14.3 (new: ordinary parts are lattice-independent), Proposition 14.4, Definition 14.5, Proposition 14.6, Theorem 14.7, Lemma 14.8, Proposition 14.9, Prediction 14.10 ((U15) [proved here], (U16), (U17) [conjectural]), Remark 14.11, the code map. Bibliography: `greenberg_stevens` (not read). Hida, Invent. 85: Lemma 6.4 (pp. 587–588), Cor. 4.5 (p. 570), Prop. 4.7 (p. 572), Thm. 6.6 (p. 590) read on the page. No statement, label or status tag of 0.4.0 changed; the status paragraph of §1 gains one sentence.

Wording repair at 0.6.0-pre.2 (part 4) (2026-10-08, slot 22 DOC22; not signed). Part 4's "Not in this part" sentence listed "the LL = 4 cells of [L37]" among the non-étale components. Those cells (type eis$(a, a{+}5)$, $\ell = 37$) are non-ordinary (U_{37} nilpotent, [L37] `prop:nonord`; indeed $U_{37} \in \mathfrak{m}_A^2$, [L37] `prop:Uposition`), so they are not objects of [U]; the entry is removed and the point stated in a parenthesis. Source: `docs/implementation/slot22/nonreduced_ordinary_survey.md`. Nothing else changed.

Release candidate 0.6.0 (2026-10-12, slot 22 CLOSE-T; **owner sign-off pending**). The text of 0.6.0-pre.1 to pre.3 as stated in their entries (part 5 refereed by F-U5, `P10C_F4_referee_report.md`; the pre.2 and pre.3 wording points read by `docs/implementation/slot22/F-T1_release_referee.md`); only the version is changed. Companions [C] v1.22.0, [E] 1.22.0.

Changelog of part 5. v0.6.0-pre.1 as inserted (2026-10-06, slot 22, **a pre-release, not signed**): this part is the refereed text v0.2 (whose entries follow, labelled by the text's own revision names); in this document it is v0.6.0-pre.1. Inserted verbatim except two cross-document references to [C] (`\eqref{eq:MHchi}` in Proposition 15.4's status, `[C] \S\ref{sec:nebwildsec}` in the entry below) written as `\HA{...}`, since a `\ref` into [C] is undefined here; the addendum of `docs/implementation/slot22/P10C_addenda.md` (2) in Definition 14.5 (labelling only, verified on Hida p. 593 by F-U5); the pointer in Remark 14.11; one sentence in the status paragraph of §1; bibliography key `miyake` (Thm. 4.6.17, locator not verified). No statement, label, number or status tag of 0.5.0 changed. **v0.6.0-pre.2** (2026-10-06, slot 22, **a pre-release, not signed**; the Fable referee F-U5, `docs/implementation/slot22/P10C_F4_referee_report.md`, ACCEPT WITH REPAIRS, 0 blocking / 3 major / 13 minor, all applied): **M1** [C] [C, `lem:nebshapiro`]'s C-action on P through the inverse of the quotient map (the application unchanged); **M2** Theorem 15.5(2) restated as a module isomorphism, Cor. 4.5's $e \circ \mathrm{res}$ is $e \circ \mathrm{dg}$ in [C]'s convention; **M3** (1), (5), Proposition 15.6, (U18)(a)–(c), (U20), [C, `NB8`] tagged [proved here] *given* Proposition 15.4; the route of record for (1) is the Thm. 1.2 + Nakayama argument with the named [standard:] input (ordinary = old); **m1–m3** the orientation remark reworded (no formal cancellation; the computation decides); **m8** the boundary systems cite [standard:] not [C, `lem:bdrytorus`]; **m10** Proposition 15.7(1) proved directly. Hida pp. 549, 552, 553, 562, 570, 571, 589, 590, 593 re-read on the page images by the referee. No statement of 0.5.x changed. **v0.6.0-pre.1** (2026-10-06, slot 22 P10-C, **a pre-release, not signed**; for the Fable referee F-U5): new §15 — Definitions 15.1, 15.3, Lemma 15.2, Proposition 15.4, Theorem 15.5, Propositions 15.6, 15.7, Corollary 15.8, Prediction 15.9 ((U18)–(U21) [proved here], (U22) [conjectural]), Remarks 15.10, 15.11, the code map; one pointer in Remark 14.11. Hida, Invent. 85, pp. 549, 552–553, 562, 570, 593 read on the page. Companion: [C, `sec:nebwildsec`] (v1.22.0-pre.1). No statement, label or status tag of 0.5.x changed.

Changelog of part 6. v0.7.0-pre.1 (2026-11-06, slot 25 U6, **a pre-release, not signed**; for an adversarial referee): new §16 — Lemma 16.1, Remark 16.2, Proposition 16.3, Proposition 16.4, Remark 16.5 (the owner-signed P-K2 of 2026-11-05), Prediction 16.6 ((U23) [proved here] given data, (U24) [conjectural]), Remark 16.7 ([verified computationally]), the

code map; the mathematics of the slot-24 HeckeZ stage 05 v0.3.1 §1.2, §2.2–§2.4 (Lemma H0, Props. H1.1, H1.2), re-proved here with two changes: Proposition 16.4(3) is proved from the fibre product directly (the census row (D4) becomes a check), and (6) adds the sign ε ($H^{\varepsilon} \cong \mathbf{T}_{486}(a)_{\mathfrak{m}}$). Elsewhere: Remark 5.10 credits Ohta's adaptation of Harder–Pink and Kurihara (P-K1, owner-signed 2026-11-05); Proposition 8.5(3) gains the Bellaïche–Pollack citation (phenomenology catalogue §11 item 4, **not yet seen by the owner**); bibliography keys `kurihara1993`, `calegari_eisdef`, `bellaiche_pollack`, `bruns_herzog_cm`. No statement, label, number or status tag of 0.6.0 moved. Report: `docs/implementation/slot25/U6_report.md`. **v0.7.0-pre.2** (2026-11-06, slot 25 U6 fix round, **a pre-release, not signed**): the referee `docs/implementation/slot25/U6_referee.md` (ACCEPT WITH MINORS, F1–F14, no MAJOR) applied; no result changes. (U23): the degree-2 clause on all of S_{k^*} is [verified computationally] (the referee's computation), the block statements stay [proved here] (F1); (U24) restated as "$\mathbf{M}(a)^{\pm}_{\mathfrak{m}}$ free over $\mathbf{h}(a)_{\mathfrak{m}}$, both signs" (F2); Proposition 16.4, Step 1 reordered, $g_a = 2$ from Corollary 5.2 and conjugation, (D2) a check (F3); Remark 16.2 gains the block-dimension condition (F4) and the locator Thm. 2.6 (F12); Lemma 16.1(3), proof: t translated first (F5); Proposition 16.4's tag completed, $\varepsilon = +$ via Lemma 14.2 (F6); Remark 16.5: the owner-signed scope limited to the one sentence, pp. 282–283, "unconditionally" (F7); the notation of $\mathfrak{m}$ (F9); Remark 16.7: "two of the three kinds" (F10); Proposition 8.5(3): the tag of the inserted sentences and "arXiv v1, p. 2" (F11). **Released front matter, not yet seen by the owner**: the status paragraph of §1 no longer calls part 5 a pre-release ("since v0.6.0 ... released with the rest", F8). With Z0 on `hecke-extensions`: [C, `prop:heckeZgen`], [C, `sec:heckeZ`] cited by label; the bibliography key `bruns_herzog_cm` replaced by [C]'s `bruns_herzog`. No statement number or label of 0.7.0-pre.1 moves. **v0.7.0-pre.3** (2026-11-06, slot 25, the H1 merge after its audit `docs/implementation/slot25/H1_audit.md`, PASS_WITH_MINORS; **a pre-release, not signed, not yet seen by the owner**): Prediction 16.6 (U23) gains the explicit anchor $x^2 + 56702273x + 83587893 \bmod 547^3$, $v(\mathrm{disc}) = 2$, before H1-R runs (F9); new Remark 16.8, Steps 1–2 of Proposition 16.4 at any $g_a \geq 2$ (F5), placed last so that no statement number moves; the code map *Built*. No result, label or status tag of pre.2 moves.

A.3 The title page of 0.7.0-pre.3: author line and abstract, verbatim

Orbifold Cohomology of GL(2)
companion to `gl2_contract.tex` [C] and `hecke_extensions.tex` [E]

Let $\ell \geq 5$. At tame level 1, Hida's ordinary Hecke algebra $\mathbf{T}$ is a finite torsion-free algebra over $\Lambda = \mathbb{Z}_\ell[[X]]$, split into the even weight classes a modulo $\ell - 1$; its irreducible components are one Eisenstein component per class and the components of the cuspidal algebra $\mathbf{h}(a)$. This document (i) fixes these objects and reads them at finite weight (§2–§4): every quotient $M/(P_k, \ell^m)M$ of a finitely generated Λ-module depends only on the weight tower Tw_m of [C], so the characteristic polynomials of the T_p on the ordinary cusp forms modulo ℓ^m, the mod-ℓ local algebras and the truncated congruence lengths are tower invariants, and a single squarefree characteristic polynomial modulo ℓ proves the whole class *étale* over Λ; (ii) proves that the Eisenstein component meets the cuspidal locus in a class $a \not\equiv 0$ exactly at the irregular pairs, with the congruence length L_k at weight k squeezed between $v_\ell(B_k/k)$ and v_ℓ of the Eisenstein value of every cuspidal characteristic polynomial, and — through Emerton's theorem, matched to these objects — equal to $v_\ell(B_k/k)$, the Λ-adic congruence module being $\Lambda/(G_a)$ (§5); (iii) proves that **no component is CM** at tame level 1 (§6); (iv) links saturation, Gorensteinness and Loewy length (§8). On the project's published tower data and one new cell it determines the component structure of every class of $\ell \in \{5, 7, 11, 13\}$ and of six classes at $\ell = 691$, and proves $L_k = 1$ at $(k, \ell) = (12, 691)$, $(200, 691)$, $(702, 691)$. §9 lists the checkable

predictions (U1)–(U7) for the weight ladders $n + j(\ell - 1)\ell^{m-1}$.